Sangfor Cloud Platform Open-API Specification Documents

1. Read Me

1.1. Overview

Sangfor Cloud Platform (SCP) manages compute, network, storage and other resources. This document mainly introduces the functions and calling methods of the APIs provided by this platform, so that third-party users can use these APIs to manage SCP.

1.2. About API Calls

The APIs introduced in this document conform to REST (Representational State Transfer) style. You can call them via HTTPS requests.

> **Note**
>
> Turn off SSL certificate verification if you need to call APIs via HTTPS requests.

1.3. Precautions

1. The APIs mentioned in this document are only supported by HCI resource pools of version 5.8.8R6 or above.
2. Applicable roles: administrator, tenant, and tenant co-admin with corresponding permissions.
3. Ensure that relevant resources are sufficient before creating resources for tenants or by tenants themselves.

2. How to Call APIs

2.1. Request Method

Table 2.1.1 Request Method

Request Method	Description
POST	Request for creation.
PUT	Request for update.
DELETE	Request for deletion.
GET	Request for query.

2.2. Interface Flow Control Strategy

The flow control policy restricts the maximum number of times the API can be accessed within a certain period (usually 1 minute). The flow control policy of the API is as follows:

Table 2.2.1 Interface Flow Control Strategy¶

Request Method	Times/Minute
GET	600
POST	100
PUT	100
DELETE	100
DEFAULT	100

Note

When the number of API requests from a client exceeds the above limit, the client's IP address will be prohibited from calling the API for a certain period (about 1 minute).

2.3. Request headers

- Request header parameters

Table 2.3.1 Request header parameters¶

Parameter	Type	Required	Remarks
Content-Type	string	Yes	Requested message type. Default: application/json
Cookie	string	Yes	SDK-API所需的验证信Authentication information required for SDK-API, where the value of aCMPAuthToken must be specified (format: aCMPAuthToken={any string}). The string is usually the UUID. Example: aCMPAuthToken=56debefee2c54355a49fc73dfaaa9466. Note: For SCP 6.3.70 and above, there is no need to pass in Cookies.
Authorization	string	No	Authentication parameter. Currently, AWS4-HMAC-SHA256 is supported. Token-based authentication method is used. Format: *{Authentication Method} + Space + {Authentication Parameters}*
X-Amz-Date	string	No	Request for the current time, which can be used to verify the signature. ISO 8601 time format is used, such as "2020-12-01T03:17:42Z".

Authorization field example:

- AWS4-HMAC-SHA256 authentication example:

  AWS4-HMAC-SHA256 Credential=be763c9968a143a99e1998ccce7594c7/20210104/cn-east-1/dsp-api/aws4_request, SignedHeaders=path;x-amz-date, Signature=b8701b453ea3c8faaf566fee220c11f2fd9cc4bbdc74b4cc927b9c31b89e8708
- Token authentication example:

  Token f6a5bfca121a401f877af196dde17ed5

2.4. Request Body

The content format of API request and response is application/json.

2.4.1. Common Request Parameter Format

2.4.2. Common Path Parameter Format

Table 2.4.1 Common Path Parameter Format¶

Parameter	Type	Required	Description
sort	string	N	Sort field. Format: sort=field1[,field2]. Example: sort=cup_ratio,mem_ratio
order_by	string	N	Sort field. Format: ASC (default), DESC. Example: ASC, DESC
fields	string	N	Expected fields. Format: fields=field1[,field2]. Example: fields=server_id,status
page_num	int	N	Number of requested pages. Default: 0
page_size	int	N	Page size. Default: 100. Maximum: 100

Note

By default, the list query conforms to this parameter format. Otherwise, it will be specially noted in the corresponding API document.

2.5. Response Format

The content format of API request and response is application/json.

2.5.1. Response Code

Table 2.5.1 Response Code¶

Response Code	Description	Remarks
200	OK	Request for query has succeeded.
201	Created	Request for creation of a resource has succeeded.
204	No Content	Request for deletion or update of a resource has succeeded.
400	Bad Request	Incorrect parameters or resource dependencies.
401	Unauthorized	Not authenticated.
403	Forbbiden	Insufficient permissions
404	Not Found	Resource not found.
409	Conflict	Resource created repeatedly.
429	Too Many Requests	The number of requests for the same interface reaches the upper limit or the idempotence check fails.
500	Internal Server Error	Internal error. Please contact technical support for troubleshooting.

2.5.2. Common Response Format

Table 2.5.2 Common Response Format¶

Parameter	Type	Required	Description
message	string	Yes	Error description displayed when a call fails.
code	int	Yes	Operation status code. Success value: 0.
data	object	Yes	Data actually returned by the API call.

- Example:

```
{
    "message": "",
    "code": 0,
    "data": {}
}
```

2.5.3. Common Response Parameter Format

Table 2.5.3 Common Response Parameter Format¶

Parameter	Type	Description
total_size	int	Total number of resources found.
page_num	int	Number of the currently requested pages.
page_size	int	Currently requested page size.
next_page_num	string	Next page number. Integer string. “” means that the current page is the last page.
data	list(dict)	Specific resource list.

Note

1. By default, the common response parameter format is included in the “data” parameter of the common response parameters.
2. By default, the list query conforms to this parameter format. Otherwise, it will be specially noted in the corresponding API document.

- Example:

```
{
    "message": "",
    "code": 0,
    "data": {
        "total_size": 0,
        "page_num": 0,
        "page_size": 100,
        "next_page_num": "",
        "data": []
    }
}
```

2.6. Authentication Methods

Open APIs support EC2 authentication and token-based authentication. We recommend using EC2 authentication.

2.6.1. EC2 Authentication

2.6.1.1. Preparation Before Authentication

2.6.1.1.1. AK/SK Request

Currently, there is no GUI for requesting and managing EC2 certificates, so you need to contact SCP technical support to manually generate the certificates in the backend. This can be completed on the GUI in future versions. The following is an example of the generated AK/SK:

- AK (access_key): 750e97ada0ac4683a13e709ab57e7f72
- SK (secret_key): e841cfc0de79410ebdd9feabfad77f9e

2.6.1.1.2. Signature Tools

A signature tool class is needed to generate signature information for each request. We have provided signature tools written in Python and Java. The tool script is in “API Document Attachment.zip”.

2.6.1.2. Request Signature Example (Python 2.7)

After completing the above preparations, sign your request. The code of the signature is as follows, taking Python as an example. View “API Document Attachment.zip” for details.

```
# -*- coding: utf-8 -*-
import datetime
import hashlib
from requests import auth
from urlparse import urlparse

from utils import Ec2Signer


class EC2RequestAuth(auth.AuthBase):
    u"""EC2 signature"""

    def __init__(self, access_key, secret_key, region, service):
        self.access_key = access_key
        self.secret_key = secret_key
        self.region = region
        self.service = service

    def __call__(self, r, *args, **kwargs):
        ec2_headers = self.get_ec2_request_headers_handler(r)
        r.headers.update(ec2_headers)
        return r

    def get_ec2_request_headers_handler(self, r):
        u"""Obtain handle of EC2 request header

        If get_ec2_request_headers fails to meet the requirements, inherit from the EC2 sign
ature and rewrite the
        function.
        """
        return self.get_ec2_request_headers(r=r,
                                            access_key=self.access_key,
                                            secret_key=self.secret_key)

    def get_ec2_request_headers(self, r, access_key, secret_key):
        u"""Obtain EC2 request header

        :param r: request instance
        :param access_key: AK
        :param secret_key: SK
        :return: EC2 signature header. Example:
            {
                'Authorization': 'AWS4-HMAC-SHA256 Credential={AK}/20210105/cn-south-1/sdk/a
ws4_request, '
                                 'SignedHeaders=path;x-amz-date, '
                                 'Signature=fff5b016763f84b5feb10d84296da4251b919c2e10e703a7
9dfea2a1fe1f151d',
                'x-amz-date': '20210105T112300Z'
            }
        """
        signer = Ec2Signer(secret_key)
        algorithm = 'AWS4-HMAC-SHA256'
        t = datetime.datetime.utcnow()
        amzdate = t.strftime('%Y%m%dT%H%M%SZ')
        datestamp = t.strftime('%Y%m%d')
        headers = r.headers.copy()

        # Note: All fields in the headers, excluding the host, can be used as the signature
header.
        signed_headers = 'cookie;x-amz-date'
        credential_scope = '/'.join([
            datestamp, self.region, self.service, 'aws4_request'
        ])
        authorization_header = '{} Credential={}/{}, SignedHeaders={}'.format(
            algorithm, access_key, credential_scope, signed_headers)
        headers.update({
            'Authorization': authorization_header,
            'X-Amz-Date': amzdate
        })

        _, host, path, _, _, _ = urlparse(r.url)

        body = r.body or ''
        body_hash = hashlib.sha256(body).hexdigest()
        credentials = {
```

```
        credentials = {
            'params': {},
            'verb': r.method,
            'host': host,
            'path': path,
            'headers': headers,
            'body_hash': body_hash
        }

        signature = signer.generate(credentials)
        authorization_header += ', Signature=%s' % signature
        headers.update({
            'Authorization': authorization_header
        })

        return headers


if __name__ == '__main__':
    import requests
    import uuid

    url = 'https://{SCP_IP}/janus/20180725/vpcs'
    auth = EC2RequestAuth(
        access_key='3d6f221410da409b9c72e7abc9c913c9',
        secret_key='30c23bde9afe439588e7af63a0b6920b',
        region='cn-south-1',
        service='sdk-api'
    )
    # For SCP 6.3.0, all requests must contain the aCMPAuthToken field, which is not require
d for SCP 6.3.70 and above.
    headers = {'Cookie': 'aCMPAuthToken=%s' % uuid.uuid4().hex}
    with requests.Session() as session:
        resp = session.request('GET', url,
                               auth=auth, headers=headers, verify=False)
    print(resp.json())
```

2.6.2. Token-Based Authentication

2.6.2.1. Token-Based Authentication

In addition to EC2 authentication, SDK-API also supports token-based authentication. The specific process is as follows:

- Get the public key modulus. View: Get public key modulus for details.
- Use the public key modulus to encrypt the token, and request the token-based authentication API to get the token;
- Add the obtained token to the authentication header of each request to call the API.

Note

- The token will expire, in 24 hours currently. If it expires, you need to request it again.
- You can also use the tenant authentication method described in the OpenStack-API document to obtain tokens. If you use this method, you can skip the subsequent authentication steps.

2.6.2.1.1. Encryption Method

The method of password encryption using the public key modulus is as follows (Python 2 and Python 3 supported).

```python
#!-*- coding:utf8 -*-
from binascii import a2b_hex, b2a_hex
from Crypto.PublicKey import RSA
from Crypto.Cipher import PKCS1_v1_5
from Crypto.Util.number import bytes_to_long


def encrypt_with_modulus(content, modulus=None):
    u"""Use modulus to encrypt text
    :param content: Content to be encrypted, such as the password
    :param modulus: Public key modulus, which must be obtained by calling the get-public-key
interface
    :return: Encrypted content
    """
    content = content.encode('utf-8')
    e = int(0x10001)
    n = bytes_to_long(a2b_hex(modulus))
    rsa_key = RSA.construct((n, e))
    # generate/export public key
    public_key = rsa_key.publickey()
    cipher = PKCS1_v1_5.new(public_key)
    content = cipher.encrypt(content)
    content = b2a_hex(content)
    return str(content.decode('utf-8'))
```

2.6.2.1.2. Request signature example

- SCP versions supported: 6.3.0 and above
- Address Requested

  https://{SCP_IP}/janus/authenticate

- Request Method

  POST

- Request Header

  - Content-Type:application/json
  - Cookie:aCMPAuthToken=56debefee2c54355a49fc73dfaaa9466

- Request Parameters

```json
{
    "auth": {
        "passwordCredentials": {
            "username": "admin",
            "password": "encrypted_password"
        }
    }
}
```

- Response Content

```
{
    "message": "",
    "code": 0,
    "data": {
        "access": {
            "token": {
                "issued_at": "2021-01-04T06:31:41.000000Z",
                "expires": "2021-01-05T06:31:41.000000Z",
                "id": "f6a5bfca121a401f877af196dde17ed5",
                "tenant": {
                    "id": "cf92f8d651c24ed6938b186e252ee0bc",
                    "enabled": true,
                    "description": "Bootstrap project for initializing the c
loud.",
                    "name": "admin"
                },
                "audit_ids": [
                    "JVwi5-PWRd-wRDd3kS3FIQ"
                ]
            },
            "user": {
                "username": "admin",
                "roles_links": [],
                "name": "admin",
                "roles": [
                    {
                        "name": "admin"
                    }
                ],
                "id": "349c19d7653349e1a8918326aacb4430"
            }
        }
    }
}
```


Note

The token ID in the response content is the token required for authentication.

2.7. Idempotence

At present, the POST and PUT interfaces of SDK-API starting with janus provide idempotent verification, which is valid within 15 minutes.

2.7.1. About Idempotence

Idempotence is the property of certain operations in mathematics and computer science whereby they can be applied multiple times without changing the result beyond the initial application.

In the context of REST APIs, when making multiple identical requests has the same effect as making a single request, then that REST API is called idempotent. Notes: the idempotence here refers to the effect on resources.

For example, the GET method in the RESTFul API is to query resource information and will not affect resources, so it is idempotent. But the result returned by each call of the GET method may be different, maybe because a certain attribute of the resource has been modified by other methods before the GET method is called.

In fact, the idempotence of APIs in a distributed architecture is not only for the RESTFul APIs, but applies to all types of interfaces. The purpose is to ensure that the impact on resources is the same when the interface is called 1 or N times.

In addition, this can be used together with 15-minute request timeout of EC2 authentication, so that it can prevent the replay attack, a type of man-in-the-middle attack, and further enhance API security.

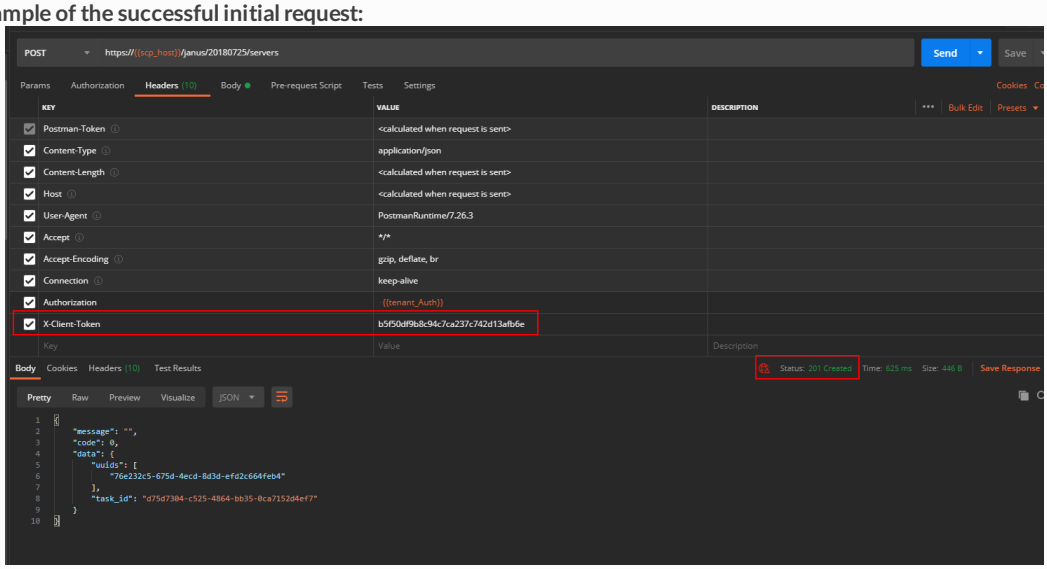
2.7.2. Idempotence

2.7.2.1. Scenario 1

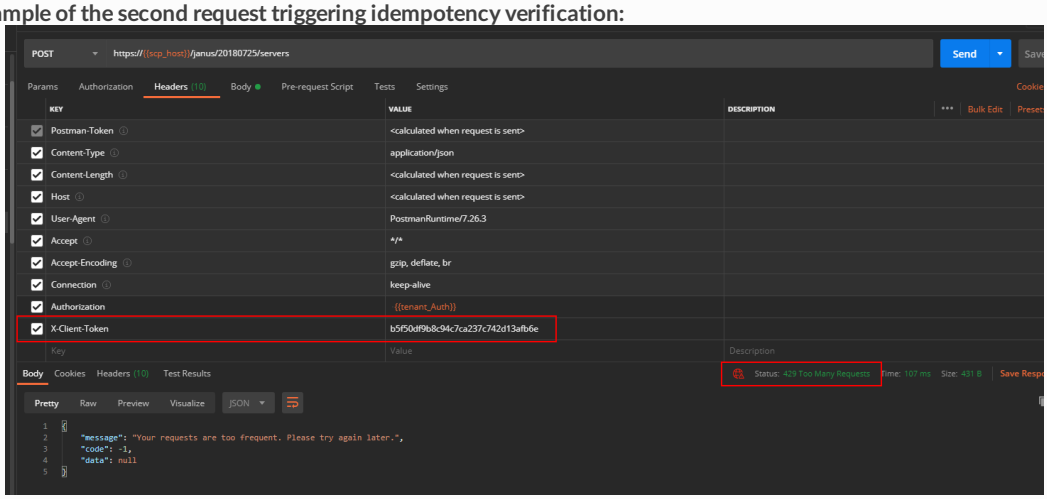
After the client has sent the request to create a VM, and the server has successfully created the VM, if the request was interrupted due to network problems, and the client did not receive a response, the request will be identified as failed and a new request with the same parameters will be sent again. In this case, another VM will be created, and there will be 2 VMs in the system, which is not expected by the client.

To solve this problem, the system provides an idempotent verification mechanism. In this way, you only need to add the “X-Client-Token” field to the request header, and the value is a random number which is advised to be generated with UUID4. Then, if the client uses the same request parameters, the server will return an error, indicating “429 Too Many Requests”.

Example of the successful initial request:



Example of the second request triggering idempotency verification:



3. API Request Examples

3.1. Create a Virtual Machine

> **Note**
>
> Here we will introduce how to use Postman to create a VM. Using postman needs to disable SSL certificate verification.
>
> Scenario: Call APIs as a tenant
>
> Version: SCP6.3.0

3.1.1. Preparations

To use SCP functionalities by APIs, perform the following operations as an admin or platform administrator:

1. Create a tenant on SCP and get its tenant name, login name and password.

2. Associate resource pools for the tenant so that the tenant will be able to create resources in the associated resource pools.
3. Configure the quota for CPU, memory, and storage resources in the specified resource pool for the tenant.
4. Create a VPC network that VMs can access.
5. Upload ISO images or built-in images, and associate them with the resource pools of the tenant, ensuring that the VM image is available when the tenant creates a VM.

3.1.2. Authentication

There are two authentication methods: EC2 and token.

- EC2 authentication:
- For SCP 6.3.0, AK/SK key pairs for tenants must be generated in the backend by the technical support.
- For SCP 6.3.70, tenants can request their own AK/SK key pairs on the interface. The admin or co-administrator must contact the technical support to request in the backend.
- For SCP 6.3.80 and above, the admin and tenants can request AK/SK key pairs on the interface. The co-administrator must contact the technical support to request in the backend.

Take the following AK/SK as an example:

- AK: e96a96326f2841ba8bdf2242f6cb832e
- SK: 68176b3ec9d64394bdfcbe124b3e6078

Request signature definitions:

- Service: open-api
- Region: cn-south-1
- EC2_ALGORITHM: AWS4-HMAC-SHA256

- Request structure

Fill in the following content in the Pre-request Script part of the Postman request:

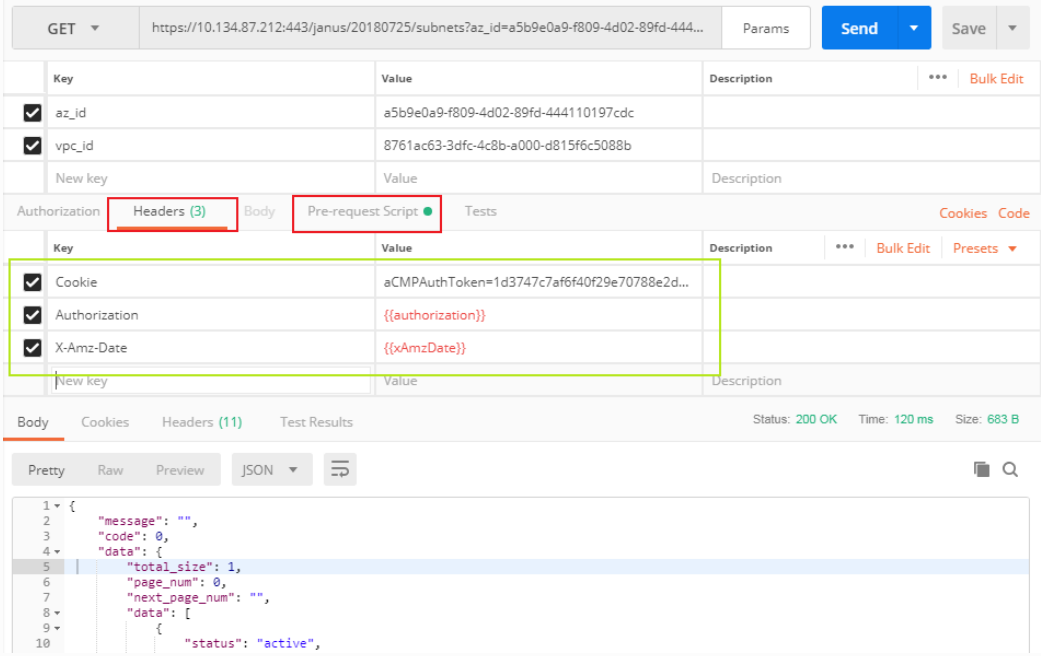
```
//Configure AK/SK for the corresponding account
var ACCESS_KEY = 'e96a96326f2841ba8bdf2242f6cb832e';
var SECRET_KEY = '68176b3ec9d64394bdfcbe124b3e6078';

var REGION = 'cn-south-1';
var SERVICE = 'open-api';
var ALGORITHM = 'AWS4-HMAC-SHA256';

var date = new Date().toISOString();
var amzdate = date.replace(/[:\-]|\.\d{3}/g, "");
var dateStamp = amzdate.slice(0, -8);


pm.environment.set('authorization', getAuthHeader(request.method, request.url, request.d
ata));
pm.environment.set('xAmzDate', amzdate);

function getPath(url) {
    var pathRegex = /.+?\:\/\/.+?(\/.+?)(?:#|\?|$)/;
    var result = url.match(pathRegex);
    return result && result.length > 1 ? result[1] : '/';
}

function getQueryString(url) {
    var arrSplit = url.split('?');
    return arrSplit.length > 1 ? url.substring(url.indexOf('?') + 1) : '';
}

function getSignatureKey(secretKey, dateStamp, regionName, serviceName) {
    var kDate = sign("AWS4" + secretKey, dateStamp);
    var kRegion = sign(kDate, regionName);
    var kService = sign(kRegion, serviceName);
    var kSigning = sign(kService, "aws4_request");
    return kSigning;
}

function sign(key, message) {
    return CryptoJS.HmacSHA256(message, key);
}

function getAuthHeader(httpMethod, requestUrl, requestBody) {
    var canonicalUri = getPath(requestUrl);
    var canonicalQueryString = ''


    if (httpMethod == 'GET' || !requestBody || Object.keys(requestBody).length === 0) {
        requestBody = '';
    }

    var hashedPayload = CryptoJS.enc.Hex.stringify(CryptoJS.SHA256(requestBody));

    // For date and time only
    var canonicalHeaders = 'x-amz-date:' + amzdate + '\n';
    // Signature field
    var signedHeaders = 'path;x-amz-date';
    var canonicalRequestData = [httpMethod, canonicalUri, canonicalQueryString, canonica
lHeaders, signedHeaders, hashedPayload].join("\n");
    var hashedRequestData = CryptoJS.enc.Hex.stringify(CryptoJS.SHA256(canonicalRequestD
ata));

    var credentialScope = dateStamp + '/' + REGION + '/' + SERVICE + '/' + 'aws4_request
';
    var stringToSign = ALGORITHM + '\n' + amzdate + '\n' + credentialScope + '\n' + hash
edRequestData;

    var signingKey = getSignatureKey(SECRET_KEY, dateStamp, REGION, SERVICE);
    var signature = CryptoJS.HmacSHA256(stringToSign, signingKey).toString(CryptoJS.enc.
Hex);


    var authHeader = ALGORITHM + ' ' + 'Credential=' + ACCESS_KEY + '/' + credentialScop
e + ', ' + 'SignedHeaders=' + signedHeaders + ', ' + 'Signature=' + signature;

    return authHeader;
}
```

Fill in the following content in the Headers part:

- Cookie:aCMPAuthToken=1d3747c7af6f40f29e70788e2d1fbe2a
- Authorization:{{authorization}}
- X-Amz-Date:{{xAmzDate}}

> **Note**
>
> For SCP 6.3.70 and above, there is no need to pass in Cookies.

Then, you can access each API.



To facilitate demonstration and testing, here we use the token authentication method described in the API document to perform operations.

Perform operations according to the API document to get the token which will be used as the request header.

3.1.3. Get Resource Pool IDs Associated with Tenants

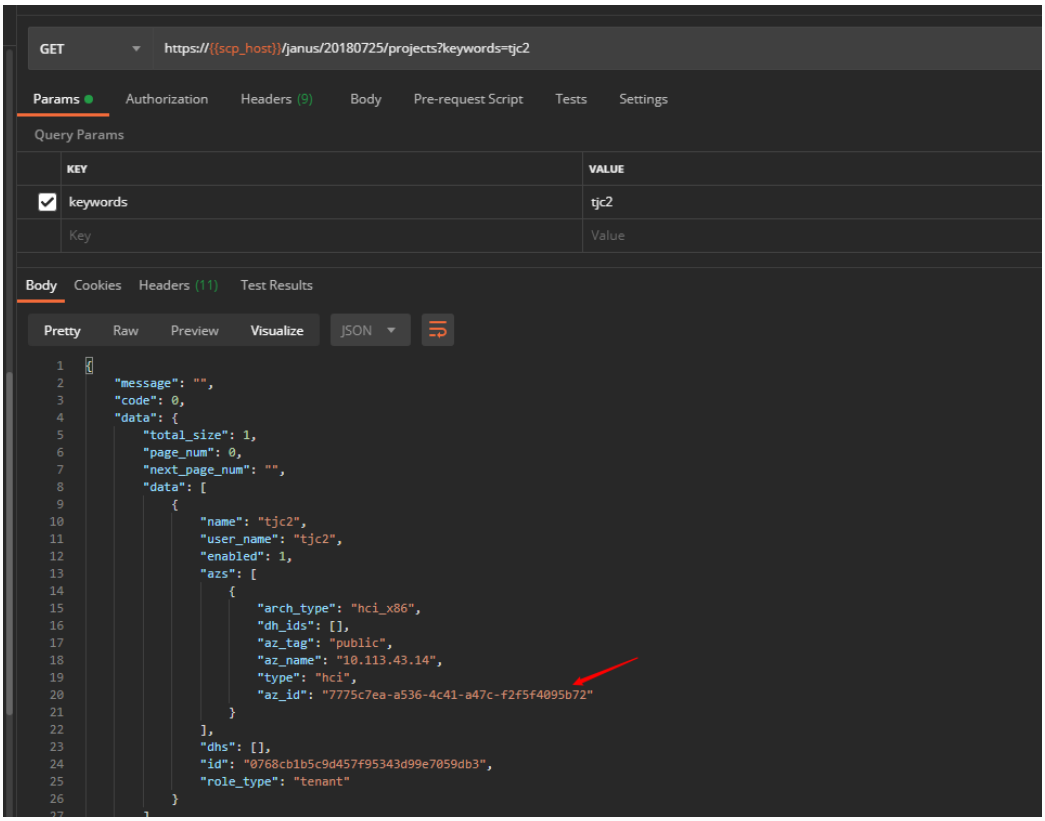
API for querying resource pools associated with a tenant is not available in SCP6.3.0. You can call the API for querying tenants as an admin, or log in to SCP as the tenant to capture packets and view the related data.

For SCP 6.3.70 and above, an interface for querying the list of resource pools associated with tenants has been provided. (View :ref: *list-projects* or :ref: *get-project* for details.). Therefore, there is no need to refer to the following method.

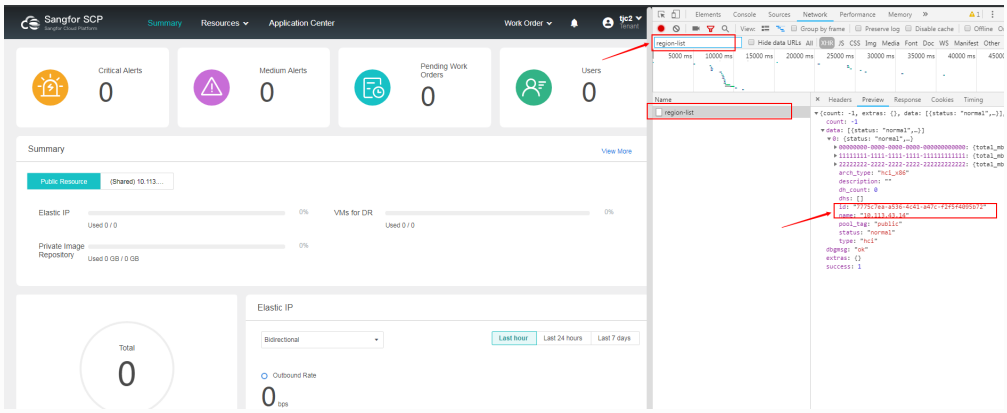
Current Methods Provided by SCP6.3.0:

- Call an API as an admin to query the tenant list.

After being authenticated with the admin account, call the tenant list API to get tenant information, including the associated resource pools.



- A tenant logs in to SCP to capture packets and get the associated resource pools.



Get the resource pool ID with the above method: 7775c7ea-a536-4c41-a47c-f2f5f4095b72

3.1.4. Get Storage Tags Available To Tenants in Resource Pools

Tenants can only create VMs on datastores with storage tags and corresponding quotas. Therefore, before creating a VM, you need to get a list of storage tags.

GET https://{{scp_host}}/janus/20180725//storages/tags?az_id=7775c7ea-a536-4c41-a47c-f2f5f4095b72

Params Authorization Headers (9) Body Pre-request Script Tests Settings

Query Params

KEY	VALUE
az_id	7775c7ea-a536-4c41-a47c-f2f5f4095b72
Key	Value

Body Cookies Headers (11) Test Results

Pretty Raw Preview Visualize JSON


```
{
    "message": "",
    "code": 0,
    "data": [
        {
            "ratio": 0.79,
            "description": "Highest IO speed, to tag SSD to create high-performance virtual machine",
            "total_mb": 11339056,
            "project_id": "0768cb1b5c9d457f95343d99e7059db3",
            "used_mb": 8994775,
            "id": "00000000-0000-0000-0000-000000000000",
            "name": "High Performance Storage"
        },
        {
            "ratio": 0.18,
            "description": "High IO speed, to tag aging storage",
            "total_mb": 1552384,
            "project_id": "0768cb1b5c9d457f95343d99e7059db3",
            "used_mb": 279132,
            "id": "11111111-1111-1111-1111-111111111111",
            "name": "Performance Sensitive Storage"
        },
        {
            "ratio": 0.04,
            "description": "Fair IO speed with high capacity and safety and cost-effective",
            "total_mb": 2093056,
            "project_id": "0768cb1b5c9d457f95343d99e7059db3",
            "used_mb": 91826,
            "id": "22222222-2222-2222-2222-222222222222",
            "name": "Capacity Sensitive Storage"
        }
    ]
}
```


According to the returned data in the figure, the storage tag available to the current tenant is performance-sensitive storage with the storage tag ID (storage_tag_id) 11111111-1111-1111-1111-111111111111.

3.1.5. Get Tenants' Network Information in Resource Pool (Optional)

First, get the tenant's VPC network information in the resource pool.

GET https://{{scp_host}}/janus/20180725/vpcs?az_id=7775c7ea-a536-4c41-a47c-f2f5f4095b72

Params Authorization Headers (9) Body Pre-request Script Tests Settings

Query Params

KEY	VALUE
az_id	7775c7ea-a536-4c41-a47c-f2f5f4095b72
Key	Value

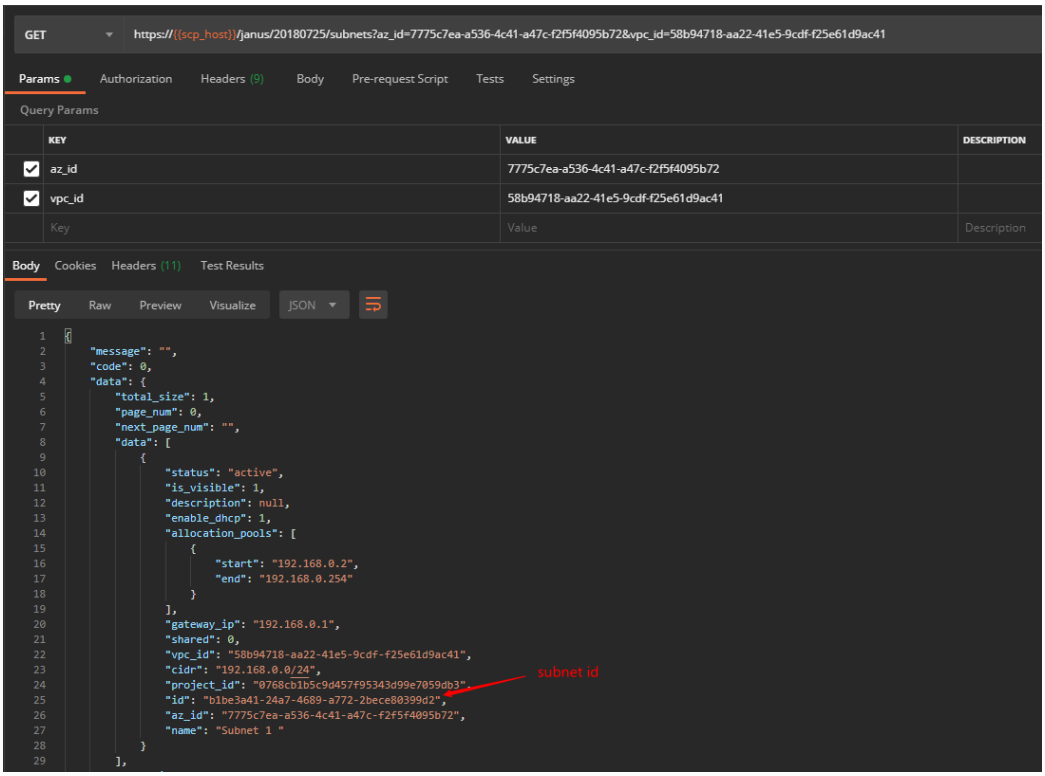
Body Cookies Headers (11) Test Results

Pretty Raw Preview Visualize JSON


```
{
    "message": "",
    "code": 0,
    "data": {
        "total_size": 1,
        "page_num": 0,
        "next_page_num": "",
        "data": [
            {
                "status": "active",
                "description": null,
                "shared": 0,
                "project_id": "0768cb1b5c9d457f95343d99e7059db3",
                "type": "vpc",
                "id": "58b94718-aa22-41e5-9cdf-f25e61d9ac41",
                "az_id": "7775c7ea-a536-4c41-a47c-f2f5f4095b72",
                "name": "VPC"
            }
        ],
        "page_size": 100
    }
}
```


Get the ID of the VPC network (vpc_id): 58b94718-aa22-41e5-9cdf-f25e61d9ac41

Then, get the tenant's subnet information in VPC.



Get the subnet ID (subnet_id): b1be3a41-24a7-4689-a772-2bece80399d2

3.1.6. Get VM Images

Related APIs: Query Images

There are two types of VM images according to disk_format: ISO and aCloud.

- The ISO image is mounted from the optical drive, and the user needs to install the operating system manually.
- The aCloud image refers to the built-in image on the SCP platform and its system disk is pre-configured, so that the VM already has its operating system when it is created.

Below describes how to query images of different types.

Use the tenant authentication information to call an API to get the current ISO image list.

GET https://{{scp_host}}/janus/20180725/images?disk_format=ISO&image_type=public

Params Authorization Headers (9) Body Pre-request Script Tests Settings

Query Params

KEY	VALUE
disk_format	ISO
image_type	public

Body Cookies Headers (11) Test Results

Pretty Raw Preview Visualize JSON


```
"next_page_num": "",
"data": [
    {
        "status": "active",
        "size_mb": 348.20703125,
        "virtual_size_mb": 0,
        "description": "",
        "task_id": "f3ccf589-74d0-4e93-a94c-50f6aecbefa6",
        "support_versions": [
            "all"
        ],
        "image_type": "public",
        "os_type": "linux-centos",
        "disks": [],
        "disk_format": "ISO",
        "owner_id": "78ab2481ebd146ed9c3e15ad571e1284",
        "visibility": "public",
        "has_gpu": 0,
        "is_vmtools_installed": 0,
        "image_members": null,
        "message": "",
        "azs": [
            {
                "is_available": 1,
                "id": "7775c7ea-a536-4c41-a47c-f2f5f4095b72",
                "arch_type": "hci_x86"
            }
        ],
        "id": "970c03af-4910-4a27-8fcd-99468f099132",
        "name": "20170817_174244"
    }
],
```


image id

Use disk_format=aCloud to filter built-in images, as shown below:

GET https://{{scp_host}}/janus/20180725/images?disk_format=aCloud&image_type=public

Params Authorization Headers (9) Body Pre-request Script Tests Settings

Query Params

KEY	VALUE
disk_format	aCloud
image_type	public

Body Cookies Headers (11) Test Results

Pretty Raw Preview Visualize JSON


```
        ],
        "image_type": "public",
        "os_type": "linux-centos",
        "disks": [
            {
                "size_mb": 81920.0,
                "preallocate": "off",
                "storage_id": "1213f1a7_vs_vol_rep3",
                "id": "ide0",
                "tpl_config": {
                    "mem_hotplug": 0,
                    "cpu_hotplug": 0,
                    "machine_type": "i440fx"
                }
            }
        ],
        "disk_format": "aCloud",
        "owner_id": "78ab2481ebd146ed9c3e15ad571e1284",
        "visibility": "public",
        "has_gpu": 0,
        "is_vmtools_installed": 1,
        "image_members": null,
        "message": "",
        "azs": [
            {
                "is_available": 1,
                "id": "7775c7ea-a536-4c41-a47c-f2f5f4095b72",
                "arch_type": "hci_x86"
            }
        ],
        "id": "524e4ea5-4b6e-4cfa-bc12-5cef2dfa19b1",
        "name": "11_10.113.43.14_498d926e_10.113.43.14_c7a26d48_10.113.43.14_524e4ea5"
```


system disk info

image id

3.1.7. Use Built-in Images to Create VMs

Related APIs: Create a VM

- Use a built-in image to create a VM with the following parameters:

```
{
    "az_id": "a5b9e0a9-f809-4d02-89fd-444110197cdc",
    "location": {
        "id": "cluster",
    },
    "storage_tag_id": "11111111-1111-1111-1111-111111111111",
    "image_id": "5dc95d56-2c19-4ccd-b94b-eafa5088dab1",
    "cores": 1,
    "sockets": 1,
    "memory_mb": 1024,
    "count": 1,
    "name": "create_vm_with_vpc",
    "description": "",
    "advance_param": {
        "boot_order": "c",
        "onboot": 0,
        "schedopt": 0,
        "abnormal_recovery": 1,
        "cpu_hotplug": 0,
        "mem_hotplug": 0,
        "balloon_memory": 0,
        "hugepage_memory": 0
    },
    "disks": [
        {
            "id": "ide0",
            "type": "derive_disk",
            "preallocate": 0,
            "size_mb": 81920
        }
    ],
    "networks": [
        {
            "vif_id": "net0",
            "connect": 1,
            "model": "virtio",
            "host_tso": 0,
            "vpc_id": "8761ac63-3dfc-4c8b-a000-d815f6c5088b",
            "subnet_id": "5b154c54-472e-4260-a3bd-af042cac5d13"
        }
    ],
    "power_on": 0
}
```

Note

During VM creation, the parameters of Disk ide0 (the system disk) are different between a built-in image and an ISO image.

- For the VM created with a built-in image, its system disk type must be derive_disk, and its size_mb and preallocate must be consistent with those in the built-in image.
- For the VM created with an ISO image, its system disk type is new_disk, but its other parameters can be specified as required.

POST https://{{scp_host}}/janus/20180725/servers

Params Authorization Headers (11) Body Pre-request Script Tests Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON


```
{
    "az_id": "7775c7ea-a536-4c41-a47c-f2f5f4095b72",
    "location": {
        "id": "cluster"
    },
    "storage_tag_id": "00000000-0000-0000-0000-000000000000",
    "image_id": "524e4ea5-4b6e-4cfa-bc12-5cef2dfa19b1",
    "cores": 1,
    "sockets": 1,
    "memory_mb": 1024,
    "count": 1,
    "name": "create_vm_with_vpc",
    "description": "",
    "advance_param": {
        "boot_order": "c",
        "onboot": 0,
        "schedopt": 0,
        "abnormal_recovery": 1,
        "cpu_hotplug": 0,
```


Body Cookies Headers (10) Test Results

Pretty Raw Preview Visualize JSON

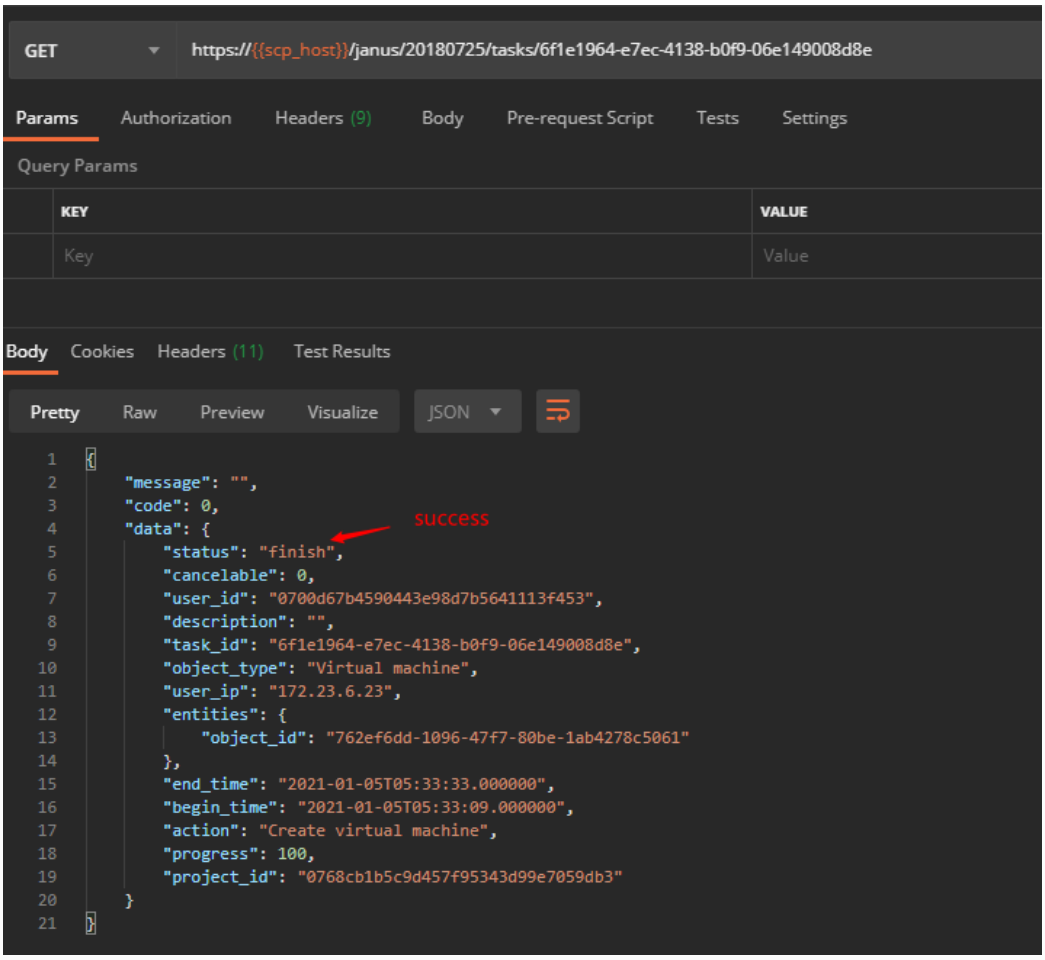

```
{
    "message": "",
    "code": 0,
    "data": {
        "uuids": [
            "762ef6dd-1096-47f7-80be-1ab4278c5061"
        ],
        "task_id": "6f1e1964-e7ec-4138-b0f9-06e149008d8e"
    }
}
```


server id

If an API is called successfully, server_id and task_id will be returned. Task_id can be used to query the task progress and confirm the operation result.

3.1.8. Query VM Creation Status

Related APIs: Query Task Information



Task status values are provided: status=finish indicates that the task is successful and the VM has been created successfully.

3.1.9. Other Operations

After a VM is successfully created, you can perform more operations on it. Examples:

- Bind elastic IP addresses (the admin needs to configure the elastic IP pool)
- Add disks
- Power on/off the VM

4. API

4.1. System

4.1.1. Common System Parameters

4.1.1.1. Task status

Table 4.1.1 Task status

Value	Description
waiting	Queuing, the corresponding progress is -2
failure	Failed, the corresponding progress is -1
doing	Executing, the corresponding progress can be 0-99
finish	Completed, the corresponding progress is 100
canceling	Canceling
canceled	Canceled

4.1.2. Get public key modulus

> **Note**
>
> The public key modulus is used for password data transmission such as user password encryption during authentication, VM password encryption during VM creation.

4.1.2.1. Scenario

- The public key modulus is a password-free interface, and the request header only needs to carry the matching cookie information to obtain the corresponding data, view: Request headers for details.
- SCP versions supported: 6.3.0 and above.

4.1.2.2. URI

Table 4.1.2 API Path and Request Method

REST VERB	PATH	Description
GET	/janus/public-key	Get public key modulus

- Parameter Description:

  None

4.1.2.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/public-key

4.1.2.4. Response

- Response Parameter

Table 4.1.3 Response Parameter

Parameter	Type	Description	Value Returned
public_key	string	Public key modulus	Yes

> **Note**
>
> 1. The returned string of public key modulus contains \n at the end, which needs to be removed when using the public key modulus.
> 2. For common response parameters, view Response Format for details

- Response Example:

```
{
    "public_key": "BDEC925242D4C028EAD2E457B519BE1A0D542C9993A7B46423D214BC0
3A8F179BF00928777EE9DBCCD6F1925650F6D783B1946A91BE9AB678E41D660741EF0C9D4AA7
E959A734004AD47F2FF7F39FD3248ED83036C6918477ED13DCD50099D1ED04C026905C65357D
FB52C937D41E4C6A739709207026D918BBAC057697B01CE66D8060B6B0321655A47225E031EB
C57C2301230BEF98CBBE02BD401657BD94D23ED71F95E791B8BCFEDE4D79AE6937FB67B41993
6CECA359817CEC15E8D9A829161F2459122AB6287E4A2B9A11BE8EEC5D86B87E48C097D6CD7B
20D23687090CDCF10E2CA747D0EBCAFF7BF57141162C01FC3AE14A373C8F482BDF39B0DAE99\
n"
}
```


4.1.2.5. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.1.3. Query Task Information

4.1.3.1. Scenario

- Scenario: Monitor the asynchronous task progress after the task ID is returned.
- Applicable roles: Super administrator and tenant
- SCP versions supported: 6.3.0 and above

4.1.3.2. URI

Table 4.1.4 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/tasks/{task_id}	Query Task Information

- Parameter Description:

Table 4.1.5 Path Parameters¶

Parameter	Location	Type	Description	Required
task_id	path	string	Task ID	Yes

4.1.3.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/tasks/3e82b989-5c9c-45a9-91a2-c673da42e65f

4.1.3.4. Response

- Response Parameter

Table 4.1.6 Response Parameter¶

Parameter	Type	Description	Value Returned
task_id	string	Task ID	Yes
description	string	Description of Tasks	Yes
project_id	string	Tenant ID	Yes
user_id	string	User ID	Yes
user_ip	string	IP address of operator	Yes
action	string	Action	Yes
object_type	string	Operation object type	Yes
object_id	string	Operation object ID	No
status	string	Operation log status, view: Task status for details.	Yes
cancelable	int	Whether the log can be canceled. It cannot be canceled by default.	Yes
progress	int	-2 indicates queuing,-1 indicates failure, values range from 0 to 100, 0 to 99 indicates executing, 100 indicates success	Yes
begin_time	string	Local time when the log-related task started	Yes
end_time	string	Local time when the log-related task ended	Yes
entities	object	Subtaks information, view 1. Parameter Description When the Return Value Involves a Single Asynchronous Task； 2. Parameter Description When the Return Value involves Multiple Asynchronous Tasks for details.	Yes

- Parameter description when the return value involves a single asynchronous task:

Table 4.1.7 Parameter Description When the Return Value Involves a Single Asynchronous Task¶

Parameter	Type	Note
object_id	string	Operation resource ID

- Parameter description when the return value involves multiple asynchronous tasks:

Table 4.1.8 Parameter Description When the Return Value involves Multiple Asynchronous Tasks¶

Parameter	Type	Note
object_id	string	Operation object ID
sub_tasks_total	int	Total number of subtasks
sub_tasks	list(dict)	List of subtasks, view: Parameter Description - sub_tasks for details.

- Parameter Description - sub_tasks:

Table 4.1.9 Parameter Description - sub_tasks¶

Parameter	Type	Note
task_id	string	Subtask ID
az_id	string	Resource pool ID
description	string	Description of Tasks
action	string	Action
object_type	string	Operation object type
status	string	Operation log status, view: Task status for details.
cancelable	int	Whether the log can be canceled. It cannot be canceled by default.
progress	int	-2 indicates queuing,-1 indicates failure, values range from 0 to 100, 0 to 99 indicates executing, 100 indicates success
begin_time	string	Local time when the log-related task started
end_time	string	Local time when the log-related task ended
entities	dict	Subtask information
entities.object_id	string	Operation resource ID

Note

For common response parameters, view Response Format for details

- Response Example:

  1. Example for the return value when a single asynchronous task is involved:

```
{
    "status": "finish",
    "cancelable": 0,
    "user_id": "bed4c431f29e42fea1dd991bc218cf48",
    "description": "",
    "task_id": "4a88bb77-07af-4369-b72e-04897feded33",
    "object_type": "VM",
    "user_ip": "172.23.6.23",
    "entities": {
        "object_id": "9fa0beed-b169-4dde-a530-4c253fac762c"
    },
    "end_time": "2020-12-15T19:22:46.000000",
    "begin_time": "2020-12-15T19:22:25.000000",
    "action": "Create VM",
    "progress": 100,
    "project_id": "9d14d076b59045b1b0aaca04cfaaa657"
}
```

  2. Example for the return value when multiple asynchronous tasks are involved:

```
{
    "status": "failure",
    "cancelable": 0,
    "user_id": "fd4394d204d241deb2cf555384620c8b",
    "description": "Failed to activate shared service. Please try agai
n or contact Sangfor technical support representative.",
    "task_id": "3e82b989-5c9c-45a9-91a2-c673da42e65f",
    "object_type": "Shared service",
    "user_ip": "172.23.13.131",
    "entities": {
        "sub_tasks_total": 1,
        "sub_tasks": [
            {
                "status": "failure",
                "description": "Failed to activate shared service. Ple
ase try again or contact Sangfor technical support representative.",
                "task_id": "3e82b989-5c9c-45a9-91a2-c673da42e65f",
                "object_type": "Shared service",
                "cancelable": 0,
                "entities": {
                    "object_id": "6a760d53-7587-4140-814f-51d7a5fef133
"
                },
                "end_time": "2020-12-10T10:13:08.000000",
                "begin_time": "2020-12-10T10:13:07.000000",
                "action": "Activate shared service",
                "progress": 50,
                "az_id": "62a4411d-2b3e-4d2d-95a4-feb7d2dbdb7b"
            }
        ]
    },
    "end_time": "2020-12-10T10:13:08.000000",
    "begin_time": "2020-12-10T10:13:07.000000",
    "action": "Activate shared service",
    "progress": 50,
    "project_id": "78ad8e9c9c7e487186f448bc81f050a6"
}
```


4.1.3.5. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.2. Platform Resource Management

4.2.1. Common Parameters of Resources

4.2.1.1. Resouce Pool Type

Table 4.2.1 Resouce Pool Type

Value	Description
hci	HCI
vmware	VMware

4.2.1.2. Resource Pool Tag

Table 4.2.2 Resource Pool Tag

Value	Description
public	Shared
private	Dedicated

4.2.1.3. Resource Pool Status

Table 4.2.3 Resource Pool Status

Value	Description
normal	Normal
offline	Offline
unauthorized	Not licensed
master_offline	The cluster controller is offline, but this does not mean that the cluster is offline because the cluster IP can still be assigned to other nodes.
password_error	Password is incorrect.

4.2.1.4. Architecture Type

Table 4.2.4 Architecture Type

Value	Description
hci_x86	HCI x86
hci_arm	HCI ARM
vmware	VMware

4.2.1.5. Physical Machine Status

Table 4.2.5 Physical Machine Status

Value	Description
running	Running
offline	Offline

4.2.2. Query Resource Pools

4.2.2.1. Scenario

- Scenario: Query resource pools
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Resource Pool Read permission
- Versions supported: SCP 6.3.0 and above

4.2.2.2. URI

Table 4.2.6 API Path and Request Method

REST VERB	PATH	Description
GET	/janus/20180725/azs	Query Resource Pools

- Parameter Description:

Table 4.2.7 Path Parameters¶

Parameter	Location	Type	Description	Required
type	path	string	Resource pool type, view Resouce Pool Type for details	No
tag	path	string	Resource pool tag, view Resource Pool Tag for details	No

Note

For common path parameters, view Common Path Parameter Format for details

4.2.2.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/azs?type=hci

4.2.2.4. Response

- Response Parameter

Table 4.2.8 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Resource pool ID	Yes
name	string	Resource pool name	Yes
description	string	Resource pool description	Yes
status	string	Resource pool status, view Resource Pool Status for details	Yes
type	string	Resource pool type, view Resouce Pool Type for details	Yes
tag	string	Resource pool tag, view Resource Pool Tag for details	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
[
    {
        "status": "normal",
        "description": "",
        "tag": "public",
        "version": "6.2.0",
        "type": "hci",
        "id": "0b1695d7-8aa8-4efb-9d25-3c0312f0c5a1",
        "name": "AZ8224"
    },
    {
        "status": "normal",
        "description": "",
        "tag": "public",
        "version": "5.8.8.R6",
        "type": "hci",
        "id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
        "name": "AZ1612"
    }
]
```

4.2.2.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.2.3. Query Resource Pool Info

4.2.3.1. Scenario

- Scenario: Query resource pool's information
- Roles Supported: Super admin, system admin, and co-administrator with the Resource Pool Read permission
- Versions supported: SCP6.3.70 and above

4.2.3.2. URI

Table 4.2.9 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/azs/{az_id}	Query Resource Pool Info

- Parameter Description:

Table 4.2.10 Path Parameters¶

Parameter	Location	Type	Description	Required
az_id	path	string	Resource pool ID	No

4.2.3.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/azs/e0403ff4-b1fc-4af4-9a4c-dac8c241eb11

4.2.3.4. Response

- Response Parameter

Table 4.2.11 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Resource pool ID	Yes
name	string	Resource pool name	Yes
description	string	Resource pool description	Yes
status	string	Resource Pool Status	Yes
type	string	Resouce Pool Type	Yes
tag	string	Resource pool tag: Shared or dedicated	Yes
virtual_resources	list(dict)	Virtual resources (resource pool quota), view Parameter Description - virtual_resources for details	Yes
physical_resources	list(dict)	Physical resources, view Parameter Description - physical_resources for details	Yes

- Parameter Description - virtual_resources:

Table 4.2.12 Parameter Description - virtual_resources¶

Parameter	Type	Required	Note
name	string	Yes	Resource name, including CPU, memory, storage (resources here indicate quota items on SCP)
limit	int	Yes	Upper limit of resources, the value -1 indicates that there is no limit
total	int	Yes	Total amount of resource
allocated	int	Yes	Allocated resource
occupied	int	Yes	Occupied resource
unit	int	Yes	Unit
extra	dict	No	Other information

- Parameter Description - physical_resources:

Table 4.2.13 Parameter Description - physical_resources¶

Parameter	Type	Required	Note
name	string	Yes	Resource name, including CPU, memory, storage, GPU, GPU_mem (resources here indicate all physical resources in the HCI clusters)
total	int	Yes	Total amount of resource
used	int	Yes	Used resource
unit	int	Yes	Unit
extra	dict	No	Other information

- Response Example:

```
{
    "status": "normal",
    "virtual_resources": [
        {
            "name": "High_Performance_Storage",
            "limit": -1,
            "allocated": 0,
            "total": 0,
            "occupied": 0,
            "unit": "MB"
        },
        {
            "name": "Performance_Sensitive_Storage",
            "limit": -1,
            "allocated": 16666624,
            "total": 1044480,
            "occupied": 4700160,
            "unit": "MB"
        },
        {
            "name": "Capacity_Sensitive_Storage",
            "limit": -1,
            "allocated": 17063936,
            "total": 6645760,
            "occupied": 4939776,
            "unit": "MB"
        },
        {
            "name": "CPU",
            "limit": 2000,
            "allocated": 356,
            "total": 0.0,
            "occupied": 209,
            "unit": "Core"
        },
        {
            "name": "Memory",
            "limit": 2048000,
            "allocated": 605184,
            "total": 0.0,
            "occupied": 292864,
            "unit": "MB"
        }
    ],
    "description": "",
    "physical_resources": [
        {
            "total": 7690419.0,
            "used": 5227252.0,
            "name": "storage",
            "unit": "mb"
        },
        {
            "total": 79224.0,
            "used": 39347.92,
            "name": "cpu",
            "unit": "mhz"
        },
        {
            "total": 98304.0,
            "used": 56171.7,
            "name": "memory",
            "unit": "mb"
        },
        {
            "total": 0,
            "used": 0.0,
            "name": "gpu",
            "unit": ""
        },
        {
            "total": 0,
            "used": 0,
            "name": "gpu_mem",
            "unit": "mb"
        }
    ],
    "tag": "public",
    "version": "5.8.8.R6",
    "type": "hci",
    "id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
    "name": "AZ1612"
}
```

4.2.3.5. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.2.4. Query Physical Machines

> **Note**
>
> This interface needs to support ranking the total capacity and usage of CPU, memory and storage of physical machines.

4.2.4.1. Location

- Administrator portal: Resources > Clusters > Cluster Details > Node

4.2.4.2. Scenario

- Scenario: Query physical machines
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Resource Pool Read permission
- Versions supported: SCP6.3.70 and above

4.2.4.3. URI

Table 4.2.14 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/hosts	Query Physical Machines

- Parameter Description:

Table 4.2.15 Path Parameters¶

Parameter	Location	Type	Description	Required
az_id	path	string	Resource pool ID	No

> **Note**
>
> For common path parameters, view Common Path Parameter Format for details

4.2.4.4. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/hosts?az_id=e0403ff4-b1fc-4af4-9a4c-dac8c241eb11

4.2.4.5. Response

- Response Parameter

Table 4.2.16 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Node ID	Yes
name	string	Name	Yes
description	string	Description	Yes
type	string	Type	Yes
status	string	Status, view Physical Machine Status for details	Yes
ip	string	Node IP	Yes
az_id	string	Resource pool ID	Yes
is_use_vs	int	Whether to use VS	Yes
alarm_count	int	Alert count	Yes
reserves	dict	Reserved memory, view Parameter Description - virtual_resources for details	No
cpu	dict	CPU information, view Parameter Description - cpu for details	Yes
memory	dict	Memory information, view Parameter Description - memory for details	Yes
storage	dict	Storage information, view Parameter Description - storage for details	Yes
gpu	dict	GPU information, view Parameter Description - gpu for details	Yes

- Parameter Description - reserves:

Table 4.2.17 Parameter Description - virtual_resources¶

Parameter	Type	Required	Note
memory_mb	int	No	Reserved memory (MB)

- Parameter Description - cpu:

Table 4.2.18 Parameter Description - cpu¶

Parameter	Type	Required	Note
allocation_max	int	Yes	Max available resources
allocation_ratio	float	Yes	Overcommitment
total_mhz	float	Yes	Totol size (MHZ)
used_mhz	float	Yes	Used size (MHZ)
ratio	float	Yes	CPU Usage
socket_count	int	Yes	Socket count
cores_per_socket	int	Yes	CPU cores per socket
core_count	int	Yes	CPU cores
type	string	Yes	CPU model

Table 4.2.19 Parameter Description - memory¶

Parameter	Type	Required	Note
allocation_max	float	Yes	Max available resources
allocation_ratio	float	Yes	Overcommitment
total_mb	float	Yes	Total size (MB)
used_mb	float	Yes	Used size (MB)
ratio	float	Yes	Memory usage

Table 4.2.20 Parameter Description - storage¶

Parameter	Type	Required	Note
total_mb	float	Yes	Total size (capacity of all disks, MB)

Table 4.2.21 Parameter Description - gpu¶

Parameter	Type	Required	Note
total	int	Yes	Total GPUs
used	float	Yes	Used GPUs
ratio	float	Yes	GPU usage
memory_total_mb	float	Yes	Graphics memory (MB)
memory_used_mb	float	Yes	Used graphics memory (MB)
memory_ratio	float	Yes	Graphics memory usage

Note

For common response parameters, view Response Format for details

- Response Example:


```
[
    {
        "status": "running",
        "description": "",
        "ip": "10.132.16.6",
        "storage": {
            "total_mb": 4105435.9936523438
        },
        "cpu": {
            "used_mhz": 11883.6,
            "allocation_ratio": 2.5,
            "ratio": 0.45,
            "allocation_max": 0.0,
            "cores_per_socket": 4,
            "socket_count": 1,
            "total_mhz": 26408.0,
            "type": "Intel(R) Xeon(R) CPU E3-1230 v3 @ 3.30GHz",
            "core_count": 4
        },
        "reserves": {
            "memory_mb": 4096
        },
        "memory": {
            "total_mb": 32768.0,
            "allocation_ratio": 1.0,
            "allocation_max": 0.0,
            "used_mb": 17039.36,
            "ratio": 0.52
        },
        "gpu": {
            "used": 0.0,
            "ratio": 0.0,
```

```
            "memory_used_mb": 0.0,
            "memory_ratio": 0.0,
            "total": 0,
            "memory_total_mb": 0.0
        },
        "alarm_count": 0,
        "is_use_vs": 1,
        "type": "h",
        "id": "host-a0369f033a73",
        "az_id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
        "name": "10.132.16.6"
    },
    {
        "status": "running",
        "description": "",
        "ip": "10.132.16.5",
        "storage": {
            "total_mb": 5059305.704589844
        },
        "cpu": {
            "used_mhz": 12939.92,
            "allocation_ratio": 2.5,
            "ratio": 0.49,
            "allocation_max": 0.0,
            "cores_per_socket": 4,
            "socket_count": 1,
            "total_mhz": 26408.0,
            "type": "Intel(R) Xeon(R) CPU E3-1230 V2 @ 3.30GHz",
            "core_count": 4
        },
        "reserves": {
            "memory_mb": 0
        },
        "memory": {
            "total_mb": 32768.0,
            "allocation_ratio": 1.0,
            "allocation_max": 0.0,
            "used_mb": 19333.12,
            "ratio": 0.59
        },
        "gpu": {
            "used": 0.0,
            "ratio": 0.0,
            "memory_used_mb": 0.0,
            "memory_ratio": 0.0,
            "total": 0,
            "memory_total_mb": 0.0
        },
        "alarm_count": 0,
        "is_use_vs": 1,
        "type": "h",
        "id": "host-001b21d61e70",
        "az_id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
        "name": "10.132.16.5"
    }
]
```


4.2.4.6. Response Code

- OK: 200
- Errors: 400, 403, 500

4.3. Tenants

4.3.1. Common Parameters of Tenants

4.3.1.1. Resouce Pool Type

Table 4.3.1 Resouce Pool Type¶

Value	Description
hci	HCI resource pool
vmware	VMware resource pool

4.3.1.2. Resource Pool Tag

Table 4.3.2 Resource Pool Tag

Value	Description
private	Dedicated resource pool
public	Shared resource pool

4.3.1.3. Dedicated Server Group Tag

Table 4.3.3 Dedicated Server Group Tag

Value	Description
private	Allocated to a single tenant
public	Shared by multiple tenants

4.3.1.4. Roles

Table 4.3.4 Roles

Value	Description
admin	Super administrator
tenant	Tenant administrator
member	Tenant user
custom_admin	Platform administrator
custom_tenant	Tenant co-administrator
msp	MSP administrator

4.3.2. Get Tenants

4.3.2.1. Scenario

- **Scenario:**
  - Administrator queries tenant list.
  - MSP administrator queries tenant list.
  - Tenants query tenant list (only return their own tenants).

- Applicable roles: Super administrator, system administrator, tenant and other administrators with permission to read users.
- Versions supported: SCP 6.3.0 and above

4.3.2.2. URI

Table 4.3.5 API Path and Request Method

REST VERB	PATH	Description
GET	/janus/20180725/projects	Get Tenants

- Parameter Description:

Table 4.3.6 Path Parameters¶

Parameter	Location	Type	Description	Required
keywords	path	string	Search tenant by name	No
az_id	path	string	Resource pool ID	No
az_type	path	string	Resource pool type, view Resouce Pool Type for details	No
dh_id	path	string	Dedicated server group ID	No
enabled	path	int	Whether the tenant is enabled [0, 1]	No

4.3.2.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/projects?keywords=tjc

4.3.2.4. Response

- Response Parameter

Table 4.3.7 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Tenant ID	No
name	string	Tenant name	No
user_name	string	Username	No
role_type	string	User role	No
azs	list	Associated resource pools, view Parameter Description - azs for details	No
dhs	list	Associated dedicated servers, view Description of the parameter dhs for details	No

Note

For common response parameters, view Response Format for details

- Parameter Description - azs

Table 4.3.8 Parameter Description - azs¶

Parameter	Type	Description	Value Returned
az_id	string	Resource pool ID	Yes
type	string	Resource pool type, view Resouce Pool Type for details	Yes
az_name	string	Resource pool name	Yes
az_tag	string	Resource pool tag, view Resource Pool Tag for details	Yes
dh_ids	list	Dedicated server group IDs	Yes
arch_type	string	Architecture Type	No

- Description of the parameter dhs

Table 4.3.9 Description of the parameter dhs¶

Parameter	Type	Description	Value Returned
az_id	string	Resource pool ID	Yes
dh_id	string	Dedicated server ID	Yes
dh_name	string	Dedicated server name	Yes
tag	string	Dedicated server tag, view Dedicated Server Group Tag for details	Yes

- Response Example:

```
[
    {
        "name": "lxm",
        "user_name": "lxm",
        "enabled": 1,
        "azs": [
            {
                "dh_ids": [],
                "az_tag": "public",
                "type": "hci",
                "az_name": "1.75",
                "az_id": "611f1b34-c727-4925-854e-07bc8607f681"
            }
        ],
        "dhs": [],
        "id": "513a35b3b2274a24bac71083909a4eec",
        "role_type": "tenant"
    }
]
```

4.3.2.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.3.3. Get Tenants

4.3.3.1. Scenario

- **Scenario:**
    - Administrator queries tenant list.
    - MSP administrator queries tenant list.

    - Tenants query tenant list (only return their own tenants).

- Applicable roles: Super administrator, system administrator, tenant and other administrators with permission to read users.
- Versions supported: SCP 6.8.0 and above

4.3.3.2. URI

Table 4.3.10 API Path and Request Method

REST VERB	PATH	Description
GET	/janus/20190725/projects	Get Tenants

- Parameter Description:

Table 4.3.11 Parameter Description of Request Path - 20190725

Parameter	Location	Type	Description	Required
keywords	path	string	Search tenant by name	No
az_id	path	string	Resource pool ID	No
az_type	path	string	Resource pool type, view Resouce Pool Type for details	No
dh_id	path	string	Dedicated server group ID	No
enabled	path	int	Whether the tenant is enabled [0, 1]	No
sort	path	string	Sorting field: sort=filed1[,filed2]	No
order_by	path	string	Sorting method: ASC (default) and DESC	No
fields	path	string	Expected field	No
page_num	path	int	Number of requested pages	No
page_size	path	int	Page size	No

4.3.3.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20190725/projects?keywords=tjc&page_num=2&page_size=50

4.3.3.4. Response

- Response Parameter

Table 4.3.12 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Tenant ID	No
name	string	Tenant name	No
user_name	string	Username	No
role_type	string	User role	No
azs	list	View Parameter Description - azs for the list of associated resource pools.	No
dhs	list	View Description of the parameter dhs for the list of associated dedicated server groups.	No
msp_id	string	ID of the MSP admin, which can be empty	No

Note

For common response parameters, view Response Format for details

- Parameter Description - azs

Table 4.3.13 Parameter Description - azs¶

Parameter	Type	Note
id	string	Resource pool ID
type	string	Resource pool type, view Resouce Pool Type for details
az_name	string	Resource pool name
az_tag	string	Resource pool tag, view Resource Pool Tag for details
dh_ids	list	Dedicated server group IDs

- Description of the parameter dhs

Table 4.3.14 Description of the parameter dhs¶

Parameter	Type	Note
az_id	string	Resource pool ID
dh_id	string	Dedicated server ID
dh_name	string	Dedicated server name
tag	string	Dedicated server tag, view Dedicated Server Group Tag for details

- Response Example:

```
[
    {
        "name": "lxm",
        "user_name": "lxm",
        "enabled": 1,
        "azs": [
            {
                "dh_ids": [],
                "az_tag": "public",
                "type": "hci",
                "az_name": "1.75",
                "az_id": "611f1b34-c727-4925-854e-07bc8607f681"
            }
        ],
        "dhs": [],
        "id": "513a35b3b2274a24bac71083909a4eec",
        "role_type": "tenant",
        "msp_id": "b1d29a3fd94048c08758ab12d34a908f"
    }
]
```

4.3.3.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.3.4. Query Tenant

4.3.4.1. Scenario

- Scenario: Get tenant's detailed information
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the User Read permission
- Versions supported: SCP6.3.70 and above

> **Note**
>
> Tenants can only obtain their own information.

4.3.4.2. URI

Table 4.3.15 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/projects/{project_id}	Query Tenant

- Parameter Description:

Table 4.3.16 Path Parameters¶

Parameter	Location	Type	Description	Required
project_id	path	string	Tenant ID	Yes

4.3.4.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/projects/8e9fbbc04d7b40f6bc3bbe549fd1655a

4.3.4.4. Response

- Response Parameter

Table 4.3.17 Response Parameter

Parameter	Type	Description	Value Returned
id	string	Tenant ID	No
name	string	Tenant name	No
enabled	int	Whether the tenant is enabled	No
role_type	string	Role type, view Roles for details	No
parent_id	string	Parent user ID (MSP administrator for current version). This parameter is an empty string by default, meaning that the tenant belongs to administrator	Yes
project_admin	dict	Tenant administrator, view Parameter Description - project_admin for details	Yes
azs	list	Resource pools, view Parameter Description - azs for details	No
dhs	list	Dedicated server groups, view Description of the parameter dhs for details	No
description	string	Description	Yes

Note

If the parameter parent_id is empty, the parent user is administrator. If not, its parent user is MSP administrator.

Note

For common response parameters, view Response Format for details

- Parameter Description - project_admin

Table 4.3.18 Parameter Description - project_admin

Parameter	Type	Description	Value Returned
user_name	string	Username of tenant	Yes
user_id	string	User ID	Yes
mail	string	Email address	Yes
phone	string	Mobile phone number	Yes
auth_driver	string	Authentication type, Enum['local','ldap']	No

- Parameter Description - azs

Table 4.3.19 Parameter Description - azs¶

Parameter	Type	Description	Value Returned
az_id	string	Resource pool ID	Yes
type	string	Resource pool type, view Resouce Pool Type for details	Yes
az_name	string	Resource pool name	Yes
az_tag	string	Resource pool tag, view Resource Pool Tag for details	Yes
dh_ids	list	Dedicated server group IDs	Yes
arch_type	string	Architecture Type	No

- Description of the parameter dhs

Table 4.3.20 Description of the parameter dhs¶

Parameter	Type	Description	Value Returned
az_id	string	Resource pool ID	Yes
dh_id	string	Dedicated server ID	Yes
dh_name	string	Dedicated server name	Yes
tag	string	Dedicated server tag, view Dedicated Server Group Tag for details	Yes

- Response Example:


```
{
    "parent_id": "3b3cefbc02d543868d4e68f3d8137fde",
    "enabled": 1,
    "role_type": "tenant",
    "id": "8e9fbbc04d7b40f6bc3bbe549fd1655a",
    "name": "aaa",
    "description": "new",
    "auth_driver": "local",
    "project_admin": {
        "user_name": "tenant_admin",
        "user_id": "dea0d2db3e9e4a5ca4f64d35813ab173",
        "mail": "aaa@123.com",
        "phone": "13412345678",
        "auth_driver": "local"
    },
    "azs": [
        {
            "dh_ids": [],
            "az_tag": "public",
            "type": "hci",
            "az_id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
            "az_name": "1.75"
        }
    ],
    "dhs": [
        {
            "dh_id": "5ec0ddb0-d651-4aab-aaef-e2a178652e02",
            "dh_name": "DH01",
            "tag": "public",
            "az_id": "611f1b34-c727-4925-854e-07bc8607f681"
        }
    ]
}
```


4.3.4.5. Response Code

- OK: 200

- Errors: 400, 403, 404, 500

4.3.5. Create a Tenant

4.3.5.1. Scenario

- Scenario: Create a tenant
- Roles Supported: Super admin, system admin, MSP admin, and co-administrator with the User Configuration permission
- Versions supported: SCP6.3.70 and above

4.3.5.2. URI

Table 4.3.21 API Path and Request Method¶

REST VERB	PATH	Description
POST	/janus/20180725/projects	Create a Tenant

- Parameter Description:

  None

4.3.5.3. Request

- Request Parameter

Table 4.3.22 Body Parameter¶

Parameter	Location	Type	Description	Required
project	body	dict	Tenant's information, view Parameter Description - project for details	Yes

- Parameter Description - project

Table 4.3.23 Parameter Description - project¶

Parameter	Location	Type	Description	Required
id	body	string	Tenant ID	No
name	body	string	Tenant name	Yes
description	body	string	Description	No
parent_id	body	string	MSP administrator ID, only available in MSP mode	No
project_admin	body	dict	User information, view Parameter Description - project-project_admin for details	Yes

- Parameter Description - project-project_admin

Table 4.3.24 Parameter Description - project-project_admin¶

Parameter	Location	Type	Description	Required
user_id	body	string	User ID	No
user_name	body	string	Username of tenant	Yes
mail	body	string	Email address	Yes
phone	body	string	Mobile phone number	Yes
password	body	string	Encrypted password, view encryption method in Token-Based Authentication for details	Yes

- Request Example

  https://{SCP_IP}/janus/20180725/projects

```
{
    "project": {
        "id": "8e9fbbc04d7b40f6bc3bbe549fd1655a",
        "name": "tenant",
        "parent_id": "3b3cefbc02d543868d4e68f3d8137fde",
        "description": "",
        "project_admin": {
            "user_name": "tenant_admin",
            "mail": "aaa@123.com",
            "phone": "13412345678",
            "password": "xxxxxx"
        }
    }
}
```

4.3.5.4. Response

- Response Parameter

Table 4.3.25 Response Parameter¶

Parameter	Type	Description	Value Returned
project	dict	Tenant information, view Parameter Description - project for details	Yes

Note

For common response parameters, view Response Format for details

- Parameter Description - project

Table 4.3.26 Parameter Description - project¶

Parameter	Type	Description	Value Returned
id	string	Tenant's project ID	Yes
name	string	Tenant name	Yes
description	string	Description	Yes
enabled	int	Check whether the tenant is available	Yes
parent_id	string	Tenant's parent_id	Yes
created_at	string	ISO 8601, for example: "2020-12-01T03:17:42Z"	Yes
updated_at	string	ISO 8601, for example: "2020-12-01T03:17:42Z"	Yes
project_admin	dict	User information, view Parameter Description - project_admin for details	Yes

- Parameter Description - project_admin

Table 4.3.27 Parameter Description - project_admin¶

Parameter	Type	Description	Value Returned
user_name	string	Username of tenant	Yes
user_id	string	User ID	Yes
mail	string	Email address	Yes
phone	string	Mobile phone number	Yes

- Response Example:

```
{
    "project": {
        "name": "tenant",
        "id": "8e9fbbc04d7b40f6bc3bbe549fd1655a",
        "parent_id": "",
        "enabled": 1,
        "description": "",
        "project_admin": {
            "user_name": "tenant_admin",
            "user_id": "dea0d2db3e9e4a5ca4f64d35813ab173",
            "mail": "aaa@123.com",
            "phone": "13412345678"
        },
        "created_at": "2020-12-21T03:07:40Z",
        "updated_at": ""
    }
}
```

4.3.5.5. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.3.6. Edit Tenant

4.3.6.1. Scenario

- Scenario: Edit tenant information
- Roles Supported: Super admin, system admin, MSP admin, and co-administrator with the User Configuration permission
- Versions supported: SCP6.3.70 and above

4.3.6.2. URI

Table 4.3.28 API Path and Request Method

REST VERB	PATH	Description
PUT	/janus/20180725/projects/{project_id}	Edit Tenant

- Parameter Description:

Table 4.3.29 Path Parameters

Parameter	Location	Type	Description	Required
project_id	path	string	Tenant ID	Yes

4.3.6.3. Request

- Request Parameter

Table 4.3.30 Body Parameter

Parameter	Location	Type	Description	Required
name	body	string	Tenant name	No
phone	body	string	Tenant's phone number	No
mail	body	string	Tenant's email address	No
description	body	string	Description	No
parent_id	body	string	The parent tenant ID to be changed, available in MSP mode.	No

- Request Example

  https://{SCP_IP}/janus/20180725/projects/e3c6b8e870c84a90be87eddfbf7ae12f

```
{
    "project": {
        "name": "aaa",
        "mail": "aaa@123.com",
        "phone": "13412345678",
        "description": "desc",
        "parent_id": "3b3cefbc02d543868d4e68f3d8137fde",
    }
}
```

4.3.6.4. Response

- Response Parameter

Table 4.3.31 Response Parameter

Parameter	Type	Description	Value Returned
project	dict	Tenant information, view Parameter Description - project for details	Yes

- Parameter Description - project

Table 4.3.32 Parameter Description - project¶

Parameter	Type	Description	Value Returned
id	string	Tenant's project ID	Yes
name	string	Tenant name	Yes
description	string	Description	Yes
enabled	int	Check whether the tenant is available	Yes
parent_id	string	Tenant's parent_id	Yes
created_at	string	ISO 8601, for example: "2020-12-01T03:17:42Z"	Yes
updated_at	string	ISO 8601, for example: "2020-12-01T03:17:42Z"	Yes
project_admin	dict	User information, view Parameter Description - project_admin for details	Yes

- Parameter Description - project_admin

Table 4.3.33 Parameter Description - project_admin¶

Parameter	Type	Description	Value Returned
user_name	string	Username of tenant	Yes
user_id	string	User ID	Yes
mail	string	Email address	Yes
phone	string	Mobile phone number	Yes

- Response Example:

```
{
    "project": {
        "name": "tenant",
        "id": "8e9fbbc04d7b40f6bc3bbe549fd1655a",
        "parent_id": "",
        "enabled": 1,
        "description": "",
        "project_admin": {
            "user_name": "tenant_admin",
            "user_id": "dea0d2db3e9e4a5ca4f64d35813ab173",
            "mail": "aaa@123.com",
            "phone": "13412345678"
        },
        "created_at": "2020-12-21T03:07:40Z",
        "updated_at": ""
    }
}
```

4.3.6.5. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.3.7. Delete Tenant

4.3.7.1. Scenario

- Scenario: Delete the specified tenant
- Roles Supported: Super admin, system admin, MSP admin, and co-administrator with the User Configuration permission
- Versions supported: SCP6.3.70 and above

Note

1. The admin and the platform administrator with permission to configure users can delete tenants on the platform.
2. MSP administrators can only delete their own tenants.
3. Before deleting a tenant, you need to clean up all the resources of the tenant, including VMs, network devices, elastic IPs, etc., otherwise the deletion will fail.

4.3.7.2. URI

Table 4.3.34 API Path and Request Method

REST VERB	PATH	Description
DELETE	/janus/20180725/projects/{project_id}	Delete Tenant

- Parameter Description:

Table 4.3.35 Path Parameters

Parameter	Location	Type	Description	Required
project_id	path	string	Tenant ID	Yes

4.3.7.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/projects/8e9fbbc04d7b40f6bc3bbe549fd1655a

4.3.7.4. Response

- Response Parameter

  None, no return value for successful request.

- Response Example:

  None

4.3.7.5. Response Code

- Normal: 204
- Errors: 400, 403, 404, 500

4.3.8. Associate Resource Pool for Tenants

4.3.8.1. Scenario

- Scenario: Associate resource pools or dedicated server groups for tenants
- Roles Supported: Super admin, system admin, MSP admin, and co-administrator with the User Configuration and Resource Pool Configuration permissions
- Versions supported: SCP6.3.70 and above

4.3.8.2. URI

Table 4.3.36 API Path and Request Method

REST VERB	PATH	Description
POST	/janus/20180725/projects/{project_id}/azs	Associate Resource Pool for Tenants

- Parameter Description:

Table 4.3.37 Path Parameters

Parameter	Location	Type	Description	Required
project_id	path	string	Tenant ID	Yes

4.3.8.3. Request

- Request Parameter

Table 4.3.38 Body Parameter

Parameter	Location	Type	Description	Required
azs	body	list	Resource pool list, view Parameter Description - azs for details	Yes

- Parameter Description - azs

Table 4.3.39 Parameter Description - azs

Parameter	Location	Type	Description	Required
az_id	body	string	Resource pool ID	No
dh_id	body	string	Dedicated host group ID	No

- Request Example

  https://{SCP_IP}/janus/20180725/projects/8e9fbbc04d7b40f6bc3bbe549fd1655a/azs

```
{
    "azs": [
        {
            "az_id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
            "dh_id": "027ab8a7-07eb-4819-94db-40ea593b6695"
        }
    ]
}
```

4.3.8.4. Response

- Response Parameter

Table 4.3.40 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Asynchronous task ID	Yes

- Response Example:

```
{
    "task_id": "d5ae0d43-4afc-4f27-a851-9989a7083d9d"
}
```

4.3.8.5. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.3.9. Disassociate Resource Pool from Tenant

4.3.9.1. Scenario

- Scenario: Disassociate resource pools from the tenant
- Roles Supported: Super admin, system admin, MSP admin, and co-administrator with the User Configuration and Resource Pool Configuration permissions
- Versions supported: SCP6.3.70 and above

> **Note**
>
> It is required to check whether all tenant's resources in the resource pool are cleaned up.

4.3.9.2. URI

Table 4.3.41 API Path and Request Method¶

REST VERB	PATH	Description
DELETE	/janus/20180725/projects/{project_id}/azs/{az_id}	Disassociate Resource Pool from Tenant

- Parameter Description:

Table 4.3.42 Path Parameters¶

Parameter	Location	Type	Description	Required
project_id	path	string	Tenant ID	Yes
az_id	path	string	Resource pool ID	Yes

4.3.9.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/projects/8e9fbbc04d7b40f6bc3bbe549fd1655a/azs/1c13ef3e-a049-4fc8-bc01-a2b5b279d46b

4.3.9.4. Response

- Response Parameter

None

- Response Example:

  None

4.3.9.5. Response Code

- Normal: 204
- Errors: 400, 403, 404, 500

4.4. Quota

4.4.1. Common Quota Parameters

4.4.1.1. Types of Quota Items

Table 4.4.1 Types of Quota Items¶

Value	Description
global	Global resource types, such as VM image size, bandwidth and number of elastic IPs, number of VMs for DR
pool	Types of resource pool, such as computing and storage
NFV	NFV types such as ADC, NGAF, IAG
cssp_nfv	Third party NFV devices such as OSM, BVT, EDR, LAS
dmp	DMP resources, such as MySQL database and Oracle database.
appcenter	Application Center resources, such as Redis, Kafka, Mongo, RabbitMQ

4.4.1.2. Quota Provider Types

Table 4.4.2 Quota Provider Types¶

Value	Description
hci	Sangfor HCI resource
vmware	VMware resource
appcenter	appcenter resource
cssp	cssp resource

4.4.1.3. Resource Quota Specifications

Table 4.4.3 Resource Quota Specifications¶

resource	Chinese Name	Unit	namespace	provider
ac_branch	Branches	-	viam	hci
ac_branch	Branches	-	vac	hci
ac_lines	Lines	-	vac	hci
ac_lines	Lines	-	viam	hci
ac_users	Mobile users	-	viam	hci
ac_users	Mobile users	-	vac	hci
	2 CPU Cores,			

ad_100Mbps_performance	4 GB Memory, 100 Mbps	-	vad	hci
ad_10Gbps_performance	16 CPU Cores, 32 GB Memory, 10 Gbps	-	vad	hci
ad_10Mbps_performance	2 CPU Cores, 4 GB Memory, 100 Mbps	-	vad	hci
ad_1Gbps_performance	2 CPU Cores, 4 GB Memory, 1 Gbps	-	vad	hci
ad_200Mbps_performance	2 CPU Cores, 4 GB Memory, 200 Mbps	-	vad	hci
ad_3Gbps_performance	4 CPU Cores, 8 GB Memory, 3 Gbps	-	vad	hci
ad_500Mbps_performance	2 CPU Cores, 4 GB Memory, 500 Mbps	-	vad	hci
ad_50Mbps_performance	2 CPU Cores, 4 GB Memory, 50 Mbps	-	vad	hci
ad_5Gbps_performance	8 CPU Cores, 16 GB Memory, 5 Gbps	-	vad	hci
ad_basic_module	WAN links	-	vad	hci
ad_DNS_module	Intelligent DNS	-	vad	hci
ad_http_cache_module	HTTP Caching	-	vad	hci
ad_SSL_module	SSL Offloading	-	vad	hci
af_bandwidth_1.6G	8 CPU Cores, 16 GB Memory, 1.6 Gbps	-	vaf	hci
af_bandwidth_100M	2 CPU Cores, 4 GB Memory, 100 Mbps	-	vaf	hci
af_bandwidth_200M	2 CPU Cores, 4 GB Memory, 200 Mbps	-	vaf	hci
af_bandwidth_400M	4 CPU Cores, 8 GB Memory, 400 Mbps	-	vaf	hci
af_bandwidth_800M	8 CPU Cores, 16 GB Memory, 800 Mbps	-	vaf	hci
af_branch	Branches	-	vaf	hci

af_ssl_vpn	SSL VPN Users	-	vaf	hci
af_tamper	Anti Web Defacement Licenses	-	vaf	hci
af_users	Mobile users	-	vaf	hci
backup_usage	Backup Repository	Bytes	storage	hci
container_cloud_service	Containers	-	compute	hci
cores	CPU	Core	compute	hci
cores	CPU	Core	compute	vmware
dblb_1Gbps_performance	Database Load Balancer Components	-	vdblb	hci
dr_vm_count	VM for DR	-	protection	hci
eip_bandwidth::{linetype_id}, obtain linetype_id from Query Line Types	Bandwidth	Kbps	network	hci
eip_count::{linetype_id}, obtain linetype_id from Query Line Types	Elastic IP	-	network	hci
gpu_slice_count	vGPU	-	gpu	hci
image_usage	Private Image Repository	Bytes	storage	hci
kafka_cluster_2c_4g	Cluster Version - 2 CPU Cores, 4 GB Memory	-	kafka	appcenter
kafka_cluster_4c_8g	Cluster Version - 4 CPU Cores, 8 GB Memory	-	kafka	appcenter
kafka_cluster_8c_16g	Cluster Version - 8 CPU Cores, 16 GB Memory	-	kafka	appcenter
kafka_single_2c_4g	Standalone Version - 2 CPU Cores, 4 GB Memory	-	kafka	appcenter
kafka_single_4c_8g	Standalone Version - 4 CPU Cores, 8 GB Memory	-	kafka	appcenter
kafka_single_8c_16g	Standalone Version - 8 CPU Cores, 16 GB Memory	-	kafka	appcenter
mongo_single_2c_4g	Standalone Version - 2 CPU Cores, 4 GB Memory	-	mongo	appcenter
mongo_single_4c_8g	Standalone Version - 4 CPU Cores, 8 GB Memory	-	mongo	appcenter
mongo_single_8c_16g	Standalone Version - 8 CPU Cores,	-	mongo	appcenter

	16 GB Memory			
mongo_single_8c_32g	Standalone Version - 8 CPU Cores, 32 GB Memory	-	mongo	appcenter
nfv_cores	NFV	Core	compute	hci
nfv_ram	NFV memory	MB	compute	hci
nvidia_m10_ram	Graphics Card Tesla M10	MB	gpu	hci
nvidia_m60_ram	Graphics Card Tesla M60	MB	gpu	hci
nvidia_p100_ram	Graphics Card Tesla P100	MB	gpu	hci
nvidia_p40_ram	Graphics Card Tesla P40	MB	gpu	hci
nvidia_rtx2080tia_ram	Graphics Card GeForce RTX 2080 TiA	MB	gpu	hci
nvidia_rtx2080ti_ram	Graphics Card GeForce RTX 2080Ti	MB	gpu	hci
nvidia_rtx6000_ram	Graphics Card Quadro RTX 6000	MB	gpu	hci
nvidia_t4_ram	Graphics Card Tesla T4	MB	gpu	hci
nvidia_v100dx_ram	Graphics Card Tesla V100DX	MB	gpu	hci
nvidia_v100d_ram	Graphics Card Tesla V100D	MB	gpu	hci
nvidia_v100s_ram	Graphics Card Tesla V100S	MB	gpu	hci
nvidia_v100_ram	Graphics Card Tesla V100	MB	gpu	hci
rabbitmq_single_16c_32g	Standalone Version - 16 CPU Cores, 32 GB Memory	-	rabbitmq	appcenter
rabbitmq_single_2c_4g	Standalone Version - 2 CPU Cores, 4 GB Memory	-	rabbitmq	appcenter
rabbitmq_single_4c_8g	Standalone Version - 4 CPU Cores, 8 GB Memory	-	rabbitmq	appcenter
rabbitmq_single_8c_16g	Standalone Version - 8 CPU Cores, 16 GB Memory	-	rabbitmq	appcenter
ram	Memory	MB	compute	hci
ram	Memory	MB	compute	vmware
	Cluster Version - 2			

redis_cluster_2c_16g	Version - 2 CPU Cores, 16 GB Memory	-	redis	appcenter
redis_cluster_2c_4g	Cluster Version - 2 CPU Cores, 4 GB Memory	-	redis	appcenter
redis_cluster_2c_8g	Cluster Version - 2 CPU Cores, 8 GB Memory	-	redis	appcenter
redis_sentinel_2c_16g	Sentinel Version - 2 CPU Cores, 16 GB Memory	-	redis	appcenter
redis_sentinel_2c_4g	Sentinel Version - 2 CPU Cores, 4 GB Memory	-	redis	appcenter
redis_sentinel_2c_8g	Sentinel Version - 2 CPU Cores, 8 GB Memory	-	redis	appcenter
redis_single_2c_16g	Standalone Version - 2 CPU Cores, 16 GB Memory	-	redis	appcenter
redis_single_2c_4g	Standalone Version - 2 CPU Cores, 4 GB Memory	-	redis	appcenter
redis_single_2c_8g	Standalone Version - 2 CPU Cores, 8 GB Memory	-	redis	appcenter
ssl_branch	Branches	-	vssl	hci
ssl_emmbasicuser	EMM Basic Licenses	-	vssl	hci
ssl_emmordinaryuser	EMM Standard Licenses	-	vssl	hci
ssl_line	Lines	-	vssl	hci
ssl_remoteappuser	Remote App Users	-	vssl	hci
ssl_ssl_vpn	Mobile Users	-	vssl	hci
storage_tag::00000000-0000-0000-0000-000000000000	High Performance Storage	MB	storage	hci
storage_tag::00000000-0000-0000-0000-000000000000	High Performance Storage	MB	storage	vmware
storage_tag::11111111-1111-1111-1111-111111111111	Performance Sensitive Storage	MB	storage	hci
storage_tag::11111111-1111-1111-1111-111111111111	Performance Sensitive Storage	MB	storage	vmware
storage_tag::22222222-2222-2222-2222-222222222222	Capacity Sensitive	MB	storage	hci

---- ---- ----------	Storage			
storage_tag::22222222-2222-2222-2222-222222222222	Capacity Sensitive Storage	MB	storage	vmware
vAC-100	1 CPU Core, 2 GB Memory, 100 Mbps	-	vac	hci
vAC-1000	8 CPU Cores, 16 GB Memory, 1 Gbps	-	vac	hci
vAC-10000	8 CPU Cores, 16 GB Memory, 10 Gbps	-	vac	hci
vAC-200	2 CPU Cores, 4 GB Memory, 200 Mbps	-	vac	hci
vAC-2000	8 CPU Cores, 16 GB Memory, 2 Gbps	-	vac	hci
vAC-300	4 CPU Cores, 8 GB Memory, 300 Mbps	-	vac	hci
vAC-3000	8 CPU Cores, 16 GB Memory, 3 Gbps	-	vac	hci
vAC-50	1 CPU Core, 2 GB Memory, 50 Mbps	-	vac	hci
vAC-500	4 CPU Cores, 8 GB Memory, 500 Mbps	-	vac	hci
vAC-5000	8 CPU Cores, 16 GB Memory, 5 Gbps	-	vac	hci
vAC-700	4 CPU Cores, 8 GB Memory, 700 Mbps	-	vac	hci
vBVT-100	4 CPU Cores, 8 GB Memory, 300 GB Disk Space, 100 Assets	-	vbvt	cssp
vBVT-200	8 CPU Cores, 16 GB Memory, 300 GB Disk Space, 200 Assets	-	vbvt	cssp
vBVT-300	8 CPU Cores, 32 GB Memory, 00 GB Disk Space, 200 Assets	-	vbvt	cssp

vBVT-50	4 CPU Cores, 8 GB Memory, 300 GB Disk Space, 50 Assets	-	vbvt	cssp
vBVT-500	16 CPU Cores, 64 GB Memory, 300 GB Disk Space, 500 Assets	-	vbvt	cssp
vBVT-test-10	4 CPU Cores, 8 GB Memory, 300 GB Disk Space, 10 Assets (trial)	-	vbvt	cssp
vDAS-100	4 CPU Cores, 8 GB Memory, 100 GB Data Disk, 300 Mbps	-	vdas	hci
vDAS-200	8 CPU Cores, 16 GB Memory, 300 GB Data Disk, 500 Mbps	-	vdas	hci
vEDR-100	2 CPU Cores, 4 GB Memory, 330 GB Disk Space, 100 Endpoints	-	vedr	cssp
vEDR-1000	4 CPU Cores, 8 GB Memory, 330 GB Disk Space, 1000 Endpoints	-	vedr	cssp
vEDR-200	2 CPU Cores, 4 GB Memory, 330 GB Disk Space, 200 Endpoints	-	vedr	cssp
vEDR-2000	8 CPU Cores, 16 GB Memory, 330 GB Disk Space, 2000 Endpoints	-	vedr	cssp
vEDR-50	2 CPU Cores, 4 GB Memory, 330 GB Disk Space, 50 Endpoints	-	vedr	cssp
vEDR-test-50	2 CPU Cores, 4 GB Memory, 300 GB Disk Space, 50 Endpoints (trial)	-	vedr	cssp
vIAG-100	1 CPU Core, 2 GB Memory, 100 Mbps	-	viam	hci

vIAG-1000	8 CPU Cores, 16 GB Memory, 1 Gbps	-	viam	hci
vIAG-10000	8 CPU Cores, 16 GB Memory, 10 Gbps	-	viam	hci
vIAG-200	2 CPU Cores, 4 GB Memory, 200 Mbps	-	viam	hci
vIAG-2000	8 CPU Cores, 16 GB Memory, 2 Gbps	-	viam	hci
vIAG-300	4 CPU Cores, 8 GB Memory, 300 Mbps	-	viam	hci
vIAG-3000	8 CPU Cores, 16 GB Memory, 3 Gbps	-	viam	hci
vIAG-50	1 CPU Core, 2 GB Memory, 50 Mbps	-	viam	hci
vIAG-500	4 CPU Cores, 8 GB Memory, 500 Mbps	-	viam	hci
vIAG-5000	8 CPU Cores, 16 GB Memory, 5 Gbps	-	viam	hci
vIAG-700	4 CPU Cores, 8 GB Memory, 700 Mbps	-	viam	hci
vLAS-100	8 CPU Cores, 16 GB Memory, 100 Assets	-	vlas	cssp
vLAS-20	4 CPU Cores, 8 GB Memory, 20 Assets	-	vlas	cssp
vLAS-200	8 CPU Cores, 16 GB Memory, 200 Assets	-	vlas	cssp
vLAS-300	12 CPU Cores, 16 GB Memory, 300 Assets	-	vlas	cssp
vLAS-50	4 CPU Cores, 8 GB Memory, 50 Assets	-	vlas	cssp
vLAS-500	20 CPU Cores, 32 GB Memory, 500 Assets	-	vlas	cssp

vLAS-test-10	4 CPU Cores, 8 GB Memory, 10 Assets (trial)	-	vlas	cssp
vMySQL	MySQL RDS	-	vmysql	hci
vMySQL_16c32g_performance	16 CPU Cores, 32 GB Memory	-	vmysql	hci
vMySQL_16c64g_performance	16 CPU Cores, 64 GB Memory	-	vmysql	hci
vMySQL_2c4g_performance	2 CPU Cores, 4 GB Memory	-	vmysql	hci
vMySQL_32c128g_performance	32 CPU Cores, 128 GB Memory	-	vmysql	hci
vMySQL_4c16g_performance	4 CPU Cores, 16 GB Memory	-	vmysql	hci
vMySQL_4c8g_performance	4 CPU Cores, 8 GB Memory	-	vmysql	hci
vMySQL_8c16g_performance	8 CPU Cores, 16 GB Memory	-	vmysql	hci
vMySQL_8c32g_performance	8 CPU Cores, 32 GB Memory	-	vmysql	hci
vMySQL_8c64g_performance	8 CPU Cores, 64 GB Memory	-	vmysql	hci
vm_count	Virtual machine	-	compute	vmware
vm_count	Virtual machine	-	compute	hci
vOracle	Oracle RDS	-	voracle	hci
vOracle_16c32g_performance	16 CPU Cores, 32 GB Memory	-	voracle	hci
vOracle_16c64g_performance	16 CPU Cores, 64 GB Memory	-	voracle	hci
vOracle_2c4g_performance	2 CPU Cores, 4 GB Memory	-	voracle	hci
vOracle_32c128g_performance	32 CPU Cores, 128 GB Memory	-	voracle	hci
vOracle_4c16g_performance	4 CPU Cores, 16 GB Memory	-	voracle	hci
vOracle_4c8g_performance	4 CPU Cores, 8 GB Memory	-	voracle	hci
vOracle_8c16g_performance	8 CPU Cores, 16 GB Memory	-	voracle	hci
vOracle_8c32g_performance	8 CPU Cores, 32 GB Memory	-	voracle	hci
	8 CPU Cores,			

vOracle_8c64g_performance	64 GB Memory	-	voracle	hci
vOSM-10	2 CPU Cores, 4 GB Memory, 10 Assets	-	vosm	cssp
vOSM-100	2 CPU Cores, 4 GB Memory, 100 Assets	-	vosm	cssp
vOSM-20	2 CPU Cores, 4 GB Memory, 20 Assets	-	vosm	cssp
vOSM-200	2 CPU Cores, 4 GB Memory, 20 Assets	-	vosm	cssp
vOSM-300	2 CPU Cores, 8 GB Memory, 300 Assets	-	vosm	cssp
vOSM-50	2 CPU Cores, 4 GB Memory, 50 Assets	-	vosm	cssp
vOSM-500	2 CPU Cores, 16 GB Memory, 500 Assets	-	vosm	cssp
vOSM-test-50	2 CPU Cores, 4 GB Memory, 50 Assets (trial)	-	vosm	cssp
vSSL-100	2 CPU Cores, 2 GB Memory, 100 Mbps	-	vssl	hci
vSSL-1000	8 CPU Cores, 8 GB Memory, 600 Mbps	-	vssl	hci
vSSL-1200	8 CPU Cores, 16 GB Memory, 900 Mbps	-	vssl	hci
vSSL-200	2 CPU Cores, 4 GB Memory, 200 Mbps	-	vssl	hci
vSSL-400	4 CPU Cores, 4 GB Memory, 300 Mbps	-	vssl	hci
vSSL-800	4 CPU Cores, 8 GB Memory, 400 Mbps	-	vssl	hci
woc_ipsec_vpn	Third Party IPsec Devices	-	vwoc	hci
woc_standard_set	1 CPU Core, 2 GB Memory, 25 Mbps	-	vwoc	hci

woc_standard_set_200	2 CPU Cores, 4 GB Memory, 50 Mbps	-	vwoc	hci
woc_standard_set_400	4 CPU Cores, 8 GB Memory, 200 Mbps	-	vwoc	hci
woc_vpn_line	Lines	-	vwoc	hci

4.4.2. Query Quota Items

4.4.2.1. Scenario

- Scenario: Obtain the allocation and usage of tenant quotas, which can be used to check available quotas for corresponding resources.
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Quota Read permission
- Versions supported: SCP 6.3.0 and above

4.4.2.2. URI

Table 4.4.4 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/quotas	Query Quota Items

- Parameter Description:

Table 4.4.5 Path Parameters¶

Parameter	Location	Type	Description	Required
category	path	string	Types of quota items, view Types of Quota Items for details.	No
provider	path	string	Types of quota provider, view Quota Provider Types for details.	No
namespace	path	string	Types of quota, view Resource Quota Specifications for details.	No
resource	path	string	Quota name, view Resource Quota Specifications	No
az_id	path	string	Resource pool ID	No
project_id	path	string	Tenant ID	No
hard_limit_gt	path	int	Quota items with maximum value greater than this	No
hard_limit	path	int	Quota items with this maximum value	No
hard_limit_lt	path	int	Quota items with maximum value lower than this	No

Note

For common path parameters, view Common Path Parameter Format for details

4.4.2.3. Request

- Request Parameter

None

- Request Example

https://{SCP_IP}/janus/20180725/quotas?resource=vMySQL

4.4.2.4. Response

- Response Parameter

Table 4.4.6 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Quota item ID	No
category	string	Types of quota items, view Types of Quota Items for details.	No
provider	string	Types of quota provider, view Quota Provider Types for details.	No
namespace	string	Types of quota, view Resource Quota Specifications for details.	No
resource	string	Quota name, view Resource Quota Specifications	No
name	string	Quota Chinese name, view Resource Quota Specifications for details.	No
project_id	string	Tenant ID	No
unit	string	Quota unit, view Resource Quota Specifications for details.	No
separated_quotas	list	Maximum quota for different resource pools, view Parameter Description - az_quotas for details.	No
az_quotas	list	Maximum quota for different resource pools, view Parameter Description - az_quotas for details.	No

Note

View Common Response Parameter Format for public response parameters in the API list.

- Parameter Description - separated_quotas

Table 4.4.7 Parameter Description - separated_quotas¶

Parameter	Type	Required	Note
az_id	string	Yes	Resource pool ID
in_use	int	Yes	Quota usage
reserved	int	Yes	Quota reserved
allocation	int	Yes	Allocated quota

- Parameter Description - az_quotas

Table 4.4.8 Parameter Description - az_quotas¶

Parameter	Type	Required	Note
az_id	string	Yes	Resource pool ID
hard_limit	int	Yes	Maximum quota, -1 indicates no limit

- Response Example:

```
[
    {
        "category": "dmp",
        "resource": "vMySQL",
        "name": "MySQL RDS",
        "separated_quotas": [
            {
                "reserved": 0,
                "az_id": "611f1b34-c727-4925-854e-07bc8607f681",
                "in_use": 0
            }
        ],
        "namespace": "vmysql",
        "az_quotas": [
            {
                "hard_limit": 11,
                "az_id": "611f1b34-c727-4925-854e-07bc8607f681"
            }
        ],
        "provider": "hci",
        "project_id": "623f7f8f6e0b42c2af429ed831da92d0",
        "id": "a8d69bf3a159681ccc9c8b6b0515a6123199aa1366b8e26ee70eee6a800fc
391",
        "unit": ""
    }
]
```

4.4.2.5. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.4.3. Set Quota

4.4.3.1. Scenario

- **Scenario:**
    - Admin sets quotas for MSP or tenants.
    - MSP sets quota for tenants.

- Roles Supported: Super admin, MSP admin, and co-administrator with the Quota Configuration permission
- Versions supported: SCP6.3.70 and above

4.4.3.2. URI

Table 4.4.9 API Path and Request Method¶

REST VERB	PATH	Description
POST	/janus/20180725/quotas	Set Quota

- Parameter Description:

  None

4.4.3.3. Request

- Request Parameter

Table 4.4.10 Description of Body Parameter¶

Parameter	Location	Type	Description	Required
project_id	body	string	MSP administrator or tenant ID	Yes
quota_sets	body	list	Quota specifications, view Description of parameter quota-sets for details.	Yes

- Description of parameter quota-sets

Table 4.4.11 Description of parameter quota-sets¶

Parameter	Type	Required	Note
provider	string	Yes	View Quota Provider Types for the quota provider type.
namespace	string	Yes	Types of quota, view Resource Quota Specifications for details.
resource	string	Yes	Quota name, view Resource Quota Specifications for details.
value	int	Yes	Specified quota value
az_id	string	Yes	Resource pool ID, the global ID is 10000000-0000-0000-0000-000000000000

Note

The resource pool ID of global resources is 10000000-0000-0000-0000-000000000000, and the global resources include: container_cloud_service, eip_bandwidth, eip_count, dr_vm_count and image_usage.

- Request Example

https://{SCP_IP}/janus/20180725/quotas

```
{
    "project_id": "8e9fbbc04d7b40f6bc3bbe549fd1655a",
    "quota_sets": [
        {
            "provider": "hci",
            "namespace": "storage",
            "resource": "image_usage",
            "value": 51,
            "az_id": "10000000-0000-0000-0000-000000000000"
        },
        {
            "provider": "hci",
            "namespace": "vmysql",
            "resource": "vMySQL_2c4g_performance",
            "value": 2,
            "az_id": "d4a1bfa5-01aa-489a-8655-4427ca6e7e11"
        }
    ]
}
```

4.4.3.4. Response

- Response Parameter

None

- Response Example:

None

4.4.3.5. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.5. Virtual Machines

4.5.1. Common VM Parameters

4.5.1.1. Disk Provisioning Method

Table 4.5.1 Disk Provisioning Method¶

Value	Description
off	Thin provisioning
full	Pre-allocating
metadata	Dynamic provisioning

4.5.1.2. vGPU Type

Table 4.5.2 vGPU Type¶

Value	Description
PASS_THROUGH	Passthrough
vGPU	vGPU

4.5.1.3. Scheduling Method

Table 4.5.3 Scheduling Method¶

Value	Description
best_effort	Best effort
equal_share	Equal share
fixed_share	Fixed share

4.5.1.4. VM Type

Table 4.5.4 VM Type¶

Value	Description
vm	Ordinary VM
placeholder	Placeholder VM
dmp_mysql	Database VM
service	Service VM
app	VM created in Application Center

4.5.1.5. VM Disk Type

Table 4.5.5 VM Disk Type¶

Value	Description
new_disk	New disk, except for the disk types listed below, other disks for VM creation are usually of new_disk type.
derive_disk	System disk. When a VM is created using a VMA image, the disk type is system disk.
physical_disk	Physical disk, such as iSCSI, FC datastores. To use physical disks, the storage_id parameter is required, which can be obtained from Get Physical Disks
exist_disk	Existing disk
share_disk	Shared disk, requires storage_id parameter. Creating VM using shared disk is not supported.
rdm_disk	RDM disk

4.5.1.6. VM Disk Cache Mode

Table 4.5.6 VM Disk Cache Mode¶

Value	Description
none	Ordinary
writethrough	Write through
writeback	Write back
unsafe	Unsafe
directsync	Directsync

4.5.1.7. VM NIC Type

Table 4.5.7 VM NIC Type¶

Value	Description
e1000	e1000
rtl8139	rtl8139
virtio	virtio
e1000e	e1000e
vmxnet	vmxnet(vmware)
vmxnet2	vmxnet2(vmware)
vmxnet3	vmxnet3(vmware)
flexible	flexible

4.5.1.8. VM Disk Encryption Granularity

Table 4.5.8 VM Disk Encryption Granularity¶

Value	Description
vm	Encrypt the VM
disk	Encrypt a single disk

4.5.1.9. VM Disk Encryption Algorithm

Table 4.5.9 VM Disk Encryption Algorithm¶

Value	Description
aes_256	aes_256
sm4_128	sm4_128

4.5.1.10. VM CPU Type

Table 4.5.10 VM CPU Type¶

Value	Description
core2duo	core2duo
host	host
phenom	phenom

4.5.1.11. VM Working Location Type (VMware)

Table 4.5.11 VM Working Location Type (VMware)¶

Value	Description
vim.ClusterComputeResource	-
vim.HostSystem	-
vim.ResourcePool	-
vim.VirtualApp	-

4.5.1.12. VM Network Type

Table 4.5.12 VM Network Type¶

Value	Description
classic	Classic Network
vpc	VPC

4.5.1.13. VM Platform Type

Table 4.5.13 VM Platform Type¶

Value	Description
hci	hci
vmware	vmware
XYclouds	XYclouds
Aliyun	Aliyun
AWS	AWS
Qcloud	Qcloud

4.5.1.14. VM Power Status

Table 4.5.14 VM Power Status¶

Value	Description
off	Powered-off
on	Powered-on

4.5.2. VM Lifecycle Management

4.5.2.1. Get VM List Information

4.5.2.1.1. Scenario

- Scenario: Query VM list
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the VM Read permission
- Versions supported: SCP 6.3.0 and above

Note

1. Tenants can only query their own VMs.
2. Admin can query all VMs on the platform.

4.5.2.1.2. URI

Table 4.5.15 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/servers	Get VM List Information

- Parameter Description:

Table 4.5.16 Path Parameters¶

Parameter	Location	Type	Description	Required
vmtype	path	string	VM types, view VM Type for details	No
network_type	path	string	Network types, view VM Network Type for details	No
vpc_id	path	string	VPC network ID	No
subnet_id	path	string	Subnet ID	No
floating_ip_address	path	string	Elastic IP address	No
floatingip_id	path	string	Elastic IP ID	No
type	path	string	Platform type, view VM Platform Type for details	No
status	path	string	VM status	No
project_id	path	string	Tenant ID	No
user_id	path	string	User ID	No
az_id	path	string	Resource pool ID	No
host_id	path	string	Physical host ID	No
group_id	path	string	Group ID	No
storage_id	path	string	ID of the storage where VM system disk resides	No
os_type	path	string	Operating system type	No
has_tag	path	int	VM tag. The options are 0 and 1	No
tag_id	path	string	VM tag ID	No
in_protection	path	int	Whether the VM is protected by a DR task.	No
protection_id	path	string	Disaster recovery task ID	No
protection_type	path	string	Disaster recovery task type	No
power_state	path	string	VM power status, view VM Power Status for details	No
scale_group_id	path	string	AS Group ID	No
expire_time	path	string	VM expiration time, parameter format: %Y-%m-%d %H:%M:%S Example: 2021-10-25 23:59:59	No
has_gpu	path	int	Whether it is a GPU-enabled VM	No
gpu_type	path	string	Graphic Card Type	No

Note

For common path parameters, view Common Path Parameter Format for details

4.5.2.1.3. Request

- Request Parameter

  None

- Request Example

https://{SCP_IP}/janus/20180725/servers?page_num=0&page_size=1&project_id=9d14d076b59045b1b0aaca04cfaaa657

4.5.2.1.4. Response

- Response Parameter

Table 4.5.17 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Virtual machine ID	No
name	string	VM name	No
type	string	The type of private cloud platform where VM is located	No
status	string	VM status	No
os_type	string	Operating system type	No
os_name	string	OS	No
description	string	Description	No
uptime	int	Uptime (second)	No
shutdown_duration	int	Downtime (second)	No
ips	list	VM IP information	No
user_id	string	ID of the tenant user where the VM belongs	No
user_name	string	Name of the tenant user where the VM belongs	No
project_id	string	ID of the tenant where the VM belongs	No
project_name	string	Name of the tenant where the VM belongs	No
group_id_path	string	Use ID to indicate node path, separated by /	No
group_name_path	string	Use NAME to indicate node path, separated by /	No
cores_per_socket	int	CPU cores	No
sockets	int	CPU sockets	No
cores	int	CPU cores	No
mhz	int	CPU frequency	No
memory_mb	int	Memory size	No
disks	list	Disk information, view Parameter Description - disks for details	No
networks	list	Network information, view Parameter Description - networks for details	No
az_id	string	Resource pool ID	No
az_name	string	Resource pool name	No
dh_id	string	Scheduling domain ID in the cluster	No
dh_name	string	Scheduling domain name in the cluster	No
host_id	string	ID of the physical host where the VM is running	No
host_name	string	Physical host name	No
image_id	string	Image ID	No
image_name	string	Image name	No

storage_id	string	ID of the storage where VM system disk resides	No
storage_name	string	Name of the storage where VM system disk is located	No
storage_tag_id	string	Storage tag ID	No
group_id	string	VM group ID	No
group_name	string	VM group name	No
cpu_status	dict	CPU status information, view Parameter Description - cpu_status for details	No
memory_status	dict	Memory status information, view Parameter Description - memory_status for details	No
storage_status	dict	Storage status information, view Parameter Description - storage_status for details	No
io_status	dict	IO status information, view Parameter Description - io_status for details	No
network_status	dict	Network status information, view Parameter Description - cpu_status for details	No
advance_param	dict	HCI-specific information, view Parameter Description - advance_param for details	No
os_installed	int	Whether an operating system is installed, 0 indicates not installed, 1 indicates installed	No
vtool_installed	int	Installation status of vmTools. 1 indicates installed, 0 indicates not installed, -1 indicates installation failed, -2 indicates installation is not supported.	No
tags	list	VM tag information, view Parameter Description - tags for details	No
alarm	dict	VM alert information, view Parameter Description - alarm for details	No
warning	dict	Alert summary information, view Paramter Description - warning for details	No
in_protection	int	Whether added to a DR task. 1 indicates yes, 0 indicates no.	No
protection_id	string	Disaster recovery task ID	No
protection_name	string	Disaster recovery task name	No
protection_type	string	Disaster recovery task type	No
protection_enable	int	Whether to enable disaster recovery	No
backup_file_count	int	Number of VM backup files	No
vmtype	string	VM Type	No
iolog	list	IOlog information, view Paramter Description - iolog for details	No
power_state	string	Power status	No
is_stopped	int	VM status: 1 indicates powered-off, 0 indicates others	No
schedule_policies	list	VM scheduling policy information, view Paramter Description - schedule_policies for details	No
location	int	Whether run location of VM is fixed, 1 indicates fixed, 0 indicates unfixed	No
floatingip	dict	Elastic IP information, view Paramter Description - floatingip for details	No

instant_vm	int	Whether the VM is being rapidly recovered	No
storage_mb	int	Total disk size	No
expire_time	string	Expiration date	No
network_type	list	Type of VM network	No
encrypted	int	Whether it is encrypted	No
cipher	string	Encryption algorithm	No
gpu_conf	dict	GPU configuration information, view Parameter Description - gpu_conf for details	No
gpu_status	dict	GPU status information, view Parameter Description - gpu_status for details	No
has_gpu	int	Whether it is a GPU-enabled VM	No
keypair	dict	Key pair information, view Parameter Description - keypair for details	No

Note

View Common Response Parameter Format for public response parameters in the API list.

- Parameter Description - disks

Table 4.5.18 Parameter Description - disks¶

Parameter	Type	Required	Note
size_mb	int	Yes	Disk size (MB), 1024 (1 GB) < size_mb < 66060288 (63 TB)
preallocate	string	No	Disk provisioning method, view Disk Provisioning Method
eagerly_scrub	int	No	Whether to reset now
id	string	No	Disk ID, for example: ide0,ide1...iden
physical_disk_type	string	No	Disk type, view VM Disk Type for details
storage_file	string	No	Storage path
storage_id	string	No	Storage ID
storage_name	string	No	Storage name
storage_tag_id	string	No	Storage Tag ID

- Parameter Description - networks

Table 4.5.19 Parameter Description - networks¶

Parameter	Type	Required	Note
connect	int	Yes	Whether the network is connected.
device_id	string	No	The connected network device ID in classic network
model	string	Yes	NIC type, view VM NIC Type for details
ip_address	string	No	IPv4 address
ipv6_address	string	No	IPv6 address
mac_address	string	Yes	MAC address
vif_id	string	Yes	Network ID (net0, net1, net2, ... netn)
port_id	string	No	Port
name	string	No	Name of the connected port
vpc_id	string	No	VPC ID of the connected tenant
vpc_name	string	No	VPC network name
network_type	string	No	Network type, classic or VPC
subnet_id	string	No	Connected VPC subnet ID
subnet_name	string	No	Name of the connected VPC subnet
cidr	string	No	Subnet CIDR
subnet_gateway_ip	string	No	Gateway of the connected VPC subnet
custom_gateway_ip	string	No	Custom gateway IP

- Parameter Description - cpu_status

Table 4.5.20 Parameter Description - cpu_status¶

Parameter	Type	Value Returned	Note
total_mhz	float	No	Total CPU cores
used_mhz	float	No	Used CPU cores
ratio	float	No	CPU usage

- Parameter Description - memory_status

Table 4.5.21 Parameter Description - memory_status¶

Parameter	Type	Value Returned	Note
total_mb	float	No	Total memory (MB)
used_mb	float	No	Used memory (MB)
ratio	float	No	CPU usage

- Parameter Description - storage_status

Table 4.5.22 Parameter Description - storage_status¶

Parameter	Type	Value Returned	Note
total_mb	float	No	Total storage (MB)
used_mb	float	No	Used memory (MB)
ratio	float	No	CPU usage
storage_file_size_mb	float	No	Storage file size, the actual storage space used (MB)

- Parameter Description - io_status

Table 4.5.23 Parameter Description - io_status¶

Parameter	Type	Value Returned	Note
read_speed_byteps	float	No	IO read speed (byte/s)
write_speed_byteps	float	No	IO write speed (byte/s)
read_iops	float	No	IO reads (times/s)
write_iops	float	No	I/O writes (times/s)

- Parameter Description - cpu_status

Table 4.5.24 Parameter Description - cpu_status¶

Parameter	Type	Value Returned	Note
read_speed_bitps	float	No	Network read speed (bit/s)
write_speed_bitps	float	No	Network write speed (bit/s)

- Parameter Description - advance_param

Table 4.5.25 Parameter Description - advance_param¶

Parameter	Type	Value Returned	Note
balloon_memory	int	No	Whether to enable the memory recycling mechanism: 0 indicates not enabled, 1 indicates enabled
onboot	int	No	Whether to run automatically at startup. 0 indicates no, 1 indicates yes.
schedopt	int	No	Whether it is a critical VM. 0 indicates no, 1 indicates yes.
abnormal_recovery	int	No	Whether to reboot if fault occurs. 0 indicates no, 1 indicates yes.
vga	string	No	Types of built-in graphic card in VM

- Parameter Description - tags

Table 4.5.26 Parameter Description - tags¶

Parameter	Type	Value Returned	Note
id	string	No	Tag ID
name	string	No	Tag name

- Parameter Description - alarm

Table 4.5.27 Parameter Description - alarm¶

Parameter	Type	Value Returned	Note
alarm	int	No	Alert status, 0 indicates no alert, 1 indicates an alert exists
alarm_info	list	No	Alert details, view Parameter Description - alarm-alarm_info

- Parameter Description - alarm-alarm_info

Table 4.5.28 Parameter Description - alarm-alarm_info

Parameter	Type	Value Returned	Note
alarm_role_type	string	No	User role of alert
severity_level	string	No	Severity
object_type	string	No	Alert object type
sub_object_type	string	No	Special alert object type
cluster_id	string	No	Cluster ID of alert
az_name	string	No	Resource pool name
az_id	string	No	Resource pool ID
id	string	No	Alert ID
user_id	string	No	User ID of alert
title	string	No	Alert event
object_id	string	No	Alert object ID
cluster_name	string	No	Cluster name of alert
sub_object_name	string	No	Special alert object name
object_name	string	No	Alert object name
project_id	string	No	Tenant ID
status	string	No	Status
resource_project_name	string	No	Tenant name of alert object
description	string	No	Description
root_id	string	No	Alert group ID
resource_project_id	string	No	Tenant ID of alert object
begin_time	string	No	Alert start time
alarm_id	string	No	Alert ID
end_time	string	No	Alert end time
resource_user_id	string	No	User ID of alert object

- Paramter Description - warning

Table 4.5.29 Paramter Description - warning

Parameter	Type	Value Returned	Note
warning	int	No	Alert status, 0 indicates no alert, 1 indicates an alert exists
warning_type	string	No	Alert type
warning_info	string	No	Alert description

- Paramter Description - iolog

Table 4.5.30 Paramter Description - iolog¶

Parameter	Type	Value Returned	Note
size_byte	float	No	Actual file size of iolog (byte)
storage_id	string	No	Storage ID

- Paramter Description - schedule_policies

Table 4.5.31 Paramter Description - schedule_policies¶

Parameter	Type	Value Returned	Note
id	string	No	Scheduling policy ID
az_id	string	No	Resource pool ID
project_id	string	No	Tenant ID
name	string	No	Scheduling policy name
type	string	No	Scheduling policy type
resource_id	string	No	External ID of Scheduling policy
status	string	No	Scheduling policy status
enable	int	No	Whether the scheduling policy is enabled
force	int	No	Whether the scheduling policy is forcibly enabled

- Paramter Description - floatingip

Table 4.5.32 Paramter Description - floatingip¶

Parameter	Type	Value Returned	Note
floatingip_id	string	No	Elastic IP ID
floating_ip_address	string	No	Elastic IP address
bind_status	string	No	Elastic IP binding status
line_type	string	No	Line type associated with elastic IP
line_type_id	string	No	ID of line type ID associated with elastic IP
bandwidth	int	No	Bandwidth associated with elastic IP
sharedbandwidth_name	string	No	Shared bandwidth name associated with elastic IP
sharedbandwidth_id	string	No	Shared bandwidth ID

- Parameter Description - gpu_conf

Table 4.5.33 Parameter Description - gpu_conf¶

Parameter	Type	Value Returned	Note
auth_require	string	No	Licensing mode
gpu_cores	int	No	Number of cores per GPU
gpu_mem_size_single_core	string	No	Graphics memory size per GPU core
gpu_name	string	No	Graphic card name
gpu_type	string	No	Graphic Card Type
total_count	int	No	Total number of graphics cards used by the VM
vgpu_mem_size	int	No	Total graphics memory used by the VM
vgpu_type	string	No	vGPU type
schedulers	list	No	Graphics card scheduler

- Parameter Description - gpu_status

Table 4.5.34 Parameter Description - gpu_status¶

Parameter	Type	Value Returned	Note
graphics_count	int	No	Number of GPUs
graphics_ratio	int	No	GPU usage
graphics_mem_ratio	float	No	Graphics memory usage
graphics_mem_total	float	No	Total Graphics memory size
graphics_mem_used	float	No	Used graphics memory size

- Parameter Description - keypair

Table 4.5.35 Parameter Description - keypair¶

Parameter	Type	Value Returned	Note
id	string	No	Key pair UUID
name	string	No	Key pair name
type	string	No	Key pair type
user_id	string	No	User ID
public_key	string	No	Public key information
created_at	string	No	Time created
fingerprint	string	No	Key fingerprint information

- Response Example:

```
[
    {
        "network_status": {
            "write_speed_bitps": 0,
```

```
            "read_speed_bitps": 0
        },
        "expire_time": "unlimited",
        "cpu_status": {
            "total_mhz": 4202,
            "ratio": 0,
            "used_mhz": 0
        },
        "storage_mb": 113664,
        "group_name": "Database",
        "os_installed": 2,
        "keypair": null,
        "warning": {
            "warning_info": "",
            "warning": 0,
            "warning_type": ""
        },
        "gpu_conf": null,
        "az_name": "1.75",
        "instant_vm": "",
        "networks": [
            {
                "ipv6_address": "",
                "name": null,
                "vif_id": "net0",
                "subnet_id": "",
                "vpc_name": "",
                "custom_gateway_ip": null,
                "cidr": "",
                "subnet_gateway_ip": "",
                "subnet_name": "",
                "connect": 1,
                "port_id": "",
                "mac_address": "FE:FC:FE:B2:C2:D8",
                "vpc_id": "",
                "model": "virtio",
                "ip_address": "172.32.0.110",
                "network_type": "",
                "device_id": null
            },
            {
                "ipv6_address": "",
                "name": "Subnet1",
                "vif_id": "net1",
                "subnet_id": "",
                "vpc_name": "",
                "custom_gateway_ip": null,
                "cidr": "",
                "subnet_gateway_ip": "",
                "subnet_name": "",
                "connect": 1,
                "port_id": "",
                "mac_address": "FE:FC:FE:73:EB:9C",
                "vpc_id": "",
                "model": "virtio",
                "ip_address": "192.168.0.113",
                "network_type": "",
                "device_id": "dvs3fbedac"
            }
        ],
        "az_id": "611f1b34-c727-4925-854e-07bc8607f681",
        "storage_tag_id": "22222222-2222-2222-2222-222222222222",
        "is_stopped": 1,
        "uptime": 0,
        "user_id": null,
        "vmtype": "dmp_mysql",
        "floatingip": {
            "sharedbandwidth_name": "",
            "floatingip_id": "",
            "bandwidth": 0,
            "line_type": "",
            "bind_status": "",
            "floating_ip_address": "",
            "line_type_id": "",
            "sharedbandwidth_id": ""
        },
        "protection_enable": "",
        "image_id": "7965a8e1-4cd8-4b62-988d-18c6ee17c4bf",
        "tags": [],
        "storage_name": "data-from-60",
        "gpu_status": {
            "graphics_mem_total": 0,
            "graphics_count": 0,
            "graphics_mem_used": 0,
            "graphics_mem_ratio": 0,
            "graphics_ratio": 0
        },
        "id": "d6ae5e1f-9ebf-42e2-90d5-2e0e0881ad95"
```

```
        "id": "d6ee5e1f-9abf-43e3-90d5-2a0e9881ad95",
        "vtool_installed": 1,
        "advance_param": {
            "schedopt": 0,
            "vga": "cirrus",
            "abnormal_recovery": 0,
            "onboot": 0,
            "balloon_memory": 1
        },
        "location": 0,
        "power_state": "off",
        "storage_id": "1SANGFOR_ASAN-00c-b4f6-42b8-938c-00a7c09d6f72",
        "in_protection": 0,
        "project_id": "9d14d076b59045b1b0aaca04cfaaa657",
        "user_name": "",
        "schedule_policies": [],
        "io_status": {
            "read_speed_byteps": 0,
            "read_iops": 0,
            "write_speed_byteps": 0,
            "write_iops": 0
        },
        "project_name": "tjc_tenant",
        "sockets": 2,
        "description": "",
        "group_id_path": "f3f9d08f-f5ec-4b85-bb7f-c542eee0e02c",
        "memory_status": {
            "used_mb": 0,
            "total_mb": 2048,
            "ratio": 0
        },
        "group_name_path": "Database",
        "image_name": "DBVM-1207",
        "has_gpu": 0,
        "dh_id": "",
        "protection_name": "",
        "protection_type": "",
        "protection_id": "",
        "shutdown_duration": 357762,
        "host_id": "cluster",
        "status": "stopped",
        "dh_name": "",
        "name": "dbvm-test-cjl-1212",
        "backup_file_count": 0,
        "ips": [
            "172.32.0.110",
            "192.168.0.113"
        ],
        "type": "hci",
        "alarm": {
            "alarm_info": [],
            "alarm": 0
        },
        "disks": [
            {
                "storage_tag_id": "22222222-2222-2222-2222-222222222222",
                "preallocate": 0,
                "storage_name": "data-from-60",
                "eagerly_scrub": 0,
                "size_mb": 102400,
                "storage_id": "1SANGFOR_ASAN-00c-b4f6-42b8-938c-00a7c09d6f72
",
                "storage_file": "vm-disk-1.qcow2",
                "id": "ide0",
                "physical_disk_type": ""
            },
            {
                "storage_tag_id": "22222222-2222-2222-2222-222222222222",
                "preallocate": 0,
                "storage_name": "data-from-60",
                "eagerly_scrub": 0,
                "size_mb": 11264,
                "storage_id": "1SANGFOR_ASAN-00c-b4f6-42b8-938c-00a7c09d6f72
",
                "storage_file": "vm-disk-2.qcow2",
                "id": "ide1",
                "physical_disk_type": ""
            }
        ],
        "cores_per_socket": 1,
        "network_type": [
            "classic"
        ],
        "memory_mb": 2048,
        "storage_status": {
            "used_mb": 3035.59,
            "total_mb": 113664,
            "storage_file_size_mb": 611
```

```
            "storage_file_size_mb": 611,
            "ratio": 0.03
        },
        "cipher": "",
        "encrypted": 0,
        "host_name": "<Auto>",
        "cores": 2,
        "os_type": "l2664",
        "group_id": "f3f9d08f-f5ec-4b85-bb7f-c542eee0e02c",
        "mhz": 4202,
        "os_name": "",
        "iolog": []
    }
]
```

- OK: 200
- Errors: 400, 403, 500

4.5.2.2. Get VM List Information

4.5.2.2.1. Scenario

- Scenario: Query VM list
- Applicable roles: Super administrator, system administrator, tenant, and roles with VM read permission.
- Versions supported: SCP 6.8.0 and above

> **Note**
>
> 1. Tenants can only query their own VMs.
> 2. Admin can query all VMs on the platform.

4.5.2.2.2. URI

Table 4.5.36 API Path and Request Method

REST VERB	PATH	Description
GET	/janus/20190725/servers	Get VM List Information

- Parameter Description:

Table 4.5.37 Path Parameters¶

Parameter	Location	Type	Description	Required
vmtype	path	string	VM types, view VM Type for details	No
network_type	path	string	Network types, view VM Network Type for details	No
vpc_id	path	string	VPC network ID	No
subnet_id	path	string	Subnet ID	No
floating_ip_address	path	string	Elastic IP address	No
floatingip_id	path	string	Elastic IP ID	No
type	path	string	Platform type, view VM Platform Type for details	No
status	path	string	VM status	No
project_id	path	string	Tenant ID	No
user_id	path	string	User ID	No
az_id	path	string	Resource pool ID	No
host_id	path	string	Physical host ID	No
group_id	path	string	Group ID	No
storage_id	path	string	ID of the storage where VM system disk resides	No
os_type	path	string	Operating system type	No
has_tag	path	string	VM tag	No
tag_id	path	string	VM tag ID	No
in_protection	path	int	Whether the VM is protected by a DR task.	No
protection_id	path	string	Disaster recovery task ID	No
protection_type	path	string	Disaster recovery task type	No
power_state	path	string	VM power status, view VM Power Status for details	No
scale_group_id	path	string	AS Group ID	No
expire_time	path	string	VM expiration time, parameter format: %Y-%m-%d %H:%M:%S Example: 2021-10-25 23:59:59	No
has_gpu	path	int	Whether it is a GPU-enabled VM	No
gpu_type	path	string	Graphic Card Type	No
vm_ids	path	string	List of VM IDs (This parameter was added on July 25, 2019 for parameter query.)	No
storage_ids	path	string	List of storage IDs (This parameter was added on July 25, 2019 for parameter query.)	No
ids	path	string	List of VM IDs where IDs are separated by commas (,) (This parameter was added on July 25, 2019 for parameter query.)	No

4.5.2.2.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20190725/servers?page_num=0&page_size=1&project_id=9d14d076b59045b1b0aaca04cfaaa657

4.5.2.2.4. Response

- Response Parameter

Table 4.5.38 Response Parameter

Parameter	Type	Description	Value Returned
id	string	Virtual machine ID	No
name	string	VM name	No
type	string	The type of private cloud platform where VM is located	No
status	string	VM status	No
os_type	string	Operating system type	No
os_name	string	OS	No
description	string	Description	No
uptime	int	Uptime (second)	No
shutdown_duration	int	Downtime (second)	No
ips	list	VM IP information	No
user_id	string	ID of the tenant user where the VM belongs	No
user_name	string	Name of the tenant user where the VM belongs	No
project_id	string	ID of the tenant where the VM belongs	No
project_name	string	Name of the tenant where the VM belongs	No
group_id_path	string	Use ID to indicate node path, separated by /	No
group_name_path	string	Use NAME to indicate node path, separated by /	No
cores_per_socket	int	CPU cores	No
sockets	int	CPU sockets	No
cores	int	CPU cores	No
mhz	int	CPU frequency	No
memory_mb	int	Memory size	No
disks	list	View Parameter Description - disks for disk information.	No
networks	list	View Parameter Description - networks for network information.	No
az_id	string	Resource pool ID	No
az_name	string	Resource pool name	No
dh_id	string	Scheduling domain ID in the cluster	No
dh_name	string	Scheduling domain name in the cluster	No

host_id	string	ID of the physical host where the VM is running	No
host_name	string	Physical host name	No
image_id	string	Image ID	No
image_name	string	Image name	No
storage_id	string	ID of the storage where VM system disk resides	No
storage_name	string	Name of the storage where VM system disk is located	No
storage_tag_id	string	Storage tag ID	No
group_id	string	VM group ID	No
group_name	string	VM group name	No
cpu_status	dict	View Parameter Description - cpu_status for CPU status information.	No
memory_status	dict	View Parameter Description - memory_status for memory status information.	No
storage_status	dict	View Parameter Description - storage_status for storage status information.	No
io_status	dict	View Parameter Description - io_status for I/O status information.	No
network_status	dict	View Parameter Description - cpu_status for network status information.	No
advance_param	dict	View Parameter Description - advance_param for HCI unique feature information.	No
os_installed	int	Whether an operating system is installed, 0 indicates not installed, 1 indicates installed	No
vtool_installed	int	Installation status of vmTools. 1 indicates installed, 0 indicates not installed, -1 indicates installation failed, -2 indicates installation is not supported.	No
tags	list	View Parameter Description - tags for VM tag information.	No
alarm	dict	View Parameter Description - alarm for VM alert information.	No
warning	dict	View Paramter Description - warning for alert overview information.	No
in_protection	int	Whether added to a DR task. 1 indicates yes, 0 indicates no.	No
protection_id	string	Disaster recovery task ID	No
protection_name	string	Disaster recovery task name	No
protection_type	string	Disaster recovery task type	No
protection_enable	int	Number of VM backup files	No
backup_file_count	int	Number of VM backup files	No
vmtype	string	VM Type	No
iolog	list	View Paramter Description - iolog for I/O log information.	No
power_state	string	Power status	No
is_stopped	int	VM status: 1 indicates powered-off, 0 indicates others	No
schedule_policies	list	View Paramter Description - schedule_policies for VM scheduling policy information.	No

location	int	Whether run location of VM is fixed, 1 indicates fixed, 0 indicates unfixed	No
floatingip	dict	View Paramter Description - floatingip for Elastic IP information.	No
instant_vm	int	Whether the VM is being rapidly recovered	No
storage_mb	int	Total disk size	No
expire_time	string	Expiration date	No
network_type	list	Type of VM network	No
encrypted	int	Whether it is encrypted	No
cipher	string	Encryption algorithm	No
gpu_conf	dict	View Parameter Description - gpu_conf for GPU configuration details.	No
gpu_status	dict	View Parameter Description - gpu_status for GPU status information.	No
has_gpu	int	Whether it is a GPU-enabled VM	No
keypair	dict	View Parameter Description - keypair for key pair information.	No
os_option	string	View Parameter Description - os_option for operating system information.	No
template_id	string	Template ID	No
storage_policy_id	string	Storage policy ID	No
vm_id	string	VM ID before conversion	No
quotas	list	View Parameter Description - quota_info for the list of dynamic quotas.	No

Note

For common response parameters, view Response Format for details

- Parameter Description - disks

Table 4.5.39 Parameter Description - disks¶

Parameter	Type	Required	Note
size_mb	int	Yes	Disk size (MB), 1024 (1 GB) < size_mb < 66060288 (63 TB)
preallocate	string	No	Disk provisioning method, view Disk Provisioning Method
eagerly_scrub	int	No	Whether to reset now
id	string	No	Disk ID, for example: ide0,ide1...iden
type	string	No	Disk type, view VM Disk Type for details
storage_file	string	No	Storage path
storage_id	string	No	Storage ID
storage_name	string	No	Storage name
storage_tag_id	string	No	Storage Tag ID

- Parameter Description - networks

Table 4.5.40 Parameter Description - networks

Parameter	Type	Required	Note
connect	int	Yes	Whether the network is connected.
device_id	string	No	The connected network device ID in classic network
model	string	No	NIC type, view VM NIC Type for details
ip_address	string	No	IPv4 address
ipv6_address	string	No	IPv6 address
mac_address	string	No	MAC address, which can be automatically generated without transmission
vif_id	string	No	Network ID (net0, net1, net2, ... netn)
port_id	string	No	Port, connected to classic network device, currently set to 12345678
name	string	No	Name of the connected port
vpc_id	string	No	VPC network ID of the connected tenant, which can be obtained from Query VPC s
vpc_name	string	No	VPC network name
network_type	string	No	Network type, classic or VPC
subnet_id	string	No	Connected VPC subnet ID
subnet_name	string	No	Name of the connected VPC subnet
cidr	string	No	Subnet CIDR
subnet_gateway_ip	string	No	Gateway of the connected VPC subnet
custom_gateway_ip	string	No	Custom gateway IP

- Parameter Description - cpu_status

Table 4.5.41 Parameter Description - cpu_status

Parameter	Type	Value Returned	Note
total_mhz	float	No	Total CPU cores
used_mhz	float	No	Used CPU cores
ratio	float	No	CPU usage

- Parameter Description - memory_status

Table 4.5.42 Parameter Description - memory_status

Parameter	Type	Value Returned	Note
total_mb	float	No	Total memory (MB)
used_mb	float	No	Used memory (MB)
ratio	float	No	CPU usage

- Parameter Description - storage_status

Table 4.5.43 Parameter Description - storage_status

Parameter	Type	Value Returned	Note
total_mb	float	No	Total storage (MB)
used_mb	float	No	Used memory (MB)
ratio	float	No	CPU usage
storage_file_size_mb	float	No	Storage file size, the actual storage space used (MB)

- Parameter Description - io_status

Table 4.5.44 Parameter Description - io_status

Parameter	Type	Value Returned	Note
read_speed_byteps	float	No	IO read speed (byte/s)
write_speed_byteps	float	No	IO write speed (byte/s)
read_iops	float	No	IO reads (times/s)
write_iops	float	No	IO reads (times/s)

- Parameter Description - cpu_status

Table 4.5.45 Parameter Description - cpu_status

Parameter	Type	Value Returned	Note
read_speed_bitps	float	No	Network read speed (bit/s)
write_speed_bitps	float	No	Network write speed (bit/s)

- Parameter Description - advance_param

Table 4.5.46 Parameter Description - advance_param

Parameter	Type	Value Returned	Note
balloon_memory	int	No	Whether to enable the memory recycling mechanism: 0 indicates not enabled, 1 indicates enabled
onboot	int	No	Whether to run automatically at startup. 0 indicates no, 1 indicates yes.
schedopt	int	No	Whether it is a critical VM. 0 indicates no, 1 indicates yes.
abnormal_recovery	int	No	Whether to reboot if fault occurs. 0 indicates no, 1 indicates yes.
vga	string	No	Types of built-in graphic card in VM

- Parameter Description - tags

Table 4.5.47 Parameter Description - tags¶

Parameter	Type	Value Returned	Note
id	string	No	Tag ID
name	string	No	Tag name

- Parameter Description - alarm

Table 4.5.48 Parameter Description - alarm¶

Parameter	Type	Value Returned	Note
alarm	int	No	Alert status, 0 indicates no alert, 1 indicates an alert exists
alarm_info	list	No	View Parameter Description - alarm-alarm_info for alert details.

- Parameter Description - alarm-alarm_info

Table 4.5.49 Parameter Description - alarm-alarm_info

Parameter	Type	Value Returned	Note
alarm_role_type	string	No	User role of alert
severity_level	string	No	Severity
object_type	string	No	Alert object type
sub_object_type	string	No	Special alert object type
cluster_id	string	No	Cluster ID of alert
az_name	string	No	Resource pool name
az_id	string	No	Resource pool ID
id	string	No	Alert ID
user_id	string	No	User ID of alert
title	string	No	Alert event
object_id	string	No	Alert object ID
cluster_name	string	No	Cluster name of alert
sub_object_name	string	No	Special alert object name
object_name	string	No	Alert object name
project_id	string	No	Tenant ID
status	string	No	Status
resource_project_name	string	No	Tenant name of alert object
description	string	No	Description
root_id	string	No	Alert group ID
resource_project_id	string	No	Tenant ID of alert object
begin_time	string	No	Alert start time
alarm_id	string	No	Alert ID
end_time	string	No	Alert end time
resource_user_id	string	No	User ID of alert object

- Paramter Description - warning

Table 4.5.50 Paramter Description - warning

Parameter	Type	Value Returned	Note
warning	string	No	Alert status, 0 indicates no alert, 1 indicates an alert exists
warning_type	string	No	Alert type
warning_info	string	No	Alert description

- Paramter Description - iolog

Table 4.5.51 Paramter Description - iolog¶

Parameter	Type	Value Returned	Note
size_byte	float	No	Actual file size of iolog (byte)
storage_id	string	No	Storage ID

- Paramter Description - schedule_policies

Table 4.5.52 Paramter Description - schedule_policies¶

Parameter	Type	Value Returned	Note
id	string	No	Scheduling policy ID
az_id	string	No	Resource pool ID
project_id	string	No	Tenant ID
name	string	No	Scheduling policy name
type	string	No	Scheduling policy type
resource_id	string	No	External ID of Scheduling policy
status	string	No	Scheduling policy status
enable	int	No	Whether the scheduling policy is enabled
force	int	No	Whether the scheduling policy is forcibly enabled

- Paramter Description - floatingip

Table 4.5.53 Paramter Description - floatingip¶

Parameter	Type	Value Returned	Note
floatingip_id	string	No	Elastic IP ID
floating_ip_address	string	No	Bind Elastic IP
bind_status	string	No	Elastic IP binding status
line_type	string	No	Line type associated with elastic IP
line_type_id	string	No	ID of line type ID associated with elastic IP
bandwidth	int	No	Bandwidth associated with elastic IP
sharedbandwidth_name	string	No	Shared bandwidth name associated with elastic IP
sharedbandwidth_id	string	No	Shared bandwidth ID

- Parameter Description - gpu_conf

Table 4.5.54 Parameter Description - gpu_conf¶

Parameter	Type	Value Returned	Note
auth_require	string	No	Licensing mode
gpu_cores	int	No	Number of cores per GPU
gpu_mem_size_single_core	string	No	Graphics memory size per GPU core
gpu_name	string	No	Graphic card name
gpu_type	string	No	Graphic Card Type
total_count	int	No	Total number of graphics cards used by the VM
vgpu_mem_size	int	No	Total graphics memory used by the VM
vgpu_type	string	No	vGPU type
schedulers	list	No	Graphics card scheduler

- Parameter Description - gpu_status

Table 4.5.55 Parameter Description - gpu_status¶

Parameter	Type	Value Returned	Note
graphics_count	int	No	Number of GPUs
graphics_ratio	int	No	GPU usage
graphics_mem_ratio	float	No	Graphics memory usage
graphics_mem_total	float	No	Total Graphics memory size
graphics_mem_used	float	No	Used graphics memory size

- Parameter Description - keypair

Table 4.5.56 Parameter Description - keypair¶

Parameter	Type	Value Returned	Note
id	string	No	Key pair UUID
name	string	No	Key pair name
type	string	No	Key pair type
user_id	string	No	User ID
public_key	string	No	Public key information
created_at	string	No	Time created
fingerprint	string	No	Key fingerprint information

- Parameter Description - os_option

Table 4.5.57 Parameter Description - os_option¶

Parameter	Type	Value Returned	Note
kernel_name	string	No	Kernel name
distribution_name	string	No	Release version
arch	string	No	CPU architecture system

- Parameter Description - quota_info

Table 4.5.58 Parameter Description - quota_info¶

Parameter	Type	Value Returned	Note
resource	string	Yes	IDs of the quota resources, which are separated by colons (;)
value	int	Yes	Quantity

- Response Example:


```
[
    {
        "network_status": {
            "write_speed_bitps": 0,
            "read_speed_bitps": 0
        },
        "expire_time": "unlimited",
        "cpu_status": {
            "total_mhz": 4202,
            "ratio": 0,
            "used_mhz": 0
        },
        "storage_mb": 113664,
        "group_name": "Database",
        "os_installed": 2,
        "keypair": null,
        "warning": {
            "warning_info": "",
            "warning": 0,
            "warning_type": ""
        },
        "gpu_conf": null,
        "az_name": "1.75",
        "instant_vm": "",
        "networks": [
            {
                "ipv6_address": "",
                "name": null,
                "vif_id": "net0",
                "subnet_id": "",
                "vpc_name": "",
                "custom_gateway_ip": null,
                "cidr": "",
                "subnet_gateway_ip": "",
                "subnet_name": "",
                "connect": 1,
                "port_id": "",
                "mac_address": "FE:FC:FE:B2:C2:D8",
                "vpc_id": "",
                "model": "virtio",
                "ip_address": "172.32.0.110",
                "network_type": "",
                "device_id": null
            },
            {
                "ipv6_address": "",
                "name": "Subnet 1",
                "vif_id": "net1",
                "subnet_id": "",
                "vpc_name": "",
                "custom_gateway_ip": null,
                "cidr": "",
                "subnet_gateway_ip": "",
                "subnet_name": "",
                "connect": 1,
```

```
            "port_id": "",
            "mac_address": "FE:FC:FE:73:EB:9C",
            "vpc_id": "",
            "model": "virtio",
            "ip_address": "192.168.0.113",
            "network_type": "",
            "device_id": "dvs3fbedac"
        }
    ],
    "az_id": "611f1b34-c727-4925-854e-07bc8607f681",
    "storage_tag_id": "22222222-2222-2222-2222-222222222222",
    "is_stopped": 1,
    "uptime": 0,
    "user_id": null,
    "vmtype": "dmp_mysql",
    "floatingip": {
        "sharedbandwidth_name": "",
        "floatingip_id": "",
        "bandwidth": 0,
        "line_type": "",
        "bind_status": "",
        "floating_ip_address": "",
        "line_type_id": "",
        "sharedbandwidth_id": ""
    },
    "protection_enable": "",
    "image_id": "7965a8e1-4cd8-4b62-988d-18c6ee17c4bf",
    "tags": [],
    "storage_name": "data-from-60",
    "gpu_status": {
        "graphics_mem_total": 0,
        "graphics_count": 0,
        "graphics_mem_used": 0,
        "graphics_mem_ratio": 0,
        "graphics_ratio": 0
    },
    "id": "d6ee5e1f-9abf-43e3-90d5-2a0e9881ad95",
    "vtool_installed": 1,
    "advance_param": {
        "schedopt": 0,
        "vga": "cirrus",
        "abnormal_recovery": 0,
        "onboot": 0,
        "balloon_memory": 1
    },
    "location": 0,
    "power_state": "off",
    "storage_id": "1SANGFOR_ASAN-00c-b4f6-42b8-938c-00a7c09d6f72",
    "in_protection": 0,
    "project_id": "9d14d076b59045b1b0aaca04cfaaa657",
    "user_name": "",
    "schedule_policies": [],
    "io_status": {
        "read_speed_byteps": 0,
        "read_iops": 0,
        "write_speed_byteps": 0,
        "write_iops": 0
    },
    "project_name": "tjc_tenant",
    "sockets": 2,
    "description": "",
    "group_id_path": "f3f9d08f-f5ec-4b85-bb7f-c542eee0e02c",
    "memory_status": {
        "used_mb": 0,
        "total_mb": 2048,
        "ratio": 0
    },
    "group_name_path": "Database",
    "image_name": "DBVM-1207",
    "has_gpu": 0,
    "dh_id": "",
    "protection_name": "",
    "protection_type": "",
    "protection_id": "",
    "shutdown_duration": 357762,
    "host_id": "cluster",
    "status": "stopped",
    "dh_name": "",
    "name": "dbvm-test-cjl-1212",
    "backup_file_count": 0,
    "ips": [
        "172.32.0.110",
        "192.168.0.113"
    ],
    "type": "hci",
    "alarm": {
        "alarm_info": [],
        "alarm": 0
```

```
            "alarm": 0
        },
        "disks": [
            {
                "storage_tag_id": "22222222-2222-2222-2222-222222222222",
                "preallocate": 0,
                "storage_name": "data-from-60",
                "eagerly_scrub": 0,
                "size_mb": 102400,
                "storage_id": "1SANGFOR_ASAN-00c-b4f6-42b8-938c-00a7c09d6f72
",
                "storage_file": "vm-disk-1.qcow2",
                "id": "ide0",
                "physical_disk_type": ""
            },
            {
                "storage_tag_id": "22222222-2222-2222-2222-222222222222",
                "preallocate": 0,
                "storage_name": "data-from-60",
                "eagerly_scrub": 0,
                "size_mb": 11264,
                "storage_id": "1SANGFOR_ASAN-00c-b4f6-42b8-938c-00a7c09d6f72
",
                "storage_file": "vm-disk-2.qcow2",
                "id": "ide1",
                "physical_disk_type": ""
            }
        ],
        "cores_per_socket": 1,
        "network_type": [
            "classic"
        ],
        "memory_mb": 2048,
        "storage_status": {
            "used_mb": 3035.59,
            "total_mb": 113664,
            "storage_file_size_mb": 611,
            "ratio": 0.03
        },
        "cipher": "",
        "encrypted": 0,
        "host_name": "<Auto select>",
        "cores": 2,
        "os_type": "l2664",
        "group_id": "f3f9d08f-f5ec-4b85-bb7f-c542eee0e02c",
        "mhz": 4202,
        "os_name": "",
        "iolog": [],
        "os_option": {
            "kernel_name": "",
            "distribution_name": "",
            "arch": "",
        },
        "template_id": "23879b10-019d-4a3a-a514-809b4abb1912",
        "storage_policy_id": "2325526a-f793-4b47-97e6-2cde1e4e9892",
        "vm_id": "e4e784b7-7906-4107-b6ba-7fc535c90208",
        "quotas": [
            {
                "resource": "",
                "value": 5,
            }
        ]
    }
]
```

- OK: 200
- Errors: 400, 403, 500

4.5.2.3. Get VM List Information

4.5.2.3.1. Scenario

- Scenario: Query VM list
- Applicable roles: Super administrator, system administrator, tenant, and roles with VM read permission.
- Versions supported: SCP 6.8.0 and above

Note

1. Tenants can only query their own VMs.
2. Admin can query all VMs on the platform.

4.5.2.3.2. URI

Table 4.5.59 API Path and Request Method

REST VERB	PATH	Description
GET	/janus/20200725/servers	Get VM List Information

- Parameter Description:

Table 4.5.60 Path Parameters¶

Parameter	Location	Type	Description	Required
vmtype	path	string	VM types, view VM Type for details	No
network_type	path	string	Network types, view VM Network Type for details	No
vpc_id	path	string	VPC network ID	No
subnet_id	path	string	Subnet ID	No
floating_ip_address	path	string	Elastic IP address	No
floatingip_id	path	string	Elastic IP ID	No
type	path	string	Platform type, view VM Platform Type for details	No
status	path	string	VM status	No
project_id	path	string	Tenant ID	No
user_id	path	string	User ID	No
az_id	path	string	Resource pool ID	No
host_id	path	string	Physical host ID	No
group_id	path	string	Group ID	No
storage_id	path	string	ID of the storage where VM system disk resides	No
os_type	path	string	Operating system type	No
has_tag	path	string	VM tag	No
tag_id	path	string	VM tag ID	No
in_protection	path	int	Whether the VM is protected by a DR task.	No
protection_id	path	string	Disaster recovery task ID	No
protection_type	path	string	Disaster recovery task type	No
power_state	path	string	VM power status, view VM Power Status for details	No
scale_group_id	path	string	AS Group ID	No
expire_time	path	string	VM expiration time, parameter format: %Y-%m-%d %H:%M:%S Example: 2021-10-25 23:59:59	No
has_gpu	path	int	Whether it is a GPU-enabled VM	No
gpu_type	path	string	Graphic Card Type	No
vm_ids	path	string	List of VM IDs (This parameter was added on July 25, 2020 for parameter query.)	No
storage_ids	path	string	List of storage IDs (This parameter was added on July 25, 2020 for parameter query.)	No
ids	path	string	List of VM IDs where IDs are separated by commas (,) (This parameter was added on July 25, 2020 for parameter query.)	No

4.5.2.3.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20200725/servers?page_num=0&page_size=1&project_id=9d14d076b59045b1b0aaca04cfaaa657

4.5.2.3.4. Response

- Response Parameter

Table 4.5.61 Response Parameter

Parameter	Type	Description	Value Returned
id	string	Virtual machine ID	No
name	string	VM name	No
type	string	The type of private cloud platform where VM is located	No
status	string	VM status	No
os_type	string	Operating system type	No
os_name	string	OS	No
description	string	Description	No
uptime	int	Uptime (second)	No
shutdown_duration	int	Downtime (second)	No
ips	list	VM IP information	No
user_id	string	ID of the tenant user where the VM belongs	No
user_name	string	Name of the tenant user where the VM belongs	No
project_id	string	ID of the tenant where the VM belongs	No
project_name	string	Name of the tenant where the VM belongs	No
group_id_path	string	Use ID to indicate node path, separated by /	No
group_name_path	string	Use NAME to indicate node path, separated by /	No
cores_per_socket	int	CPU cores	No
sockets	int	CPU sockets	No
cores	int	CPU cores	No
mhz	int	CPU frequency	No
memory_mb	int	Memory size	No
disks	list	View Parameter Description - disks for disk information.	No
networks	list	View Parameter Description - networks for network information.	No
az_id	string	Resource pool ID	No
az_name	string	Resource pool name	No
dh_id	string	Scheduling domain ID in the cluster	No
dh_name	string	Scheduling domain name in the cluster	No

host_id	string	ID of the physical host where the VM is running	No
host_name	string	Physical host name	No
image_id	string	Image ID	No
image_name	string	Image name	No
storage_id	string	ID of the storage where VM system disk resides	No
storage_name	string	Name of the storage where VM system disk is located	No
storage_tag_id	string	Storage tag ID	No
group_id	string	VM group ID	No
group_name	string	VM group name	No
cpu_status	dict	View Parameter Description - cpu_status for CPU status information.	No
memory_status	dict	View Parameter Description - memory_status for memory status information.	No
storage_status	dict	View Parameter Description - storage_status for storage status information.	No
io_status	dict	View Parameter Description - io_status for I/O status information.	No
network_status	dict	View Parameter Description - cpu_status for network status information.	No
advance_param	dict	View Parameter Description - advance_param for HCI unique feature information.	No
os_installed	int	Whether an operating system is installed, 0 indicates not installed, 1 indicates installed	No
vtool_installed	int	Installation status of vmTools. 1 indicates installed, 0 indicates not installed, -1 indicates installation failed, -2 indicates installation is not supported.	No
tags	list	View Parameter Description - tags for VM tag information.	No
alarm	dict	View Parameter Description - alarm for VM alert information.	No
warning	dict	View Paramter Description - warning for alert overview information.	No
in_protection	int	Whether added to a DR task. 1 indicates yes, 0 indicates no.	No
protection_id	string	Disaster recovery task ID	No
protection_name	string	Disaster recovery task name	No
protection_type	string	Disaster recovery task type	No
protection_enable	int	Number of VM backup files	No
backup_file_count	int	Number of VM backup files	No
vmtype	string	VM Type	No
iolog	list	View Paramter Description - iolog for I/O log information.	No
power_state	string	Power status	No
is_stopped	int	VM status: 1 indicates powered-off, 0 indicates others	No
schedule_policies	list	View Paramter Description - schedule_policies for VM scheduling policy information.	No

location	int	Whether run location of VM is fixed, 1 indicates fixed, 0 indicates unfixed	No
floatingip	dict	View Paramter Description - floatingip for Elastic IP information.	No
instant_vm	int	Whether the VM is being rapidly recovered	No
storage_mb	int	Total disk size	No
expire_time	string	Expiration date	No
network_type	list	Type of VM network	No
encrypted	int	Whether it is encrypted	No
cipher	string	Encryption algorithm	No
gpu_conf	dict	View Parameter Description - gpu_conf for GPU configuration details.	No
gpu_status	dict	View Parameter Description - gpu_status for GPU status information.	No
has_gpu	int	Whether it is a GPU-enabled VM	No
keypair	dict	View Parameter Description - keypair for key pair information.	No
os_option	string	View Parameter Description - os_option for operating system information.	No
template_id	string	Template ID	No
storage_policy_id	string	Storage policy ID	No
vm_id	string	VM ID before conversion	No
quotas	list	View Parameter Description - quota_info for the list of dynamic quotas.	No
real_ips	list	IP address used by the VM	No

Note

For common response parameters, view Response Format for details

- Parameter Description - disks

Table 4.5.62 Parameter Description - disks¶

Parameter	Type	Required	Note
size_mb	int	Yes	Disk size (MB), 1024 (1 GB) < size_mb < 66060288 (63 TB)
preallocate	string	No	Disk provisioning method, view Disk Provisioning Method
eagerly_scrub	int	No	Whether to reset now
id	string	No	Disk ID, for example: ide0,ide1...iden
type	string	No	Disk type, view VM Disk Type for details
storage_file	string	No	Storage path
storage_id	string	No	Storage ID
storage_name	string	No	Storage name
storage_tag_id	string	No	Storage Tag ID

- Parameter Description - networks

Table 4.5.63 Parameter Description - networks

Parameter	Type	Required	Note
connect	int	Yes	Whether the network is connected.
device_id	string	No	The connected network device ID in classic network
model	string	No	NIC type, view VM NIC Type for details
ip_address	string	No	IPv4 address
ipv6_address	string	No	IPv6 address
mac_address	string	No	MAC address, which can be automatically generated without transmission
vif_id	string	No	Network ID (net0, net1, net2, ... netn)
port_id	string	No	Port, connected to classic network device, currently set to 12345678
name	string	No	Name of the connected port
vpc_id	string	No	VPC network ID of the connected tenant, which can be obtained from Query VPC s
vpc_name	string	No	VPC network name
network_type	string	No	Network type, classic or VPC
subnet_id	string	No	Connected VPC subnet ID
subnet_name	string	No	Name of the connected VPC subnet
cidr	string	No	Subnet CIDR
subnet_gateway_ip	string	No	Gateway of the connected VPC subnet
custom_gateway_ip	string	No	Custom gateway IP
real_ip	string	No	IPv4 address used by the VM
real_ipv6	string	No	IPv6 address used by the VM

- Parameter Description - cpu_status

Table 4.5.64 Parameter Description - cpu_status

Parameter	Type	Value Returned	Note
total_mhz	float	No	Total CPU cores
used_mhz	float	No	Used CPU cores
ratio	float	No	CPU usage

- Parameter Description - memory_status

Table 4.5.65 Parameter Description - memory_status

Parameter	Type	Value Returned	Note
total_mb	float	No	Total memory (MB)
used_mb	float	No	Used memory (MB)
ratio	float	No	CPU usage

- Parameter Description - storage_status

Table 4.5.66 Parameter Description - storage_status¶

Parameter	Type	Value Returned	Note
total_mb	float	No	Total storage (MB)
used_mb	float	No	Used memory (MB)
ratio	float	No	CPU usage
storage_file_size_mb	float	No	Storage file size, the actual storage space used (MB)

- Parameter Description - io_status

Table 4.5.67 Parameter Description - io_status¶

Parameter	Type	Value Returned	Note
read_speed_byteps	float	No	IO read speed (byte/s)
write_speed_byteps	float	No	IO write speed (byte/s)
read_iops	float	No	IO reads (times/s)
write_iops	float	No	IO reads (times/s)

- Parameter Description - cpu_status

Table 4.5.68 Parameter Description - cpu_status¶

Parameter	Type	Value Returned	Note
read_speed_bitps	float	No	Network read speed (bit/s)
write_speed_bitps	float	No	Network write speed (bit/s)

- Parameter Description - advance_param

Table 4.5.69 Parameter Description - advance_param¶

Parameter	Type	Value Returned	Note
balloon_memory	int	No	Whether to enable the memory recycling mechanism: 0 indicates not enabled, 1 indicates enabled
onboot	int	No	Whether to run automatically at startup. 0 indicates no, 1 indicates yes.
schedopt	int	No	Whether it is a critical VM. 0 indicates no, 1 indicates yes.
abnormal_recovery	int	No	Whether to reboot if fault occurs. 0 indicates no, 1 indicates yes.
vga	string	No	Types of built-in graphic card in VM

- Parameter Description - tags

Table 4.5.70 Parameter Description - tags¶

Parameter	Type	Value Returned	Note
id	string	No	Tag ID
name	string	No	Tag name

- Parameter Description - alarm

Table 4.5.71 Parameter Description - alarm¶

Parameter	Type	Value Returned	Note
alarm	int	No	Alert status, 0 indicates no alert, 1 indicates an alert exists
alarm_info	list	No	View Parameter Description - alarm-alarm_info for alert details.

- Parameter Description - alarm-alarm_info

Table 4.5.72 Parameter Description - alarm-alarm_info

Parameter	Type	Value Returned	Note
alarm_role_type	string	No	User role of alert
severity_level	string	No	Severity
object_type	string	No	Alert object type
sub_object_type	string	No	Special alert object type
cluster_id	string	No	Cluster ID of alert
az_name	string	No	Resource pool name
az_id	string	No	Resource pool ID
id	string	No	Alert ID
user_id	string	No	User ID of alert
title	string	No	Alert event
object_id	string	No	Alert object ID
cluster_name	string	No	Cluster name of alert
sub_object_name	string	No	Special alert object name
object_name	string	No	Alert object name
project_id	string	No	Tenant ID
status	string	No	Status
resource_project_name	string	No	Tenant name of alert object
description	string	No	Description
root_id	string	No	Alert group ID
resource_project_id	string	No	Tenant ID of alert object
begin_time	string	No	Alert start time
alarm_id	string	No	Alert ID
end_time	string	No	Alert end time
resource_user_id	string	No	User ID of alert object

- Paramter Description - warning

Table 4.5.73 Paramter Description - warning

Parameter	Type	Value Returned	Note
warning	string	No	Alert status, 0 indicates no alert, 1 indicates an alert exists
warning_type	string	No	Alert type
warning_info	string	No	Alert description

- Paramter Description - iolog

Table 4.5.74 Paramter Description - iolog¶

Parameter	Type	Value Returned	Note
size_byte	float	No	Actual file size of iolog (byte)
storage_id	string	No	Storage ID

- Paramter Description - schedule_policies

Table 4.5.75 Paramter Description - schedule_policies¶

Parameter	Type	Value Returned	Note
id	string	No	Scheduling policy ID
az_id	string	No	Resource pool ID
project_id	string	No	Tenant ID
name	string	No	Scheduling policy name
type	string	No	Scheduling policy type
resource_id	string	No	External ID of Scheduling policy
status	string	No	Scheduling policy status
enable	int	No	Whether the scheduling policy is enabled
force	int	No	Whether the scheduling policy is forcibly enabled

- Paramter Description - floatingip

Table 4.5.76 Paramter Description - floatingip¶

Parameter	Type	Value Returned	Note
floatingip_id	string	No	Elastic IP ID
floating_ip_address	string	No	Bind Elastic IP
bind_status	string	No	Elastic IP binding status
line_type	string	No	Line type associated with elastic IP
line_type_id	string	No	ID of line type ID associated with elastic IP
bandwidth	int	No	Bandwidth associated with elastic IP
sharedbandwidth_name	string	No	Shared bandwidth name associated with elastic IP
sharedbandwidth_id	string	No	Shared bandwidth ID

- Parameter Description - gpu_conf

Table 4.5.77 Parameter Description - gpu_conf¶

Parameter	Type	Value Returned	Note
auth_require	string	No	Licensing mode
gpu_cores	int	No	Number of cores per GPU
gpu_mem_size_single_core	string	No	Graphics memory size per GPU core
gpu_name	string	No	Graphic card name
gpu_type	string	No	Graphic Card Type
total_count	int	No	Total number of graphics cards used by the VM
vgpu_mem_size	int	No	Total graphics memory used by the VM
vgpu_type	string	No	vGPU type
schedulers	list	No	Graphics card scheduler

- Parameter Description - gpu_status

Table 4.5.78 Parameter Description - gpu_status¶

Parameter	Type	Value Returned	Note
graphics_count	int	No	Number of GPUs
graphics_ratio	int	No	GPU usage
graphics_mem_ratio	float	No	Graphics memory usage
graphics_mem_total	float	No	Total Graphics memory size
graphics_mem_used	float	No	Used graphics memory size

- Parameter Description - keypair

Table 4.5.79 Parameter Description - keypair¶

Parameter	Type	Value Returned	Note
id	string	No	Key pair UUID
name	string	No	Key pair name
type	string	No	Key pair type
user_id	string	No	User ID
public_key	string	No	Public key information
created_at	string	No	Time created
fingerprint	string	No	Key fingerprint information

- Parameter Description - os_option

Table 4.5.80 Parameter Description - os_option¶

Parameter	Type	Value Returned	Note
kernel_name	string	No	Kernel name
distribution_name	string	No	Release version
arch	string	No	CPU architecture system

- Parameter Description - quota_info

Table 4.5.81 Parameter Description - quota_info¶

Parameter	Type	Value Returned	Note
resource	string	Yes	IDs of the quota resources, which are separated by colons (;)
value	int	Yes	Quantity

- Response Example:


```
[
    {
        "network_status": {
            "write_speed_bitps": 0,
            "read_speed_bitps": 0
        },
        "expire_time": "unlimited",
        "cpu_status": {
            "total_mhz": 4202,
            "ratio": 0,
            "used_mhz": 0
        },
        "storage_mb": 113664,
        "group_name": "Database",
        "os_installed": 2,
        "keypair": null,
        "warning": {
            "warning_info": "",
            "warning": 0,
            "warning_type": ""
        },
        "gpu_conf": null,
        "az_name": "1.75",
        "instant_vm": "",
        "networks": [
            {
                "ipv6_address": "",
                "name": null,
                "vif_id": "net0",
                "subnet_id": "",
                "vpc_name": "",
                "custom_gateway_ip": null,
                "cidr": "",
                "subnet_gateway_ip": "",
                "subnet_name": "",
                "connect": 1,
                "port_id": "",
                "mac_address": "FE:FC:FE:B2:C2:D8",
                "vpc_id": "",
                "model": "virtio",
                "ip_address": "172.32.0.110",
                "network_type": "",
                "device_id": null,
                "real_ip": "10.10.10.10",
                "real_ipv6": ""
            },
            {
                "ipv6_address": "",
                "name": "Subnet 1",
                "vif_id": "net1",
                "subnet_id": "",
                "vpc_name": "",
                "custom_gateway_ip": null,
                "cidr": "",
                "subnet_gateway_ip": "",
```

```
            "subnet_name": "",
            "connect": 1,
            "port_id": "",
            "mac_address": "FE:FC:FE:73:EB:9C",
            "vpc_id": "",
            "model": "virtio",
            "ip_address": "192.168.0.113",
            "network_type": "",
            "device_id": "dvs3fbedac",
            "real_ip": "",
            "real_ipv6": "2022:3CA1:010F:001A:121B:0000:0000:0010"
        }
    ],
    "az_id": "611f1b34-c727-4925-854e-07bc8607f681",
    "storage_tag_id": "22222222-2222-2222-2222-222222222222",
    "is_stopped": 1,
    "uptime": 0,
    "user_id": null,
    "vmtype": "dmp_mysql",
    "floatingip": {
        "sharedbandwidth_name": "",
        "floatingip_id": "",
        "bandwidth": 0,
        "line_type": "",
        "bind_status": "",
        "floating_ip_address": "",
        "line_type_id": "",
        "sharedbandwidth_id": ""
    },
    "protection_enable": "",
    "image_id": "7965a8e1-4cd8-4b62-988d-18c6ee17c4bf",
    "tags": [],
    "storage_name": "data-from-60",
    "gpu_status": {
        "graphics_mem_total": 0,
        "graphics_count": 0,
        "graphics_mem_used": 0,
        "graphics_mem_ratio": 0,
        "graphics_ratio": 0
    },
    "id": "d6ee5e1f-9abf-43e3-90d5-2a0e9881ad95",
    "vtool_installed": 1,
    "advance_param": {
        "schedopt": 0,
        "vga": "cirrus",
        "abnormal_recovery": 0,
        "onboot": 0,
        "balloon_memory": 1
    },
    "location": 0,
    "power_state": "off",
    "storage_id": "1SANGFOR_ASAN-00c-b4f6-42b8-938c-00a7c09d6f72",
    "in_protection": 0,
    "project_id": "9d14d076b59045b1b0aaca04cfaaa657",
    "user_name": "",
    "schedule_policies": [],
    "io_status": {
        "read_speed_byteps": 0,
        "read_iops": 0,
        "write_speed_byteps": 0,
        "write_iops": 0
    },
    "project_name": "tjc_tenant",
    "sockets": 2,
    "description": "",
    "group_id_path": "f3f9d08f-f5ec-4b85-bb7f-c542eee0e02c",
    "memory_status": {
        "used_mb": 0,
        "total_mb": 2048,
        "ratio": 0
    },
    "group_name_path": "Database",
    "image_name": "DBVM-1207",
    "has_gpu": 0,
    "dh_id": "",
    "protection_name": "",
    "protection_type": "",
    "protection_id": "",
    "shutdown_duration": 357762,
    "host_id": "cluster",
    "status": "stopped",
    "dh_name": "",
    "name": "dbvm-test-cjl-1212",
    "backup_file_count": 0,
    "ips": [
        "172.32.0.110",
        "192.168.0.113"
```

```
        ],
        "type": "hci",
        "alarm": {
            "alarm_info": [],
            "alarm": 0
        },
        "disks": [
            {
                "storage_tag_id": "22222222-2222-2222-2222-222222222222",
                "preallocate": 0,
                "storage_name": "data-from-60",
                "eagerly_scrub": 0,
                "size_mb": 102400,
                "storage_id": "1SANGFOR_ASAN-00c-b4f6-42b8-938c-00a7c09d6f72
",
                "storage_file": "vm-disk-1.qcow2",
                "id": "ide0",
                "physical_disk_type": ""
            },
            {
                "storage_tag_id": "22222222-2222-2222-2222-222222222222",
                "preallocate": 0,
                "storage_name": "data-from-60",
                "eagerly_scrub": 0,
                "size_mb": 11264,
                "storage_id": "1SANGFOR_ASAN-00c-b4f6-42b8-938c-00a7c09d6f72
",
                "storage_file": "vm-disk-2.qcow2",
                "id": "ide1",
                "physical_disk_type": ""
            }
        ],
        "cores_per_socket": 1,
        "network_type": [
            "classic"
        ],
        "memory_mb": 2048,
        "storage_status": {
            "used_mb": 3035.59,
            "total_mb": 113664,
            "storage_file_size_mb": 611,
            "ratio": 0.03
        },
        "cipher": "",
        "encrypted": 0,
        "host_name": "<Auto select>",
        "cores": 2,
        "os_type": "l2664",
        "group_id": "f3f9d08f-f5ec-4b85-bb7f-c542eee0e02c",
        "mhz": 4202,
        "os_name": "",
        "iolog": [],
        "os_option": {
            "kernel_name": "",
            "distribution_name": "",
            "arch": "",
        },
        "template_id": "23879b10-019d-4a3a-a514-809b4abb1912",
        "storage_policy_id": "2325526a-f793-4b47-97e6-2cde1e4e9892",
        "vm_id": "e4e784b7-7906-4107-b6ba-7fc535c90208",
        "quotas": [
            {
                "resource": "",
                "value": 5,
            }
        ]
        "real_ips": ["10.10.10.10", "2022:3CA1:010F:001A:121B:0000:0000:0010
"]
    }
]
```

- OK: 200
- Errors: 400, 403, 500

4.5.2.4. Get VM Configuration

4.5.2.4.1. Scenario

- Scenario: Query VM configuration

- Roles Supported: Super admin, system admin, tenant, and co-administrator with the VM Status Read permission
- Versions supported: SCP 6.3.0 and above

> **Note**
>
> Tenants can only obtain their own VM configuration details.

4.5.2.4.2. URI

Table 4.5.82 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/servers/{server_id}	Get VM Configuration

- Parameter Description:

Table 4.5.83 Path Parameters¶

Parameter	Location	Type	Description	Required
server_id	path	string	Virtual machine ID	Yes

4.5.2.4.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/servers/68f6a32a-912a-4fc6-a3ef-17abac27f947

4.5.2.4.4. Response

- Response Parameter

Table 4.5.84 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	VM UUID	Yes
status	string	View Parameter Description - status for VM status.	Yes
name	string	VM name	Yes
vm_type	string	VM type, view VM Type for details	No
description	string	VM description	No
group_id	string	VM group ID	Yes
group_id_path	string	VM group ID path	Yes
group_name	string	Group name	Yes
storage_name	string	Storage name	No
host_name	string	Host name	No

project_id	string	Tenant ID	Yes
project_name	string	Tenant name	Yes
user_id	string	User ID	Yes
cores	int	CPU cores	Yes
sockets	int	Socket number	Yes
memory_mb	int	Memory size (MB)	Yes
location	dict	Run location, view Parameter Description - location for details	Yes
storage_type	string	Storage type	No
storage_location	string	Storage location	No
storage_tag_id	string	Storage Tag ID	Yes
storage_tag_name	string	Storage tag name	Yes
az_id	string	Resource pool ID	No
az_name	string	Resource pool name	No
pool_tag	string	Usage tag of the resource pool	No
dh_id	string	Dedicated server group ID	No
dh_tag	string	Usage tag of the dedicated server group	No
disks	list	Disk information, view Parameter Description - disks for details	Yes
networks	list	Network information, view Parameter Description - networks for details	Yes
cdroms	list	CD/DVD drive information, view Parameter Description - cdroms for details	No
usbs	list	USB information, view Parameter Description - usbs for details	No
os_type	string	Operating system type	No
os_name	string	OS	No
image_name	string	Image name	No
os_installed	int	Whether an operating system is installed	No
vtool_installed	int	Whether vmTools is installed	Yes
ha	int	Whether it is a HA-enabled VM	No
snap	int	Whether a snapshot exists	No
backup	int	Whether a backup exists	No
derived	int	Whether it is a deployed VM	No
hotplug_sup_info	dict	Hot add information, view Parameter Description - hotplug_sup_info for details	No
floatingip	dict	Elastic IP information, view Paramter Description - floatingip for details	No
advance_param	dict	HCI-specific parameter, view Parameter Description - advance_param for details	No
instant_vm	string	Whether the VM is rapidly recovered	No
encrypted	int	Whether it is encrypted	Yes
cipher	string	Encryption algorithm	Yes
		GPU configuration information, view Parameter	

gpu	dict	GPU configuration information, view Parameter Description - gpu	No
machine_type	string	Motherboard type	No
arch_type	string	Cluster architecture type	No

Note

For common response parameters, view Response Format for details

- Parameter Description - status

Table 4.5.85 Parameter Description - status

status	Note
running	Running
suspend	Suspended
stopped	Powered off
alert	Alert
overdue	Expired
lost	Storage missing
paused_io_error	IO error occurred
starting	Powering on
stoping	Shutting down
tpl_lost	Template missing
clone	Cloning
backup	Backing up
recovery	Recovering
run_backup	Backing up
resetting	Restarting
suspending	Suspending
updating	Updating
deleting	Deleting
snapshotting	Taking snapshot
creating	Creating
migrate	Migrating
deriving	Deploying
tpl_updating	Updating
tpl_moving	Preparing for reverse copy
tpl_syncing	Reverse copy
offline	Offline

- Parameter Description - location

Table 4.5.86 Parameter Description - location¶

Parameter	Type	Value Returned	Note
id	string	Yes	Run location, such as host-xx, cluster
type	string	No	Type of run location, which is specific to VMware, view VM Working Location Type (VMware) for details
location	int	No	Whether the VM's run location is fixed.
policy_type	string	No	Type of VM scheduling policy

- Parameter Description - disks

Table 4.5.87 Parameter Description - disks¶

Parameter	Type	Value Returned	Note
size_mb	int	Yes	Disk size (MB), 1024 (1 GB) < size_mb < 66060288 (63 TB)
is_old_disk	int	No	Whether the disk exists before the VM is edited
preallocate	string	Yes	Disk provisioning method, view Disk Provisioning Method
eagerly_scrub	int	No	Whether to reset now
id	string	Yes	Disk ID, for example: ide0,ide1...iden
type	string	No	Disk type, view VM Disk Type for details
storage_file	string	No	Storage path
storage_id	string	No	Storage ID
storage_name	string	No	Storage name
backing_file	string	No	Backend image
label	string	No	Display name of disks, such as disk 1, disk 2
use_virtio	int	No	Whether the virtio disk is used.
storage_policy	string	No	Storage policy ID

- Parameter Description - networks

Table 4.5.88 Parameter Description - networks¶

Parameter	Type	Value Returned	Note
connect	int	Yes	Whether the network is connected.
device_id	string	No	The connected network device ID in classic network
model	string	Yes	NIC type, view VM NIC Type for details
label	string	No	Display name of NIC, such as network connection 1
ip_address	string	No	IPv4 address
ipv6_address	string	No	IPv6 address
mac_address	string	Yes	MAC address
vif_id	string	Yes	Network ID (net0, net1, net2, ... netn)
port_id	string	No	Port
host_tso	int	No	Whether to enable TSO
name	string	No	Name of the connected port
vpc_id	string	No	VPC ID of the connected tenant
vpc_name	string	No	VPC network name
network_type	string	No	Network type, classic or VPC
subnet_id	string	No	Connected VPC subnet ID
subnet_name	string	No	Name of the connected VPC subnet
cidr	string	No	Subnet CIDR
ip_info	dict	No	View Parameter Description - ip_info for IP address information.
ipv6_info	dict	No	View Parameter Description - ipv6_info for IPv6 address information.
subnet_gateway_ip	string	No	Gateway of the connected VPC subnet
custom_gateway_ip	string	No	Custom gateway IP

- Parameter Description - ip_info

Table 4.5.89 Parameter Description - ip_info¶

Parameter	Type	Value Returned	Note
ip_address	string	Yes	Specified IP Address
netmask	string	No	Netmask
gateway	string	No	Gateway
dns	list	No	DNS

- Parameter Description - ipv6_info

Table 4.5.90 Parameter Description - ipv6_info¶

Parameter	Type	Value Returned	Note
ipv6_address	string	Yes	Specified IPv6 address
prefixlen	string	No	Prefix length
gateway	string	No	Gateway
dns	list	No	DNS

- Parameter Description - cdroms

Table 4.5.91 Parameter Description - cdroms¶

Parameter	Type	Value Returned	Note
id	string	Yes	Slot number of CD/DVD drive, such as ide0, ide1
image_id	string	Yes	ISO image ID
location	string	Yes	ISO location
storage_name	string	No	Storage name
image_name	string	No	Name of the mounted image

- Parameter Description - usbs

Table 4.5.92 Parameter Description - usbs¶

Parameter	Type	Value Returned	Note
id	string	Yes	USB ID(udisk0,udisk1,udisk2...udiskn)
usb_device	string	No	USB device
controller_type	string	No	Type of USB controller, such as 2.0, 3.0
online	int	No	Online or not
description	string	No	Description of USB device

- Parameter Description - hotplug_sup_info

Table 4.5.93 Parameter Description - hotplug_sup_info¶

Parameter	Type	Value Returned	Note
cpu	int	No	Whether CPU hot add is supported
memory	int	No	Whether memory hot add is supported

- Paramter Description - floatingip

Table 4.5.94 Paramter Description - floatingip¶

Parameter	Type	Value Returned	Note
floatingip_id	string	No	Elastic IP ID
floatingip_ip	string	No	Elastic IP address
line_type_name	string	No	Line type name
line_type_id	string	No	Line type ID
bandwidth	int	No	Elastic IP bandwidth
sharedbandwidth_name	string	No	Line type name
sharedbandwidth_id	string	No	Shared bandwidth ID

- Parameter Description - advance_param

Table 4.5.95 Parameter Description - advance_param¶

Parameter	Type	Value Returned	Note
cpu_type	string	No	CPU type, view VM CPU Type for details
onboot	int	No	Whether the VM is started automatically when the physical host is powered on.
use_uuid	int	No	Whether to enable UUID.
uuid	string	No	UUID
abnormal_recovery	int	No	Whether to reboot if fault occurs.
hugepage_memory	int	No	Whether to use huge-page memory.
balloon_memory	int	No	Whether to enable memory reclaiming。
schedopt	int	No	Whether it is a critical VM.
cpu_hotplug	int	No	Whether to enable CPU hot add.
mem_hotplug	int	No	Whether to enable memory hot add.
boot_order	string	No	Boot sequence, c indicates cdrom, d indicates disk.
boot_disk	string	No	Boot disk
dir	string	No	HCI group
use_vblk	int	No	Whether to enable vblk.
real_use_vblk	int	No	Whether to really use vblk.
storage_policy	string	No	Storage policy

- Parameter Description - gpu

Table 4.5.96 Parameter Description - gpu¶

Parameter	Type	Required	Note
vendor	string	Yes	GPU manufacturers, such as NVIDIA and AMD. Only NVIDIA is currently supported.
conf	dict	Yes	GPU configuration, which differs as per different manufacturers. Only NVIDIA is supported now, view Parameter Description - gpu.GpuNvidia for details

- Parameter Description - gpu-GpuNvidia

Table 4.5.97 Parameter Description - gpu.GpuNvidia¶

Parameter	Type	Required	Note
total_count	int	No	Total number of graphic cards
enable_count	int	No	Number of graphic cards enabled
gpu_type	string	Yes	Graphics Card Model
slice_type	string	Yes	vGPU type, view vGPU Type for details
schedulers	list	No	List of vGPU schedulers, view Scheduling Method for details

- Response Example:


```
{
    "usbs": [],
    "encrypted": 0,
    "networks": [
        {
            "ip_info": {
                "ip_address": "172.32.0.121"
            },
            "vif_id": "net0",
            "subnet_id": "fec18705-62ce-4cbc-85a7-95c17980298b",
            "host_tso": 0,
            "custom_gateway_ip": null,
            "network_type": "vpc",
            "model": "virtio",
            "subnet_name": "NFV_management_subnet",
            "connect": 1,
            "mac_address": "fe:fc:fe:85:cc:fd",
            "vpc_id": "12af9eee-8f4c-4ece-af86-4ea76edbb6f9",
            "cidr": "172.32.0.0/24",
            "port_id": "12345678",
            "subnet_gateway_ip": "172.32.0.1"
        },
        {
            "ip_info": {
                "ip_address": "192.168.0.127"
            },
            "vif_id": "net1",
            "subnet_id": "7a606fd1-fd9b-4d44-8753-a76e26ca1ca0",
            "host_tso": 0,
            "custom_gateway_ip": "",
            "network_type": "vpc",
            "model": "virtio",
            "subnet_name": "Subnet1",
            "connect": 1,
            "mac_address": "fe:fc:fe:47:5a:58",
            "vpc_id": "12af9eee-8f4c-4ece-af86-4ea76edbb6f9",
            "cidr": "192.168.0.0/24",
            "port_id": "12345678",
            "subnet_gateway_ip": "192.168.0.1"
        }
    ],
    "group_name": "Database",
    "os_installed": 2,
    "storage_type": "iscsi",
    "project_name": "tjc_tenant",
    "az_name": "1.75",
    "sockets": 2,
    "az_id": "611f1b34-c727-4925-854e-07bc8607f681",
    "storage_tag_id": "00000000-0000-0000-0000-000000000000",
    "cdroms": [
        {
            "image_id": "",
            "image_name": "",
            "location": "",
            "id": "ide2"
        }
    ],
    "user_id": null,
    "vm_type": "dmp_mysql",
    "floatingip": {
        "sharedbandwidth_name": "",
        "line_type_name": "Shared_Services",
        "floatingip_id": "0d98d028-b6eb-41d6-8d44-94f214139727",
```

```
        "floatingip_id": "0d98d028-b6eb-41d6-8d44-94f214139727",
        "bandwidth": 100000,
        "sharedbandwidth_id": "",
        "line_type_id": "14d9a0b8-caac-48f3-b66c-19e58b5d88f5",
        "floatingip_ip": "9.9.9.103"
    },
    "id": "68f6a32a-912a-4fc6-a3ef-17abac27f947",
    "vtool_installed": 1,
    "advance_param": {
        "schedopt": 0,
        "hugepage_memory": 0,
        "use_vblk": 1,
        "boot_order": "dc",
        "cpu_hotplug": 1,
        "balloon_memory": 1,
        "use_uuid": 0,
        "abnormal_recovery": 0,
        "mem_hotplug": 1,
        "cpu_type": "core2duo",
        "real_use_vblk": 1,
        "onboot": 0,
        "dir": "832e7d558801",
        "boot_disk": "ide0"
    },
    "location": {
        "location": 0,
        "policy_type": "",
        "id": "cluster"
    },
    "snap": 0,
    "gpu": null,
    "project_id": "9d14d076b59045b1b0aaca04cfaaa657",
    "status": "stopped",
    "storage_location": "1SANGFOR_ASAN-00e-eb0f-48d0-a8bd-7f86ade50a97",
    "description": "",
    "group_id_path": "f3f9d08f-f5ec-4b85-bb7f-c542eee0e02c",
    "derived": 1,
    "image_name": "DBVM-1207",
    "dh_id": null,
    "cipher": "",
    "machine_type": "i440fx",
    "ha": 1,
    "storage_tag_name": "High_Performance_Storage",
    "name": "66666-caiyi-6602",
    "storage_name": "backup-from-60",
    "disks": [
        {
            "preallocate": 0,
            "storage_name": "backup-from-60",
            "id": "ide0",
            "size_mb": 102400,
            "use_virtio": 1,
            "storage_file": "1SANGFOR_ASAN-00e-eb0f-48d0-a8bd-7f86ade50a97:vm-disk-1.qcow2",
            "type": "derive_disk",
            "is_old_disk": 1
        },
        {
            "preallocate": 0,
            "storage_name": "backup-from-60",
            "id": "ide1",
            "size_mb": 10240,
            "use_virtio": 1,
            "storage_file": "1SANGFOR_ASAN-00e-eb0f-48d0-a8bd-7f86ade50a97:vm-disk-2.qcow2",
            "type": "new_disk",
            "is_old_disk": 1
        }
    ],
    "hotplug_sup_info": {
        "cpu": 1,
        "memory": 1
    },
    "memory_mb": 2048,
    "host_name": "<Auto>",
    "cores": 2,
    "os_type": "l2664",
    "group_id": "f3f9d08f-f5ec-4b85-bb7f-c542eee0e02c",
    "backup": 0,
    "os_name": ""
}
```


4.5.2.4.5. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.5.2.5. Create a VM

4.5.2.5.1. Scenario

- **Scenario:**
  - admin creates a VM for tenants or users.
  - Tenant creates its own VM or for its users.
  - Create a VM and connect it to a device in VPC subnet or classic network.
  - Create a VM via ISO/VMA image.
  - Create a GPU-enabled VM

- Roles Supported: Super admin, system admin, tenant, and co-administrator with the VM Creation permission
- Versions supported: SCP 6.3.0 and above

4.5.2.5.2. URI

Table 4.5.98 API Path and Request Method¶

REST VERB	PATH	Description
POST	/janus/20180725/servers	Create a VM

- Parameter Description:

None

4.5.2.5.3. Request

- Request Parameter

Table 4.5.99 Description of Body Parameter¶

Parameter	Location	Type	Description	Required
name	body	string	VM name	Yes
cores	body	int	Total number of CPU cores. Note that this is the total number, not the number of cores per socket. The value must be greater than or equal to sockets.	Yes
sockets	body	int	Socket number	Yes
memory_mb	body	int	Memory size (MB)	Yes
disks	body	list	Disk information, view Parameter Description - disks for details.	Yes
networks	body	list	Network information, view Parameter Description - networks for details.	Yes
image_id	body	string	Image ID, which can be obtained from Query Images	Yes
az_id	body	string	Resource pool ID	Yes
			Dedicated server group ID. When the specified resource pool ID is associated with a dedicated server	

dh_id	body	string	associated with a dedicated server group, this parameter must be passed in; otherwise, the VM will fail to power on.	No
description	body	string	VM description, up to 100 characters	No
group_id	body	string	VM group	No
count	body	int	VM count	No
type	body	string	VM types, view VM Type for details	No
uuids	body	list	VM UUID list, which indicates creating VM with UUID specified. Its length corresponds to the count parameter. If the parameter is not passed in, UUID will be automatically generated.	No
storage_tag_id	body	string	Storage tag ID, you can pass either this ID or storage_id, or both of them. It can be obtained from Get a list of storage tags	No
storage_id	body	string	Storage ID, which indicates creating VM with the specified storage, and can be obtained from Query Storage List	No
location	body	dict	Run location, view Parameter Description - location for details.	No
power_on	body	int	Whether to power on VM after it is created. If the parameter is not passed in, then the VM is powered off by default after it is created.	No
cdroms	body	list	CD/DVD drive information, view Parameter Description - cdroms for details	No
usbs	body	list	USB information, view Parameter Description - usbs for details	No
advance_param	body	dict	HCI advanced parameter, view Parameter Description - advance_param for details	No
password	body	string	VM password, up to 1024 characters. This parameter does not take effect if the image format is ISO, and the password needs to be encrypted before being passed in. For the encryption method, view Token-Based Authentication for details	No
floatingip	body	dict	Information about creating elastic public IP address, view Paramter Description - floatingip for details	No
hostname	body	string	VM hostname, up to 100 characters	No
lifespan	body	string	VM expiration date, up to 100 characters. The parameter format must be %Y-%m-%d %H:%M:%S, for example: 2021-10-25 23:59:59	No
hostname_index	body	int	Start value of VM host name suffix, the maximum is 9999	No
encryption	body	dict	Encryption information, view Parameter Description - encryption for details	No
gpu	body	dict	GPU configuration information, view Parameter Description - gpu for details	No
keypair	body	dict	Key information, view Parameter Description - keypair for details	No

project_id	body	string	Tenant ID. When a VM is created for tenant, user_id will not take effect unless project_id has been passed in as well.	No
user_id	body	string	User ID, indicating a VM is created for users; in this case, both project_id and user_id need to be passed in.	No

- Parameter Description - disks

Table 4.5.100 Parameter Description - disks¶

Parameter	Type	Required	Note
size_mb	int	Yes	Disk size (MB), 1024 (1 GB) < size_mb < 66060288 (63 TB)
is_old_disk	int	No	Whether the disk exists before the VM is edited
preallocate	string	No	Disk provisioning method, view Disk Provisioning Method
eagerly_scrub	int	No	Whether to reset now
id	string	Yes	Disk ID. Note: ide2 indicates the system disk, which is already in use. Other disks can be used. The value can be ide0, ide1, ..., iden (n ≤ 15)
type	string	No	Disk type, view VM Disk Type for details
storage_file	string	No	Storage path
storage_id	string	No	Storage ID. When type=physical_disk, this parameter needs to be passed in
backing_file	string	No	Backend image
copy	string	No	Existing disk path
cache	string	No	Cache mode, view VM Disk Cache Mode for details
cache_size	int	No	Cache size
forecast	string	No	Whether to enable cache.
use_virtio	int	No	Whether the virtio disk is used.
storage_policy	string	No	Storage policy ID

- Parameter Description - networks

Table 4.5.101 Parameter Description - networks¶

Parameter	Type	Required	Note
connect	int	Yes	Whether the network is connected.
device_id	string	No	The connected network device ID in classic network
model	string	No	NIC type, view VM NIC Type for details
mac_address	string	No	MAC address, which can be automatically generated without transmission
vif_id	string	Yes	Network ID (net0, net1, net2, ... netn)
port_id	string	No	Port, connected to classic network device, currently set to 12345678
host_tso	int	No	Whether to enable TSO.
name	string	No	Name of the connected port
vpc_id	string	No	VPC network ID of the connected tenant, which can be obtained from Query VPC s
subnet_id	string	No	Connected VPC subnet ID
ip_info	dict	No	IP address information, view Parameter Description - network-ip_info for details
custom_gateway_ip	string	No	Custom gateway IP

- Parameter Description - network-ip_info

Table 4.5.102 Parameter Description - network-ip_info¶

Parameter	Type	Required	Note
ip_address	string	Yes	Specified IP Address
netmask	string	No	Netmask
gateway	string	No	Gateway
dns	list	No	DNS address list, multiple addresses can be passed in.

- Parameter Description - location

Table 4.5.103 Parameter Description - location¶

Parameter	Type	Required	Note
id	string	Yes	Run location, such as host-xx, cluster
type	string	No	Type of run location, which is specific to VMware, view VM Working Location Type (VMware) for details
location	int	No	Whether the VM's run location is fixed.
policy_type	string	No	Type of VM scheduling policy

- Parameter Description - cdroms

Table 4.5.104 Parameter Description - cdroms¶

Parameter	Type	Required	Note
id	string	Yes	Slot number of CD/DVD drive, such as ide0, ide1
image_id	string	No	ISO image ID
location	string	No	ISO location
image_name	string	No	Name of the mounted image

- Parameter Description - usbs

Table 4.5.105 Parameter Description - usbs¶

Parameter	Type	Required	Note
id	string	Yes	USB ID(udisk0,udisk1,udisk2...udiskn)
usb_device	string	No	USB device
controller_type	string	No	Type of USB controller, such as 2.0, 3.0

- Parameter Description - advance_param

Table 4.5.106 Parameter Description - advance_param¶

Parameter	Type	Required	Note
cpu_type	string	No	CPU type, view VM CPU Type for details
onboot	int	No	Whether the VM is started automatically when the physical host is powered on.
use_uuid	int	No	Whether to enable UUID.
uuid	string	No	UUID
regen_uuid	int	No	Whether to regenerate a UUID. The options are 0 and 1
abnormal_recovery	int	No	Whether to reboot if fault occurs.
hugepage_memory	int	No	Whether to use huge-page memory.
balloon_memory	int	No	Whether to enable memory reclaiming。
schedopt	int	No	Whether it is a critical VM.
cpu_hotplug	int	No	Whether to enable CPU hot add.
mem_hotplug	int	No	Whether to enable memory hot add.
boot_order	string	No	Boot sequence, c indicates cdrom, d indicates disk.
boot_disk	string	No	Boot disk
dir	string	No	HCI group
use_vblk	int	No	Whether to enable vblk.
real_use_vblk	int	No	Whether to really use vblk.
storage_policy	string	No	Storage policy
invtsc	int	No	Whether to enable high-performance clock. (This parameter can be edited only when VM is shut down.) The options are 0 and 1. Note: Supported in SCP 6.7.35 and above.

- Paramter Description - floatingip

Table 4.5.107 Paramter Description - floatingip¶

Parameter	Type	Required	Note
floatingip_id	string	No	Elastic IP ID
bandwidth	string	No	Elastic IP bandwidth
line_type	string	No	Elastic IP line type ID
sharedbandwidth_id	string	No	Shared bandwidth ID

- Parameter Description - encryption

Table 4.5.108 Parameter Description - encryption¶

Parameter	Type	Required	Note
type	string	Yes	Encryption method, view VM Disk Encryption Granularity for details
properties	dict	Yes	Encryption property, view Parameter Description - encryption.properties for details

- Parameter Description - encryption-properties

Table 4.5.109 Parameter Description - encryption.properties¶

Parameter	Type	Required	Note
cipher	string	Yes	Encryption algorithm, view VM Disk Encryption Algorithm for details

- Parameter Description - gpu

Table 4.5.110 Parameter Description - gpu¶

Parameter	Type	Required	Note
vendor	string	Yes	GPU manufacturers, such as NVIDIA and AMD. Only NVIDIA is currently supported.
conf	dict	Yes	GPU configuration, which differs as per different manufacturers. Only NVIDIA is supported now, view Parameter Description - gpu.GpuNvidia for details

- Parameter Description - gpu-GpuNvidia

Table 4.5.111 Parameter Description - gpu.GpuNvidia¶

Parameter	Type	Required	Note
total_count	int	No	Total number of graphic cards
enable_count	int	No	Number of graphic cards enabled
gpu_type	string	Yes	Graphics Card Model
slice_type	string	Yes	vGPU type, view vGPU Type for details
schedulers	list	No	List of vGPU schedulers, view Scheduling Method for details

- Parameter Description - keypair

Table 4.5.112 Parameter Description - keypair¶

Parameter	Type	Required	Note
key_name	string	Yes	Key pair name
user_id	string	No	Key pair user

- Request Example

  https://{SCP_IP}/janus/20180725/servers

```
{
    "advance_param": {
        "abnormal_recovery": 0,
        "balloon_memory": 1,
        "boot_disk": "ide0",
        "boot_order": "c",
        "cpu_hotplug": 1,
        "hugepage_memory": 0,
        "mem_hotplug": 1,
        "onboot": 0,
        "schedopt": 0
    },
    "az_id": "611f1b34-c727-4925-854e-07bc8607f681",
    "cores": 2,
    "disks": [
        {
            "id": "ide0",
            "preallocate": 0,
            "size_mb": 102400,
            "type": "derive_disk"
        },
        {
            "id": "ide1",
            "preallocate": 0,
            "size_mb": 102400,
            "type": "new_disk"
        }
    ],
    "image_id": "354ed5ba-2b9f-4d65-b4ea-6526d8ab9b97",
    "memory_mb": 4096,
    "name": "test1210-1",
    "networks": [
        {
            "connect": 1,
            "host_tso": 0,
            "model": "virtio",
            "vif_id": "net0"
        },
        {
            "connect": 1,
            "host_tso": 0,
            "model": "virtio",
            "vif_id": "net1"
        }
    ],
    "power_on": 0,
    "sockets": 2,
    "storage_tag_id": "22222222-2222-2222-2222-222222222222",
    "type": "vm"
}
```

4.5.2.5.4. Response

- Response Parameter

Table 4.5.113 Response Parameter¶

Parameter	Type	Description	Value Returned
task_id	string	Asynchronous task ID	Yes
uuids	list	VM ID list	Yes

> **Note**
>
> For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "d5ae0d43-4afc-4f27-a851-9989a7083d9d",
    "uuids": ["server_uuid1", "server_uuid2", "...", "server_uuidn"]
}
```

4.5.2.5.5. Example of Creating a VM

- Example 1: Create a VM without network connection

```
{
    "az_id": "611f1b34-c727-4925-854e-07bc8607f681",
    "location": {
        "id": "cluster",
        "location": 0
    },
    "storage_tag_id": "22222222-2222-2222-2222-222222222222",
    "image_id": "354ed5ba-2b9f-4d65-b4ea-6526d8ab9b97",
    "cores": 1,
    "sockets": 1,
    "memory_mb": 1024,
    "count": 1,
    "name": "create_vm_with_lifespan",
    "description": "",
    "advance_param": {
        "boot_order": "c",
        "onboot": 0,
        "schedopt": 0,
        "abnormal_recovery": 1,
        "cpu_hotplug": 0,
        "mem_hotplug": 0,
        "balloon_memory": 0,
        "hugepage_memory": 0
    },
    "disks": [
        {
            "id": "ide0",
            "type": "derive_disk",
            "preallocate": 0,
            "size_mb": 81920
        }
    ],
    "networks": [
        {
            "vif_id": "net0",
            "mac_address": "fe:fc:fe:25:3a:2d",
            "connect": 0,
            "model": "virtio",
            "host_tso": 0
        }
    ],
    "usbs": [],
    "power_on": 0,
    "lifespan": "2021-05-24 23:59:59"
}
```

- Example 2: Create a VM for tenant and connect the VM to a classic network

```
{
    "az_id": "611f1b34-c727-4925-854e-07bc8607f681",
    "location": {
        "id": "cluster",
        "location": 0
    },
    "storage_tag_id": "22222222-2222-2222-2222-222222222222",
    "image_id": "354ed5ba-2b9f-4d65-b4ea-6526d8ab9b97",
    "cores": 1,
    "sockets": 1,
    "memory_mb": 1024,
    "count": 1,
    "name": "create_vm_with_classic",
    "description": "",
    "advance_param": {
        "boot_order": "c",
        "onboot": 0,
        "schedopt": 0,
        "abnormal_recovery": 1,
        "cpu_hotplug": 0,
        "mem_hotplug": 0,
        "balloon_memory": 0,
        "hugepage_memory": 0
    },
    "disks": [
        {
            "id": "ide0",
            "type": "derive_disk",
            "preallocate": 0,
            "size_mb": 81920
        }
    ],
    "networks": [
        {
            "vif_id": "net0",
            "connect": 1,
            "model": "virtio",
            "host_tso": 0,
            "name": "Default_classic_network_gateway1-Switch",
            "device_id": "dvsa734f95",
            "port_id": "12345678"
        }
    ],
    "usbs": [],
    "power_on": 0,
    "lifespan": "2021-05-24 23:59:59",
    "project_id": "c5f9ed4abab94b46ad0e7a1783a7e2ba"
}
```


- Example 3: Create a VM for tenant and connect the VM to VPC

```
{
    "az_id": "611f1b34-c727-4925-854e-07bc8607f681",
    "location": {
        "id": "cluster",
        "location": 0
    },
    "storage_tag_id": "22222222-2222-2222-2222-222222222222",
    "image_id": "354ed5ba-2b9f-4d65-b4ea-6526d8ab9b97",
    "cores": 1,
    "sockets": 1,
    "memory_mb": 1024,
    "count": 1,
    "name": "create_vm_with_vpc",
    "description": "",
    "advance_param": {
        "boot_order": "c",
        "onboot": 0,
        "schedopt": 0,
        "abnormal_recovery": 1,
        "cpu_hotplug": 0,
        "mem_hotplug": 0,
        "balloon_memory": 0,
        "hugepage_memory": 0
    },
    "disks": [
        {
            "id": "ide0",
            "type": "derive_disk",
            "preallocate": 0,
            "size_mb": 81920
        }
    ],
    "networks": [
        {
            "vif_id": "net0",
            "connect": 1,
            "model": "virtio",
            "host_tso": 0,
            "vpc_id": "b4076571-7662-45e7-853c-3049a6e3b703",
            "subnet_id": "1b5f2c05-ef57-4803-b6f1-2068084f555c"
        }
    ],
    "usbs": [],
    "power_on": 0,
    "lifespan": "2021-05-24 23:59:59",
    "project_id": "c5f9ed4abab94b46ad0e7a1783a7e2ba"
}
```


- **Example 4: Create a VM and add an ordinary disk and a physical disk in addition to the system disk**

```
{
    "az_id": "611f1b34-c727-4925-854e-07bc8607f681",
    "location": {
        "id": "cluster",
        "location": 0
    },
    "storage_tag_id": "22222222-2222-2222-2222-222222222222",
    "image_id": "354ed5ba-2b9f-4d65-b4ea-6526d8ab9b97",
    "cores": 1,
    "sockets": 1,
    "memory_mb": 1024,
    "count": 1,
    "name": "create_vm_with_extra_disks",
    "description": "",
    "advance_param": {
        "boot_order": "c",
        "onboot": 0,
        "schedopt": 0,
        "abnormal_recovery": 1,
        "cpu_hotplug": 0,
        "mem_hotplug": 0,
        "balloon_memory": 0,
        "hugepage_memory": 0
    },
    "disks": [
        {
            "id": "ide0",
            "type": "derive_disk",
            "preallocate": 0,
            "size_mb": 81920
        },
        {
            "id": "ide1",
            "type": "new_disk",
            "preallocate": 0,
            "size_mb": 102400
        },
        {
            "id": "ide3",
            "type": "physical_disk",
            "preallocate": 0,
            "size_mb": 20480,
            "storage_id": "1SANGFOR_ASAN-006-10c5-4901-a143-45449cba0c85"
        }
    ],
    "networks": [
        {
            "vif_id": "net0",
            "connect": 1,
            "model": "virtio",
            "host_tso": 0
        }
    ],
    "usbs": [],
    "power_on": 0,
    "lifespan": "2021-05-24 23:59:59"
}
```


- **Example 5: Create a VM and customize password**

```
{
    "az_id": "611f1b34-c727-4925-854e-07bc8607f681",
    "location": {
        "id": "cluster",
        "location": 0
    },
    "storage_tag_id": "22222222-2222-2222-2222-222222222222",
    "image_id": "354ed5ba-2b9f-4d65-b4ea-6526d8ab9b97",
    "cores": 1,
    "sockets": 1,
    "memory_mb": 1024,
    "count": 1,
    "name": "create_vm_with_password",
    "description": "",
    "advance_param": {
        "boot_order": "c",
        "onboot": 0,
        "schedopt": 0,
        "abnormal_recovery": 1,
        "cpu_hotplug": 0,
        "mem_hotplug": 0,
        "balloon_memory": 0,
        "hugepage_memory": 0
    },
    "disks": [
        {
            "id": "ide0",
            "type": "derive_disk",
            "preallocate": 0,
            "size_mb": 81920
        }
    ],
    "networks": [
        {
            "vif_id": "net0",
            "connect": 1,
            "model": "virtio",
            "host_tso": 0
        }
    ],
    "usbs": [],
    "power_on": 0,
    "password": "9d8ed8fc1f898636dc10cba2d67d66ae0dc801dfc5a8c43cbc6f448de41
f858e7ca16cf4d60045c7dfb3bb040902a61d904174a9adbd6adfd6f060978ca45328ce2162a
f410c79c164ad1ac437261716e832f7169075495aa11745becaeba553cd44895580808bf51dd
c53aefeab1c5c70886e1198e8c7d7cb10a09a8e96a5cf0937a78ed4899ac6f266559e06afd22
76c301715c66ae0f4b72624355fdadb352ac734096a13fd4e15af79522d4f6d03bccd8764cc0
b230ae5256cb79be0d21f7192780ac11cf5003e54e84bff5cc14ee530407037da89749622331
8df1c1f4e0529fa07b117bb8e81cbb20c98c28846d8f4d451c36857ca2e418546e8f5804b",
    "lifespan": "2021-05-24 23:59:59"
}
```


4.5.2.5.6. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.5.2.6. Edit VM

4.5.2.6.1. Scenario

- **Scenario:**
    - Expand VM memory and update CPU configuration.
    - Modify VM name, description, group and other information.
    - Add or delete disks for a VM.
    - Change network connection for a VM.
    - Add or delete CD/DVD drive or USB drive for a VM.

- Roles Supported: Super admin, system admin, tenant, and co-administrator with the VM Configuration permission
- Versions supported: SCP 6.3.0 and above

4.5.2.6.2. URI

Table 4.5.114 API Path and Request Method

REST VERB	PATH	Description
PUT	/janus/20180725/servers/{server_id}	Edit VM

- Parameter Description:

Table 4.5.115 Path Parameters

Parameter	Location	Type	Description	Required
server_id	path	string	VM UUID	Yes

4.5.2.6.3. Request

- Request Parameter

Table 4.5.116 Description of Body Parameter

Parameter	Location	Type	Description	Required
name	body	string	VM name, up to 100 characters	No
description	body	string	VM description, up to 100 characters	No
group_id	body	string	VM group	No
cores	body	int	Total number of CPU cores. Note that this is the total number, not the number of cores per socket. The value must be greater than or equal to sockets.	No
sockets	body	int	Socket number	No
memory_mb	body	int	Memory size (MB)	No
os_type	body	string	Operating system type	No
storage_location	body	string	Storage location	No
location	body	dict	Run location, view Parameter Description - location for details	No
hard_delete_disks	body	list	List of permanently deleted disks	No
data_erase_disks	body	list	List of disks with data erased	No
disks	body	dict	Disk information, view Parameter Description - disks for details	No
networks	body	dict	Network information, view Parameter Description - networks for details	No
cdroms	body	dict	CD/DVD drive information, view Parameter Description - cdroms for details	No
usbs	body	dict	USB device information, view Parameter Description - usbs for details	No
advance_param	body	dict	HCI advanced parameter, view Parameter Description - advance_param for details	No
gpu	body	dict	GPU configuration information, view Parameter Description - gpu for details	No

Note

When a VM is edited, only parameters and the corresponding values to be updated need to be passed in. Parameters not passed in remain unchanged.

- Parameter Description - location

Table 4.5.117 Parameter Description - location

Parameter	Type	Required	Note
id	string	Yes	Run location, such as host-xx, cluster
type	string	No	Type of run location, which is specific to VMware, view VM Working Location Type (VMware) for details
location	inst	No	Whether the VM's run location is fixed.
policy_type	string	No	Type of VM scheduling policy

- Parameter Description - disks

Table 4.5.118 Parameter Description - disks¶

Parameter	Type	Required	Note
size_mb	int	Yes	Disk size (MB), 1024 (1 GB) < size_mb < 66060288 (63 TB)
is_old_disk	int	No	Whether the disk exists before the VM is edited. 1 indicates yes.
preallocate	string	Yes	Disk provisioning method, view Disk Provisioning Method
eagerly_scrub	int	No	Whether to reset now
id	string	Yes	Disk ID. Note: ide2 indicates the system disk, which is already in use. Other disks can be used. The value can be ide0, ide1, ..., iden (n ≤ 15)
type	string	Yes	Disk type, view VM Disk Type for details
storage_file	string	No	Storage path
storage_id	string	No	Storage ID. When type=physical_disk or share_disk, this parameter needs to be passed in.
backing_file	string	No	Backend image
copy	string	No	Existing disk path
cache	string	No	Cache mode, view VM Disk Cache Mode for details
cache_size	int	No	Cache size
forecast	string	No	Whether to enable cache.
use_virtio	int	No	Whether the virtio disk is used.
storage_policy	string	No	Storage policy ID

- Parameter Description - networks

Table 4.5.119 Parameter Description - networks¶

Parameter	Type	Required	Note
connect	int	Yes	Whether the network is connected.
device_id	string	No	The connected network device ID in classic network
model	string	No	NIC type, view VM NIC Type for details
mac_address	string	No	MAC address
vif_id	string	Yes	Network ID (net0, net1, net2, ... netn)
port_id	string	No	Port
host_tso	int	No	Whether to enable TSO.
name	string	No	Name of the connected port
vpc_id	string	No	VPC network ID of the connected tenant, which can be obtained from Query VPC s
subnet_id	string	No	Connected VPC subnet ID
ip_info	dict	No	IP address information, view Parameter Description - network-ip_info for details
custom_gateway_ip	string	No	Custom gateway IP

- Parameter Description - network-ip_info

Table 4.5.120 Parameter Description - network-ip_info¶

Parameter	Type	Required	Note
ip_address	string	Yes	Specified IP Address
netmask	string	No	Netmask
gateway	string	No	Gateway
dns	list	No	DNS address list, multiple addresses can be passed in.

- Parameter Description - cdroms

Table 4.5.121 Parameter Description - cdroms¶

Parameter	Type	Required	Note
id	string	Yes	Slot number of CD/DVD drive, such as ide0, ide1
image_id	string	No	ISO image ID
location	string	No	ISO location
image_name	string	No	Name of the mounted image

- Parameter Description - usbs

Table 4.5.122 Parameter Description - usbs¶

Parameter	Type	Required	Note
id	string	Yes	USB ID(udisk0,udisk1,udisk2...udiskn)
usb_device	string	No	USB device
controller_type	string	No	Type of USB controller, such as 2.0, 3.0

- Parameter Description - advance_param

Table 4.5.123 Parameter Description - advance_param

Parameter	Type	Required	Note
cpu_type	string	No	CPU type, view VM CPU Type for details
onboot	int	No	Whether the VM is started automatically when the physical host is powered on.
use_uuid	int	No	Whether to enable UUID.
uuid	string	No	UUID
regen_uuid	int	No	Whether to regenerate a UUID. The options are 0 and 1
abnormal_recovery	int	No	Whether to reboot if fault occurs.
hugepage_memory	int	No	Whether to use huge-page memory.
balloon_memory	int	No	Whether to enable memory reclaiming。
schedopt	int	No	Whether it is a critical VM.
cpu_hotplug	int	No	Whether to enable CPU hot add.
mem_hotplug	int	No	Whether to enable memory hot add.
boot_order	string	No	Boot sequence, c indicates cdrom, d indicates disk.
boot_disk	string	No	Boot disk
dir	string	No	HCI group
use_vblk	int	No	Whether to enable vblk.
real_use_vblk	int	No	Whether to really use vblk.
storage_policy	string	No	Storage policy
invtsc	int	No	Whether to enable high-performance clock. The options are 0 and 1. Note: Supported in SCP 6.7.35 and above.

Note

You need to specify the boot_disk parameter when editing the use_uuid parameter.

- Parameter Description - gpu

Table 4.5.124 Parameter Description - gpu

Parameter	Type	Required	Note
vendor	string	Yes	GPU manufacturers, such as NVIDIA and AMD. Only NVIDIA is currently supported.
conf	dict	Yes	GPU configuration, which differs as per different manufacturers. Only NVIDIA is supported now, view Parameter Description - gpu.GpuNvidia for details

- Parameter Description - gpu-GpuNvidia

Table 4.5.125 Parameter Description - gpu.GpuNvidia¶

Parameter	Type	Required	Note
total_count	int	No	Total number of graphic cards
enable_count	int	No	Number of graphic cards enabled
gpu_type	string	Yes	Graphics Card Model
slice_type	string	Yes	vGPU type, view vGPU Type for details
schedulers	list	No	List of vGPU schedulers, view Scheduling Method for details

- Request Example

https://{SCP_IP}/janus/20180725/servers/661ce7a4-5abe-4191-832c-7d1dc6c5fd9a

```
{
    "disks": [
        {
            "id": "ide0",
            "type": "derive_disk",
            "preallocate": 0,
            "size_mb": 81920
        }
    ],
    "networks": [
        {
            "device_id": "dvs11e2c5f",
            "name": "Default_classic_network_gateway1-Switch",
            "vif_id": "net0",
            "connect": 1,
            "model": "virtio",
            "port_id": "12345678",
            "ip_info": {
                "netmask": "255.255.255.0",
                "ip_address": "192.168.33.100",
                "dns": []
            }
        }
    ]
}
```

4.5.2.6.4. Response

- Response Parameter

Table 4.5.126 Response Parameter¶

Parameter	Type	Description	Value Returned
task_id	string	Asynchronous task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "d5ae0d43-4afc-4f27-a851-9989a7083d9d"
}
```

4.5.2.6.5. Response Code

- OK: 200

- Errors: 400, 403, 404, 500

4.5.2.7. Delete VM

4.5.2.7.1. Scenario

- **Scenario:**
  - Move VM to Recycle Bin.
  - Delete VM directly without moving it to Recycle Bin.

- Roles Supported: Super admin, system admin, tenant, and co-administrator with the VM Deletion permission
- Versions supported: SCP 6.3.0 and above

4.5.2.7.2. URI

Table 4.5.127 API Path and Request Method

REST VERB	PATH	Description
DELETE	/janus/20180725/servers/{server_id}	Delete VM

- Parameter Description:

Table 4.5.128 Path Parameters

Parameter	Location	Type	Description	Required
server_id	path	string	Virtual machine ID	Yes

4.5.2.7.3. Request

- Request Parameter

Table 4.5.129 Description of Body Parameter

Parameter	Location	Type	Description	Required
force	body	int	Whether to delete VM permanently. 1 indicates permanent delete, 0 indicates soft delete.	No

Note

Soft delete indicates moving VM to Recycle Bin, and permanent delete indicates deleting VM without moving it to Recycle Bin. Soft delete is performed by default.

- Request Example

  https://{SCP_IP}/janus/20180725/servers/661ce7a4-5abe-4191-832c-7d1dc6c5fd9a

```
{
    "force": 1
 }
```

4.5.2.7.4. Response

- Response Parameter

Table 4.5.130 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Asynchronous task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "d5ae0d43-4afc-4f27-a851-9989a7083d9d"
}
```

4.5.2.7.5. Response Code

- Normal: 204
- Errors: 400, 403, 404, 500

4.5.3. VM Operations & Maintenance

4.5.3.1. Enable VM

4.5.3.1.1. Scenario

- Scenario: Enable VM
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the VM Power Control permission
- Versions supported: SCP 6.3.0 and above

4.5.3.1.2. URI

Table 4.5.131 API Path and Request Method

REST VERB	PATH	Description
POST	/janus/20180725/servers/{server_id}/start	Enable VM

- Parameter Description:

Table 4.5.132 Path Parameters

Parameter	Location	Type	Description	Required
server_id	path	string	Virtual machine ID	Yes

4.5.3.1.3. Request

- **Request Parameter**
  None

- Request Example

https://{SCP_IP}/janus/20180725/servers/661ce7a4-5abe-4191-832c-7d1dc6c5fd9a/start

4.5.3.1.4. Response

- Response Parameter

Table 4.5.133 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Asynchronous task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "d5ae0d43-4afc-4f27-a851-9989a7083d9d"
}
```

4.5.3.1.5. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.5.3.2. Shut Down VM

4.5.3.2.1. Scenario

- Scenario: Shut down VM
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the VM Power Control permission
- Versions supported: SCP 6.3.0 and above

4.5.3.2.2. URI

Table 4.5.134 API Path and Request Method

REST VERB	PATH	Description
POST	/janus/20180725/servers/{server_id}/stop	Shut Down VM

- Parameter Description:

Table 4.5.135 Path Parameters

Parameter	Location	Type	Description	Required
server_id	path	string	Virtual machine ID	Yes

4.5.3.2.3. Request

- Request Parameter

Table 4.5.136 Description of Body Parameter

Parameter	Location	Type	Description	Required
force	body	int	Whether to force shutdown (forced shutdown will power off the VM). 1 indicates yes, 0 indicates no	No

Note

The default is not to force shutdown.

- Request Example

  https://{SCP_IP}/janus/20180725/servers/661ce7a4-5abe-4191-832c-7d1dc6c5fd9a/stop

```
{
  "force": 1
}
```

4.5.3.2.4. Response

- Response Parameter

Table 4.5.137 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Asynchronous task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "d5ae0d43-4afc-4f27-a851-9989a7083d9d"
}
```

4.5.3.2.5. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.5.3.3. Restart VM

4.5.3.3.1. Scenario

- Scenario: Restart the VM
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the VM Power Control permission
- Versions supported: SCP 6.3.0 and above

4.5.3.3.2. URI

Table 4.5.138 API Path and Request Method

REST VERB	PATH	Description
POST	/janus/20180725/servers/{server_id}/reboot	Restart VM

- Parameter Description:

Table 4.5.139 Path Parameters

Parameter	Location	Type	Description	Required
server_id	path	string	Virtual machine ID	Yes

4.5.3.3.3. Request

- Request Parameter

Table 4.5.140 Description of Body Parameter

Parameter	Location	Type	Description	Required
force	body	int	Whether to force restart (forced restart will power on the VM). 1 indicates forcing restart, 0 indicates restarting the VM internally.	No

Note

The default is to restart the VM internally.

- Request Example

  https://{SCP_IP}/janus/20180725/servers/661ce7a4-5abe-4191-832c-7d1dc6c5fd9a/reboot

```
{
  "force": 1
}
```

4.5.3.3.4. Response

- Response Parameter

Table 4.5.141 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Asynchronous task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "d5ae0d43-4afc-4f27-a851-9989a7083d9d"
}
```

4.5.3.3.5. Response Code

- Success: 201
- Errors: 400, 403, 404，500

4.5.3.4. Allocate VM

4.5.3.4.1. Scenario

- **Scenario:**
  - Allocate VM from the cloud platform to tenant.
  - Allocate VM from the cloud platform to user.
  - Allocate VM from the tenant to user.

- Roles Supported: Super admin, system admin, tenant, and co-administrator with the VM Allocation/Deallocation permission
- Versions supported: SCP 6.3.0 and above

4.5.3.4.2. URI

Table 4.5.142 API Path and Request Method

REST VERB	PATH	Description
POST	/janus/20180725/servers/{server_id}/allocation	Allocate VM

- Parameter Description:

Table 4.5.143 Path Parameters

Parameter	Location	Type	Description	Required
server_id	path	string	Virtual machine ID	Yes

4.5.3.4.3. Request

- Request Parameter

Table 4.5.144 Description of Body Parameter

Parameter	Location	Type	Description	Required
project_id	body	string	Tenant ID	Yes
user_id	body	string	User ID	No

Note

Passing user_id indicates the VM is allocated to a tenant's user

- Request Example

  https://{SCP_IP}/janus/20180725/servers/661ce7a4-5abe-4191-832c-7d1dc6c5fd9a/allocation

```
{
    "project_id": "d58775da112543679725a95deaa1f8ed",
    "user_id": "0f744f996ee5484285d0ca0c74d655f3"
}
```

4.5.3.4.4. Response

- Response Parameter

Table 4.5.145 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Asynchronous task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "d5ae0d43-4afc-4f27-a851-9989a7083d9d"
}
```

4.5.3.4.5. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.5.3.5. Allocate VM

4.5.3.5.1. Scenario

- **Scenario:**
    - Allocate VM from the cloud platform to tenant.
    - Allocate VM from the cloud platform to user.
    - Allocate VM from the tenant to user.

- Applicable roles: Super administrator, system administrator, tenant, and roles with VM allocation and deallocation permissions
- Versions supported: SCP 6.8.0 and above

4.5.3.5.2. URI

Table 4.5.146 API Path and Request Method

REST VERB	PATH	Description
POST	/janus/20190725/servers/{server_id}/allocation	Allocate VM

- Parameter Description:

Table 4.5.147 Path Parameters¶

Parameter	Location	Type	Description	Required
server_id	path	string	Virtual machine ID	Yes

4.5.3.5.3. Request

- Request Parameter

Table 4.5.148 Description of Body Parameter¶

Parameter	Location	Type	Description	Required
project_id	body	string	Tenant ID	Yes
user_id	body	string	User ID	No

Note

Passing user_id indicates the VM is allocated to a tenant's user

- Request Example

https://{SCP_IP}/janus/20180725/servers/661ce7a4-5abe-4191-832c-7d1dc6c5fd9a/allocation

```
{
    "project_id": "d58775da112543679725a95deaa1f8ed",
    "user_id": "0f744f996ee5484285d0ca0c74d655f3"
}
```

4.5.3.5.4. Response

- Response Parameter

Table 4.5.149 Response Parameter¶

Parameter	Type	Description	Value Returned
task_id	string	Asynchronous task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "d5ae0d43-4afc-4f27-a851-9989a7083d9d"
}
```

4.5.3.5.5. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.5.3.6. Clone VM

4.5.3.6.1. Scenario

- **Scenario:**
  - Clone from the current state of a VM.
  - Clone from the specified snapshot of a VM.

- Roles Supported: Super admin, system admin, tenant, and co-administrator with the VM Clone permission
- Versions supported: SCP6.3.70 and above

> **Note**
>
> 1. Cloning VM supports cloning from snapshot. Cloned VM is disconnected from network by default.
> 2. Tenants can only clone from their own VMs or snapshots.
> 3. admin can clone from any VMs on the platform, but the cloned VM belongs to admin only. If admin clones a tenant's VM, the tenant can view this operation in Tasks.

4.5.3.6.2. URI

Table 4.5.150 API Path and Request Method

REST VERB	PATH	Description
POST	/janus/20180725/servers/{server_id}/clone	VM clone

- Parameter Description:

Table 4.5.151 Path Parameters

Parameter	Location	Type	Description	Required
server_id	path	string	VM UUID	Yes

4.5.3.6.3. Request

- Request Parameter

Table 4.5.152 Description of Body Parameter

Parameter	Location	Type	Description	Required
name	body	string	VM name, 1-70 characters	Yes
power_on	body	int	Whether to power on the cloned VM immediately.	Yes
count	body	int	Number of clones, the maximum is 20.	Yes
description	body	string	Description information, 1-100 characters	No
storage_tag_id	body	string	Storage Tag ID	No
compute_location_id	body	string	Run location ID. The default is cluster. To specify run location, view Query Physical Machines	No
storage_location	body	string	Storage location, view Query Storage List	No
group_id	body	string	Group where the clone locates	No
advance_param	body	dict	Additional parameter, view Parameter Description - advance_param	No
clone_type	body	string	View clone-type-table for clone method. Note: This parameter is supported only in SCP 6.3.83 above	No

Note

1. When storage_tag_id is specified, the compute_location_id and storage_location parameters do not need to be specified.
2. To clone VM from a snapshot, you need to pass in advance_param.snapshot_id; otherwise it will clone from the current state of the VM by default.

- Parameter Description - advance_param

Table 4.5.153 Parameter Description - advance_param

Parameter	Location	Type	Description	Required
clone_share_disk	body	int	Whether to clone shared disk.	No
snapshot_id	body	string	Snapshot ID, indicating the VM is cloned from a specified snapshot, which can be obtained from Get VM snapshots	No
return_uuids	body	int	Whether to return the cloned list of VM IDs. The options are 0 and 1	No

- Request Example

https://{SCP_IP}/janus/20180725/servers/bb22a4b2-41e7-4404-89c1-745182f790d6/clone

```
{
    "name": "Test cloned VM",
    "description": "",
    "storage_tag_id": "00000000-0000-0000-0000-000000000000",
    "power_on": 0,
    "advance_param": {
        "return_uuids": 1
    },
    "count": 1
}
```

4.5.3.6.4. Response

- Response Parameter

Table 4.5.154 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Asynchronous task ID	Yes
uuids	list	Cloned list of VM UUIDs	No

- Response Example:

```
{
    "uuids": [
        "aaea7006-f4e8-4914-92e8-e61ad07ca909"
    ],
    "task_id": "df1ecb44-817b-4544-b83a-92ea89973918"
}
```

4.5.3.6.5. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.5.3.7. Suspend VM

4.5.3.7.1. Scenario

- Scenario: Suspend VM
- Roles Supported: Super admin and co-administrator with the VM Power Operation permission, tenant and the tenant co-administrator with the VM Power Operation permission
- Versions supported: SCP 6.3.83 and above

4.5.3.7.2. URI

Table 4.5.155 API Path and Request Method

REST VERB	PATH	Description
POST	/janus/20180725/servers/{server_id}/suspend	Suspend VM

- Parameter Description:

Table 4.5.156 Path Parameters

Parameter	Location	Type	Description	Required
server_id	path	string	Virtual machine ID	Yes

4.5.3.7.3. Request

- **Request Parameter**
  None

- Request Example

  https://{SCP_IP}/janus/20180725/servers/68c8e770-b9f6-4435-9951-59fd7dac3aa2/suspend

4.5.3.7.4. Response

- Response Parameter

Table 4.5.157 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Asynchronous task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "a3bb9e04-82b0-4d3d-ab0e-1e714c941559"
}
```

4.5.3.7.5. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.5.4. VM Snapshot Managment

4.5.4.1. Get VM snapshots

4.5.4.1.1. Scenario

- Scenario: Get VM snapshots
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the VM Snapshot permission
- Versions supported: SCP6.3.70 and above

Note

1. Only VM in the HCI resource pool can manage snapshots.
2. In the resource pool of HCI 6.2.0 and earlier, the snapshot number of a single VM cannot exceed 7. This restriction does not exist in later versions.
3. Tenants can only query their own VM snapshots.
4. admin can query the snapshots of any VMs on the platform.

4.5.4.1.2. URI

Table 4.5.158 API Path and Request Method

REST VERB	PATH	Description
GET	/janus/20180725/servers/{server_id}/snapshots	Get VM snapshots

- Parameter Description:

Table 4.5.159 Path Parameters

Parameter	Location	Type	Description	Required
server_id	path	string	VM UUID	Yes

4.5.4.1.3. Request

- Request Parameter

None

- Request Example

  https://{SCP_IP}/janus/20180725/servers/bb22a4b2-41e7-4404-89c1-745182f790d6/snapshots

4.5.4.1.4. Response

- Response Parameter

Table 4.5.160 Response Parameter

Parameter	Type	Description	Value Returned
name	string	Snapshot name	Yes
id	string	Snapshot ID	Yes
description	string	Snapshot description	Yes
parent_id	string	Previous snapshot ID, indicating that a previous snapshot exists before this snapshot is created. When the value is empty, it indicates the first snapshot.	Yes
created_at	string	Snapshot creation time in ISO 8601 format. Example: 2020-12-01T03:17:42Z	Yes
is_latest	int	Whether it is the latest snapshot of the VM, which mostly resembles the current VM status. It is usually the last snapshot taken by the VM or the last snapshot to which the VM is recovered.	Yes

> **Note**
>
> For common response parameters, view Response Format for details

- Response Example:

```
[
    {
        "name": "Snapshot1",
        "create_at": "2020-12-01T03:17:42Z",
        "id": "a047380ba-21b1-4437-8b0c-f113f9054fc8a",
        "parent_id": null,
        "is_latest": 0,
        "description": "Initial snapshot"
    }
]
```

4.5.4.1.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.5.4.2. Get VM snapshots

4.5.4.2.1. Scenario

- Scenario: Get VM snapshots
- Applicable roles: Super administrator, system administrator, tenant, and roles with permissions related to VM snapshots
- Versions supported: SCP 6.8.0 and above

> **Note**
>
> 1. Only VM in the HCI resource pool can manage snapshots.
> 2. In the resource pool of HCI 6.2.0 and earlier, the snapshot number of a single VM cannot exceed 7. This restriction does not exist in later versions.
> 3. Tenants can only query their own VM snapshots.
> 4. admin can query the snapshots of any VMs on the platform.

4.5.4.2.2. URI

Table 4.5.161 API Path and Request Method

REST VERB	PATH	Description
GET	/janus/20190725/servers/{server_id}/snapshots	Get VM snapshots

- Parameter Description:

Table 4.5.162 Path Parameters

Parameter	Location	Type	Description	Required
server_id	path	string	VM UUID	Yes

4.5.4.2.3. Request

- Request Parameter

None

- Request Example

  https://{SCP_IP}/janus/20190725/servers/bb22a4b2-41e7-4404-89c1-745182f790d6/snapshots

4.5.4.2.4. Response

- Response Parameter

Table 4.5.163 Response Parameter

Parameter	Type	Description	Value Returned
name	string	Snapshot name	Yes
id	string	Snapshot ID	Yes
description	string	Snapshot description	Yes
parent_id	string	Previous snapshot ID, indicating that a previous snapshot exists before this snapshot is created. When the value is empty, it indicates the first snapshot.	Yes
created_at	string	Snapshot creation time in ISO 8601 format. Example: 2020-12-01T03:17:42Z	Yes
is_latest	int	Whether it is the latest snapshot of the VM, which mostly resembles the current VM status. It is usually the last snapshot taken by the VM or the last snapshot to which the VM is recovered.	Yes
snapshot_type	string	View server-snapshot-type-table for snapshot types.	Yes
is_protected	int	Whether the image is protected	Yes
trigger	string	Trigger type	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
[
    {
        "name": "auto_2022-04-28-05-12-21",
        "trigger": "auto",
        "created_at": "2022-04-27T21:17:27Z",
        "snapshot_type": "vs",
        "parent_id": "",
        "is_protected": 0,
        "is_latest": 0,
        "id": "33d587d2-f7cf-4683-8835-6caadd495831",
        "description": "auto_2022-04-28-05-12-21"
    }
]
```

4.5.4.2.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.5.4.3. Create a VM Snapshot

4.5.4.3.1. Scenario

- Scenario: Create a snapshot of the VM
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the VM Snapshot permission
- Versions supported: SCP6.3.70 and above

Note

1. Only VM in the HCI resource pool can manage snapshots.
2. Snapshot number of a single VM cannot exceed 7.

4.5.4.3.2. URI

Table 4.5.164 API Path and Request Method

REST VERB	PATH	Description
POST	/janus/20180725/servers/{server_id}/snapshots	Create a VM Snapshot

- Parameter Description:

Table 4.5.165 Path Parameters

Parameter	Location	Type	Description	Required
server_id	path	string	VM UUID	Yes

4.5.4.3.3. Request

- Request Parameter

Table 4.5.166 Description of Body Parameter

Parameter	Location	Type	Description	Required
name	body	string	Snapshot name, 'current' is not allowed because it is reserved.	Yes
description	body	string	Snapshot description	No

- Request Example

https://{SCP_IP}/janus/20180725/servers/bb22a4b2-41e7-4404-89c1-745182f790d6/snapshots

```
{
    "name": "Snapshot1",
    "description": "Initial snapshot"
}
```

4.5.4.3.4. Response

- Response Parameter

Table 4.5.167 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Asynchronous task ID	Yes

> **Note**
>
> For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "d5ae0d43-4afc-4f27-a851-9989a7083d9d"
}
```

4.5.4.3.5. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.5.4.4. Create a VM Snapshot

4.5.4.4.1. Scenario

- Scenario: Create a snapshot of the VM
- Applicable roles: Super administrator, system administrator, tenant, and roles with permissions related to VM snapshots
- Versions supported: SCP 6.8.0 and above

> **Note**
>
> 1. Only VM in the HCI resource pool can manage snapshots.
> 2. Snapshot number of a single VM cannot exceed 7.

4.5.4.4.2. URI

Table 4.5.168 API Path and Request Method

REST VERB	PATH	Description
POST	/janus/20190725/servers/{server_id}/snapshots	Create a VM Snapshot

- Parameter Description:

Table 4.5.169 Path Parameters

Parameter	Location	Type	Description	Required
server_id	path	string	VM UUID	Yes

4.5.4.4.3. Request

- Request Parameter

Table 4.5.170 Description of Body Parameter

Parameter	Location	Type	Description	Required
name	body	string	Snapshot name, 'current' is not allowed because it is reserved.	Yes
description	body	string	Snapshot description	No
snap_type	body	string	View server-snapshot-type-table for snapshot types.	No

- Request Example

  https://{SCP_IP}/janus/20180725/servers/bb22a4b2-41e7-4404-89c1-745182f790d6/snapshots

```
{
    "name": "Snapshot 1",
    "description": "Initial snapshot",
    "snap_type": "vs"
}
```

4.5.4.4.4. Response

- Response Parameter

Table 4.5.171 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Asynchronous task ID	Yes
resource_id	string	Resource ID	No

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "d5ae0d43-4afc-4f27-a851-9989a7083d9d",
    "resource_id": "1fc31330-20b9-4d0e-ae21-25041eeb0f03"
}
```

4.5.4.4.5. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.5.4.5. Edit VM Snapshot

4.5.4.5.1. Scenario

- Scenario: Update snapshot information of the VM
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the VM Snapshot permission
- Versions supported: SCP6.3.70 and above

> **Note**
>
> Only VM in the HCI resource pool supports snapshot management.

4.5.4.5.2. URI

Table 4.5.172 API Path and Request Method

REST VERB	PATH	Description
PUT	/janus/20180725/servers/{server_id}/snapshots/{snapshot_id}	Edit VM Snapshot

- Parameter Description:

Table 4.5.173 Path Parameters

Parameter	Location	Type	Description	Required
server_id	path	string	VM UUID	Yes
snapshot_id	path	string	Snapshot ID	Yes

4.5.4.5.3. Request

- Request Parameter

Table 4.5.174 Description of Body Parameter

Parameter	Location	Type	Description	Required
name	body	string	Snapshot name, which cannot exceed 90 characters. 'current' is not allowed because it is reserved.	Yes
description	body	string	Snapshot description, which cannot exceed 90 characters.	No
is_protected	body	int	Protected snapshot	No

- Request Example

  https://{SCP_IP}/janus/20180725/servers/bb22a4b2-41e7-4404-89c1-745182f790d6/snapshots/a047380ba-21b1-4437-8b0c-f113f9054fc8a

```
{
    "name": "Snapshot 1",
    "description": "First snapshot",
    "is_protected": 1
}
```

4.5.4.5.4. Response

- Response Parameter

Table 4.5.175 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Asynchronous task ID	Yes

> **Note**
>
> For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "d5ae0d43-4afc-4f27-a851-9989a7083d9d"
}
```

4.5.4.5.5. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.5.4.6. Delete VM Snapshot

4.5.4.6.1. Scenario

- Scenario: Delete specified VM snapshots
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the VM Snapshot permission
- Versions supported: SCP6.3.70 and above

> **Note**
>
> 1. Tenants can only delete snapshots of their own VMs.
> 2. admin can delete snapshots of any VMs on the platform.

4.5.4.6.2. URI

Table 4.5.176 API Path and Request Method

REST VERB	PATH	Description
DELETE	/janus/20180725/servers/{server_id}/snapshots/{snapshot_id}	Delete VM Snapshot

- Parameter Description:

Table 4.5.177 Path Parameters

Parameter	Location	Type	Description	Required
server_id	path	string	VM UUID	Yes
snapshot_id	path	string	Snapshot ID	Yes

4.5.4.6.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/servers/bb22a4b2-41e7-4404-89c1-

745182f790d6/snapshots/a047380ba-21b1-4437-8b0c-f113f9054fc8a

4.5.4.6.4. Response

- Response Parameter

Table 4.5.178 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Asynchronous task ID	Yes

> **Note**
>
> For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "d5ae0d43-4afc-4f27-a851-9989a7083d9d"
}
```

4.5.4.6.5. Response Code

- Normal: 204
- Errors: 400, 403, 404, 500

4.5.4.7. Recover VM from Snapshot

4.5.4.7.1. Scenario

- Scenario: Recover VM from specified snapshots
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the VM Snapshot permission
- Versions supported: SCP6.3.70 and above

> **Note**
>
> Only VM in the HCI resource pool supports snapshot management.

4.5.4.7.2. URI

Table 4.5.179 API Path and Request Method

REST VERB	PATH	Description
POST	/janus/20180725/servers/{server_id}/snapshots/{snapshot_id}/recovery	Recover VM from Snapshot

- Parameter Description:

Table 4.5.180 Path Parameters

Parameter	Location	Type	Description	Required
server_id	path	string	VM UUID	Yes
snapshot_id	path	string	Snapshot ID	Yes

4.5.4.7.3. Request

- Request Parameter

Table 4.5.181 Description of Body Parameter

Parameter	Location	Type	Description	Required
backup	body	int	Whether to create a snapshot automatically before snapshot recovery.	Yes
power_on	body	int	Whether to power on after recovery	No

- Request Example

  https://{SCP_IP}/janus/20180725/servers/bb22a4b2-41e7-4404-89c1-745182f790d6/snapshots/a047380ba-21b1-4437-8b0c-f113f9054fc8a/recovery

```
{
    "backup": 1,
    "power_on":  1
}
```

4.5.4.7.4. Response

- Response Parameter

Table 4.5.182 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Asynchronous task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "d5ae0d43-4afc-4f27-a851-9989a7083d9d"
}
```

4.5.4.7.5. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.5.4.8. Clone VM from Snapshot

View Clone VM for details

4.5.5. VM Group Mangement

4.5.5.1. Create a VM Group

4.5.5.1.1. Scenario

- Scenario: Create a VM group
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the VM Grouping and Labelling permissions
- Versions supported: SCP 6.3.0 and above

4.5.5.1.2. URI

Table 4.5.183 API Path and Request Method

REST VERB	PATH	Description
POST	/janus/20180725/servers/groups	Create a VM Group

- Parameter Description:

  None

4.5.5.1.3. Request

- Request Parameter

Table 4.5.184 Description of Body Parameter

Parameter	Location	Type	Description	Required
name	body	string	Group name	Yes
az_id	body	string	Resource pool ID	Yes
project_id	body	string	Tenant ID	Yes
parent_id	body	string	Parent group ID	No

- Request Example

  https://{SCP_IP}/janus/20180725/servers/groups

```
{
    "name": "Database",
    "az_id": "70172f8b-ad3d-47c3-9427-308a4c5052d0",
    "project_id": "51b7ce9a856b48908b0b1afd143b95e0"
}
```

4.5.5.1.4. Response

- Response Parameter

Table 4.5.185 Response Parameter

Parameter	Type	Description	Value Returned
id	string	Group ID	Yes
name	string	Group name	Yes
az_id	string	Resource pool ID	Yes
project_id	string	Tenant ID	No
parent_id	string	Parent group ID	No
type	string	Group type	No

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "parent_id": null,
    "project_id": "51b7ce9a856b48908b0b1afd143b95e0",
    "id": "83452197-2c50-439c-aa2f-cf2cdce3ced1",
    "az_id": "70172f8b-ad3d-47c3-9427-308a4c5052d0",
    "name": "Database"
}
```

4.5.5.1.5. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.5.5.2. Update VM Group

4.5.5.2.1. Scenario

- **Scenario:**
  - Modify the name of VM group
  - Move VM to group

- Roles Supported: Super admin, system admin, tenant, and co-administrator with the VM Grouping and Labelling permissions
- Versions supported: SCP 6.3.0 and above

4.5.5.2.2. URI

Table 4.5.186 API Path and Request Method

REST VERB	PATH	Description
PUT	/janus/20180725/servers/groups/{group_id}	Update VM Group

- Parameter Description:

Table 4.5.187 Path Parameters

Parameter	Location	Type	Description	Required
group_id	path	string	Group ID	Yes

4.5.5.2.3. Request

- Request Parameter

Table 4.5.188 Description of Body Parameter

Parameter	Location	Type	Description	Required
server_ids	body	list	VMs which need to be moved to a group	No
name	body	string	The updated group name	No

- Request Example

  https://{SCP_IP}/janus/20180725/servers/groups/661ce7a4-5abe-4191-832c-7d1dc6c5fd9a

```
{
    "server_ids": [
        "b2321502-933a-4193-bb2e-8a094d76cefd"
    ],
    "name": "tjc_test"
}
```

4.5.5.2.4. Response

- Response Parameter

Table 4.5.189 Response Parameter

Parameter	Type	Description	Value Returned
id	string	Group ID	Yes
name	string	Group name	Yes
az_id	string	Resource pool ID	Yes
project_id	string	Tenant ID	No
parent_id	string	Parent group ID	No
server_update_summary	dict	VM migration result	No
type	string	Group type	No

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "parent_id": null,
    "project_id": "51b7ce9a856b48908b0b1afd143b95e0",
    "id": "83452197-2c50-439c-aa2f-cf2cdce3ced1",
    "az_id": "70172f8b-ad3d-47c3-9427-308a4c5052d0",
    "name": "tjc_test",
    "server_update_summary": {
        "succeeded": ["b2321502-933a-4193-bb2e-8a094d76cefd"],
        "failed": []
    }
}
```

4.5.5.2.5. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.6. Images

4.6.1. Common Image Parameters

4.6.1.1. Image Type

Table 4.6.1 Image Type¶

Value	Description
private	Private VM images, which are used to create virtual machines and only visible to users themselves, and consume the quota of users' private images
public	Public image, used to create virtual machines and visible to all users on the platform
service	(Removed)
vnetdev	Network device images, required by ADC, NGAF, IAG and other network devices
app-center	Application Center image, which is required by devices in Application Center

4.6.1.2. Image Status

Table 4.6.2 Image Status¶

Value	Description
active	Available
deactivated	Unavailable
saving	Saving
error	Abnormal

4.6.1.3. Image OSes

Table 4.6.3 Image OSes¶

Value	OS
linux-centos	CentOS
linux-suse	SUSE Linux
linux-oracle	Oracle Linux
l26	Linux 3.X/2.6 Kernel (32 bit)
l2664	Linux 3.X/2.6 Kernel (64 bit)
l24	Linux 2.4 Kernel (32-bit)
l2464	Linux 2.4 Kernel (64 bit)
linux-freebsd	FreeBSD
linux-redflag	Red Flag Linux
linux-neokylin	NeoKylin
linux-neoshine	NeoShine Linux
linux-deepin	Deepin Linux
linux-ubuntu	Ubuntu
linux-debian	Debian GNU/Linux
linux-redhat	Red Hat Enterprise Linux
linux-solaris	Solaris
linux-fedora	Fedora
linux-kylinos	Kylin OS
linux-arm	Linux ARM
ws1964	Windows Server 2019 (64 bit)
ws1664	Windows Server 2016 (64 bit)
ws1264	Windows Server 2012 (64 bit)
ws08	Windows Server 2008 (32 bit)
ws0864	Windows Server 2008 (64 bit)
ws03	Windows Server 2003 (32 bit)
ws0364	Windows Server 2003 (64 bit)
win10	Microsoft Windows 10 (32 bit)
win1064	Microsoft Windows 10 (64 bit)
win8	Microsoft Windows 8 (32 bit)
win864	Microsoft Windows 8 (64 bit)
win7	Microsoft Windows 7 (32 bit)
win764	Microsoft Windows 7 (64 bit)
wxp	Microsoft Windows XP (32 bit)
wxp64	Microsoft Windows XP (64 bit)
w2k	Microsoft Windows 2000
other	32-bit OS
other64	64-bit OS

4.6.1.4. Image Format

Table 4.6.4 Image Format¶

Value	Image Type
ISO	ISO image
aCloud	Built-in image (HCI)
vmware	VMware image

4.6.1.5. Image Visibility

Table 4.6.5 Image Visibility¶

Value	Description
private	Private image
public	Public image
shared	Shared image
community	Community image

4.6.1.6. Image Sharing Status

Table 4.6.6 Image Sharing Status¶

Value	Description
accepted	Accepted
rejected	Rejected
pending	Pending

4.6.1.7. Service Image Type

Table 4.6.7 Service Image Type¶

Value	Description
cssp	(Removed)
dsp-db	Database image (for cloud database service)
dsp-dmp	Database Management Platform image (for cloud database platform)
cbsp	aSecurity service image
edr-mgr	VM security protection image

4.6.2. Query Images

4.6.2.1. Scenario

- Scenario: Query images
- Applicable roles: Super administrator, system administrator, tenant and other roles with permission to read images.
- Versions supported: SCP 6.3.0 and above; SCP 6.8.0 and above support filtering by image_ids

> **Note**
>
> Tenants can only view private images and public images.

4.6.2.2. URI

Table 4.6.8 API Path and Request Method

REST VERB	PATH	Description
GET	/janus/20180725/images	Get images

- Parameter Description:

Table 4.6.9 Path Parameters

Parameter	Location	Type	Description	Required
name	path	string	Image name	No
image_type	path	string	Image type, view Image Type for details	No
status	path	string	Image status, view Image Status for details	No
os_type	path	string	Image OS, view Image OSes for details	No
disk_format	path	string	Image format, view Image Format for details	No
az_id	path	string	Associated resource pool ID	No
owner_id	path	string	Tenant ID	No
image_ids	path	string	List of image IDs where IDs are separated by commas (,)	No

> **Note**
>
> For common path parameters, view Common Path Parameter Format for details

4.6.2.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/images?page_num=0&page_size=100&image_type=service

4.6.2.4. Response

- Response Parameter

Table 4.6.10 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Image ID	No
name	string	Image name	No
description	string	Description of service image	No
image_type	string	Image type, view Image Type for details	No
status	string	Image status, view Image Status for details	No
os_type	string	Image OS, view Image OSes for details	No
visibility	string	Image visibility, view Image Visibility for details	No
disk_format	string	Image format, view Image Format for details	No
owner_id	string	Tenant ID	No
size_mb	float	Image size (MB)	No
virtual_size_mb	float	Image disk capacity (MB)	No
category	string	Image type. normal indicates common images, and service indicates service images	No
is_vmtools_installed	int(0, 1)	Whether vmtools is installed. 0: Not installed, 1: Installed	No
has_gpu	int(0, 1)	Whether image supports GPU. 0: Do not support, 1: Support	No
message	string	Image related message (error message)	No
task_id	string	Task ID, used to query progress when the image is applied	No
support_versions	list(string)	Supported cluster versions, all means that image supports all cluster versions	No
azs	list(dict)	Associated resource pool information, view Parameter Description - azs for details	No
disks	list(dict)	Built-in disk information of the image, view Parameter Description - disks for details	No
image_members	list(dict) or None	Image users, view Parameter Description - image_members for details	No

- Parameter Description - azs:

Table 4.6.11 Parameter Description - azs¶

Parameter	Type	Description	Value Returned
id	string	Resource pool ID	Yes
is_available	string	Whether the image is available in the resource pool, 0: Unavailable, 1: Available	Yes
arch_type	string	Architecture type, view Architecture Type for details	Yes

- Parameter Description - disks:

Table 4.6.12 Parameter Description - disks¶

Parameter	Type	Description	Value Returned
id	string	Disk ID	Yes
storage_id	string	Storage ID	Yes
size_mb	float	Image size (MB)	Yes
preallocate	string	Disk provisioning method, view Disk Provisioning Method for details	Yes
tpl_config	dict	Template configuration, view Parameter Description - disks tpl_config for details	No

- Parameter Description - disks tpl_config:

Table 4.6.13 Parameter Description - disks tpl_config¶

Parameter	Type	Description	Value Returned
mem_hotplug	int(0, 1)	Whether to enable memory hotplug	Yes
cpu_hotplug	int(0, 1)	Whether to enable CPU hotplug	Yes
machine_type	string	Motherboard information	No

- Parameter Description - image_members:

Table 4.6.14 Parameter Description - image_members¶

Parameter	Type	Description	Value Returned
member_id	string	Image user ID (tenant ID)	Yes
status	string	Image sharing status, view Image Sharing Status for details	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
[
    {
        "status": "error",
        "size_mb": 7.62939453125e-06,
        "virtual_size_mb": 0,
        "description": "",
        "task_id": "",
        "support_versions": [
            "all"
        ],
        "image_type": "public",
        "os_type": "",
        "disks": [],
        "disk_format": "ISO",
        "owner_id": "b9f644e5d4ed42358d287b19ea944e71",
        "visibility": "public",
        "has_gpu": 0,
        "is_vmtools_installed": 0,
        "image_members": null,
        "message": "",
        "azs": [],
        "id": "71eed759-4745-4a6b-97b7-844a9ea77198",
        "name": "wzg test_1612-588R6_b7240d86 _1612-588R6_71eed759"
    },
    {
        "status": "active",
        "size_mb": 2304.0,
        "virtual_size_mb": 81920.0,
        "description": "",
        "task_id": "57fd00dd-487d-4f49-9951-455755087f84",
        "support_versions": [
            "5.8.8.R6"
        ],
        "image_type": "public",
        "os_type": "l2664",
        "disks": [
            {
                "size_mb": 81920.0,
                "preallocate": "off",
                "storage_id": "71295054_vs_vol_rep2",
                "id": "ide0",
                "tpl_config": {
                    "mem_hotplug": 0,
                    "cpu_hotplug": 0,
                    "machine_type": "i440fx"
                }
            },
            {
                "size_mb": 81920.0,
                "preallocate": "off",
                "storage_id": "1SANGFOR_ASAN-001-983f-4930-8000-2d33c04ffc92
",
                "id": "ide0",
                "tpl_config": {
                    "mem_hotplug": 0,
                    "cpu_hotplug": 0,
                    "machine_type": "i440fx"
                }
            }
        ],
        "disk_format": "aCloud",
        "owner_id": "b9f644e5d4ed42358d287b19ea944e71",
        "visibility": "public",
        "has_gpu": 0,
        "is_vmtools_installed": 1,
        "image_members": null,
        "message": "",
        "azs": [
            {
                "is_available": 1,
                "id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
                "arch_type": "hci_x86"
            }
        ],
        "id": "d2f37876-8e5d-4d12-8de3-07e3c23ca54b",
        "name": "centos-6.5-X64-20190426"
    }
]
```


4.6.2.5. Response Code

- OK: 200

- Errors: 400, 403, 500

4.6.3. Query Service Images

4.6.3.1. Scenario

- Scenario: Query service images
- Applicable roles: Super administrator, system administrator and other administrators with permission to read images.
- Versions supported: SCP 6.3.0 and above; SCP 6.8.0 and above support filtering by image_ids

4.6.3.2. URI

Table 4.6.15 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/service-images	Get service images

- Parameter Description:

Table 4.6.16 Path Parameters¶

Parameter	Location	Type	Description	Required
name	path	string	Service image name	No
service_type	path	string	Service image type, view Service Image Type for details	No
service_tag	path	string	Service image tag	No
status	path	string	Image status, view Image Status for details	No
os_type	path	string	Image OS, view Image OSes for details	No
disk_format	path	string	Image format, view Image Format for details	No
az_id	path	string	Associated resource pool ID	No
owner_id	path	string	Tenant ID	No
image_ids	path	string	List of image IDs where IDs are separated by commas (,) (This parameter was added on July 25, 2019 for image filtering.	No

Note

For common path parameters, view Common Path Parameter Format for details

4.6.3.2.1. Request

- Request Parameter

None

- Request Example

https://{SCP_IP}/janus/20180725/service-images?page_num=0&page_size=100&owner_id=78ad8e9c9c7e487186f448bc81f050a6

4.6.3.3. Response

- Response parameter:

Table 4.6.17 Response Parameter

Parameter	Type	Description	Value Returned
id	string	Image ID	No
name	string	Image name	No
description	string	Description of service image	No
image_type	string	Image type, view Image Type for details	No
service_type	string	Service image type, view Service Image Type for details	No
service_tag	string	Service image tag	No
service_version	string	Service image version	No
category	string	Image type. normal indicates common images, and service indicates service images	No
status	string	Image status, view Image Status for details	No
os_type	string	Image OS, view Image OSes for details	No
visibility	string	Image visibility, view Image Visibility for details	No
disk_format	string	Image format, view Image Format for details	No
owner_id	string	Tenant ID	No
size_mb	float	Image size (MB)	No
virtual_size_mb	float	Image disk capacity (MB)	No
is_vmtools_installed	int(0, 1)	Whether vmtools is installed. 0: Not installed, 1: Installed	No
has_gpu	int(0, 1)	Whether image supports GPU. 0: Do not support, 1: Support	No
message	string	Image related message (error message)	No
task_id	string	Task ID, used to query progress when the image is applied	No
support_versions	list(string)	Supported cluster versions, all means that image supports all cluster versions	No
azs	list(dict)	Associated resource pool information, view Parameter Description - azs for details	No
disks	list(dict)	Built-in disk information of the image, view Parameter Description - disks for details	No
image_members	list(dict) or None	Image users, view Parameter Description - image_members for details	No

- Parameter Description - azs:

Table 4.6.18 Parameter Description - azs¶

Parameter	Type	Description	Value Returned
id	string	Resource pool ID	Yes
is_available	string	Whether the image is available in the resource pool, 0: Unavailable, 1: Available	Yes
arch_type	string	Architecture type, view Architecture Type for details	Yes

- Parameter Description - disks:

Table 4.6.19 Parameter Description - disks¶

Parameter	Type	Description	Value Returned
id	string	Disk ID	Yes
storage_id	string	Storage ID	Yes
size_mb	float	Image size (MB)	Yes
preallocate	string	Disk provisioning method, view Disk Provisioning Method for details	Yes
tpl_config	dict	View Parameter Description - disks_tpl_config for template configuration details.	No

- Parameter Description - disks_tpl_config:

Table 4.6.20 Parameter Description - disks_tpl_config¶

Parameter	Type	Description	Value Returned
mem_hotplug	int(0, 1)	Whether to enable memory hotplug	Yes
cpu_hotplug	int(0, 1)	Whether to enable CPU hotplug	Yes
machine_type	string	Motherboard information	No

- Parameter Description - image_members:

Table 4.6.21 Parameter Description - image_members¶

Parameter	Type	Description	Value Returned
member_id	string	Image user ID (tenant ID)	Yes
status	string	Image sharing status, view Image Sharing Status for details	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
[
    {
        "status": "active",
        "size_mb": 5761,
        "disks": [
```

```
            {
                "size_mb": 102400,
                "preallocate": false,
                "storage_id": "36589cfc000000c6d33d91ecdeeaf8cac",
                "id": "ide0",
                "tpl_config": {
                    "mem_hotplug": 1,
                    "cpu_hotplug": 1,
                    "machine_type": ""
                }
            }
        ],
        "virtual_size_mb": 102400,
        "description": "",
        "task_id": "204f02fb-9901-45c0-82bb-b1a8cbcc9165",
        "service_version": "",
        "image_type": "service",
        "os_type": "l2664",
        "support_versions": [
            "6.3.0"
        ],
        "disk_format": "aCloud",
        "owner_id": "78ad8e9c9c7e487186f448bc81f050a6",
        "visibility": "public",
        "has_gpu": 0,
        "is_vmtools_installed": 1,
        "image_members": null,
        "service_type": "dsp-db",
        "message": "",
        "azs": [
            {
                "is_available": 1,
                "id": "30457665-e8ca-4821-bfd9-d29dc2006834"
            }
        ],
        "id": "557f70a2-674b-4fc6-ab21-2f48494cc052",
        "name": "DBVM-1201-wz"
    },
    {
        "disks": [
            {
                "size_mb": 102400.0,
                "preallocate": "off",
                "storage_id": "1SANGFOR_ASAN-00c-b4f6-42b8-938c-00a7c09d6f72
",
                "id": "ide0",
                "tpl_config": {
                    "mem_hotplug": 0,
                    "cpu_hotplug": 0,
                    "machine_type": "i440fx"
                }
            }
        ],
        "service_tag": "",
        "is_vmtools_installed": 1,
        "message": "",
        "id": "f7ff2ca9-14ae-4b12-bd9f-9d423159b7bb",
        "support_versions": [
            "6.3.0"
        ],
        "image_type": "service",
        "disk_format": "aCloud",
        "size_mb": 8065.0,
        "service_type": "dsp-dmp",
        "owner_id": "cf92f8d651c24ed6938b186e252ee0bc",
        "status": "active",
        "virtual_size_mb": 102400.0,
        "description": "",
        "visibility": "public",
        "has_gpu": 0,
        "name": "DMP2.0.2-202103260325_1.73",
        "task_id": "98371524-9b55-4f27-9f4f-b68bccced2b7",
        "service_version": "",
        "image_members": null,
        "os_type": "l2664",
        "azs": [
            {
                "is_available": 1,
                "id": "bb912da8-6bb7-4bc7-b581-6f965de2edc2",
                "arch_type": "hci_x86"
            }
        ]
    }
]
```

4.6.3.4. Response Code

- OK: 200
- Errors: 400, 403, 500

4.6.4. Query Virtual Network Device Images

4.6.4.1. Scenario

- Scenario: Query virtual network device images
- Roles Supported: Super admin, system admin, and co-administrator with the Image Read permission
- Versions supported: SCP 6.3.0 and above

4.6.4.2. URI

Table 4.6.22 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/vnetdev-images	Query Virtual Network Device Images

- Parameter Description:

Table 4.6.23 Path Parameters¶

Parameter	Location	Type	Description	Required
name	path	string	Image name	No
type	path	string	Virtual network device image type	No
status	path	string	Image status, view Image Status for details	No
os_type	path	string	Image OS, view Image OSes for details	No
disk_format	path	string	Image format, view Image Format for details	No
az_id	path	string	Associated resource pool ID	No
owner_id	path	string	Tenant ID	No

Note

For common path parameters, view Common Path Parameter Format for details

4.6.4.3. Request

- Request Parameter

  None

- Request Example

  https://{{scp_host}}/janus/20180725/vnetdev-images

4.6.4.4. Response

- Response parameter:

Table 4.6.24 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Image ID	No
name	string	Image name	No
description	string	Image description	No
image_type	string	Image type, view Image Type for details	No
type	string	Virtual network device image type	No
type_name	string	Virtual network device type and name	No
version	string	Image version	No
build_time	string	Image creation time, which is a string in ISO format	No
status	string	Image status, view Image Status for details	No
os_type	string	Image OS, view Image OSes for details	No
visibility	string	Image visibility, view Image Visibility for details	No
disk_format	string	Image format, view Image Format for details	No
owner_id	string	Tenant ID	No
size_mb	float	Image size (MB)	No
virtual_size_mb	float	Image disk capacity (MB)	No
is_vmtools_installed	int(0, 1)	Whether vmtools is installed. 0: Not installed, 1: Installed	No
has_gpu	int(0, 1)	Whether image supports GPU. 0: Do not support, 1: Support	No
category	string	Image type. normal indicates common images, and service indicates service images	No
message	string	Image related message (error message)	No
task_id	string	Task ID, used to query progress when the image is applied	No
support_versions	list(string)	Supported cluster versions, all means that image supports all cluster versions	No
azs	list(dict)	Associated resource pool information, view Parameter Description - azs for details	No
disks	list(dict)	Built-in disk information of the image, view Parameter Description - disks for details	No
image_members	list(dict) or None	Image users, view Parameter Description - image_members for details	No

- Parameter Description - azs:

Table 4.6.25 Parameter Description - azs¶

Parameter	Type	Description	Value Returned
id	string	Resource pool ID	Yes
is_available	string	Whether the image is available in the resource pool, 0: Unavailable, 1: Available	Yes
arch_type	string	Architecture type, view Architecture Type for details	No

- Parameter Description - disks:

Table 4.6.26 Parameter Description - disks¶

Parameter	Type	Description	Value Returned
id	string	Disk ID	Yes
storage_id	string	Storage ID	Yes
size_mb	float	Image size (MB)	Yes
preallocate	string	Disk provisioning method, view Disk Provisioning Method for details	Yes
tpl_config	dict	Template configuration, view Parameter Description - disks tpl_config for details	No

- Parameter Description - disks tpl_config:

Table 4.6.27 Parameter Description - disks tpl_config¶

Parameter	Type	Description	Value Returned
mem_hotplug	int(0, 1)	Whether to enable memory hotplug	Yes
cpu_hotplug	int(0, 1)	Whether to enable CPU hotplug	Yes
machine_type	string	Motherboard information	No

- Parameter Description - image_members:

Table 4.6.28 Parameter Description - image_members¶

Parameter	Type	Description	Value Returned
member_id	string	Image user ID (tenant ID)	Yes
status	string	Image sharing status, view Image Sharing Status for details	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
[
    {
        "status": "active",
        "size_mb": 2580.5,
        "virtual_size_mb": null,
        "description": "vnet_dev",
        "task_id": "bedcf6fd-c30a-4f6e-b839-d39555343a49",
        "support_versions": [
            "6.3.0"
        ],
        "image_type": "vnetdev",
        "os_type": null,
        "disks": [
            {
                "size_mb": 81920,
                "preallocate": false,
                "storage_id": "1SANGFOR_ASAN-00c-b4f6-42b8-938c-00a7c09d6f72
",
                "id": "ide0",
                "tpl_config": {
                    "mem_hotplug": 0,
                    "cpu_hotplug": 0,
                    "machine_type": ""
                }
            }
        ],
        "disk_format": null,
        "owner_id": null,
        "visibility": "public",
        "has_gpu": 0,
        "is_vmtools_installed": 0,
        "image_members": null,
        "message": "",
        "azs": [
            {
                "is_available": 1,
                "id": "611f1b34-c727-4925-854e-07bc8607f681"
            }
        ],
        "type": "DBLB",
        "id": "77222661-22f7-40a0-b3ea-d3578f4bf792",
        "name": "NFV_vDBLB_7.0.8R3_20201110"
    }
]
```


4.6.4.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.6.5. Create an Image from a Virtual Machine

4.6.5.1. Scenario

- Scenario: Create an image from a virtual machine
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the VM Image Creation permission
- Versions supported: SCP6.3.70 and above

4.6.5.1.1. URI

Table 4.6.29 API Path and Request Method¶

REST VERB	PATH	Description
POST	/janus/20180725/images/action	Create an Image from a Virtual Machine

- Parameter Description:

  None

4.6.5.1.2. Request

- Request Parameter

Table 4.6.30 Response Parameter¶

Parameter	Location	Type	Description	Required
server_id	body	string	Virtual machine ID	Yes
name	body	string	Image name	Yes
description	body	string	Image description	No
os_type	body	string	OS type, view Image OSes for details	Yes
azs	body	list(string)	Resource pool IDs	Yes
gpu_conf	body	dict	GPU parameters, view Parameter Description - gpu_conf for details	No

- Parameter Description - gpu_conf:

Table 4.6.31 Parameter Description - gpu_conf¶

Parameter	Type	Required	Note
slice_type	string	Yes	vGPU type, view vGPU Type for details
gpu_types	list(string)	Yes	GPU types
schedulers	list(string)	Yes	GPU scheduling, view Scheduling Method for details

- Request Example

```
{
    "server_id": "5eefb8f7-b6c8-436d-a883-e92c0d978f4c",
    "name": "Create_Image_from_VM",
    "os_type": "l2664",
    "azs": [
        "611f1b34-c727-4925-854e-07bc8607f681"
    ],
    "description": "Create image from VM"
}
```

4.6.5.1.3. Response

- Response Parameter

Table 4.6.32 Response Parameter¶

Parameter	Type	Description	Value Returned
task_id	string	Task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "d5ae0d43-4afc-4f27-a851-9989a7083d9d"
}
```

Note

The ID of the image being created cannot be obtained immediately. For image information, search images by image name (Query Images).

4.6.5.1.4. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.6.6. Create an Image

4.6.6.1. Scenario

- Scenario: To upload an image to the platform, you need to create an image first.
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Image Write permission
- Versions supported: SCP6.3.70 and above

Note

After an image is created, it will take effect only after it is uploaded to the platform. View Upload an Image for details.

4.6.6.1.1. URI

Table 4.6.33 API Path and Request Method

REST VERB	PATH	Description
POST	/janus/20180725/images	Used to create image

- Parameter Description:

  None

4.6.6.1.2. Request

- Request Parameter

Table 4.6.34 Response Parameter¶

Parameter	Location	Type	Description	Required
image_type	body	string	Image Type	Yes
name	body	string	Image name	No
azs	body	list(string)	Resource pool IDs	Yes
disk_format	body	string	Image format. The options are ISO and aCloud	Yes
description	body	string	Image description	No
os_type	body	string	Image OS, view Image OSes for details	No
owner_id	body	string	Image owner ID	No
service_type	body	string	Service image type, view Service Image Type for details	No
service_tag	body	string	Image tag	No

- Request Example

  https://{SCP_IP}/janus/20180725/images

  - Create a public image:

```
{
    "image_type": "public",
    "disk_format": "aCloud",
    "name": "Sangfor_SCP_6.2.0(20201016)",
    "os_type": "linux-centos",
    "description": "Public Image",
    "azs": [
        "611f1b34-c727-4925-854e-07bc8607f681"
    ]
}
```

  - Create a private image:

```
{
    "image_type": "private",
    "disk_format": "ISO",
    "name": "Sangfor_SCP_6.2.0(20201016)",
    "os_type": "linux-centos",
    "description": "Private image",
    "azs": [
        "611f1b34-c727-4925-854e-07bc8607f681"
    ],
    "owner": "c5f9ed4abab94b46ad0e7a1783a7e2ba"
}
```

  - Create a network device image:

```
{
    "image_type": "vnetdev",
    "disk_format": "aCloud",
    "azs": [
        "611f1b34-c727-4925-854e-07bc8607f681"
    ]
}
```

  - Create a service image:

```
{
    "image_type": "service",
    "disk_format": "aCloud",
    "name": "Sangfor_SCP_6.2.0(20201016)",
    "service_type": "cssp",
    "description": "Service image",
    "azs": [
        "611f1b34-c727-4925-854e-07bc8607f681"
    ]
}
```

4.6.6.1.3. Response

- Response Parameter

Table 4.6.35 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Task ID	Yes
id	string	Image ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "24591d7e-170c-4bc0-a1e1-d2cddfe8d4cb",
    "id": "882a56d4-0304-4904-a030-2d851bb36231"
}
```

4.6.6.1.4. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.6.7. Upload an Image

Note

1. Upload will fail if it is interrupted.
2. After the image is uploaded, it is not available until it has been distributed to a specified resource pool or the task “creating image” is completed.

4.6.7.1. Scenario

- Scenario:
    - Administrator uploads virtual machine images.
    - Administrator uploads private images.
    - Administrator uploads network device images.
    - (Removed)
    - Tenant uploads private images.
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Image Write permission
- Versions supported: SCP6.3.70 and above

4.6.7.1.1. URI

Table 4.6.36 API Path and Request Method¶

REST VERB	PATH	Description
PUT	/janus/20180725/images/{image_id}/upload	Upload an Image

- Parameter Description:

Table 4.6.37 Path Parameters¶

Parameter	Location	Type	Description	Required
image_id	path	string	Image ID	Yes

4.6.7.1.2. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/images/a043b00e-0b79-4199-b301-d424a35feeb6/upload

4.6.7.1.3. Response

- Response Parameter

Table 4.6.38 Response Parameter¶

Parameter	Type	Description	Value Returned
task_id	UUIDType	Task ID, the ID for distributing image	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "d5ae0d43-4afc-4f27-a851-9989a7083d9d"
}
```

4.6.7.1.4. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.6.8. Edit an Image

4.6.8.1. Scenario

- **Scenario:**
  - Modify an image.

- Associate the image with a specified resource pool.

- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Image Write permission
- Versions supported: SCP6.3.70 and above

4.6.8.1.1. URI

Table 4.6.39 API Path and Request Method¶

REST VERB	PATH	Description
PUT	/janus/20180725/images/{image_id}	Edit an Image

- Parameter Description:

Table 4.6.40 Path Parameters¶

Parameter	Location	Type	Description	Required
image_id	path	string	Image ID	Yes

4.6.8.1.2. Request

- Request Parameter

Table 4.6.41 Response Parameter¶

Parameter	Location	Type	Description	Required
name	body	string	Image name	No
description	body	string	Image description	No
os_type	body	string	OS type, view Image OSes for details	No
azs	body	list(string)	Resource pool IDs	No
gpu_conf	body	dict	GPU parameter description, view Parameter Description - gpu_conf for details	No

- Parameter Description - gpu_conf:

Table 4.6.42 Parameter Description - gpu_conf¶

Parameter	Type	Required	Note
slice_type	string	Yes	vGPU type, view vGPU Type for details
gpu_types	list(string)	Yes	GPU types
schedulers	list(string)	Yes	GPU scheduling, view Scheduling Method for details

- Request Example

https://{SCP_IP}/janus/20180725/images/e3c6b8e870c84a90be87eddfbf7ae12f

```
{
    "name": "centos7-test",
    "description": "",
    "os_type": "linux-centos",
    "azs": [
        "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b"
    ]
}
```

4.6.8.1.3. Response

- Response Parameter

Parameter	Type	Description	Value Returned
task_id	UUIDType	Asynchronous task ID	Yes

- Response Example:

```
{
    "task_id": "d5ae0d43-4afc-4f27-a851-9989a7083d9d"
}
```

4.6.8.1.4. Response Code

- OK: 200
- Errors: 400, 403, 404，500

4.6.9. Delete an Image

4.6.9.1. Scenario

- Scenario: Delete an image.
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the VM Image Creation permission
- Versions supported: SCP6.3.70 and above

Note

Before deleting an image, check the following:

- Whether the image has been associated with auto scaling.
- (Removed)

If the above situation exists, the image cannot be deleted.

4.6.9.1.1. URI

Table 4.6.43 API Path and Request Method¶

REST VERB	PATH	Description
DELETE	/janus/v20180725/images/{image_id}	Delete an image

- Parameter Description:

Table 4.6.44 Path Parameters¶

Parameter	Location	Type	Description	Required
image_id	path	strng	Image ID	Yes

4.6.9.1.2. Request

- Request Parameter

  None

- Request Example

  None

4.6.9.1.3. Response

- Response Parameter

  None

- Response Example:

  None

4.6.9.1.4. Response Code

- Normal: 204
- Errors: 400, 403, 404, 500

4.7. Billing

4.7.1. Query Tenants' Bills

4.7.1.1. Scenario

- Scenario: Query tenants' bills
- Roles Supported: Super admin, system admin, MSP admin, tenant, and co-administrator with the Billing Read permission
- Versions supported: SCP6.3.70 and above

4.7.1.2. URI

Table 4.7.1 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/projects/{project_id}/bills/fee-records	Query tenants' bills

- Parameter Description:

Table 4.7.2 Path Parameters¶

Parameter	Location	Type	Description	Required
project_id	path	string	MSP administrator/tenant ID	Yes
start_time	path	string	Start time, in ISO 8601 format, example: “2020-12-01T03:17:42Z”	Yes
end_time	path	string	End time (up to 1 year from the start time)	Yes
az_id	path	string	Resource pool ID	No

Note

For common path parameters, view Common Path Parameter Format for details

4.7.1.3. Request

- Request Parameter

  None

- Request Example

  https://{{SCP_Endpoint}}/janus/20180725/projects/88eb9b4ab11f4f1eb3f1e1e6e7ff522b/bills/fee-records?start_time=2020-12-01T03:17:42Z&end_time=2021-12-01T03:17:42Z

4.7.1.4. Response

- Response Parameter

Table 4.7.3 Response Parameter¶

Parameter	Type	Description	Value Returned
total_cost	float	Total costs	Yes
bills	list(dict)	Bills, view Parameter Description - bills for details	Yes

- Parameter Description - bills:

Table 4.7.4 Parameter Description - bills¶

Parameter	Type	Description	Value Returned
id	string	Bill ID	No
use_period	string	Time range	No
resource_id	string	Resource ID	No
price	string	Unit price (with unit)	No
use_time	string	Service duration	No
cost	string	Total cost	No
az_id	string	Resource pool ID	No
resource_name	string	Resource name	No
resource_type	string	Resource type	No
desc	string	Pricing of resources	No

> **Note**
>
> For common response parameters, view Response Format for details

- Response Example:


```
{
    "total": "72.5350",
    "bills": [
        {
            "resource_name": "-",
            "use_period": "from 2020-12-21 00:00:00 to 2020-12-22 23:59:59",
            "resource_id": "84b657b4d2329d758cf08dd0d63b9bdc2dadcd7ea948bee8
19c6cd256ce0bc61",
            "price": "0.0000 CNY/GB/hr",
            "use_time": "1 day 23 hours 59 minutes",
            "cost": "0.0000",
            "az_id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
            "id": "74",
            "resource_type": "backup repository",
            "desc": "500 GB"
        },
        {
            "resource_name": "user_vm222",
            "use_period": "from 2020-12-18 16:50:28 up to now",
            "resource_id": "db11ce3f-046c-4784-a7a1-c24096e72a8b",
            "price": "0.0000 CNY/VM/hr",
            "use_time": "4 days 0 hour 30 minutes",
            "cost": "0.0000",
            "az_id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
            "id": "68_65_69",
            "resource_type": "VM",
            "desc": "Powered-off, 1 CPU core, 1 GB Memory, Performance Sensi
tive Storage: 80 GB"
        }
    ]
}
```


4.7.1.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.8. Storage

4.8.1. Description of Common Storage Management Parameters

4.8.1.1. storage_type

Table 4.8.1 Storage type¶

Value	Description
local	Local storage
vs	Virtual storage
fc	FC storage
iscsi	iSCSI storage
VMFS	VMFS storage
vmfs	VMFS storage
vsan	vSAN storage
cifs	CIFS storage
nfs	NFS storage
asan	aSAN storage
samba_share	samba_share storage

4.8.1.2. storage_status

Table 4.8.2 Storage status¶

Value	Description
normal	Normal
offline	Offline
alarm	Alert
error	Abnormal
Only VS has the following statuses:	
ok	Normal
fail	Abnormal
creating	Creating
expanding	Expanding capacity
deleting	Deleting
replacing_disk	Replacing disk
replacing_host	Replacing host
replacing_backup_disk	Replacing spare disk
create_reverting	Canceling storage creation
expand_reverting	Canceling capacity expansion
initting	Initializing

4.8.2. Get Physical Disks

4.8.2.1. Scenario

- Scenario: Query the list of physical disk volumes (iSCSI/FC, etc.)
- Applicable roles: Super administrator and tenant
- Versions supported: SCP 6.3.0 and above

4.8.2.2. URI

Table 4.8.3 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/storages/physical-disks	Get Physical Disks

- Parameter Description:

Table 4.8.4 Path Parameters¶

Parameter	Location	Type	Description	Required
az_id	path	string	Resource pool ID	Yes
type	path	string	Disk type, view Storage type for details. Both iSCSI and FC can be queried.	No
in_use	path	int	Whether used by the VM. The options are 0 and 1	No

4.8.2.3. Request

- Request Parameter

None

- Request Example

  https://{SCP_IP}/janus/20180725/storages/physical-disks?az_id=611f1b34-c727-4925-854e-07bc8607f681&in_use=0

4.8.2.4. Response

- Response Parameter

Table 4.8.5 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Disk ID	Yes
hosts	list	List of hosts that can be accessed by storage, view Parameter Description - hosts for details.	Yes
name	name	Disk name	Yes
status	string	Disk status	Yes
type	string	Disk type	Yes
size_mb	int	Disk size (MB)	Yes
lun	int	Logical unit number	Yes
in_use	int	Whether the disk is being used by a VM	Yes
server	string or dict	Server information, view Parameter Description - hosts for details.	Yes
is_asan	int	Whether this is an aSAN disk	Yes
description	string	Disk description	Yes
vms	list	List of VMs that use this disk, view Parameter Description - vms for details.	Yes
server_detail	dict	Server information, view Parameter Description - hosts for details.	No
target	string	Storage information on the server	No
tid	int	Mark an iSCSI disk on an iSCSI server	No
policy_id	string	Storage policy ID	No
policy_name	string	Storage policy name	No

Note

For common response parameters, view Response Format for details

- Parameter Description - hosts

Table 4.8.6 Parameter Description - hosts¶

Parameter	Type	Description	Value Returned
status	int	Host status	Yes
ip	string	Host IP	Yes
name	string	Host name	Yes
id	string	Host ID	Yes

- Parameter Description - vms

Table 4.8.7 Parameter Description - vms¶

Parameter	Type	Description	Value Returned
id	int	VM ID on HCI	Yes
name	string	VM Name on HCI	Yes
server_id	string	VM UUID	Yes
server_name	string	VM name	Yes

- Response Example:

```
[
    {
        "status": "normal",
        "description": null,
        "in_use": 0,
        "is_asan": 0,
        "server": {
            "status": 1,
            "ip": "10.106.1.73",
            "name": "10.106.1.73",
            "id": "host-6c92bf58b580"
        },
        "size_mb": 3815447,
        "server_detail": {
            "status": 1,
            "ip": "10.106.1.73",
            "name": "10.106.1.73",
            "id": "host-6c92bf58b580"
        },
        "vms": [],
        "hosts": [
            {
                "status": 1,
                "ip": "10.106.1.73",
                "name": "10.106.1.73",
                "id": "host-6c92bf58b580"
            }
        ],
        "type": "local",
        "id": "35000cca0bcd0a3f3_6c92bf58b580",
        "lun": 0,
        "name": "HGST HUS726T4TAL_ATA_35000cca0bcd0a3f3_6c92bf58b580"
    }
]
```

4.8.2.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.8.3. Get a list of storage tags

4.8.3.1. Scenario

- Scenario: Query tags of storages for which tenants have quotas
- Applicable roles: Super administrator and tenant
- Versions supported: SCP 6.3.0 and above

Note

Only tags of storages for which tenants have quotas will be returned.

4.8.3.2. URI

Table 4.8.8 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/storages/tags	Get a list of storage tags

- Parameter Description:

Table 4.8.9 Path Parameters¶

Parameter	Location	Type	Description	Required
az_id	path	string	Resource pool ID	Yes
project_id	path	string	Tenant ID	No

4.8.3.3. Request

- Request Parameter

None

- Request Example

  https://{SCP_IP}/janus/20180725/storages/tags?az_id=611f1b34-c727-4925-854e-07bc8607f681&project_id=9d14d076b59045b1b0aaca04cfaaa657

4.8.3.4. Response

- Response Parameter

Table 4.8.10 Response Parameter¶

Parameter	Type	Description	Value Returned
data	list	Storage tag list, view Parameter Description - data for details.	Yes

Note

For common response parameters, view Response Format for details

- Parameter Description - data

Table 4.8.11 Parameter Description - data¶

Parameter	Type	Description	Value Returned
id	string	Storage tag ID	Yes
name	string	Storage tag name	Yes
description	string	Description of storage tag	Yes
total_mb	int	Total space	Yes
used_mb	int	Used space	Yes
ratio	float	CPU usage	Yes
project_id	string	Tenant ID	Yes

- Response Example:

```
[
    {
        "ratio": 0.57,
        "description": "Highest IO speed, to tag SSD to create high-performa
nce virtual machine",
        "total_mb": 507904,
        "project_id": "9d14d076b59045b1b0aaca04cfaaa657",
        "used_mb": 288275,
        "id": "00000000-0000-0000-0000-000000000000",
        "name": "High_Performance_Storage"
    },
    {
        "ratio": 0.46,
        "description": "Fair IO speed with high capacity and safety and cost
-effective",
        "total_mb": 2093056,
        "project_id": "9d14d076b59045b1b0aaca04cfaaa657",
        "used_mb": 972414,
        "id": "22222222-2222-2222-2222-222222222222",
        "name": "Capacity_Sensitive_Storage"
    }
]
```

4.8.3.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.8.4. Query Storage List

4.8.4.1. Scenario

- Scenario: Query the storage list
- Roles Supported: Super admin and system admin
- Versions supported: SCP 6.3.70 and above

4.8.4.2. URI

Table 4.8.12 API Path and Request Method

REST VERB	PATH	Description
GET	/janus/20180725/storages	Query Storage List

4.8.4.2.1. Description

Table 4.8.13 Path Parameters¶

Parameter	Location	Type	Description	Required
name	path	string	Storage name	No
type	path	string	Storage type, view：Storage type	No
status	path	string	Storage status, view：Storage status	No
storage_tag_id	path	string	Storage tag ID, corresponding to the storage tag in the Get a list of storage tags interface.	No
az_id	path	string	Resource pool ID	No
target	path	int	Storage target	No
archive_usable	path	int	Whether archive is supported	No
backup_enable	path	int	Whether backup is supported	No
shared	path	int	Whether it is shared storage	No

Note

The storage is not available until it is marked with the corresponding storage tag, that is, storage_tag_id cannot be not empty.

4.8.4.3. Request

4.8.4.3.1. Response Parameter

None

4.8.4.3.2. Request Example

- Query the list of storage in resource pools: *https://{SCP_Endpoint}/janus/20180725/storages?az_id=b9f644e5d4ed42358d287b19ea944e71*
- Query the VS storage list: *https://{SCP_Endpoint}/janus/20180725/storages?type=vs*

4.8.4.4. Response

4.8.4.4.1. Response Parameter

Table 4.8.14 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Storage ID	No
name	string	Name	No
description	string	Description	No
type	string	Storage type, view：Storage type	No
status	string	Storage status, view：Storage status	No
storage_tag_id	string	Storage tag ID, corresponding to the storage tag in the Get a list of storage tags interface.	No
az_id	string	Resource pool ID	No
target	int	Storage target	No
total_mb	int	Total space (MB)	No
used_mb	int	Used space (MB)	No
ratio	float	CPU usage	No
read_byteps	float	Read bytes per second	No
write_byteps	float	Write bytes per second	No
max_read_byteps	float	Maximum read bytes per second	No
max_write_byteps	float	Maximum write bytes per second	No

4.8.4.4.2. Response Example

```
[
    {
        "status": "offline",
        "write_byteps": 0.0,
        "ratio": 0.0,
        "description": "16.5 Local storage of the node",
        "max_write_byteps": 60030976.0,
        "read_byteps": 0.0,
        "total_mb": 0,
        "name": "1605_local",
        "used_mb": 0,
        "max_read_byteps": 132593664.0,
        "storage_tag_id": "",
        "type": "local",
        "id": "350014ee2b684e903_001b21d61e70",
        "az_id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
        "target": ""
    },
    {
        "status": "normal",
        "write_byteps": 0.0,
        "ratio": 0.0,
        "description": "",
        "max_write_byteps": 15659008.0,
        "read_byteps": 0.0,
        "total_mb": 520192,
        "name": "test",
        "used_mb": 1367,
        "max_read_byteps": 35479552.0,
        "storage_tag_id": "11111111-1111-1111-1111-111111111111",
        "type": "iscsi",
        "id": "1SANGFOR_ASAN-00a-299e-4c5c-b2d2-14c5fc6f319b",
        "az_id": "0b1695d7-8aa8-4efb-9d25-3c0312f0c5a1",
        "target": "iqn.2015-08.71295054.com.sangfor.asan:xhf-iscsi-01"
    }
]
```


4.8.4.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.8.5. Query Storage List

4.8.5.1. Scenario

- Scenario: Query the storage list
- Applicable roles: Administrator, system administrator
- Versions supported: SCP 6.8.0 and above

4.8.5.2. URI

Table 4.8.15 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20190725/storages	Query Storage List

4.8.5.2.1. Description

Table 4.8.16 Path Parameters¶

Parameter	Location	Type	Description	Required
name	path	string	Storage name	No
type	path	string	Storage type, view：Storage type	No
status	path	string	Storage status, view：Storage status	No
storage_tag_id	path	string	Storage tag ID, corresponding to the storage tag in the Get a list of storage tags interface.	No
az_id	path	string	Resource pool ID	No
target	path	int	Storage target	No
archive_usable	path	string	Whether archive is supported	No
backup_enable	path	string	Whether backup is supported	No
shared	path	string	Whether it is shared storage	No
storage_ids	path	list	List of storage IDs	No

Note

The storage is not available until it is marked with the corresponding storage tag, that is, storage_tag_id cannot be not empty.

4.8.5.3. Request

4.8.5.3.1. Response Parameter

None

4.8.5.3.2. Request Example

- Query the list of storage resources in the resource pool: *https://{SCP_Endpoint}/janus/20190725/storages?az_id=b9f644e5d4ed42358d287b19ea944e71*
- Query :term: List of storage of the *VS* type: *https://{SCP_Endpoint}/janus/20190725/storages?type=vs*

4.8.5.4. Response

4.8.5.4.1. Response Parameter

Table 4.8.17 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Storage ID	Yes
name	string	Name	Yes
description	string	Description	Yes
type	string	Storage type, view：Storage type	Yes
status	string	Storage status, view：Storage status	Yes
storage_tag_id	string	Storage tag ID, corresponding to the storage tag in the Get a list of storage tags interface.	Yes
az_id	string	Resource pool ID	Yes
target	int	Storage target	Yes
total_mb	int	Total space (MB)	Yes
used_mb	int	Used space (MB)	Yes
ratio	float	CPU usage	Yes
read_byteps	float	Read bytes per second	Yes
write_byteps	float	Write bytes per second	Yes
max_read_byteps	float	Maximum read bytes per second	Yes
max_write_byteps	float	Maximum write bytes per second	Yes
connected_hosts	int	Connected nodes	Yes
supported_allocate_types	list	Supported allocation method	Yes
archive_usable	float	Whether archive is supported	Yes
backup_enable	float	Whether backup is supported	Yes
backup_used_mb	int	Used backup	Yes
backup_total_mb	int	Total backup	Yes
backup_ratio	float	Backup usage	Yes
shared	int	Whether it is shared storage	Yes

4.8.5.4.2. Response Example

```
[
    {
        "backup_used_mb": 0,
        "backup_total_mb": 0,
        "total_mb": 15992070,
        "id": "5e8b0eec_vs_vol_rep3",
        "az_id": "466c3ea6-2537-4de1-b294-6fd96d8b89a4",
        "storage_tag_id": "00000000-0000-0000-0000-000000000000",
        "ratio": 0.72,
        "archive_usable": 0,
        "supported_allocate_types": [
            "off",
            "metadata",
            "full"
        ],
        "shared": 1,
        "used_mb": 11524259,
        "status": "ok",
        "write_byteps": 382156.0,
        "description": "",
        "backup_enable": 0,
        "max_read_byteps": 224820224.0,
        "backup_ratio": 0.0,
        "name": "Virtual Datastore 1",
        "target": "",
        "connected_hosts": 3,
        "max_write_byteps": 77648896.0,
        "type": "vs",
        "read_byteps": 16971366.0
    }
]
```


4.8.5.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.8.6. Query Storage List

4.8.6.1. Scenario

- Scenario: Query the storage list
- Applicable roles: Administrator, system administrator
- Versions supported: SCP 6.8.0 and above

4.8.6.2. URI

Table 4.8.18 API Path and Request Method

REST VERB	PATH	Description
GET	/janus/20200725/storages	Query Storage List

4.8.6.2.1. Description

Table 4.8.19 Path Parameters¶

Parameter	Location	Type	Description	Required
name	path	string	Storage name	No
type	path	string	Storage type, view: Storage type	No
status	path	string	Storage status, view: Storage status	No
storage_tag_id	path	string	Storage tag ID, corresponding to the storage tag in the Get a list of storage tags interface.	No
az_id	path	string	Resource pool ID	No
target	path	int	Storage target	No
archive_usable	path	string	Whether archive is supported	No
backup_enable	path	string	Whether backup is supported	No
shared	path	string	Whether it is shared storage	No
storage_ids	path	list	List of storage IDs	No

Note

The storage is not available until it is marked with the corresponding storage tag, that is, storage_tag_id cannot be not empty.

4.8.6.3. Request

4.8.6.3.1. Response Parameter

None

4.8.6.3.2. Request Example

- Query the list of storage resources in the resource pool: *https://{SCP_Endpoint}/janus/20200725/storages?az_id=b9f644e5d4ed42358d287b19ea944e71*
- Query :term: List of storage of the *VS* type: *https://{SCP_Endpoint}/janus/20200725/storages?type=vs*

4.8.6.4. Response

4.8.6.4.1. Response Parameter

Table 4.8.20 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Storage ID	Yes
name	string	Name	Yes

description	string	Description	Yes
type	string	Storage type, view：Storage type	Yes
status	string	Storage status, view：Storage status	Yes
storage_tag_id	string	Storage tag ID, corresponding to the storage tag in the Get a list of storage tags interface.	Yes
storage_tag_name	string	Storage tag name	Yes
storage_label	string	Storage usage of the backup repository	Yes
cluster_id	string	Cluster ID	Yes
cluster_name	string	Cluster name	Yes
cluster_type	string	Cluster type	Yes
az_id	string	Resource pool ID	Yes
target	int	Storage target	Yes
total_mb	int	Total space (MB)	Yes
used_mb	int	Used space (MB)	Yes
ratio	float	CPU usage	Yes
read_byteps	float	Read bytes per second	Yes
write_byteps	float	Write bytes per second	Yes
max_read_byteps	float	Maximum read bytes per second	Yes
max_write_byteps	float	Maximum write bytes per second	Yes
connected_hosts	int	Connected nodes	Yes
supported_allocate_types	list	Supported allocation method	Yes
archive_usable	float	Whether archive is supported	Yes
backup_enable	float	Whether backup is supported	Yes
backup_used_mb	int	Used backup	Yes
backup_total_mb	int	Total backup	Yes
backup_ratio	float	Backup usage	Yes
shared	int	Whether it is shared storage	Yes
host_ids	string	Device ID list	Yes
hosts_online_state	string	Online status of the node on which the storage depends	Yes

4.8.6.4.2. Response Example

```
[
    {
        "backup_used_mb": 0,
        "backup_total_mb": 0,
        "total_mb": 15992070,
        "cluster_id": "bca21dfb-be78-464e-be34-bd12c4a92c76",
        "id": "5e8b0eec_vs_vol_rep3",
        "az_id": "466c3ea6-2537-4de1-b294-6fd96d8b89a4",
        "storage_tag_id": "00000000-0000-0000-0000-000000000000",
        "ratio": 0.72,
        "cluster_type": "hci",
        "archive_usable": 0,
        "cluster_name": "10.132.16.86",
        "supported_allocate_types": [
            "off",
            "metadata",
            "full"
        ],
        "shared": 1,
        "used_mb": 11524259,
        "status": "ok",
        "write_byteps": 382156.0,
        "description": "",
        "storage_label": "",
        "backup_enable": 0,
        "max_read_byteps": 224820224.0,
        "storage_tag_name": "High performance storage",
        "backup_ratio": 0.0,
        "name": "Virtual Datastore 1",
        "host_ids": [
            "host-6c92bf84989a",
            "host-6c92bfc4f9fe",
            "host-6c92bfa35a90"
        ],
        "hosts_online_state": "ALL",
        "target": "",
        "connected_hosts": 3,
        "max_write_byteps": 77648896.0,
        "type": "vs",
        "read_byteps": 16971366.0
    }
]
```


4.8.6.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.8.7. Refresh Physical Disk Configuration

4.8.7.1. Scenario

- Scenario: The iSCSI/FC datastore is expanded on the storage server side, and the VM mounted with the corresponding datastore is notified to update the capacity.
- Applicable roles: Super administrator and tenant
- Versions supported: SCP 6.3.0 and above

4.8.7.2. URI

Table 4.8.21 API Path and Request Method¶

REST VERB	PATH	Description
PUT	/janus/20180725/storages/physical-disks/{volume_id}	Update physical disk

- Parameter Description:

Table 4.8.22 Path Parameters¶

Parameter	Location	Type	Description	Required
volume_id	path	string	Shared external datastore ID, which can be obtained from Get Physical Disks	Yes

4.8.7.3. Request

- Request Parameter

Table 4.8.23 Body Parameter¶

Parameter	Location	Type	Description	Required
az_id	body	string	Resource pool ID	Yes
is_update_vm	body	int	Whether to notify the VM to update the configuration	Yes

- Request Example

  https://{SCP_IP}/janus/20180725/storages/physical-disks/35000cca0bcd0a3f3_6c92bf58b580

```
{
    "az_id": "70172f8b-ad3d-47c3-9427-308a4c5052d0",
    "is_update_vm": 1
}
```

4.8.7.4. Response

- Response Parameter

Table 4.8.24 Response Parameter¶

Parameter	Type	Description	Value Returned
task_id	string	Asynchronous task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "d5ae0d43-4afc-4f27-a851-9989a7083d9d"
}
```

4.8.7.5. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.9. Networking

4.9.1. Common Parameters

4.9.1.1. Common Parameters

4.9.1.1.1. VPC Network

Table 4.9.1 VPC Network

Value	Description
creating	Creating
active	Normal
updating	Updating
deleting	Deleting
error	Abnormal

4.9.1.1.2. Network Type

Table 4.9.2 Network Type

Value	Description
vpc	VPC
classic	Classic Network

4.9.1.1.3. Device Type

Table 4.9.3 Device Type¶

Value	Description
vnet:ssl	SSL VPN
vnet:af	NGAF
vnet:iam	IAG
vnet:ad	ADC
vnet:dblb	DBLB
vnet:ac	IAG
vnet:das	DAS
vnet:apm	APM
vnet:woc	WANO
vnet:bvt	BVT
vnet:las	LAS
vnet:edr	Endpoint Secure
vnet:osm	OSM
vm	Virtual machine
bvs	BVS
dvs	DVS
vrouter	Virtual router
project_border	Project Border
project_symbol	Project Symbol
vlan	Port group of BVS
user_gw	Gateway of private cloud user
placeholder	Placeholder virtual machine
service	Service
cssp	(Removed)
dmp	Database Management Platform

4.9.1.1.4. Type of Elastic IP Associated Device

Table 4.9.4 Type of Elastic IP Associated Device¶

Value	Description
vm	Virtual machine type, including NFV devices
router_primary	Primary IP address of router, i.e., the gateway IP
router_second	Secondary IP address of router

4.9.1.1.5. Elastic IP Association Status

Table 4.9.5 Elastic IP Association Status¶

Value	Description
active	Normal
creating	Creating
updating	Updating
deleting	Deleting
errror	Abnormal

4.9.1.1.6. Subnet Status

Table 4.9.6 Subnet Status¶

Value	Description
creating	Creating
active	Normal
updating	Updating
deleting	Deleting
error	Abnormal

4.9.1.1.7. Line Type

Table 4.9.7 Line Type¶

Value	Description
inner_linetype	Built-in
user_defined	Custom
pub_service	Shared service

4.9.1.1.8. Internet Protocol Version

Table 4.9.8 Internet Protocol Version¶

Value	Description
4	IPv4
6	IPv6

4.9.1.1.9. Encryption Algorithm of NFV Devices

Table 4.9.9 Encryption Algorithm of NFV Devices¶

Value	Description
aes_256	AES 256
sm4_128	SM4 128

4.9.1.1.10. NFV Devices in VPC

Table 4.9.10 NFV Devices in VPC¶

Value	Description
vnet:ssl	SSL VPN
vnet:af	NGAF
vnet:iam	IAG
vnet:ad	ADC
vnet:dblb	DBLB
vnet:bvt	BVT
vnet:las	LAS
vnet:edr	Endpoint Secure
vnet:osm	OSM

4.9.1.1.11. NFV Interfaces in Classic Network

Table 4.9.11 NFV Interfaces in Classic Network¶

Value	Description
manage_port	Management interface
other_port	Others

4.9.1.1.12. NFV Interfaces in VPC

Table 4.9.12 NFV Interfaces in VPC¶

Value	Description
manage_port	Management interface
business_port	Edge-connected interface
heartbeat_port	Heartbeat interface
wan_port	WAN interface
lan_port	LAN interface
bridge_lan_port	The interface that the device is connected to subnet in bridge mode
bridge_wan_port	The interface that the device is connected to the external devices (such as internal router and direct line) in bridge mode
other_port	Others

4.9.1.1.13. NFV Location Deployed in VPC

Table 4.9.13 NFV Location Deployed in VPC¶

Value	Description
outer_vr_inner_vr	Between the external router and internal router
inner_vr_inner_dvs	Between the internal router and subnet
inner_dvs_private_line	Between the subnet and direct line
inner_dvs	Under the subnet

4.9.1.1.14. NFV Deployment Mode in VPC

Table 4.9.14 NFV Deployment Mode in VPC¶

Value	Description
router	Route (deployed between the internal router and external router)
bridge	Bridge (deployed between the subnet and other devices, such as internal router and direct line)
single	Single arm (deployed under the subnet)

4.9.1.1.15. VPC ACL Action

Table 4.9.15 VPC ACL Action¶

Value	Description
permit	Allow
drop	Discard
reject	Deny

4.9.1.1.16. ACL Type for VPC

Table 4.9.16 ACL Type for VPC¶

Value	Description
public_network	Public network
public_service	Shared Services
high_risk_port	High-risk port
external_traffic	External traffic
individual	Custom
traffic_and_port	High-risk port and external traffic
default	Default policy

4.9.1.1.17. Src or Dst Types of VPC

Table 4.9.17 Src or Dst Types of VPC¶

Value	Description
ip	IP address type
all	All IP address types
vif	VPC gateway
network	Specified network

4.9.2. VPC

4.9.2.1. Query VPC s

4.9.2.1.1. Scenario

- Scenario: Query VPCs
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Read permission
- Versions supported: SCP 6.3.0 and above

4.9.2.1.2. URI

Table 4.9.18 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/vpcs	Query VPCs

- Parameter Description:

Table 4.9.19 Path Parameters¶

Parameter	Location	Type	Description	Required
id	path	string	Network ID	No
name	path	string	Network name	No
az_id	path	string	Resource pool ID	No
project_id	path	string	Tenant ID	No
status	path	string	Network status, view VPC Network for details	No
keywords	path	string	Search network name by keywords	No

Note

For common path parameters, view Common Path Parameter Format for details

4.9.2.1.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_Endpoint}/janus/20180725/vpcs?project_id=78ad8e9c9c7e487186f448bc81f050a6

4.9.2.1.4. Response

- Response Parameter

Table 4.9.20 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Network ID	Yes
name	string	Network name	Yes
description	string	Network description	Yes
az_id	string	Resource pool ID	Yes
project_id	string	Tenant ID	Yes
type	string	Network type: VPC	Yes
shared	int(0, 1)	Whether VPC is visible to all tenants. 0: Invisible, 1: Visible	Yes
status	string	Network status	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
[
    {
        "status": "active",
        "description": null,
        "shared": 0,
        "project_id": "de3c453465034a31a2e054c6e4250626",
        "type": "vpc",
        "id": "6733078a-3572-4680-a750-e9d5eca630bc",
        "az_id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
        "name": "VPC"
    },
    {
        "status": "active",
        "description": null,
        "shared": 0,
        "project_id": "0885364e43854b19a2c81298ed963456",
        "type": "vpc",
        "id": "ee5ce1e7-665e-4e98-93b0-8777454e0eb4",
        "az_id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
        "name": "VPC"
    }
]
```

4.9.2.1.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.9.2.2. Create VPC

4.9.2.2.1. Scenario

- Scenario: Create VPC for tenants
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Write permission
- Versions supported: SCP6.3.70 and above

Note

1. To create network for tenants, you need to associate them with a resource pool.
2. One tenant can only create a VPC in the same resource pool.

4.9.2.2.2. URI

Table 4.9.21 API Path and Request Method¶

REST VERB	PATH	Description
POST	/janus/20180725/vpcs	Create VPC

- Parameter Description:

  None

4.9.2.2.3. Request

- Request Parameter

Table 4.9.22 Response Parameter¶

Parameter	Location	Type	Description	Required
az_id	body	string	Resource pool ID	Yes
project_id	body	string	Tenant ID	Yes
dh_id	body	string	Dedicated server group ID	No
name	body	string	VPC network name	No

- Request Example

  https://{SCP_Endpoint}/janus/20180725/vpcs

```
{
    "az_id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
    "project_id": "6f4e8e5bd6f5476a9c28a26fba03796a"
}
```

4.9.2.2.4. Response

- Response Parameter

Table 4.9.23 Response Parameter¶

Parameter	Type	Description	Value Returned
task_id	string	Task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "d5ae0d43-4afc-4f27-a851-9989a7083d9d"
}
```

4.9.2.2.5. Response Code

- Success: 201
- Errors: 400, 403, 409, 500

4.9.2.3. Delete VPC

4.9.2.3.1. Scenario

- Scenario: Delete VPC
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Write permission
- Versions supported: SCP6.3.70 and above

Warning

Deleting VPC may affect network connectivity. Confirm the following first:

1. The virtual machines and NFV resources (including related resources in the Recycle Bin) in VPC have been deleted permanently.
2. Shared bandwidth, direct line and elastic IPs have been deleted.

4.9.2.3.2. URI

Table 4.9.24 API Path and Request Method¶

REST VERB	PATH	Description
DELETE	/janus/20180725/vpcs/{vpc_id}	Delete VPC

- Parameter Description:

Table 4.9.25 Path Parameters¶

Parameter	Location	Type	Description	Required
vpc_id	path	string	VPC ID	Yes

4.9.2.3.3. Request

- Request Parameter

  None

- Request Example

  None

4.9.2.3.3.1. Response¶

- Response Parameter

Table 4.9.26 Response Parameter¶

Parameter	Type	Description	Value Returned
task_id	UUIDType	Asynchronous task ID	Yes

- Response Example:

```
{
    "task_id": "d5ae0d43-4afc-4f27-a851-9989a7083d9d"
}
```

4.9.2.3.3.2. Response Code¶

- OK: 200
- Errors: 400, 403, 404, 500

4.9.3. VPC Subnet

4.9.3.1. Query Subnets

4.9.3.1.1. Scenario

- Scenario: Query subnets
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Read permission
- Versions supported: SCP 6.3.0 and above

Warning

The cloud platform will create WAN subnet and NFV management subnet that are invisible to tenants for internal communication and management on the platform. If not necessary, do not modify or delete them.

4.9.3.1.1.1. URI¶

Table 4.9.27 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/subnets	Query Subnets

- Parameter Description:

Table 4.9.28 Path Parameters¶

Parameter	Location	Type	Description	Required
name	path	string	Subnet name	No
az_id	path	string	Resource pool ID	No
vpc_id	path	string	VPC ID	No
project_id	path	string	Tenant ID	No
keywords	path	string	Query subnet name	No
is_visible	path	int	Whether the subnet is visible to tenants	No

Note

For common path parameters, view Common Path Parameter Format for details

4.9.3.1.2. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/subnets?vpc_id=f6f18ab6-42ab-4110-82ef-e2bd17fcd7de&is_visible=1

4.9.3.1.3. Response

- Response Parameter

Table 4.9.29 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Subnet ID	No
name	string	Subnet name	No
description	string	Subnet description	No
az_id	string	Resource pool ID	No
project_id	string	Tenant ID	No
vpc_id	string	VPC ID	No
cidr	string	Subnet CIDR	No
status	string	Subnet status, view Subnet Status for details	No
gateway_ip	string	Gateway IP	No
is_visible	int(0, 1)	Whether the subnet is visible to tenants	No
enable_dhcp	int(0, 1)	Whether to enable DHCP	No
shared	int(0, 1)	Whether it is visible to all tenants	No
allocation_pools	list(dict)	Available IP ranges, view Parameter Description - allocation_pools for details	No

- Parameter Description - allocation_pools:

Table 4.9.30 Parameter Description - allocation_pools¶

Parameter	Type	Description	Value Returned
start	string	Start IP	Yes
end	string	End IP	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
[
    {
        "status": "active",
        "is_visible": 1,
        "description": null,
        "enable_dhcp": 1,
        "allocation_pools": [
            {
                "start": "192.168.0.2",
                "end": "192.168.0.254"
            }
        ],
        "gateway_ip": "192.168.0.1",
        "shared": 0,
        "vpc_id": "6733078a-3572-4680-a750-e9d5eca630bc",
        "cidr": "192.168.0.0/24",
        "project_id": "de3c453465034a31a2e054c6e4250626",
        "id": "a366a91c-f2d2-40db-a3b9-32c51b0a4cac",
        "az_id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
        "name": "Subnet1"
    },
    {
        "status": "active",
        "is_visible": 0,
        "description": null,
        "enable_dhcp": 1,
        "allocation_pools": [
            {
                "start": "172.31.0.2",
                "end": "172.31.0.254"
            }
        ],
        "gateway_ip": "172.31.0.1",
        "shared": 0,
        "vpc_id": "6733078a-3572-4680-a750-e9d5eca630bc",
        "cidr": "172.31.0.0/24",
        "project_id": "de3c453465034a31a2e054c6e4250626",
        "id": "ed45c4a3-df8a-4589-b048-a345f2c8dd56",
        "az_id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
        "name": "wan_subnet"
    },
    {
        "status": "active",
        "is_visible": 0,
        "description": null,
        "enable_dhcp": 1,
        "allocation_pools": [
            {
                "start": "172.32.0.2",
                "end": "172.32.0.254"
            }
        ],
        "gateway_ip": "172.32.0.1",
        "shared": 0,
        "vpc_id": "6733078a-3572-4680-a750-e9d5eca630bc",
        "cidr": "172.32.0.0/24",
        "project_id": "de3c453465034a31a2e054c6e4250626",
        "id": "d1cb73a2-750f-4d40-82e4-784518f4e5d3",
        "az_id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
        "name": "NFV_management_subnet"
    }
]
```


4.9.3.1.4. Response Code

- OK: 200
- Errors: 400, 403, 500

4.9.3.2. Query Subnet

4.9.3.2.1. Scenario

- Scenario: Query subnet
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Read permission
- Versions supported: SCP6.3.70 and above

4.9.3.2.2. URI

Table 4.9.31 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/subnets/{subnet_id}	Query Subnet

- Parameter Description:

Table 4.9.32 Path Parameters¶

Parameter	Location	Type	Description	Required
subnet_id	path	string	Subnet ID	Yes

4.9.3.2.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/subnets/f6f18ab6-42ab-4110-82ef-e2bd17fcd7de

4.9.3.2.4. Response

- Response Parameter

Table 4.9.33 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Subnet ID	Yes
name	string	Subnet name	Yes
description	string	Subnet description	Yes
az_id	string	Resource pool ID	Yes
project_id	string	Tenant ID	Yes
vpc_id	string	VPC ID	Yes
cidr	string	Subnet CIDR	Yes
status	string	Subnet status	Yes
gateway_ip	string	Gateway IP	Yes
is_visible	int(0, 1)	Whether the subnet is visible to tenants	Yes
enable_dhcp	int(0, 1)	Whether to enable DHCP	Yes
shared	int(0, 1)	Whether it is visible to all tenants	Yes
allocation_pools	list(dict)	Available IP ranges, view Parameter Description - allocation_pools for details	Yes

- Parameter Description - allocation_pools:

Table 4.9.34 Parameter Description - allocation_pools¶

Parameter	Type	Description	Value Returned
start	string	Start IP	Yes
end	string	End IP	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "status": "active",
    "is_visible": 1,
    "description": null,
    "enable_dhcp": 1,
    "vpc_id": "f6f18ab6-42ab-4110-82ef-e2bd17fcd7de",
    "allocation_pools": [
        {
            "start": "192.168.0.2",
            "end": "192.168.0.254"
        }
    ],
    "gateway_ip": "192.168.0.1",
    "cidr": "192.168.0.0/24",
    "shared": 0,
    "project_id": "e3c6b8e870c84a90be87eddfbf7ae12f",
    "id": "454ae1e8-eaa2-405a-922c-9a654aa2240b",
    "az_id": "30457665-e8ca-4821-bfd9-d29dc2006834",
    "name": "Subnet1"
}
```

4.9.3.2.5. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.9.3.3. Create a Subnet

4.9.3.3.1. Scenario

- Scenario: Create a subnet for tenants.
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Write permission
- Versions supported: SCP6.3.70 and above

Note

1. To create a subnet, you are recommended to use the IP ranges that start with 10, 172 or 192 as the subnet range. The following are the recommended subnet range:
    a. For large-sized enterprises, use IP range starting with 10.
    b. For medium-sized enterprises, use IP range starting with 172.
    c. For small enterprises, use IP range starting with 192.
2. The IP range 10.251.251.0/24 is reserved for NFV devices and cannot be used as elastic IP pool subnet.
3. Subnet range cannot overlap with elastic IP subnet range.
4. One tenant can create at most 20 subnets.

4.9.3.3.2. URI

Table 4.9.35 API Path and Request Method¶

REST VERB	PATH	Description
POST	/janus/20180725/subnets	Create a Subnet

- Parameter Description:

  None

4.9.3.3.3. Request

- Request Parameter

Table 4.9.36 Response Parameter¶

Parameter	Location	Type	Description	Required
name	body	string	Subnet name	Yes
description	body	string	Description	No
vpc_id	body	string	VPC ID	Yes
enable_dhcp	body	int(0, 1)	Whether to enable DHCP	No
gateway_ip	body	string	Gateway IP	Yes
cidr	body	string	Subnet	Yes
allocation_pools	body	list(dict)	Available IP ranges, view Parameter Description - allocation_pools for details	Yes

- Parameter Description - allocation_pools:

Table 4.9.37 Parameter Description - allocation_pools¶

Parameter	Type	Description	Value Returned
start	string	Start IP	Yes
end	string	End IP	Yes

- Request Example

  https://{SCP_IP}/janus/20180725/subnets

```
{
   "name": "new",
   "gateway_ip": "192.168.1.1",
   "netdomain_id": "6733078a-3572-4680-a750-e9d5eca630bc",
   "enable_dhcp": 1,
   "allocation_pools": [
       {
           "start": "192.168.1.2",
           "end": "192.168.1.254"
       }
   ],
   "cidr": "192.168.1.0/24",
   "description": "111"
}
```

4.9.3.3.4. Response

- Response Parameter

Table 4.9.38 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Subnet ID	Yes
name	string	Subnet name	Yes
description	string	Subnet description	No
status	string	Subnet status, view Subnet Status for details	Yes
is_visible	int(0, 1)	Whether the subnet is visible to the its tenant	No
vpc_id	string	VPC ID	Yes
project_id	string	Tenant ID	Yes
az_id	string	Resource pool ID	No
ip_version	int	Internet protocol version (supports IPv4)	Yes
enable_dhcp	int(0, 1)	Whether to enable DHCP	No
gateway_ip	string	Gateway IP	Yes
netmask	string	Netmask	Yes
cidr	string	Subnet CIDR	Yes
shared	int(0, 1)	Whether it is visible to all tenants	No
create_at	string	Time created	Yes
update_at	string	Time updated	Yes
qos_uplink	int(0, 10485760)	Outbound bandwidth, in Kbps	No
qos_downlink	int(0, 10485760)	Inbound bandwidth, in Kbps	No
allocation_pools	list(dict)	Available IP ranges, view Parameter Description - allocation_pools for details	Yes

- Parameter Description - allocation_pools:

Table 4.9.39 Parameter Description - allocation_pools¶

Parameter	Type	Description	Value Returned
start	string	Start IP	Yes
end	string	End IP	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "status": "active",
    "is_visible": 1,
    "description": "111",
    "enable_dhcp": 1,
    "created_at": "2020-12-28T07:14:11Z",
    "updated_at": "2020-12-28T07:14:19Z",
    "allocation_pools": [
        {
            "start": "192.168.1.2",
            "end": "192.168.1.254"
        }
    ],
    "vpc_id": "6733078a-3572-4680-a750-e9d5eca630bc",
    "gateway_ip": "192.168.1.1",
    "ip_version": 4,
    "shared": 0,
    "cidr": "192.168.1.0/24",
    "project_id": "de3c453465034a31a2e054c6e4250626",
    "netmask": "255.255.255.0",
    "id": "1e5c4b98-0955-4ace-ac71-9cf4c4d1c209",
    "az_id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
    "name": "new"
}
```


4.9.3.3.5. Response Code

- Success: 201
- Errors: 400, 403, 409, 404

4.9.3.4. Edit a Subnet

- Scenario: Edit a subnet
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Write permission
- Versions supported: SCP6.3.70 and above

4.9.3.4.1. URI

Table 4.9.40 API Path and Request Method¶

REST VERB	PATH	Description
PUT	/janus/20180725/subnets/{subnet_id}	Edit a Subnet

- Parameter Description:

Table 4.9.41 Path Parameters¶

Parameter	Location	Type	Description	Required
subnet_id	path	string	Subnet ID	Yes

4.9.3.4.2. Request

- Request Parameter

Table 4.9.42 Response Parameter¶

Parameter	Location	Type	Description	Required
name	body	string	Subnet name	No
description	body	string	Description	No
enable_dhcp	body	int(0, 1)	Whether to enable DHCP	No
allocation_pools	body	list(dict)	Available IP ranges, view Parameter Description - allocation_pools for details	No

- Parameter Description - allocation_pools:

Table 4.9.43 Parameter Description - allocation_pools¶

Parameter	Type	Description	Value Returned
start	string	Start IP	Yes
end	string	End IP	Yes

- Request Example

  https://{SCP_IP}/janus/20180725/subnets/454ae1e8-eaa2-405a-922c-9a654aa2240b

```
{
    "name": "new1",
    "enable_dhcp": 1,
    "allocation_pools": [
        {
            "start": "192.168.1.2",
            "end": "192.168.1.250"
        }
    ],
    "description": "111222"
}
```

4.9.3.4.3. Response

- Response Parameter

Table 4.9.44 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Subnet ID	Yes
name	string	Subnet name	Yes
description	string	Subnet description	No
status	string	Subnet status, view Subnet Status for details	Yes
is_visible	int(0, 1)	Whether the subnet is visible to the its tenant	No
vpc_id	string	VPC ID	Yes
project_id	string	Tenant ID	Yes
az_id	string	Resource pool ID	No
ip_version	int	Internet protocol version (supports IPv4)	Yes
enable_dhcp	int(0, 1)	Whether to enable DHCP	No
gateway_ip	string	Gateway IP	Yes
netmask	string	Netmask	Yes
cidr	string	Subnet CIDR	Yes
shared	int(0, 1)	Whether it is visible to all tenants	No
create_at	string	Time created	Yes
update_at	string	Time updated	Yes
qos_uplink	int(0, 10485760)	Outbound bandwidth, in Kbps	No
qos_downlink	int(0, 10485760)	Inbound bandwidth, in Kbps	No
allocation_pools	list(dict)	Available IP ranges, view Parameter Description - allocation_pools for details	Yes

- Parameter Description - allocation_pools:

Table 4.9.45 Parameter Description - allocation_pools¶

Parameter	Type	Description	Value Returned
start	string	Start IP	Yes
end	string	End IP	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "status": "active",
    "is_visible": 1,
    "description": "111222",
    "enable_dhcp": 1,
    "created_at": "2020-12-28T07:14:11Z",
    "updated_at": "2020-12-28T07:14:19Z",
    "allocation_pools": [
        {
            "start": "192.168.1.2",
            "end": "192.168.1.250"
        }
    ],
    "netdomain_id": "6733078a-3572-4680-a750-e9d5eca630bc",
    "gateway_ip": "192.168.1.1",
    "ip_version": 4,
    "shared": 0,
    "cidr": "192.168.1.0/24",
    "project_id": "de3c453465034a31a2e054c6e4250626",
    "netmask": "255.255.255.0",
    "id": "1e5c4b98-0955-4ace-ac71-9cf4c4d1c209",
    "az_id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
    "name": "new1"
}
```

4.9.3.4.3.1. Response Code¶

- OK: 200
- Abnormal: 400, 403, 404, 409, 500

4.9.3.5. Delete a Subnet

4.9.3.5.1. Scenario

- Scenario: Delete tenants' subnet
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Write permission
- Versions supported: SCP6.3.70 and above

Note

1. If a subnet has been used by router ACL, it cannot be deleted.

4.9.3.5.2. URI

Table 4.9.46 API Path and Request Method¶

REST VERB	PATH	Description
DELETE	/janus/20180725/subnets/{subnet_id}	Delete a Subnet

- Parameter Description:

Table 4.9.47 Path Parameters¶

Parameter	Location	Type	Description	Required
subnet_id	path	string	Subnet ID	Yes

4.9.3.5.3. Request

- Request Parameter

  None

- Request Example

  None

4.9.3.5.4. Response

- Response Parameter

  None

- Response Example:

  None

4.9.3.5.4.1. Response Code¶

- Normal: 204
- Errors: 400, 403, 404, 500

4.9.4. Virtual IP

4.9.4.1. Query Virtual IPs

4.9.4.1.1. Scenario

- Scenario: Query virtual IPs
- Roles Supported: Super admin, system admin, and co-administrator with the Network Read permission
- Versions supported: SCP6.3.70 and above

4.9.4.1.2. URI

Table 4.9.48 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/virtualips	Query Virtual IPs

- Description

Table 4.9.49 Path Parameters¶

Parameter	Location	Type	Description	Required
id	path	string	Virtual IP ID	No
project_id	path	string	Tenant ID	No
subnet_id	path	string	Subnet ID	No
vpc_id	path	string	VPC ID	No
az_id	path	string	Resource pool ID	No
port_id	path	string	Port ID	No
name	path	string	Name	No
tag	path	string	Tag	No
description	path	string	Description	No

4.9.4.1.3. Request

- Response Parameter

None

- Request Example

https://{SCP_IP}/janus/20180725/virtualips?subnet_id=1b4a29f1-6c51-441f-af36-97ba69ea1506&page_num=1&page_size=2&sort=name&order_by=DESC

4.9.4.1.4. Response

- Response Parameter

Table 4.9.50 Response Parameter

Parameter	Type	Description	Value Returned
id	string	Virtual IP ID	No
project_id	string	Tenant ID	No
vpc_id	string	VPC ID	No
subnet_id	string	Subnet ID	No
fixed_ips	list	Virtual IP, view Parameter Description - fixed_ips for details	No
name	string	Virtual IP name	No
description	string	Description	No
az_id	string	Resource pool ID of virtual IP	No
port_id	string	Port ID	No
tag	string	Virtual IP tag	No
binds	list	Association information (because binding virtual IP is not currently supported, no result will be returned.)	No
created_at	string	Time created	No
updated_at	string	Last updated	No

Note

Virtual IP cannot be associated currently, so the parameter binds will return no results.

Note

For common response parameters, view Response Format for details

- Parameter Description - fixed_ips

Table 4.9.51 Parameter Description - fixed_ips

Parameter	Type	Value Returned	Note
ip_address	string	Yes	Virtual IP address

- Response Example

```
[
    {
        "description": "",
        "subnet_id": "1b4a29f1-6c51-441f-af36-97ba69ea1506",
        "fixed_ips": [
            {
                "ip_address": "192.168.0.3"
            }
        ],
        "created_at": "2021-04-26T03:31:20.000000",
        "updated_at": "",
        "binds": [],
        "tag": "",
        "vpc_id": "c92f12ce-9088-474d-a7ea-b248131734d3",
        "project_id": "d58775da112543679725a95deaa1f8ed",
        "port_id": "79cfbd51-4b70-4836-9eee-75dae592d98a",
        "id": "d4977ac7-fc01-45a0-afc9-9d86819e7976",
        "az_id": "bb912da8-6bb7-4bc7-b581-6f965de2edc2",
        "name": "tjc1test1"
    }
]
```


4.9.4.1.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.9.4.2. Query a Specified Virtual IP

4.9.4.2.1. Scenario

- Scenario: Query a specified virtual IP
- Roles Supported: Super admin, system admin, and co-administrator with the Network Read permission
- Versions supported: SCP6.3.70 and above

4.9.4.2.2. URI

Table 4.9.52 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/virtualips/{virtualip_id}	Query a Specified Virtual IP

- Description

Table 4.9.53 Path Parameters¶

Parameter	Location	Type	Description	Required
virtualip_id	path	string	Virtual IP ID	Yes

4.9.4.2.3. Request

- Response Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/virtualips/d4977ac7-fc01-45a0-afc9-9d86819e7976

4.9.4.2.4. Response

- Response Parameter

Table 4.9.54 Response Parameter

Parameter	Type	Description	Value Returned
id	string	Virtual IP ID	Yes
subnet_id	string	Subnet ID	Yes
project_id	string	Tenant ID	Yes
fixed_ips	list	Assigned virtual IPs, view Parameter Description - fixed_ips for reference	Yes
vpc_id	string	VPC ID	Yes
az_id	string	Resource pool ID of virtual IP	Yes
port_id	string	Port ID	Yes
name	string	Virtual IP name	Yes
description	string	Description	Yes
tag	string	Virtual IP tag	Yes
binds	list	Association information (because binding virtual IP is not currently supported, no result will be returned.)	Yes
created_at	string	Time created	Yes
updated_at	string	Time updated	Yes

Note

Virtual IP cannot be associated currently, so the parameter binds will return no results.

Note

For common response parameters, view Response Format for details

- Parameter Description - fixed_ips

Table 4.9.55 Parameter Description - fixed_ips

Parameter	Type	Value Returned	Note
ip_address	string	Yes	Virtual IP address

- Response Example

```
{
    "message": "",
    "code": 0,
    "data": {
        "description": "",
        "subnet_id": "1b4a29f1-6c51-441f-af36-97ba69ea1506",
        "fixed_ips": [
            {
                "ip_address": "192.168.0.3"
            }
        ],
        "created_at": "2021-04-26T03:31:20.000000",
        "updated_at": "",
        "binds": [],
        "tag": "",
        "vpc_id": "c92f12ce-9088-474d-a7ea-b248131734d3",
        "project_id": "d58775da112543679725a95deaa1f8ed",
        "port_id": "79cfbd51-4b70-4836-9eee-75dae592d98a",
        "id": "d4977ac7-fc01-45a0-afc9-9d86819e7976",
        "az_id": "bb912da8-6bb7-4bc7-b581-6f965de2edc2",
        "name": "tjctest1"
    }
}
```


4.9.4.2.5. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.9.4.3. Create a Virtual IP

4.9.4.3.1. Scenario

- Scenario: Create a virtual IP
- Roles Supported: Co-administrator with the Network Write permission
- Versions supported: SCP6.3.70 and above

4.9.4.3.2. URI

Table 4.9.56 API Path and Request Method

REST VERB	PATH	Description
POST	/janus/20180725/virtualips	Create a Virtual IP

- Description

  None

4.9.4.3.3. Request

- Response Parameter

Table 4.9.57 Description of Body Parameter¶

Parameter	Location	Type	Description	Required
vpc_id	body	string	VPC ID, it can be obtained from Query VPC s	Yes
id	body	string	Specified ID for creating virtual IP address	No
subnet_id	body	string	Subnet ID	No
fixed_ips	body	list	Virtual IP, view Parameter Description - fixed_ips for details	No
name	body	string	Virtual IP name	No
description	body	string	Description	No
tag	body	string	Virtual IP tag	No

Note

1. At least one of the parameters subnet_id and fixed_ips must be passed. If both of them are passed, the IP range must be the same.
2. The parameter tag can only be passed by value “DMP”.

- Parameter Description - fixed_ips

Table 4.9.58 Parameter Description - fixed_ips¶

Parameter	Type	Value Returned	Note
ip_address	string	Yes	Virtual IP address

- Request Example

https://{SCP_IP}/janus/20180725/virtualips

```
{
    "vpc_id": "1752c3d2-2df9-4e2f-92c9-9e4d68c72da1",
    "fixed_ips": [
            {
                "ip_address": "192.168.0.55"
            }
        ],
    "name": "testvip1",
    "description": "Creating virtual IP for testing"
}
```

4.9.4.3.4. Response

- Response Parameter

Table 4.9.59 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Virtual IP ID	Yes
project_id	string	Tenant ID	Yes
vpc_id	string	VPC ID	Yes
subnet_id	string	Subnet ID	Yes
fixed_ips	list	Virtual IP	Yes
port_id	string	Port ID	Yes
az_id	string	Resource pool ID	Yes
name	string	Name	No
description	string	Description	No
tag	string	Tag	No

Note

For common response parameters, view Response Format for details

- Response Example

```
{
    "name": "testvip1",
    "subnet_id": "129404c4-d19a-4487-8628-349e4e44dbd1",
    "fixed_ips": [
        {
            "ip_address": "192.168.0.55"
        }
    ],
    "tag": "",
    "vpc_id": "1752c3d2-2df9-4e2f-92c9-9e4d68c72da1",
    "project_id": "a74b7c91331547d58b78d44a2821bda1",
    "port_id": "1daeb534-327f-4ca0-b729-be94a9091698",
    "id": "fcbc8cac-9abd-4a7a-9fe4-d1c133e14132",
    "az_id": "611f1b34-c727-4925-854e-07bc8607f681",
    "description": "Creating virtual IP for testing"
}
```

4.9.4.3.5. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.9.4.4. Delete a Specified Virtual IP

4.9.4.4.1. Scenario

- Scenario: Delete a specified virtual IP.
- Roles Supported: Super admin, system admin, and co-administrator with the Network Configuration permission
- Versions supported: SCP6.3.70 and above

4.9.4.4.2. URI

Table 4.9.60 API Path and Request Method¶

REST VERB	PATH	Description
DELETE	/janus/20180725/virtualips/{virtualip_id}	Delete a Specified Virtual IP

- Description

Table 4.9.61 Path Parameters¶

Parameter	Location	Type	Description	Required
virtualip_id	path	string	Virtual IP ID	Yes

4.9.4.4.3. Request

- Response Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/virtualips/d4977ac7-fc01-45a0-afc9-9d86819e7976

4.9.4.4.4. Response

- Response Parameter

  None

- Response Example

4.9.4.4.5. Response Code

- Normal: 204
- Errors: 400, 403, 404, 500

4.9.5. Ports

4.9.5.1. Query Ports

4.9.5.1.1. Scenario

- Scenario: Query ports
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Read permission
- Versions supported: SCP 6.3.0 and above

4.9.5.1.2. URI

Table 4.9.62 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/ports	Query Ports

- Parameter Description:

Table 4.9.63 Path Parameters¶

Parameter	Location	Type	Description	Required
ip_address	body	string	IP addresses to be queried	No
subnet_id	body	string	Subnet ID	No
az_id	body	string	Resource pool ID	No
project_id	body	string	Tenant ID	No
vpc_id	body	string	VPC ID	No
device_id	body	string	Device ID	No
vif_id	body	string	Network ID (net0..n or device_key)	No

Note

For common path parameters, view Common Path Parameter Format for details

4.9.5.1.3. Request

- Request Parameter

  None
- Request Example

  https://{SCP_IP}/janus/20180725/ports?device_id=c7b2db93-fdb5-438e-8012-49e4f77de06c

4.9.5.1.4. Response

- Response Parameter

Table 4.9.64 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Assigned IP	No
name	string	Port name	No
vpc_id	string	VPC ID	No
subnet_id	string	Assigned subnet ID	No
admin_status_up	string	Whether to enable the port	No
device_owner	string	Owner of the associated device	No
project_id	string	Tenant ID	No
az_id	string	Resource pool ID	No
device_id	string	Device ID	No
device_type	string	Associated device type, view Device Type for details	No
mac_address	string	MAC address of the port	No
ips_info	lis(dict)	Assigned IP addresses, view Parameter Description - ips_info for details	No

- Parameter Description - ips_info:

Table 4.9.65 Parameter Description - ips_info¶

Parameter	Type	Required	Description
ip_address	string	Assigned IP address of the same subnet	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
[
    {
        "name": null,
        "subnet_id": "39bfcb6d-b1cd-4c20-9294-0e969eaede9a",
        "ips_info": [
            {
                "ip_address": "192.168.0.1"
            }
        ],
        "admin_status_up": true,
        "device_owner": "network:router_interface",
        "device_type": "vrouter",
        "mac_address": "fa:16:3e:97:ae:89",
        "vpc_id": "19556411-8f8e-47f1-96e6-48867acbf011",
        "project_id": "d8618b1267444201be087bca0cce7209",
        "id": "320223bf-e839-4359-bce1-138ba10c1117",
        "az_id": "cd8378b2-7cd6-46e5-80eb-662111529986",
        "device_id": "2fd954f1-a76a-40e6-bbfa-b95980961423"
    }
]
```

4.9.5.1.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.9.5.2. Create a Port

4.9.5.2.1. Scenario

- **Scenario:**
    - This interface can be called when a virtual network device needs to connect to the switch on the subnet.
    - Create a port.

- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Write permission
- Versions supported: SCP 6.3.0 and above

4.9.5.2.2. URI

Table 4.9.66 API Path and Request Method¶

REST VERB	PATH	Description
POST	/janus/20180725/ports	Create a Port

- Parameter Description:

  None

4.9.5.2.3. Request

- Request Parameter

Table 4.9.67 Response Parameter

Parameter	Location	Type	Description	Required
vpc_id	body	string	Network ID	Yes
device_id	body	string	Device ID	No
device_owner	body	string	Owner of the associated device	No
device_type	body	string	Associated device type, view Device Type for details	No
subnet_id	body	string	Assigned subnet ID	No
ips_info	body	list(dict)	Assigned IPs, view Parameter Description - allocation_pools for details	No
mac_address	body	string	MAC address of the port	No
name	body	string	Port name	No
vif_id	body	string	HCI ID of the port	No
admin_status_up	body	int(0, 1)	Whether to enable this interface	No
custom_gateway_ip	body	string	Custom gateway IP	No

- Parameter Description - allocation_pools:

Table 4.9.68 Parameter Description - allocation_pools

Parameter	Type	Description	Value Returned
start	string	Start IP	Yes
end	string	End IP	Yes

- Request Example

  https://{SCP_IP}/janus/20180725/ports

```
{
    "vpc_id": "19556411-8f8e-47f1-96e6-48867acbf011",
    "device_id": "836c6ba5-5ca0-44fd-8ef6-cf316f68dd30",
    "device_owner": "",
    "device_type": "",
    "subnet_id": "836c6ba5-5ca0-44fd-8ef6-cf316f68dd30",
    "ips_info": [
        {
            "ip_address": "192.168.0.55"
        }
    ],
    "mac_address": "",
    "name": "test",
    "vif_id": "",
    "admin_status_up": "",
    "custom_gateway_ip": ""
}
```

4.9.5.2.4. Response

- Response Parameter

Table 4.9.69 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Port ID	Yes
vpc_id	string	Network ID	Yes
project_id	string	Tenant ID	Yes
subnet_id	string	Assigned subnet ID	Yes
custom_gateway_ip	string	Custom gateway IP	Yes
device_id	string	Device ID	No
device_type	string	Associated device type, view Device Type for details	No
mac_address	string	MAC address of the port	No
name	string	Port name	No
az_id	string	Resource pool ID	No
admin_status_up	string	Whether to enable this interface	No
external	string	Whether it is an external port	No
vif_id	string	HCI ID of the port	No
created_at	string	Time created	No
updated_at	string	Time updated	No
ips_info	list(dict)	Assigned IPs, view Parameter Description - ips_info for details	Yes

- Parameter Description - ips_info:

Table 4.9.70 Parameter Description - ips_info¶

Parameter	Type	Required	Description
ip_address	string	Assigned IP address of the same subnet	Yes

- Response Example:

```
{
    "message": "",
    "code": 0,
    "data": {
        "name": "eth0",
        "subnet_id": "e54eeb16-ad77-4f28-bf3f-407189dab4a8",
        "created_at": "2020-12-04T02:47:00.000000",
        "custom_gateway_ip": null,
        "updated_at": null,
        "ips_info": [
            {
                "ip_address": "192.168.0.28"
            }
        ],
        "admin_status_up": true,
        "mac_address": "fa:16:3e:15:3b:46",
        "vpc_id": "cb367624-b91a-4463-b45a-b756c54857a2",
        "project_id": "a4020e234e57481887b1e5ec72446a6f",
        "id": "1b195b06-dc6c-4ae8-85e8-64117b0938e8",
        "az_id": "cd8378b2-7cd6-46e5-80eb-662111529986",
        "device_id": "aa05bf0f-ef99-44f9-ad4a-fd62b31aded5"
    }
}
```

4.9.5.2.5. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.9.5.3. Delete a Port

4.9.5.3.1. Scenario

- **Scenario:**
  - This interface can be called when you need to disconnect a virtual network device from switch on the subnet.
  - Delete a port.

- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Write permission
- Versions supported: SCP 6.3.0 and above

4.9.5.3.2. URI

Table 4.9.71 API Path and Request Method¶

REST VERB	PATH	Description
DELETE	/janus/20180725/ports/{port_id}	Delete port/disconnect the NFV device from switch on the subnet

- Parameter Description:

Table 4.9.72 API Path and Request Method¶

Parameter	Location	Type	Description	Required
port_id	path	string	port_id	Yes

4.9.5.3.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/ports/1b195b06-dc6c-4ae8-85e8-64117b0938e8

4.9.5.3.4. Response

- Response Parameter

  None

- Response Example:

  None

4.9.5.3.5. Response Code

- Normal: 204
- Errors: 400, 403, 404, 500

4.9.6. Line Type

4.9.6.1. Query Line Types

4.9.6.1.1. Scenario

- Scenario: Query the list of line types
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Read permission
- Versions supported: SCP 6.3.0 and above

4.9.6.1.2. URI

Table 4.9.73 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/linetypes	Query Line Types

- Parameter Description:

Table 4.9.74 Path Parameters¶

Parameter	Location	Type	Description	Required
name	path	string	Name	No
type	path	string	Type, view Line Type for details	No
available_only	path	boolean	Whether to enable filtering	No
az_id	path	string	Resource pool ID	No

Warning

Common query parameters cannot be used for this interface.

Note

Whether to enable filtering. You can enable this parameter to filter out types of lines that support Elastic IP pool and quota creation.

4.9.6.1.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/linetypes?type=pub_service

4.9.6.1.4. Response

- Response Parameter

Table 4.9.75 Response Parameter

Parameter	Type	Description	Value Returned
linetypes	list(dict)	Line type, view Parameter Description - linetypes for details	Yes
total_size	int	Query results	Yes
page_num	int	Current page	Yes
page_size	int	Page size	Yes
next_page_num	int	Next page, empty string indicates that the current page is the last page	Yes

Warning

This interface cannot be used as the common response parameters. These parameters will be returned, but they do not have any meaning.

- Parameter Description - linetypes:

Table 4.9.76 Parameter Description - linetypes

Parameter	Type	Description	Required
name	string	Name	Yes
id	string	ID	Yes
type	string	Type, view Line Type for details	Yes
description	string	Description	No
created_at	string	Time created	No
updated_at	string	Time updated	No

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "linetypes": [
        {
            "description": "Predefined line type, unable to be edited",
            "created_at": "2020-11-26T11:51:13.000000",
            "updated_at": null,
            "type": "inner_linetype",
            "id": "01356005-8975-4799-9c60-0b5fa806ebf3",
            "name": "China_Mobile"
        },
        {
            "description": "Predefined line type, unable to be edited",
            "created_at": "2020-11-26T11:51:13.000000",
            "updated_at": null,
            "type": "inner_linetype",
            "id": "0779c89e-dadb-4d1f-bb87-3f18883ca066",
            "name": "CERNET"
        }
    ],
    "total_size": 0,
    "next_page_num": "",
    "page_num": 0,
    "page_size": 0
}
```

4.9.6.1.4.1. Response Code¶

- OK: 200
- Errors: 400, 403, 500

4.9.7. Elastic IP Pool

4.9.7.1. Query Elastic IP Pools

- Scenario: Query elastic IP pools
- Roles Supported: Super admin, system admin, and co-administrator with the Network Read permission
- Versions supported: SCP6.3.70 and above

4.9.7.1.1. URI

Table 4.9.77 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/floatingippools	Query Elastic IP Pools

- Parameter Description:

Table 4.9.78 Path Parameters¶

Parameter	Location	Type	Description	Required
floatingip_pool_id	path	string	Elastic IP pool ID	No
name	path	string	Elastic IP pool name	No
line_type_id	path	string	Line type ID, view Query Line Types for details	No
query	path	string	Query name	No

Note

For common path parameters, view Common Path Parameter Format for details

4.9.7.1.2. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/floatingippools?floatingip_pool_id=661ce7a4-5abe-4191-832c-7d1dc6c5fd9a

4.9.7.1.3. Response

- Response Parameter

Table 4.9.79 Response Parameter¶

Parameter	Type	Description	Required
id	string	Elastic IP pool ID	No
name	string	Name	No
line_type_id	string	Line type ID	No
description	string	Description	No
is_global	int(0, 1)	Whether it applies to the whole platform (whether it is associated with all the resource pools automatically)	No
az_ids	list(string)	Associated resource pools, when the elastic IP pool is associated with all the resource pools, the parameter will not return any result	No
vlan	int	VLAN ID	No
ip_count	int	Elastic IP count	No
qos_downlink	int(1000, 1023998976)	Inbound bandwidth of elastic IP pool (Kbps)	No
qos_uplink	int(1000, 1023998976)	Outbound bandwidth of elastic IP pool (Kbps)	No
allocation_pools	list(dict)	Available IP ranges, view Parameter Description - allocation_pools for details	No

- Parameter Description - allocation_pools:

Table 4.9.80 Parameter Description - allocation_pools¶

Parameter	Type	Description	Value Returned
start	string	Start IP	Yes
end	string	End IP	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
[
    {
        "is_global": 1,
        "az_ids": [],
        "description": "",
        "vlan": 0,
        "allocation_pools": [
            {
                "start": "100.0.0.1",
                "end": "100.0.0.100"
            }
        ],
        "ip_count": 100,
        "qos_downlink": 100000,
        "qos_uplink": 100000,
        "line_type_id": "d7125278-486a-4692-bec5-87911c9ee866",
        "id": "30b5445f-099f-4ea5-9df3-e974684bb774",
        "name": "China_Mobile_Zone1"
    },
    {
        "is_global": 0,
        "az_ids": [
            "0b1695d7-8aa8-4efb-9d25-3c0312f0c5a1"
        ],
        "description": "",
        "vlan": null,
        "allocation_pools": [],
        "ip_count": 0,
        "qos_downlink": 102400,
        "qos_uplink": 102400,
        "line_type_id": "d7125278-486a-4692-bec5-87911c9ee866",
        "id": "be777f96-b4d3-4ca0-bdf5-0aa159ad5d75",
        "name": "China_Telecom_Zone2"
    }
]
```

4.9.7.1.4. Response Code

- OK: 200
- Errors: 400, 403, 500

4.9.7.2. Query Elastic IP Pools

- Scenario: Query elastic IP pools
- Roles Supported: Super admin, system admin, and co-administrator with the Network Read permission
- Versions supported: SCP6.3.70 and above

4.9.7.2.1. URI

Table 4.9.81 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/floatingippools/{floatingippool_id}	Query Elastic IP Pools

- Parameter Description:

Table 4.9.82 Path Parameters¶

Parameter	Location	Type	Description	Required
floatingippool_id	path	string	Elastic IP ID	Yes

4.9.7.2.2. Request

- Request Parameter

None

- Request Example

  https://{SCP_IP}/janus/20180725/floatingippools/661ce7a4-5abe-4191-832c-7d1dc6c5fd9a

4.9.7.2.3. Response

- Response Parameter

Table 4.9.83 Response Parameter¶

Parameter	Type	Description	Required
id	string	Elastic IP pool ID	Yes
name	string	Name	Yes
line_type_id	string	Line type ID	Yes
description	string	Description	Yes
is_global	int(0, 1)	Whether it applies to the whole platform (whether it is associated with all the resource pools automatically)	Yes
az_ids	list(string)	Associated resource pools, when the elastic IP pool is associated with all the resource pools, the parameter will not return any result	Yes
vlan	int	VLAN ID	Yes
ip_count	int	Elastic IP count	Yes
ip_bind	int	Associated elastic IPs	Yes
ip_used	int	Used elastic IPs	Yes
allocated_bandwidth	int	Assigned bandwidth (Kbps)	Yes
qos_downlink	int(1000, 1023998976)	Inbound bandwidth of elastic IP pool (Kbps)	Yes
qos_uplink	int(1000, 1023998976)	Outbound bandwidth of elastic IP pool (Kbps)	Yes
pool_in_kbps	int	Outbound bandwidth of elastic IP pool (Kbps)	Yes
pool_out_kbps	int	Inbound bandwidth of elastic IP pool (Kbps)	Yes
allocation_pools	list(dict)	Available IP ranges, view Parameter Description - allocation_pools for details	Yes

- Parameter Description - allocation_pools:

Table 4.9.84 Parameter Description - allocation_pools¶

Parameter	Type	Description	Value Returned
start	string	Start IP	Yes
end	string	End IP	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "id": "d9387ddb-a0b7-4870-af36-a0e7630517e1",
    "name": "China_Telecom_Zone1",
    "line_type_id": "1019851c-46f1-4adb-b6be-f5fde6be9ebd",
    "description": "",
    "is_global": 1,
    "az_ids": [],
    "vlan": 0,
    "allocation_pools": [
        {
            "start": "10.10.110.1",
            "end": "10.10.110.253"
        }
    ],
    "ip_count": 253,
    "ip_bind": 0,
    "ip_used": 0,
    "allocated_bandwidth": 0,
    "qos_uplink": 20000000,
    "qos_downlink": 20000000,
    "pool_in_kbps": 0,
    "pool_out_kbps": 0
}
```

4.9.7.2.4. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.9.7.3. Create an Elastic IP Pool

- Scenario: Create an elastic IP pool
- Roles Supported: Super admin, system admin, and co-administrator with the Network Write permission
- Versions supported: SCP6.3.70 and above

> **Note**
>
> After an elastic IP pool is created, you need to call the interface for creating elastic IP pool subnet (view Create an Elastic IP Pool Subnet for details) to set IP range for the elastic IP pool. Otherwise, it cannot be used.

4.9.7.3.1. URI

Table 4.9.85 API Path and Request Method

REST VERB	PATH	Description
POST	/janus/20180725/floatingippools	Create an Elastic IP Pool

- Parameter Description:

  None

4.9.7.3.2. Request

- Request Parameter

Table 4.9.86 Response Parameter¶

Parameter	Location	Type	Description	Required
name	body	string	Elastic IP pool name	Yes
line_type_id	body	string	Line type ID, view Query Line Types for details	Yes
description	body	string	Description	Yes
is_global	body	int(0, 1)	Whether it applies to the whole platform (whether it is associated with all the resource pools automatically)	Yes
az_ids	body	list(string)	Associated resource pool, when the elastic IP pool is associated with all the resource pools, the parameter will not return any result	Yes
vlan	body	int	VLAN ID	Yes
qos_downlink	body	int(1000, 1023998976)	Inbound bandwidth of elastic IP pool (Kbps)	Yes
qos_uplink	body	int(1000, 1023998976)	Inbound bandwidth of elastic IP pool (Kbps)	Yes

Note

1. Outbound and inbound bandwidth (qos_downlink, qos_uplink) must be the same.
2. The parameter is_global and az_ids cannot be invalid (*is_global=0, az_ids=[]*) at the same time.
3. Number of elastic IPs cannot exceed 50.

- Request Example

https://{SCP_IP}/janus/20180725/floatingippools

```
{
    "name": "China_Telecom_Zone1",
    "description": "China Telecom",
    "is_global": 0,
    "az_ids": [
        "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b"
    ],
    "line_type_id": "1019851c-46f1-4adb-b6be-f5fde6be9ebd",
    "qos_uplink": 2000000,
    "qos_downlink": 2000000,
    "vlan": 0
}
```

4.9.7.3.3. Response

- Response Parameter

Table 4.9.87 Response Parameter¶

Parameter	Type	Value Returned	Description
id	string	Elastic IP pool ID	Yes
name	string	Elastic IP pool name	Yes
line_type_id	string	Line type ID	Yes
description	string	Description	Yes
is_global	int(0, 1)	Whether it applies to the whole platform (whether it is associated with all the resource pools automatically)	Yes
az_ids	list(string)	Associated resource pool, when the elastic IP pool is associated with all the resource pools, the parameter will not return any result	Yes
vlan	int	VLAN ID	Yes
qos_downlink	int(1000, 1023998976)	Inbound bandwidth of elastic IP pool (Kbps)	Yes
qos_uplink	int(1000, 1023998976)	Inbound bandwidth of elastic IP pool (Kbps)	Yes
project_id	string	User ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "id": "878c9ede-57dd-4b9d-93f8-c5781f514692",
    "name": "China_Telecom_Zone1",
    "line_type_id": "1019851c-46f1-4adb-b6be-f5fde6be9ebd",
    "description": "China Telecom",
    "is_global": 0,
    "az_ids": [
        "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b"
    ],
    "vlan": 0,
    "qos_downlink": 2000000,
    "qos_uplink": 2000000,
    "project_id": "b9f644e5d4ed42358d287b19ea944e71"
}
```

4.9.7.3.4. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.9.7.4. Edit an Elastic IP Pool

- Scenario: Edit an Elastic IP Pool
- Roles Supported: Super admin, system admin, and co-administrator with the Network Write permission
- Versions supported: SCP6.3.70 and above

4.9.7.4.1. URI

Table 4.9.88 API Path and Request Method¶

REST VERB	PATH	Description
PUT	/janus/20180725/floatingippools/{floatingippool_id}	Edit an Elastic IP Pool

- Parameter Description:

Table 4.9.89 Path Parameters¶

Parameter	Location	Type	Description	Required
floatingippool_id	path	string	Elastic IP pool ID	Yes

4.9.7.4.2. Request

- Request Parameter

Table 4.9.90 Response Parameter¶

Parameter	Location	Type	Description	Required
name	body	string	Elastic IP name	No
line_type_id	body	string	Line type ID, view Query Line Types for details	No
description	body	string	Description	No
is_global	body	int(0, 1)	Whether it applies to the whole platform (whether it is associated with all the resource pools automatically)	No
az_ids	body	list(string)	Associated resource pool, when the elastic IP pool is associated with all the resource pools, the parameter will not return any result	No
vlan	body	int	VLAN ID	No
qos_downlink	body	int(1000, 1023998976)	Inbound bandwidth of elastic IP pool (Kbps)	No
qos_uplink	body	int(1000, 1023998976)	Inbound bandwidth of elastic IP pool (Kbps)	No

Note

1. Outbound and inbound bandwidth (qos_downlink, qos_uplink) must be the same.
2. The parameter is_global and az_ids cannot be invalid (*is_global=0, az_ids=[]*) at the same time.

- Request Example

https://{SCP_IP}/janus/20180725/floatingippools/661ce7a4-5abe-4191-832c-7d1dc6c5fd9a

```
{
    "name": "China_Telecom_Zone1",
    "description": "China Telecom",
    "is_global": 0,
    "az_ids": [
        "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b"
    ],
    "line_type_id": "1019851c-46f1-4adb-b6be-f5fde6be9ebd",
    "qos_uplink": 2000000,
    "qos_downlink": 2000000,
    "vlan": 0
}
```

4.9.7.4.3. Response

- Response Parameter

Table 4.9.91 Response Parameter

Parameter	Type	Value Returned	Description
id	string	Elastic IP pool ID	Yes
name	string	Elastic IP pool name	Yes
line_type_id	string	Line type ID	Yes
description	string	Description	Yes
is_global	int(0, 1)	Whether it applies to the whole platform (whether it is associated with all the resource pools automatically)	Yes
az_ids	list(string)	Associated resource pool, when the elastic IP pool is associated with all the resource pools, the parameter will not return any result	Yes
vlan	int	VLAN ID	Yes
qos_downlink	int(1000, 1023998976)	Inbound bandwidth of elastic IP pool (Kbps)	Yes
qos_uplink	int(1000, 1023998976)	Inbound bandwidth of elastic IP pool (Kbps)	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "id": "878c9ede-57dd-4b9d-93f8-c5781f514692",
    "name": "China_Telecom_Zone1",
    "line_type_id": "1019851c-46f1-4adb-b6be-f5fde6be9ebd",
    "description": "China Telecom",
    "is_global": 0,
    "az_ids": [
        "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b"
    ],
    "vlan": 0,
    "qos_downlink": 2000000,
    "qos_uplink": 2000000
}
```

4.9.7.4.4. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.9.7.5. Delete an Elastic IP pool

- Scenario: Delete an elastic IP pool
- Roles Supported: Super admin, system admin, and co-administrator with the Network Write permission
- Versions supported: SCP6.3.70 and above

Note

1. If there exist allocated elastic IPs in the elastic IP pool, the IP pool cannot be deleted. Use the interface Query Elastic IPs to query whether there exist allocated elastic IPs.

4.9.7.5.1. URI

Table 4.9.92 API Path and Request Method

REST VERB	PATH	Description
DELETE	/janus/20180725/floatingippools/{floatingippool_id}	Delete an Elastic IP pool

- Parameter Description:

Table 4.9.93 Path Parameters

Parameter	Location	Type	Description	Required
floatingippool_id	path	string	Elastic IP pool ID	Yes

4.9.7.5.2. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/floatingippools?floatingippool_id=661ce7a4-5abe-4191-832c-7d1dc6c5fd9a

4.9.7.5.3. Response

- Response Parameter

  None

- Response Example:

  None

4.9.7.5.4. Response Code

- Normal: 204
- Errors: 400, 403, 404, 500

4.9.7.6. Query Elastic IP Pool Subnets

4.9.7.6.1. Scenario

- Scenario: Query elastic IP pool subnets
- Roles Supported: Super admin, system admin, and co-administrator with the Network Read permission
- Versions supported: SCP6.3.70 and above

4.9.7.6.2. URI

Table 4.9.94 API Path and Request Method

REST VERB	PATH	Description
GET	/janus/20180725/floatingippools/{floatingip_pool_id}/subnets	Query Elastic IP Pool Subnets

- Parameter Description:

Table 4.9.95 Path Parameters

Parameter	Location	Type	Description	Required
floatingip_pool_id	path	string	Elastic IP pool ID	No

Warning

Common query parameters cannot be used for this interface.

4.9.7.6.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/floatingippools/661ce7a4-5abe-4191-832c-7d1dc6c5fd9a/subnets

4.9.7.6.4. Response

- Response Parameter

Table 4.9.96 Response Parameter

Parameter	Type	Description	Required
subnets	list(dict)	Elastic IP pool subnets, view Response Parameter for details	Yes

- Parameter Description - subnets:

Table 4.9.97 Response Parameter¶

Parameter	Type	Required	Note
id	string	Subnet ID	Yes
name	string	Subnet name	Yes
status	string	Subnet status, view Subnet Status for details	Yes
is_visible	int(0, 1)	Whether the subnet is visible to tenants	Yes
floatingip_pool_id	string	Elastic IP pool ID	Yes
project_id	string	Tenant ID	Yes
shared	int(0, 1)	Whether it is visible to all tenants	Yes
ip_version	int	Internet protocol version, view Internet Protocol Version for details	Yes
enable_dhcp	int(0, 1)	Whether to enable DHCP	Yes
gateway_ip	string	Gateway IP	Yes
netmask	string	Netmask	Yes
cidr	string	Subnet CIDR	Yes
allocation_pools	list(dict)	Subnet range, view Parameter Description - allocation_pools for details	Yes

- Parameter Description - allocation_pools:

Table 4.9.98 Parameter Description - allocation_pools¶

Parameter	Type	Description	Value Returned
start	string	Start IP	Yes
end	string	End IP	Yes

Warning

Common response parameters cannot be used for this interface.

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "subnets": [
        {
            "status": "active",
            "is_visible": 1,
            "name": "88b47bc9-63b3-4e51-bed8-e70957da3ac3",
            "enable_dhcp": 0,
            "allocation_pools": [
                {
                    "start": "10.10.110.1",
                    "end": "10.10.110.253"
                }
            ],
            "floatingip_pool_id": "d9387ddb-a0b7-4870-af36-a0e7630517e1",
            "gateway_ip": "10.10.110.254",
            "ip_version": 4,
            "shared": 1,
            "cidr": "10.10.110.0/24",
            "project_id": "b9f644e5d4ed42358d287b19ea944e71",
            "netmask": "255.255.255.0",
            "id": "bb648d94-41bb-4741-b400-63f7c44848ba"
        }
    ]
}
```

4.9.7.6.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.9.7.7. Query Elastic IP Pool Subnet

4.9.7.7.1. Scenario

- Scenario: Query elastic IP pool subnet
- Roles Supported: Super admin, system admin, and co-administrator with the Network Write permission
- Versions supported: SCP6.3.70 and above

4.9.7.7.2. URI

Table 4.9.99 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/floatingippools/{floatingip_pool_id}/subnets/{floatingip_subnet_id}	Query Elastic IP Pool Subnet

- Parameter Description:

Table 4.9.100 API Path and Request Method¶

Parameter	Location	Type	Description	Required
floatingip_pool_id	path	string	Elastic IP pool ID	Yes
floatingip_subnet_id	path	string	Elastic IP pool subnet ID	Yes

4.9.7.7.3. Request

- Request Parameter

None

- Request Example

  https://{SCP_IP}/janus/20180725/floatingippools/6733078a-3572-4680-a750-e9d5eca630bc/subnets/1e5c4b98-0955-4ace-ac71-9cf4c4d1c209

4.9.7.7.4. Response

- Response Parameter

Table 4.9.101 Response Parameter

Parameter	Type	Description	Value Returned
id	string	Subnet ID	Yes
name	string	Subnet name	Yes
status	string	Subnet status, view Subnet Status for details	Yes
is_visible	int(0, 1)	Whether the subnet is visible to the its tenant	No
floatingip_pool_id	string	Elastic IP pool ID	Yes
project_id	string	Tenant ID	Yes
ip_version	int	Internet protocol version, view Internet Protocol Version for details	Yes
enable_dhcp	int(0, 1)	Whether to enable DHCP	No
gateway_ip	string	Gateway IP	Yes
netmask	string	Netmask	Yes
cidr	string	Subnet CIDR	Yes
shared	int(0, 1)	Whether it is visible to all tenants	No
create_at	string	Time created	Yes
update_at	string	Time updated	Yes
allocation_pools	list(dict)	Subnet range, view Parameter Description - allocation_pools for details	Yes

- Parameter Description - allocation_pools:

Table 4.9.102 Parameter Description - allocation_pools

Parameter	Type	Description	Value Returned
start	string	Start IP	Yes
end	string	End IP	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "status": "active",
    "is_visible": 1,
    "enable_dhcp": 0,
    "created_at": "2020-12-28T07:14:11Z",
    "updated_at": "2020-12-28T07:14:19Z",
    "allocation_pools": [
        {
            "start": "192.168.1.2",
            "end": "192.168.1.254"
        }
    ],
    "floatingip_pool_id": "6733078a-3572-4680-a750-e9d5eca630bc",
    "gateway_ip": "192.168.1.1",
    "ip_version": 4,
    "shared": 1,
    "cidr": "192.168.1.0/24",
    "project_id": "de3c453465034a31a2e054c6e4250626",
    "netmask": "255.255.255.0",
    "id": "1e5c4b98-0955-4ace-ac71-9cf4c4d1c209",
    "name": "7481b187-aea5-4654-b4d4-ed16739cf5b8"
}
```

4.9.7.7.5. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.9.7.8. Create an Elastic IP Pool Subnet

4.9.7.8.1. Scenario

- Scenario: Create an elastic IP pool subnet.
- Roles Supported: Super admin, system admin, and co-administrator with the Network Write permission
- Versions supported: SCP6.3.70 and above

Note

1. To create a subnet, you are recommended to use the IP ranges that start with 10, 172 or 192 as the subnet range. The following are the recommended subnet range:
    a. For large-sized enterprises, use IP range starting with 10.
    b. For medium-sized enterprises, use IP range starting with 172.
    c. For small enterprises or individuals, use IP range starting with 192.
2. The IP range 10.251.251.0/24 is reserved for NFV devices and cannot be used as elastic IP pool subnet.
3. Subnet range cannot overlap with elastic IP subnet range.
4. One tenant can create at most 20 subnets.

4.9.7.8.2. URI

Table 4.9.103 API Path and Request Method

REST VERB	PATH	Description
POST	/janus/20180725/floatingippools/{floatingip_pool_id}/subnets	Create an Elastic IP Pool Subnet

- Parameter Description:

Table 4.9.104 API Path and Request Method¶

Parameter	Location	Type	Description	Required
floatingip_pool_id	path	string	Elastic IP pool ID	Yes

4.9.7.8.3. Request

- Request Parameter

Table 4.9.105 Response Parameter¶

Parameter	Location	Type	Description	Required
name	body	string	Subnet name, the default name is UUID if not passed	No
description	body	string	Description	No
gateway_ip	body	string	Gateway IP	Yes
shared	body	int(0, 1)	Whether it is visible to all tenants (default: 1)	No
is_visible	body	int(0, 1)	Whether is is visible to all tenants (default: 1)	No
enable_dhcp	body	int(0, 1)	Whether to enable DHCP (default: 0)	No
netmask	body	string	Netmask	Yes
allocation_pools	body	list(dict)	Subnet range, view Parameter Description - allocation_pools for details	Yes

Warning

CIDR that consists of gateway_ip and netmask cannot be in the following IP ranges: *192.168.0.0/24* and *172.30.0.0/17*

- Parameter Description - allocation_pools:

Table 4.9.106 Parameter Description - allocation_pools¶

Parameter	Type	Description	Value Returned
start	string	Start IP	Yes
end	string	End IP	Yes

- Request Example

https://{SCP_IP}/janus/20180725/floatingippools/661ce7a4-5abe-4191-832c-7d1dc6c5fd9a/subnets

```
{
    "gateway_ip": "192.168.1.1",
    "allocation_pools": [
        {
        "start": "192.168.1.2",
        "end": "192.168.1.254"
        }
    ],
    "netmask": "255.255.255.0",
}
```

4.9.7.8.4. Response

- Response Parameter

Table 4.9.107 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Subnet ID	Yes
name	string	Subnet name	Yes
status	string	Subnet status, view Subnet Status for details	Yes
is_visible	int(0, 1)	Whether the subnet is visible to the its tenant	No
floatingip_pool_id	string	Elastic IP pool ID	Yes
project_id	string	Tenant ID	Yes
ip_version	int	Internet protocol version (supports IPv4)	Yes
enable_dhcp	int(0, 1)	Whether to enable DHCP	No
gateway_ip	string	Gateway IP	Yes
netmask	string	Netmask	Yes
cidr	string	Subnet CIDR	Yes
shared	int(0, 1)	Whether it is visible to all tenants	No
create_at	string	Time created	Yes
update_at	string	Time updated	Yes
allocation_pools	list(dict)	Subnet range, view Parameter Description - allocation_pools for details	Yes

- Parameter Description - allocation_pools:

Table 4.9.108 Parameter Description - allocation_pools¶

Parameter	Type	Description	Value Returned
start	string	Start IP	Yes
end	string	End IP	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "status": "active",
    "is_visible": 1,
    "enable_dhcp": 0,
    "created_at": "2020-12-28T07:14:11Z",
    "updated_at": "2020-12-28T07:14:19Z",
    "allocation_pools": [
        {
            "start": "192.168.1.2",
            "end": "192.168.1.254"
        }
    ],
    "floatingip_pool_id": "6733078a-3572-4680-a750-e9d5eca630bc",
    "gateway_ip": "192.168.1.1",
    "ip_version": 4,
    "shared": 1,
    "cidr": "192.168.1.0/24",
    "project_id": "de3c453465034a31a2e054c6e4250626",
    "netmask": "255.255.255.0",
    "id": "1e5c4b98-0955-4ace-ac71-9cf4c4d1c209",
    "name": "7481b187-aea5-4654-b4d4-ed16739cf5b8"
}
```

4.9.7.8.5. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.9.7.9. Edit an Elastic IP Pool Subnet

- Scenario: Edit an elastic IP pool subnet
- Roles Supported: Super admin, system admin, and co-administrator with the Network Write permission
- Versions supported: SCP6.3.70 and above

4.9.7.9.1. URI

Table 4.9.109 API Path and Request Method

REST VERB	PATH	Description
PUT	/janus/20180725/floatingippools/{float_ip_pool_id}/subnets/{floatingip_subnet_id}	Edit an Elastic IP Pool Subnet

- Parameter Description:

Table 4.9.110 Path Parameters

Parameter	Location	Type	Description	Required
floatingip_pool_id	path	string	Elastic IP pool ID	Yes
floatingip_subnet_id	path	string	Elastic IP pool subnet ID	Yes

4.9.7.9.2. Request

- Request Parameter

Table 4.9.111 Response Parameter¶

Parameter	Location	Type	Description	Required
allocation_pools	body	list(dict)	Available IP ranges, view Parameter Description - allocation_pools for details	Yes
name	body	string	Name	No
desc	body	string	Description	No

- Parameter Description - allocation_pools:

Table 4.9.112 Parameter Description - allocation_pools¶

Parameter	Type	Description	Value Returned
start	string	Start IP	Yes
end	string	End IP	Yes

- Request Example

  https://{SCP_IP}/janus/20180725/floatingippools/661ce7a4-5abe-4191-832c-7d1dc6c5fd9a/subnets/d5ae0d43-4afc-4f27-a851-9989a7083d9d

```
{
    "allocation_pools": [
        {
            "start": "192.168.1.2",
            "end": "192.168.1.250",
        }
    ]
}
```

4.9.7.9.3. Response

- Response Parameter

Table 4.9.113 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Subnet ID	Yes
name	string	Subnet name	Yes
status	string	Subnet status, view Subnet Status for details	Yes
is_visible	int(0, 1)	Whether the subnet is visible to the its tenant	No
floatingip_pool_id	string	Elastic IP pool ID	Yes
project_id	string	Tenant ID	Yes
ip_version	int	Internet protocol version (supports IPv4)	Yes
enable_dhcp	int(0, 1)	Whether to enable DHCP	No
gateway_ip	string	Gateway IP	Yes
netmask	string	Netmask	Yes
cidr	string	Subnet CIDR	Yes
shared	int(0, 1)	Whether it is visible to all tenants	No
create_at	string	Time created	Yes
update_at	string	Time updated	Yes
allocation_pools	list(dict)	Subnet range, view Parameter Description - allocation_pools for details	Yes

- Parameter Description - allocation_pools:

Table 4.9.114 Parameter Description - allocation_pools¶

Parameter	Type	Description	Value Returned
start	string	Start IP	Yes
end	string	End IP	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "status": "active",
    "is_visible": 1,
    "enable_dhcp": 0,
    "created_at": "2020-12-28T07:14:11Z",
    "updated_at": "2020-12-28T07:14:19Z",
    "allocation_pools": [
        {
            "start": "192.168.1.2",
            "end": "192.168.1.254"
        }
    ],
    "floatingip_pool_id": "6733078a-3572-4680-a750-e9d5eca630bc",
    "gateway_ip": "192.168.1.1",
    "ip_version": 4,
    "shared": 1,
    "cidr": "192.168.1.0/24",
    "project_id": "de3c453465034a31a2e054c6e4250626",
    "netmask": "255.255.255.0",
    "id": "1e5c4b98-0955-4ace-ac71-9cf4c4d1c209",
    "name": "7481b187-aea5-4654-b4d4-ed16739cf5b8"
}
```

4.9.7.9.4. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.9.7.10. Delete an Elastic IP Pool Subnet

- Scenario: Delete an elastic IP pool subnet.
- Roles Supported: Super admin, system admin, and co-administrator with the Network Write permission
- Versions supported: SCP6.3.70 and above

Note

1. If the IP address in this subnet has been used, the subnet cannot be deleted.

4.9.7.10.1. URI

Table 4.9.115 API Path and Request Method¶

REST VERB	PATH	Descripti
DELETE	/janus/20180725/floatingippools/{floatingip_pool_id}/subnets/{floatingip_subnet_id}	Delete an Elastic IP Pool Subnet

- Parameter Description:

Table 4.9.116 Path Parameters¶

Parameter	Location	Type	Description	Required
floatingip_pool_id	path	string	Elastic IP pool ID	Yes
floatingip_subnet_id	path	string	Subnet ID	Yes

4.9.7.10.2. Request

- Request Parameter

None

- Request Example

  https://{SCP_IP}/janus/20180725/floatingippools/661ce7a4-5abe-4191-832c-7d1dc6c5fd9a/subnets/d5ae0d43-4afc-4f27-a851-9989a7083d9d

4.9.7.10.3. Response

- Response Parameter

  None

- Response Example:

  None

4.9.7.10.4. Response Code

- Normal: 204
- Errors: 400, 403, 404, 500

4.9.8. Elastic IP

4.9.8.1. Query Elastic IPs

4.9.8.1.1. Scenario

- Scenario: Query elastic IPs
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Write, Elastic IP, and VM Creation permissions
- Versions supported: SCP6.3.70 and above

4.9.8.1.2. URI

Table 4.9.117 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/floatingips	Query Elastic IPs

- Parameter Description:

Table 4.9.118 Path Parameters

Parameter	Location	Type	Description	Required
router_id	path	string	Router ID	No
project_id	path	string	Tenant ID	No
floating_ip_pool_id	path	string	ID of the associated elastic IP pool	No
floating_ip	path	string	Assigned elastic IP address	No
bind_vpc_id	path	string	VPC ID of the associated device	No
bind_az_id	path	string	Resource pool ID of the associated device	No
bind_type	path	string	Associated device type, view Type of Elastic IP Associated Device for details	No
keywords	path	string	Search elastic IP	No

Note

For common path parameters, view Common Path Parameter Format for details

4.9.8.1.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/floatingips?floatingip_pool_id=aee07dd2-cb5a-44b2-a453-5f624996432f

4.9.8.1.4. Response

- Response Parameter

Table 4.9.119 Response Parameter

Parameter	Type	Description	Value Returned
id	object	Elastic IP ID	Yes
project_id	string	Tenant ID	Yes
floating_ip	string	Assigned elastic IP address	Yes
floatingip_pool_id	string	Associated elastic IP pool	Yes
floating_port_id	string	Elastic IP port ID	Yes
sharedbandwidth_id	string	ID of the associated shared bandwidth	Yes
bandwidth	dict	Elastic IP bandwidth, view Parameter escription - bandwidth for details	Yes
binding_info	dict	Association information, view Description of binding_info for details	No

- Parameter escription - bandwidth:

Table 4.9.120 Parameter escription - bandwidth¶

Parameter	Type	Description	Value Returned
qos_uplink_kbps	int(0, 10485760)	Outbound bandwidth of the assigned elastic IP (kbps)	Yes
qos_downlink_kbps	int(0, 10485760)	Inbound bandwidth of the assigned elastic IP (kbps)	Yes

- Description of the parameter binding_info:

Table 4.9.121 Description of binding_info¶

Parameter	Type	Value Returned	Note
router_id	string	Yes	Router ID
status	string	Yes	Association status, view Elastic IP Association Status for details
bind_type	string	Yes	Associated device type, view Type of Elastic IP Associated Device for details
ip_address	string	Yes	Internal IP
port_id	string	Yes	LAN port ID
vpc_id	string	Yes	VPC ID of the associated device
bind_id	string	Yes	Association ID
az_id	string	Yes	Resource pool ID of the associated device

Note

For common response parameters, view Response Format for details

- Response Example:

```
[
    {
        "floating_port_id": "8081d873-f1b3-4aba-bbeb-a23ae79e149c",
        "floating_ip": "10.106.123.79",
        "bandwidth": {
            "qos_downlink_kbps": 100000,
            "qos_uplink_kbps": 100000
        },
        "sharedbandwidth_id": "",
        "id": "ac7d555c-5708-4ac0-8b73-17ceb342e712",
        "floatingip_pool_id": "7af91739-e9c1-4704-8ec2-c757e264f6b3",
        "project_id": "a4020e234e57481887b1e5ec72446a6f",
        "binding_info": {
            "router_id": "23f4a574-02d2-435c-aeb0-dcad508a2a1b",
            "status": "active",
            "bind_type": "vm",
            "ip_address": "172.32.0.12",
            "bind_id": "46e76815-689c-45ca-a74a-827e0f07f738",
            "vpc_id": "cb367624-b91a-4463-b45a-b756c54857a2",
            "port_id": "7b0a9b58-1b2a-484f-a5d0-dd22d6222075",
            "az_id": "cd8378b2-7cd6-46e5-80eb-662111529986"
        }
    }
]
```

4.9.8.1.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.9.8.2. Query Elastic IP

4.9.8.2.1. Scenario

- Scenario: Query Elastic IP
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Write, Elastic IP, and VM Creation permissions
- Versions supported: SCP 6.3.0 and above

4.9.8.2.2. URI

Table 4.9.122 API Path and Request Method

REST VERB	PATH	Description
GET	/janus/20180725/floatingips/{floatingip_id}	Query Elastic IP

- Parameter Description:

Table 4.9.123 Path Parameters

Parameter	Location	Type	Description	Required
floatingip_id	path	string	Elastic IP ID	Yes

4.9.8.2.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/floatingips/d5ae0d43-4afc-4f27-a851-9989a7083d9d

4.9.8.2.4. Response

- Response Parameter

Table 4.9.124 Response Parameter

Parameter	Type	Description	Value Returned
id	string	Elastic IP ID	Yes
project_id	string	Tenant ID	Yes
subnet_id	string	Associated subnet ID	Yes
floating_ip	string	Assigned elastic IP address	Yes
floatingip_pool_id	string	ID of the associated elastic IP pool	Yes
floating_port_id	string	Elastic IP port ID	Yes
sharedbandwidth_id	string	Shared bandwidth ID	Yes
created_at	string	Time created	No
updated_at	string	Time updated	No
bandwidth	dict	Bandwidth configuration, view Parameter escription - bandwidth for details	Yes
binding_info	dict	Bandwidth configuration, view Description of binding_info for details	No

Note

1. The parameter floatingip_pool_id is called floating_ip_pool_id in the versions earlier than 6.3.70
2. Elastic IP has two statuses: Associated, not associated. If the status is Associated, then the parameter binding_info will be returned.

- Parameter escription - bandwidth:

Table 4.9.125 Parameter escription - bandwidth

Parameter	Type	Description	Value Returned
qos_uplink	int(0, 10485760)	Outbound bandwidth of elastic IP (kbps)	Yes
qos_downlink	int(0, 10485760)	Inbound bandwidth of elastic IP (kbps)	Yes

- Description of the parameter binding_info:

Table 4.9.126 Description of binding_info¶

Parameter	Type	Value Returned	Description
router_id	string	Yes	Router ID
status	string	Yes	Status, view Elastic IP Association Status for details
bind_type	string	Yes	Associated device type
ip_address	string	Yes	Internal IP address
port_id	string	Yes	Port ID of the internal IP address
vpc_id	string	Yes	VPC ID of the associated device
bind_id	string	Yes	Association ID
az_id	string	Yes	Resource pool ID of the associated device
created_at	string	No	Time created
updated_at	string	No	Time updated

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "message": "",
    "code": 0,
    "data": {
        "subnet_id": "8af58af8-51e7-469d-b169-fcba90c59bb4",
        "created_at": "2020-11-19T02:59:39.000000",
        "floating_port_id": "8081d873-f1b3-4aba-bbeb-a23ae79e149c",
        "updated_at": null,
        "floating_ip": "10.106.123.79",
        "bandwidth": {
            "qos_downlink": 100000,
            "qos_uplink": 100000
        },
        "sharedbandwidth_id": null,
        "id": "ac7d555c-5708-4ac0-8b73-17ceb342e712",
        "floatingip_pool_id": "7af91739-e9c1-4704-8ec2-c757e264f6b3",
        "project_id": "a4020e234e57481887b1e5ec72446a6f",
        "binding_info": {
            "router_id": "23f4a574-02d2-435c-aeb0-dcad508a2a1b",
            "status": "active",
            "bind_type": "vm",
            "ip_address": "172.32.0.12",
            "bind_id": "46e76815-689c-45ca-a74a-827e0f07f738",
            "vpc_id": "cb367624-b91a-4463-b45a-b756c54857a2",
            "port_id": "7b0a9b58-1b2a-484f-a5d0-dd22d6222075",
            "az_id": "cd8378b2-7cd6-46e5-80eb-662111529986"
        }
    }
}
```

4.9.8.2.5. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.9.8.3. Create an Elastic IP

4.9.8.3.1. Scenario

- **Scenario:**

- Create an elastic IP.
- Assign IP address to the specified tenant from elastic IP pools.

- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Write, Elastic IP, and VM Creation permissions
- Versions supported: SCP 6.3.0 and above

4.9.8.3.2. URI

Table 4.9.127 API Path and Request Method

REST VERB	PATH	Description
POST	/janus/20180725/floatingips	Create an Elastic IP

- Parameter Description:

  None

4.9.8.3.3. Request

- Request Parameter

Table 4.9.128 Response Parameter

Parameter	Location	Type	Description	Required
project_id	body	string	Tenant ID	Yes
az_ids	body	list(string)	IDs of the associated resource pool	No
floating_ip	body	string	Specified elastic IP	No
subnet_id	body	string	Assigned elastic subnet ID	No
line_type	body	string	Line type of elastic IP pool	No
floatingip_pool_id	body	string	Elastic IP pool ID, view Query Elastic IP Pools for details	No
sharedbandwidth_id	body	string	Shared bandwidth ID	No
bandwidth	body	dict	Bandwidth configuration, view Parameter escription - bandwidth for details	No

- Parameter escription - bandwidth:

Table 4.9.129 Parameter escription - bandwidth

Parameter	Type	Description	Value Returned
qos_uplink	int(0, 10485760)	Outbound bandwidth of elastic IP (kbps)	Yes
qos_downlink	int(0, 10485760)	Inbound bandwidth of elastic IP (kbps)	Yes

Note

1. If the parameter sharedbandwidth_id is passed, then the shared bandwidth exists and belongs to the tenant.
2. If the parameter sharedbandwidth_id is passed, then bandwidth is set to QoS of the shared bandwidth by default.
3. If the parameter sharedbandwidth_id is not passed, then you need to set bandwidth.

- Request Example

https://{SCP_IP}/janus/20180725/floatingips

```
{
    "az_ids": [],
    "floating_ip": "192.168.0.55",
    "subnet_id": "0d26c0c9-532c-4bda-b10e-e383ff1044b7",
    "line_type": "684c8345-62f4-4602-ba01-c90bc65c732b",
    "floatingip_pool_id": "26b2c0c4-1a1e-4fdc-a837-063b303a120d",
    "bandwidth": {
        "qos_uplink": 10000,
        "qos_downlink": 10000
    },
    "sharedbandwidth_id": "",
    "project_id": "306f8205727949ef98cd0d689d90332b"
}
```

4.9.8.3.4. Response

- Response Parameter

Table 4.9.130 Response Parameter

Parameter	Type	Description	Value Returned
id	string	Elastic IP ID	Yes
project_id	string	Tenant ID	Yes
subnet_id	string	Associated subnet ID	Yes
floating_ip	string	Assigned elastic IP address	Yes
floatingip_pool_id	string	ID of the associated elastic IP pool	Yes
port_id	string	Elastic IP port ID	Yes
sharedbandwidth_id	string	Shared bandwidth ID	Yes
created_at	string	Time created	No
updated_at	string	Time updated	No
bandwidth	dict	Bandwidth configuration, view Parameter escription - bandwidth for details	Yes

- Parameter escription - bandwidth:

Table 4.9.131 Parameter escription - bandwidth

Parameter	Type	Description	Value Returned
qos_uplink	int(0, 10485760)	Outbound bandwidth of elastic IP (kbps)	Yes
qos_downlink	int(0, 10485760)	Inbound bandwidth of elastic IP (kbps)	Yes

> **Note**
>
> For common response parameters, view Response Format for details

- Response Example:

```
{
    "subnet_id": "5bf5c6e5-ca0f-4641-8e2d-284fcb3f0398",
    "created_at": "2020-12-03T02:59:59.442302",
    "updated_at": null,
    "floating_ip": "10.107.124.14",
    "floatingip_pool_id": "66f33608-9c91-4e00-9aea-822fdaf9fb1d",
    "bandwidth": {
        "qos_downlink": 20000,
        "qos_uplink": 20000
    },
    "sharedbandwidth_id": "cb861399-63eb-44b5-b048-621cd0c3ef30",
    "project_id": "a4020e234e57481887b1e5ec72446a6f",
    "port_id": "1bd55d42-d064-4cf0-ac22-ad2ecc548440",
    "id": "d46425c8-ad41-406f-a4b1-4e784ac0ff2a"
}
```

4.9.8.3.4.1. Response Code¶

- Success: 201
- Errors: 400, 403, 404，500

4.9.8.4. Associate Elastic IP

4.9.8.4.1. Scenario

- Scenario: Associate elastic IP.
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Write, Elastic IP, and VM Creation permissions
- Versions supported: SCP 6.3.0 and above

4.9.8.4.2. URI

Table 4.9.132 API Path and Request Method¶

REST VERB	PATH	Description
POST	/janus/20180725/floatingips/{floatingip_id}/binding	Associate Elastic IP

- Parameter Description:

Table 4.9.133 Path Parameters¶

Parameter	Location	Type	Description	Required
floatingip_id	path	string	Elastic IP ID	Yes

4.9.8.4.2.1. Request¶

- Request Parameter

Table 4.9.134 Response Parameter¶

Parameter	Location	Type	Description	Required
vpc_id	body	string	VPC ID	Yes
port_id	body	string	Port ID of the virtual machine or NFV device	No
ip_address	body	string	IP address of the virtual machine or NFV device	No

- Request Example

  https://{SCP_IP}/janus/20180725/floatingips/d5ae0d43-4afc-4f27-a851-9989a7083d9d/binding

```
{
    "vpc_id": "cb367624-b91a-4463-b45a-b756c54857a2",
    "port_id": "41a0f6d0-ab29-40a0-8b42-797b4437b942",
    "ip_address":  "192.168.0.55"
}
```

4.9.8.4.2.2. Response¶

- Response Parameter

Table 4.9.135 Response Parameter¶

Parameter	Type	Description	Value Returned
task_id	string	Task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "a4020e234e57481887b1e5ec72446a6f"
}
```

4.9.8.4.3. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.9.8.5. Disassociate Elastic IP

4.9.8.5.1. Scenario

- Scenario: Associate elastic IP.
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Write, Elastic IP, and VM Creation permissions
- Versions supported: SCP 6.3.0 and above

4.9.8.5.2. URI

Table 4.9.136 API Path and Request Method¶

REST VERB	PATH	Description
DELETE	/janus/20180725/floatingips/{floatingip_id}/unbinding	Disassociate Elastic IP

- Parameter Description:

Table 4.9.137 Path Parameters¶

Parameter	Location	Type	Description	Required
floatingip_id	path	string	Elastic IP ID	Yes

4.9.8.5.2.1. Request¶

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/floatingips/d5ae0d43-4afc-4f27-a851-9989a7083d9d/unbinding

4.9.8.5.3. Response

- Response Parameter

Table 4.9.138 Response Parameter¶

Parameter	Type	Description	Value Returned
task_id	string	Task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "a4020e234e57481887b1e5ec72446a6f"
}
```

4.9.8.5.4. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.9.8.6. Edit Elastic IP Bandwidth

4.9.8.6.1. Scenario

- Scenario: Edit elastic IP bandwidth
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Write permission
- Versions supported: SCP6.3.70 and above

4.9.8.6.2. URI

Table 4.9.139 API Path and Request Method

REST VERB	PATH	Description
PUT	/janus/20180725/floatingips/{floatingip_id}	Edit Elastic IP Bandwidth

- Parameter Description:

Table 4.9.140 Path Parameters

Parameter	Location	Type	Description	Required
floatingip_id	path	string	Elastic IP ID	Yes

4.9.8.6.3. Request

- Request Parameter

Table 4.9.141 Response Parameter

Parameter	Location	Type	Description	Required
qos_uplink	body	int(0, 10485760)	Outbound bandwidth of the assigned elastic IP (kbps)	No
qos_downlink	body	int(0, 10485760)	Inbound bandwidth of the assigned elastic IP (kbps)	No

Note

Outbound and inbound bandwidth must be the same. If only the value for one of the two is configured, the other will be automatically set to the same value.

- Request Example

  https://{SCP_IP}/janus/20180725/floatingips/a047380ba-21b1-4437-8b0c-f113f9054fc8a

```
{
    "qos_uplink": 20000,
    "qos_downlink": 20000
}
```

4.9.8.6.4. Response

- Response Parameter

Table 4.9.142 Response Parameter

Parameter	Type	Description	Value Returned
id	string	Elastic IP ID	Yes
project_id	string	Tenant ID	Yes
subnet_id	string	Associated subnet ID	Yes
floating_ip	string	Assigned elastic IP address	Yes
floatingip_pool_id	string	ID of the associated elastic IP pool	Yes
port_id	string	Elastic IP port ID	Yes
sharedbandwidth_id	string	Shared bandwidth ID	Yes
created_at	string	Time created	No
updated_at	string	Time updated	No
bandwidth	dict	Bandwidth configuration, view Parameter escription - bandwidth for details	Yes

- Parameter escription - bandwidth:

Table 4.9.143 Parameter escription - bandwidth

Parameter	Type	Description	Value Returned
qos_uplink	int(0, 10485760)	Outbound bandwidth of elastic IP (kbps)	Yes
qos_downlink	int(0, 10485760)	Inbound bandwidth of elastic IP (kbps)	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "subnet_id": "5bf5c6e5-ca0f-4641-8e2d-284fcb3f0398",
    "created_at": "2020-12-03T02:59:59.442302",
    "updated_at": null,
    "floating_ip": "10.107.124.14",
    "floatingip_pool_id": "66f33608-9c91-4e00-9aea-822fdaf9fb1d",
    "bandwidth": {
        "qos_downlink": 20000,
        "qos_uplink": 20000
    },
    "sharedbandwidth_id": "cb861399-63eb-44b5-b048-621cd0c3ef30",
    "project_id": "a4020e234e57481887b1e5ec72446a6f",
    "port_id": "1bd55d42-d064-4cf0-ac22-ad2ecc548440",
    "id": "d46425c8-ad41-406f-a4b1-4e784ac0ff2a"
}
```

4.9.8.6.5. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.9.8.7. Delete Elastic IP

4.9.8.7.1. Scenario

- Scenario: Delete elastic IP.
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Write, Elastic IP, and VM Creation permissions
- Versions supported: SCP 6.3.0 and above

4.9.8.7.2. URI

Table 4.9.144 API Path and Request Method

REST VERB	PATH	Description
DELETE	/janus/20180725/floatingips/{floatingip_id}	Delete Elastic IP

- Parameter Description:

Table 4.9.145 Path Parameters

Parameter	Location	Type	Description	Required
floatingip_id	path	string	Elastic IP ID	Yes

4.9.8.7.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/floatingips/d5ae0d43-4afc-4f27-a851-9989a7083d9d

4.9.8.7.4. Response

- Response Parameter

  None

- Response Example:

  None

4.9.8.7.5. Response Code

- Normal: 204
- Errors: 400, 403, 404, 500

4.9.9. Network Device

4.9.9.1. Query Network Devices

- Scenario: Query network devices
- Roles Supported: Super admin, system admin, and co-administrator with the Network Read permission
- Versions supported: SCP 6.3.0 and above

4.9.9.1.1. URI

Table 4.9.146 API Path and Request Method

REST VERB	PATH	Description
GET	/janus/20180725/vnetdev/{vnetdev_id}	Query Network Devices

- Parameter Description:

Table 4.9.147 Path Parameters

Parameter	Location	Type	Description	Required
vnetdev_id	path	string	Network device ID	Yes

4.9.9.1.2. Request

- Request Parameter

  None

- Request Example

  None

4.9.9.1.3. Response

- Response Parameter

Table 4.9.148 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	NFV device ID	Yes
name	string	NFV name	No
description	string	Description	No
az_id	string	Resource pool ID	No
encrypted	int(0, 1)	Whether it is encrypted	No
cipher	string	Encryption algorithm, view Encryption Algorithm of NFV Devices for details	No
cpus	int	Unique to NFV, the occupied CPU cores when the device is running	No
project_id	string	Tenant ID	No
type	string	Supported NFV devices in VPC, view NFV Devices in VPC for details	No
vif_count	int(4, 24)	NFV device interfaces	No
host_id	string	Node ID	No
vm_type	string	Device type, view VM Type for details	No
status	string	Device status	No
linked_amount	int	Number of connected devices	No
network_type	string	Network type, view Network Type for details	No
network_id	string	Network ID	No
ip_list	string	Interface IPs, JSON string	No
classics_networks	list(dict)	Classic network, view Description of classics_network for details	No
vpc_networks	list(dict)	VPC, view Description of vpc_networks for details	No
storage_info	string	Storage information, view Parameter Description - storage_info for details	No
auth_info	dict	Licensing, view Parameter Description - auth_info for details	No
location_info	dict	Deployment location, view Parameter Description - location_info for details	No
vlinks_info	list(dict)	Virtual link, view Parameter Description - vlinks_info for details	No
extra_info	dict	Other information, view Parameter Description - extra_info for details	No

- Description of the parameter classics_network:

Table 4.9.149 Description of classics_network¶

Parameter	Type	Description	Value Returned
port_type	string	Port type, view NFV Interfaces in Classic Network for details	Yes
interface_name	string	Interface name, example: eth0	Yes
ipv4	string	Interface IP	Yes
mask	string	Interface netmask	Yes
gateway	string	Gateway	Yes

- Description of the parameter vpc_networks:

Table 4.9.150 Description of vpc_networks¶

Parameter	Type	Description	Value Returned
port_type	string	Port type, view NFV Interfaces in VPC for details	Yes
port_id	string	Interface ID, it may not exist, such as the LAN interface in NGAF	Yes
ips	list(string)	Subnet IPs	Yes
port_name	string	Subnet port name	Yes
subnet_id	string	Subnet ID	Yes
fip	string	Elastic IP address	No
fip_id	string	Elastic IP ID	No

- Parameter Description - auth_info:

Table 4.9.151 Parameter Description - auth_info¶

Parameter	Type	Description	Value Returned
is_auth	int(0, 1)	Whether the device is licensed	No
auth_resource_name	string	Licensing name	No

- Parameter Description - storage_info:

Table 4.9.152 Parameter Description - storage_info¶

Parameter	Type	Description	Value Returned
storage_id	string	Storage ID	No
disk_total_mb	int	Total disk size (MB)	No
disk_used_mb	int	Used disk size (MB)	No
mem_total_mb	int	Total memory size (MB)	No
mem_used_mb	int	Used memory size (MB)	No

- Parameter Description - location_info:

Table 4.9.153 Parameter Description - location_info¶

Parameter	Type	Description	Value Returned
location	int(0, 1)	Whether the device runs on specified node only	No
host_config	string	Run location configuration of NFV	No
dh_id	string	Dedicated server group ID	No
deploy_location	string	Deployment location, view NFV Location Deployed in VPC for details	No
deploy_mode	string	Deployment mode, view NFV Deployment Mode in VPC for details	No

- Parameter Description - vlinks_info:

Table 4.9.154 Parameter Description - vlinks_info¶

Parameter	Type	Description	Value Returned
status	string	Status	No
isbridge	int(0, 1)	Whether it is bridge	No
name	string	Interface name	No
id	string	Interface ID	No
peer_device_id	string	Connected device ID	No
peer_device_type	string	Connected device type	No

- Parameter Description - extra_info:

Table 4.9.155 Parameter Description - extra_info¶

Parameter	Type	Description	Value Returned
peer_ha_device_id	string	HA peer device ID	No
cssp_id	string	(Removed)	Yes
ha	int(0, 1)	Whether to enable VM failover	No

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "encrypted": 0,
    "cpus": 2,
    "id": "d6757edc-7bb3-495c-abd6-c5af6e96dd15",
    "az_id": "cd8378b2-7cd6-46e5-80eb-662111529986",
    "extra_info": {
        "ha": 1,
        "peer_ha_device_id": "",
        "cssp_id": ""
    },
    "ip_list": "{\"eth14\": [], \"eth1\": [\"192.168.1.16/24\"], \"eth13\": [], \"
eth12\": [], \"eth11\": [], \"eth10\": [], \"eth15\": [], \"eth0\": [\"172.32.0.34
/24\"]}",
    "storage_info": {
        "disk_used_mb": 792,
        "mem_total_mb": 4096,
        "disk_total_mb": 81920,
        "mem_used_mb": 2252,
        "storage_id": "e6fc6b25_vs_vol_rep3"
    },
    "vlinks_info": [
        {
            "status": "1",
            "name": "eth0",
            "peer_device_type": "dvs",
            "peer_device_id": "8cdf5b97-c8d8-4dfb-9d35-efb838f9362c",
            "isbridge": 1,
            "id": "net0"
        }
    ],
    "vif_count": 16,
    "project_id": "a4020e234e57481887b1e5ec72446a6f",
    "type": "vnet:dblb",
    "status": "running",
    "description": "",
    "linked_amount": 3,
    "classics_networks": [],
    "cipher": "",
    "host_id": "host-6c92bf5197e6",
    "vm_type": "vm",
    "auth_info": {
        "is_auth": 0,
        "auth_resource_name": "Database_Load_Balancer_Components"
    },
    "name": "DBLB_device2",
    "network_id": "cb367624-b91a-4463-b45a-b756c54857a2",
    "location_info": {
        "dh_id": "",
        "deploy_location": "inner_dvs",
        "host_config": "<Auto>",
        "location": false,
        "deploy_mode": "single"
    },
    "vpc_networks": [
        {
            "fip": "10.106.123.86",
            "subnet_id": "8cdf5b97-c8d8-4dfb-9d35-efb838f9362c",
            "ips": [
                "172.32.0.34"
            ],
            "port_name": "eth0",
            "port_type": "manage_port",
            "port_id": "45eaeec6-76c1-45cd-9024-436c83121ea8",
            "fip_id": "a250455a-ec14-4331-a758-6744d8b3a95b"
        }
    ],
    "network_type": "vpc"
}
```


4.9.9.1.4. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.9.9.2. Deploy a Network Device

- Scenario: Deploy a network device
- Roles Supported: Super admin, system admin, and co-administrator with the Network Write permission
- Versions supported: SCP 6.3.0 and above

4.9.9.2.1. URI

Table 4.9.156 API Path and Request Method

REST VERB	PATH	Description
POST	/janus/20180725/vnetdev	Deploy a Network Device

- Parameter Description:

  None

4.9.9.2.2. Request

- Request Parameter

Table 4.9.157 Response Parameter

Parameter	Location	Type	Description	Required
az_id	body	string	Resource pool ID	Yes
project_id	body	string	Tenant ID	Yes
name	body	string	Network device name	Yes
type	body	string	Supported NFV devices in VPC, view NFV Devices in VPC for details	Yes
auth_info	body	string	Licensing parameter, JSON string	Yes
vpc_id	body	string	VPC ID	Yes
config_info	body	dict	NFV device parameters for creating VPC, view Parameter Description - config_info for details	No
extra_info	body	dict	Configuration of other parameters, view Parameter Description - extra_info for details	No
location	body	dict	Deployment location, view Parameter Description - location for details	No

- Parameter Description - config_info:

Table 4.9.158 Parameter Description - config_info

Parameter	Type	Description	Required
vif_count	string	NFV device interfaces	No
networks	list(dict)	Network configuration, view Description of config_info networks for details	No

- Description of the parameter config_info networks:

Table 4.9.159 Description of config_info networks¶

Parameter	Type	Description	Required
port_type	string	Port type, view NFV Interfaces in VPC for details	Yes
subnet_id	string	Subnet ID	Yes
floatingip_id	object	Floating IP ID	No
ip_address	string	Elastic IP address	No
custom_gateway_ip	string	Subnet port gateway	No

- Parameter Description - extra_info:

Table 4.9.160 Parameter Description - extra_info¶

Parameter	Type	Description	Required
ha_enable	int(0, 1)	Whether to enable high availability	No
peer_ha_device_id	string	HA peer device ID	No
cssp_id	string	(Removed)	No

- Parameter Description - location:

Table 4.9.161 Parameter Description - location¶

Parameter	Type	Description	Required
dh_id	string	Dedicated server group ID	No
deploy_location	string	Deployment location, view NFV Location Deployed in VPC for details	No

- Request Example

  https://{SCP_IP}/janus/20180725/vnetdev

```
{
      "name": "DBLB_device",
      "vpc_id": "cb367624-b91a-4463-b45a-b756c54857a2",
      "auth_info": "{\"dev_type\":\"dblb_1Gbps_performance\"}",
      "config_info": {
         "vif_count": 16,
         "networks": [
             {
                 "port_type": "manage_port",
                 "subnet_id": "8cdf5b97-c8d8-4dfb-9d35-efb838f9362c",
                 "floatingip_id": "a250455a-ec14-4331-a758-6744d8b3a95b"
             }
         ]
     },
     "type": "vnet:dblb",
     "project_id": "a4020e234e57481887b1e5ec72446a6f",
     "az_id": "cd8378b2-7cd6-46e5-80eb-662111529986",
     "extra_info": {
           "ha_enable": 0,
           "cssp_id": "a250455a-ec14-4331-a758-6744d8b3a95b"
      }
      "location": {
           "deploy_location": "",
           "dh_id": ""
      }
 }
```

4.9.9.2.3. Response

- Response Parameter

Table 4.9.162 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "a4020e234e57481887b1e5ec72446a6f"
}
```

4.9.9.2.4. Response Code

- Success: 201
- Errors: 400, 403, 404

4.9.9.3. Update Network Device Information

- Scenario: Update network device information
- Roles Supported: Super admin, system admin, and co-administrator with the Network Write permission
- Versions supported: SCP 6.3.0 and above

4.9.9.3.1. URI

Table 4.9.163 API Path and Request Method

REST VERB	PATH	Description
PUT	/janus/20180725/vnetdev/{vnetdev_id}	Update Network Device Information

- Parameter Description:

Table 4.9.164 Path Parameters

Parameter	Location	Type	Description	Required
vnetdev_id	path	string	Network device ID	Yes

4.9.9.3.2. Request

- Request Parameter

Table 4.9.165 Response Parameter

Parameter	Location	Type	Description	Required
vnetdev_id	body	string	Network device ID	Yes
name	body	string	Name	No
auth_info	body	string	Licensing information, JSON string	No
recover_auth	body	int(0, 1)	Whether to re-license the device	No
peer_ha_device	body	string	HA peer device ID	No
disk_size	body	float	Total size after expanding capacity	No
cssp_id	body	string	(Removed)	No
reboot	body	int(0, 1)	Whether to restart the device	No
migrate_info	body	dict	Migration location, view Parameter Description - migrate_info for details	No

- Parameter Description - migrate_info:

Table 4.9.166 Parameter Description - migrate_info

Parameter	Type	Description	Required
project_id	string	Tenant ID	Yes
az_id	string	Resource pool ID	Yes
target_host_id	string	Destination node	Yes
target_volume_id	string	Destination datastore	Yes
target_volume_dir	string	Destination storage directory	No
priority_host_id	string	Migrate node first	No
location	int(0, 1)	Whether the device runs on specified node only	No

- Request Example

https://{SCP_IP}/janus/20180725/vnetdev/71ab08a6-5e35-449d-8e9d-ce015e127332

```
{
     "vnetdev_id":  "1420a8fb-8bca-4526-aa52-bd65bbc18e36",
     "name": "test",
     "auth_info":  "",
     "recover_auth": "",
     "peer_ha_device":  "c7b2db93-fdb5-438e-8012-49e4f77de06c",
     "migrate_info": {
         "project_id": "a4020e234e57481887b1e5ec72446a6f",
         "az_id": "cd8378b2-7cd6-46e5-80eb-662111529986",
         "target_host_id":  "",
         "target_volume_id":  "",
         "target_volume_dir": "",
         "priority_host_id": "",
         "location": 0
     },
     "disk_size": 11111112.2,
     "cssp_id":  "78842f26-a9cd-4412-9747-3d0a679cec96",
     "reboot": 0
 }
```

4.9.9.3.3. Response

- Response Parameter

Table 4.9.167 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "a4020e234e57481887b1e5ec72446a6f"
}
```

4.9.9.3.4. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.9.9.4. Power Off Multiple Network Devices

- Scenario: Power off multiple network devices
- Roles Supported: Super admin, system admin, and co-administrator with the Network Write permission
- Versions supported: SCP 6.3.0 and above

4.9.9.4.1. URI

Table 4.9.168 API Path and Request Method

REST VERB	PATH	Description
PUT	/janus/20180725/vnetdev/nfvs/stop	Power Off Multiple Network Devices

- Parameter Description:

  None

4.9.9.4.2. Request

- Request Parameter

Table 4.9.169 Response Parameter

Parameter	Location	Type	Description	Required
vnetdev_ids	body	list(string)	Network device IDs	Yes

- Request Example

  https://{SCP_IP}/janus/20180725/vnetdev/nfvs/stop

```
{
     "vnetdev_ids": [
         "71ab08a6-5e35-449d-8e9d-ce015e127332",
         "84acfc02-0cc2-4967-b79c-64bb93f30307"
     ]
 }
```

4.9.9.4.3. Response

- Response Parameter

Table 4.9.170 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "a4020e234e57481887b1e5ec72446a6f"
}
```

4.9.9.4.4. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.9.9.5. Power On Multiple Network Devices

- Scenario: Power on multiple network devices
- Roles Supported: Super admin, system admin, and co-administrator with the Network Write permission
- Versions supported: SCP6.3.70 and above

4.9.9.5.1. URI

Table 4.9.171 API Path and Request Method

REST VERB	PATH	Description
PUT	/janus/20180725/vnetdev/nfvs/start	Power On Multiple Network Devices

- Parameter Description:

  None

4.9.9.5.2. Request

- Request Parameter

Table 4.9.172 Response Parameter

Parameter	Location	Type	Description	Required
vnetdev_ids	body	list(string)	Network device IDs	Yes

- Request Example

  https://{SCP_IP}/janus/20180725/vnetdev/nfvs/stop

```
{
     "vnetdev_ids": [
         "71ab08a6-5e35-449d-8e9d-ce015e127332",
         "84acfc02-0cc2-4967-b79c-64bb93f30307"
     ]
 }
```

4.9.9.5.3. Response

- Response Parameter

Table 4.9.173 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "a4020e234e57481887b1e5ec72446a6f"
}
```

4.9.9.5.4. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.9.9.6. Delete a Network Device

- Scenario: Delete a network device
- Roles Supported: Super admin, system admin, and co-administrator with the Network Write permission
- Versions supported: SCP 6.3.0 and above

4.9.9.6.1. URI

Table 4.9.174 API Path and Request Method

REST VERB	PATH	Description
DELETE	/janus/20180725/vnetdev/{vnetdev_id}	Delete a Network Device

- Parameter Description:

Table 4.9.175 Path Parameters

Parameter	Location	Type	Description	Required
vnetdev_id	path	string	Network device ID	Yes

4.9.9.6.2. Request

- Request Parameter

  None

- Request Example

  None

4.9.9.6.3. Response

- Response Parameter

Table 4.9.176 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "a4020e234e57481887b1e5ec72446a6f"
}
```

4.9.9.6.4. Response Code

- Normal: 204
- Errors: 400, 403, 404, 500

4.9.10. Shared Services

4.9.10.1. Query Shared Services

- Scenario: Query shared services
- Roles Supported: Super admin, system admin, and co-administrator with the Network Read permission
- Versions supported: SCP 6.3.0 and above

4.9.10.1.1. URI

Table 4.9.177 API Path and Request Method

REST VERB	PATH	Description
GET	/janus/20180725/public-services	Get shared services

- Parameter Description:

Table 4.9.178 Path Parameters

Parameter	Location	Type	Description	Required
az_id	path	string	Resource pool ID	No
name	path	string	Name	No
service_ip	path	string	IP	No
service_domain	path	string	Domain name	No
service_type	path	string	Type	No
tag	path	string	Service tag	No
keywords	path	string	Search keywords. Supports searching name, domain name or IP address	No

Note

For common path parameters, view Common Path Parameter Format for details

4.9.10.1.2. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/public-services?az_id=f6f18ab6-42ab-4110-82ef-e2bd17fcd7de

4.9.10.1.3. Response

- Response Parameter

Table 4.9.179 Response Parameter

Parameter	Type	Description	Value Returned
id	string	Service ID	No
name	string	Service name	No
description	string	Service description	No
service_ip	string	Shared service IP	No
service_domain	string	Domain name	No
az_ids	list(string)	Resource pool IDs	No
is_global	int(0, 1)	Whether it is used by all resource pools	No
auto_enable	int(0, 1)	Whether to activate the service for tenants	No
protocol	string	Protocol type	No
port	string	Port number	No
status	string	Status	No
project_id	string	Tenant ID	No
tag	string	Service tag	No
service_type	string	Service type	No

Note

For common response parameters, view Response Format for details

- Response Example:

```
[
    {
        "status": null,
        "is_global": 1,
        "az_ids": [],
        "name": "Database_Services",
        "service_ip": "9.9.9.10",
        "service_domain": "dmp.local.cloud.scp",
        "port": null,
        "tag": "6c1c6b32-7f5b-4bc3-8431-3146ef50b595",
        "auto_enable": 1,
        "service_type": "dmp",
        "protocol": "all",
        "project_id": "cf92f8d651c24ed6938b186e252ee0bc",
        "id": "00512815-aa36-4c93-9c27-197bb9efbb96",
        "description": "Predefined DMP service, unable to be edited"
    }
]
```

4.9.10.1.4. Response Code

- OK: 200
- Errors: 400, 403, 500

4.9.10.2. Query Tenants Associated with Shared Services

- Scenario: Query tenants associated with shared services
- Roles Supported: Super admin, system admin, and co-administrator with the Network Read permission
- Versions supported: SCP 6.3.0 and above

4.9.10.2.1. URI

Table 4.9.180 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/public-services/{publicservice_id}/bindings	Obtain tenants associated with shared services

- Parameter Description:

Table 4.9.181 Path Parameters¶

Parameter	Location	Type	Description	Required
publicservice_id	path	string	Resource pool ID	Yes
project_id	path	string	Tenant ID	No
az_id	path	string	Resource pool ID	No
vpc_id	path	string	VPC ID	No
status	path	string	Status	No

Note

For common path parameters, view Common Path Parameter Format for details

4.9.10.2.1.1. Request¶

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/public-services/00512815-aa36-4c93-9c27-197bb9efbb96/bindings

4.9.10.2.2. Response

- Response Parameter

Table 4.9.182 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Association ID	No
az_id	string	Resource pool ID	No
project_id	string	Tenant ID	No
vpc_id	string	VPC ID	No
publicservice_id	string	Service ID	No
status	string	Status of shared service	No

> **Note**
>
> For common response parameters, view Response Format for details

- Response Example:

```
[
    {
        "status": "active",
        "publicservice_id": "00512815-aa36-4c93-9c27-197bb9efbb96",
        "vpc_id": "12af9eee-8f4c-4ece-af86-4ea76edbb6f9",
        "project_id": "9d14d076b59045b1b0aaca04cfaaa657",
        "id": "6533082a-8ca1-4280-a297-4603286a4c9c",
        "az_id": "611f1b34-c727-4925-854e-07bc8607f681"
    }
]
```

4.9.10.2.2.1. Response Code¶

- OK: 200
- Errors: 400, 403, 404

4.9.11. DNAT Management for VPC

4.9.11.1. Common Parameters for DNAT

4.9.11.1.1. Bulk Operation Identifier Type

Table 4.9.183 Bulk Operation Identifier Type¶

Value	Description
delete	Bulk delete
enable	Bulk enable
disable	Bulk disable

4.9.11.1.2. Protocols

Table 4.9.184 Protocols¶

Value	Description
tcp	tcp
udp	udp
icmp	icmp
any	All

4.9.11.2. Create DNAT Rule

4.9.11.2.1. Scenario

- Scenario: Create DNAT rule
- Roles Supported: admin, tenant, and co-administrator with the PERM_NET_WRITE permission
- Versions Supported: SCP6.7.35 and above

4.9.11.2.2. URI

Table 4.9.185 API Path and Request Method

REST VERB	PATH	Description
POST	/janus/20180725/dnat_maps	Create DNAT Rule

4.9.11.2.3. Request

- Request Parameter

Table 4.9.186 Description of Body Parameter

Parameter	Location	Type	Description	Required
vpc_id	body	string	VPC network ID	Yes
description	body	string	Description	No
enable	body	string	Whether the tenant is enabled	Yes
src_ip_range	body	list	Src IP	Yes
protocol	body	string	Protocols	Yes
map_info	body	dict	View map_info Parameter for DNAT details.	Yes

- map_info Parameter:

Table 4.9.187 map_info Parameter

Parameter	Type	Description	Required
external	dict	View external Parameter for pre-NAT information.	Yes
internet	dict	View internet Parameter for post-NAT information.	Yes

- external Parameter:

Table 4.9.188 external Parameter

Parameter	Type	Description	Required
ips	list	Pre-NAT IP address list	Yes
port_range	string	Original port	No

- internet Parameter:

Table 4.9.189 internet Parameter

Parameter	Type	Description	Required
ip	string	Mapped IP address	No
port_range	string	Mapped port	No
port_id	string	Resource ID	No

- Request Example

  https://{SCP_IP}/janus/20180725/dnat_maps

```
{
    "vpc_id": "c6abea06-e3fd-4dad-b7be-a1ea827a9799",
    "description": "22",
    "enable": 1,
    "src_ip_range": ["10.103.245.1-10.103.245.100"],
    "protocol": "tcp",
    "map_info": {
        "external": {
            "ips": ["10.132.3.1"],
            "port_range": [61,62]
        },
        "internet": {
            "ip": "10.132.5.1",
            "port_range": [40]
        }
    }
}
```

4.9.11.2.4. Response

- Response Parameter

Table 4.9.190 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "3d865258-30eb-477a-907d-ba1c1c14d3bd"
}
```

4.9.11.2.5. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.9.11.3. Edit a DNAT rule

4.9.11.3.1. Scenario

- Scenario: Edit a DNAT rule
- Roles Supported: admin, tenant, and co-administrator with the PERM_NET_WRITE permission
- Versions Supported: SCP6.7.35 and above

4.9.11.3.2. URI

Table 4.9.191 API Path and Request Method

REST VERB	PATH	Description
PUT	/janus/20180725/dnat_maps/{portmap_id}	Edit a DNAT rule

- Parameter Description:

Table 4.9.192 Path Parameters

Parameter	Location	Type	Description	Required
portmap_id	path	string	DNAT ID	Yes

4.9.11.3.3. Request

- Request Parameter

Table 4.9.193 Description of Body Parameter

Parameter	Location	Type	Description	Required
vpc_id	body	string	VPC network ID	Yes
description	body	string	Description	No
enable	body	string	Whether the tenant is enabled	No
src_ip_range	body	list	Src IP	No
protocol	body	string	Protocols	No
map_info	body	dict	View map_info Parameter for DNAT details.	No

- map_info Parameter:

Table 4.9.194 map_info Parameter

Parameter	Type	Description	Required
external	dict	View external Parameter for pre-NAT information.	Yes
internet	dict	View internet Parameter for post-NAT information.	Yes

- external Parameter:

Table 4.9.195 external Parameter

Parameter	Type	Description	Required
ips	list	Pre-NAT IP address list	Yes
port_range	string	Original port	No

- internet Parameter:

Table 4.9.196 internet Parameter

Parameter	Type	Description	Required
ip	string	Mapped IP address	No
port_range	string	Mapped port	No
port_id	string	Resource ID	No

- Request Example

  https://{SCP_IP}/janus/20180725/dnat_maps/1449c54c-357b-4aab-8790-37741398d583

```
{
    "vpc_id": "c6abea06-e3fd-4dad-b7be-a1ea827a9799",
    "description": "99",
    "src_ip_range": ["10.103.245.1-10.103.245.100"],
    "map_info": {
        "external": {
            "ips": ["10.132.4.1"],
            "port_range": [61]

        },
        "internet": {
            "ip": "192.168.0.3",
            "port_range": [70]

        }
    }
}
```

4.9.11.3.4. Response

- Response Parameter

Table 4.9.197 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "b908ebcb-e46c-4a35-a8a6-6f6c33ad1b5d"
}
```

4.9.11.3.5. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.9.11.4. Bulk Operations on DNAT Rules

4.9.11.4.1. Scenario

- Scenario: Perform bulk operations on DNAT rules
- Roles Supported: admin, tenant, and co-administrator with the PERM_NET_WRITE permission
- Versions Supported: SCP6.7.35 and above

4.9.11.4.2. URI

Table 4.9.198 API Path and Request Method

REST VERB	PATH	Description
POST	/janus/20180725/dnat_maps/action	Bulk Operations on DNAT Rules

- Parameter Description:

None

4.9.11.4.3. Request

- Request Parameter

Table 4.9.199 Description of Body Parameter¶

Parameter	Type	Description	Required
vpc_id	string	VPC network ID	Yes
portmap_ids	list(string)	DNAT ID list	Yes
action	string	View Bulk Operation Identifier Type for bulk operation identifier type.	Yes

- Request Example

https://{SCP_IP}/janus/20180725/dnat_maps/action

```
{
    "vpc_id": "3839f30b-6318-4572-b4ff-ee8f24ccff22",
    "action": "delete",
    "portmap_ids": [
        "88b04caf-87ee-4372-b5af-8a42c0c80983",
        "34de43a6-c0ee-403f-b593-79fb845cc1e7"
    ]
}
```

4.9.11.4.4. Response

- Response Parameter

Table 4.9.200 Response Parameter¶

Parameter	Type	Description	Value Returned
task_id	string	Task ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
{
    "task_id": "a9ef0fdd-e3e1-467a-9032-631436b2b820"
}
```

4.9.11.4.5. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.9.11.5. Query DNAT List

4.9.11.5.1. Scenario

- Scenario: Query DNAT list
- Roles Supported: admin, tenant, and co-administrator with the PERM_NET_Read permission

- Versions Supported: SCP6.7.35 and above

4.9.11.5.2. URI

Table 4.9.201 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/dnat_maps	Query DNAT List

- Parameter Description:

Table 4.9.202 Path Parameters¶

Parameter	Location	Type	Description	Required
enable	path	int(0, 1)	Whether the tenant is enabled	No
az_id	path	string	Resource pool ID	No
project_id	path	string	Tenant ID	No
vpc_id	path	string	VPC network ID	Yes
src_ip_range	path	string	Src IP	No
protocol	path	string	View Protocols for details.	No
external_ip	path	string	Pre-NAT IP address	No
external_port_range	path	string	Original port	No
internet_ip	path	string	Mapped IP address	No
internet_port_range	path	string	Mapped port	No
port_id	path	string	Resource ID (not supported)	No

4.9.11.5.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/dnat_maps?vpc_id=6a8bc428-6c4d-4fd0-8cb1-1ca97e8ffe0f

4.9.11.5.4. Response

- Response Parameter

Table 4.9.203 Response Parameter

Parameter	Type	Description	Value Returned
id	string	DNAT ID	Yes
enable	int(0, 1)	Whether the tenant is enabled	Yes
az_id	string	Resource pool ID	Yes
project_id	string	Tenant ID	Yes
vpc_id	string	VPC network ID	Yes
src_ip_range	list(string)	Src IP	Yes
protocol	string	View Protocols for details.	Yes
description	string	Description	Yes
map_info	dict	View map_info Parameter for DNAT details.	Yes

- map_info Parameter:

Table 4.9.204 map_info Parameter

Parameter	Type	Description	Value Returned
external	dict	View external Parameter for pre-NAT information.	Yes
internet	dict	View internet Parameter for post-DNAT information.	Yes

- external Parameter:

Table 4.9.205 external Parameter

Parameter	Type	Description	Value Returned
ips	list(string)	Pre-NAT IP address list	Yes
port_range	list(string)	Original port	Yes

- internet Parameter:

Table 4.9.206 internet Parameter

Parameter	Type	Description	Value Returned
ip	string	Mapped IP address	Yes
port_range	list(string)	Mapped port	Yes
port_id	string	Resource ID	Yes

Note

For common response parameters, view Response Format for details

- Response Example:

```
[
    {
        "enable": 1,
        "protocol": "tcp",
        "description": "1232",
        "src_ip_range": [
            "192.168.0.113"
        ],
        "map_info": {
            "external": {
                "ips": [
                    "10.168.0.7",
                    "10.168.0.8",
                    "10.168.0.10"
                ],
                "port_range": [
                    "66",
                    "67-68"
                ]
            },
            "internet": {
                "ip": "192.168.0.188",
                "port_id": "",
                "port_range": [
                    "88",
                    "97-98"
                ]
            }
        },
        "vpc_id": "6a8bc428-6c4d-4fd0-8cb1-1ca97e8ffe0f",
        "project_id": "f93eebef20c245b89550410245d2ab8c",
        "id": "5e5d8503-0c96-4c03-b59f-61f51915f40d",
        "az_id": "a4cdd017-e3da-46f4-b134-4b4fbf7c2380"
    },
    {
        "enable": 1,
        "protocol": "any",
        "description": "",
        "src_ip_range": [
            "192.168.1.111",
            "192.168.0.112/24",
            "192.168.0.113-192.168.1.116"
        ],
        "map_info": {
            "external": {
                "ips": [
                    "10.168.0.1"
                ],
                "port_range": [
                    ""
                ]
            },
            "internet": {
                "ip": "192.168.0.66",
                "port_id": "",
                "port_range": [
                    ""
                ]
            }
        },
        "vpc_id": "6a8bc428-6c4d-4fd0-8cb1-1ca97e8ffe0f",
        "project_id": "f93eebef20c245b89550410245d2ab8c",
        "id": "affba335-5a4d-401f-9aca-7dc4243eafae",
        "az_id": "a4cdd017-e3da-46f4-b134-4b4fbf7c2380"
    }
]
```


4.9.11.5.5. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.9.11.6. Query DNAT details

4.9.11.6.1. Scenario

- Scenario: Query DNAT details
- Roles Supported: admin, tenant, and co-administrator with the PERM_NET_Read permission

- Versions Supported: SCP6.7.35 and above

4.9.11.6.2. URI

Table 4.9.207 API Path and Request Method

REST VERB	PATH	Description
GET	/janus/20180725/dnat_maps/{portmap_id}	Query DNAT details

- Parameter Description:

Table 4.9.208 Path Parameters

Parameter	Location	Type	Description	Required
portmap_id	path	string	DNAT ID	Yes
vpc_id	path	string	VPC network ID	Yes

4.9.11.6.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/dnat_maps/5e5d8503-0c96-4c03-b59f-61f51915f40d?vpc_id=832818d6-b459-4e42-8745-f618893f7c2c

4.9.11.6.4. Response

- Response Parameter

Table 4.9.209 Response Parameter

Parameter	Type	Description	Value Returned
id	string	DNAT ID	Yes
enable	int(0, 1)	Whether the tenant is enabled	Yes
az_id	string	Resource pool ID	Yes
project_id	string	Tenant ID	Yes
vpc_id	string	VPC network ID	Yes
src_ip_range	list(string)	Src IP	Yes
protocol	string	View Protocols for details.	Yes
description	string	Description	Yes
map_info	dict	View map_info Parameter for DNAT details.	Yes
created_at	string or None	Time of DNAT rule creation	Yes
updated_at	string or None	Last modified time of DNAT rule	Yes
priority	int or None	Priority	Yes
status	string	Status	Yes
task_status	string	Task status	Yes

- map_info Parameter:

Table 4.9.210 map_info Parameter

Parameter	Type	Description	Value Returned
external	dict	View external Parameter for pre-NAT information.	Yes
internet	dict	View internet Parameter for post-NAT information.	Yes

- external Parameter:

Table 4.9.211 external Parameter

Parameter	Type	Description	Value Returned
ips	list(string)	Pre-NAT IP address list	Yes
port_range	list(string)	Original port	Yes

- internet Parameter:

Table 4.9.212 internet Parameter

Parameter	Type	Description	Value Returned
ip	string	Mapped IP address	Yes
port_range	list(string)	Mapped port	Yes
port_id	string	Resource ID	Yes

> **Note**
>
> For common response parameters, view Response Format for details

- Response Example:

```
{
    "status": "active",
    "enable": 1,
    "protocol": "udp",
    "description": "",
    "src_ip_range": [
        "192.168.0.113-192.168.1.116"
    ],
    "created_at": "2022-05-17T01:56:51.000000",
    "updated_at": "2022-05-19T07:56:21.000000",
    "map_info": {
        "external": {
            "ips": [
                "10.168.0.11"
            ],
            "port_range": [
                "999"
            ]
        },
        "internet": {
            "ip": "192.168.9.63",
            "port_id": "",
            "port_range": [
                "999"
            ]
        }
    },
    "priority": null,
    "project_id": "f93eebef20c245b89550410245d2ab8c",
    "vpc_id": "6a8bc428-6c4d-4fd0-8cb1-1ca97e8ffe0f",
    "task_status": "",
    "id": "577e668f-1454-4ba9-ab35-7ce6ec6e9c39",
    "az_id": "a4cdd017-e3da-46f4-b134-4b4fbf7c2380"
}
```

4.9.11.6.5. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.9.12. ACL Management for VPC

4.9.12.1. Query the List of ACL Services in VPC

4.9.12.1.1. Scenario

- Scenario: Query the list of ACL services in VPC
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Read permission
- Versions supported: SCP 6.8.0 custom version

4.9.12.1.2. URI

Table 4.9.213 API Path and Request Method

REST VERB	PATH	Description
GET	/janus/20180725/vpcs/{vpc_id}/acl-services	Query the List of ACL Services in VPC

- Parameter Description:

Table 4.9.214 Path Parameters

Parameter	Location	Type	Description	Required
vpc_id	path	string	VPC ID	Yes
name	path	string	Service name	No
group_name	path	string	Group name	No
type	path	string	Service type	No

Note

For common path parameters, view Common Path Parameter Format for details

4.9.12.1.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/vpcs/fd220b5c-f419-44b6-8943-0dcea8d99a07/acl-services

4.9.12.1.4. Response

- Response Parameter

Table 4.9.215 Response Parameter

Parameter	Location	Type	Description	Value Returned
id	body	string	Service ID	No
name	body	string	Service name	No
group_name	body	string	Group name	No
group_id	body	string	Group ID	No
type	body	string	Service type (The default value for built-in services is system.)	No
protocol_info	body	list	View Parameter Description - protocol_info for protocol details.	No

- Parameter Description - protocol_info:

Table 4.9.216 Parameter Description - protocol_info

Parameter	Type	Description	Value Returned
protocol	string	Protocol type	Yes
icmp_code	int	Code of the ICMP protocol	Yes
icmp_type	int	Type of the ICMP protocol	Yes
port_range	string	Port range	Yes

> **Note**
>
> View Common Response Parameter Format for public response parameters in the API list.

- Response example:

```
[
    {
        "name": "ssh",
        "group_name": "",
        "group_id": "",
        "type": "safe",
        "id": "8dbaf627-3a77-48e9-bb29-a964f5ccd2be",
        "protocol_info": [
            {
                "icmp_code": 0,
                "port_range": "22",
                "icmp_type": 0,
                "protocol": 6
            }
        ]
    },
    {
        "name": "smb",
        "group_name": "",
        "group_id": "",
        "type": "safe",
        "id": "2c75d860-ec2a-4109-b19f-8a840529ed33",
        "protocol_info": [
            {
                "icmp_code": 0,
                "port_range": "445",
                "icmp_type": 0,
                "protocol": 6
            }
        ]
    },
]
```

4.9.12.1.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.9.12.2. Query VPC ACL

4.9.12.2.1. Scenario

- Scenario: Query VPC ACL
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Read permission
- Versions supported: SCP 6.8.0 custom version

4.9.12.2.2. URI

Table 4.9.217 API Path and Request Method

REST VERB	PATH	Description
GET	/janus/20180725/vpcs/{vpc_id}/acls	Query VPC ACL

- Parameter Description:

Table 4.9.218 Path Parameters

Parameter	Location	Type	Description	Required
vpc_id	path	string	VPC ID	Yes
description	path	string	ACL description	No
action	path	string	View VPC ACL Action for ACL action details.	No
enable	path	int(0, 1)	Whether the tenant is enabled	No
index	path	int	ACL index	No

Note

For common path parameters, view Common Path Parameter Format for details

4.9.12.2.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/vpcs/c95cd26a-efaa-4fa8-aeb2-6b140689c16e/acls

4.9.12.2.4. Response

- Response Parameter

Table 4.9.219 Response Parameter

Parameter	Location	Type	Description	Value Returned
id	body	string	ACL ID	No
description	body	string	Description	No
action	body	string	View VPC ACL Action for ACL action details.	No
service_ids	body	list(string)	List of service IDs	No
src	body	dict	View Parameter Description - src for source information details.	No
dst	body	dict	View Parameter Description - dst for destination information details.	No
index	body	int	ACL index	No
default	body	int(0,1)	Whether the ACL is predefined	No
priority	body	int	ACL rule priority	No
enable	body	int(0,1)	Whether the tenant is enabled	No
acl_type	body	string	View ACL Type for VPC for ACL type details.	No

- Parameter Description - src:

Table 4.9.220 Parameter Description - src¶

Parameter	Type	Description	Value Returned
type	string	View Src or Dst Types of VPC for the source type details.	Yes
ip_range	list(string)	IP range	Yes
vif_ids	list(string)	List of VPC gateway IDs	Yes
network_name	string	Network name (returned for the specified network)	No

- Parameter Description - dst:

Table 4.9.221 Parameter Description - dst¶

Parameter	Type	Description	Value Returned
type	string	View Src or Dst Types of VPC for the destination type details.	Yes
ip_range	list(string)	IP range	Yes
vif_ids	list(string)	List of VPC gateway IDs	Yes
network_name	string	Network name (returned for the specified network)	No

Note

1.接口列表公共响应参数，参见：Common Response Parameter Format

- Response example:

```
[
    {
        "src": {
            "network_name": "sunhao11",
            "type": "network",
            "ip_range": [
                "172.16.0.0/24"
            ],
            "vif_ids": []
        },
        "enable": 1,
        "description": "Test priority",
        "index": 4,
        "default": 0,
        "dst": {
            "network_name": "Subnet 1 ",
            "type": "network",
            "ip_range": [
                "192.168.0.0/24"
            ],
            "vif_ids": []
        },
        "priority": 3,
        "action": "drop",
        "acl_type": "individual",
        "id": "ad2d9cc2-cca7-46a7-88ec-67fd58444157",
        "service_ids": [
            "01f152f9-73fe-4106-8bb2-78c093d0cb8d",
            "2c75d860-ec2a-4109-b19f-8a840529ed33",
            "10000000-0000-0000-0000-b5bb7c29a3ab",
            "10000000-0000-0000-0000-60550aac10a2",
            "10000000-0000-0000-0000-38d48cbcff9c",
            "10000000-0000-0000-0000-c53ac35b3008",
            "10000000-0000-0000-0000-5115cb56b4ec"
        ]
    },
    {
        "src": {
            "network_name": "Subnet 1 ",
            "type": "network",
            "ip_range": [
                "192.168.0.0/24"
            ],
            "vif_ids": []
        },
        "enable": 1,
        "description": "Test",
        "index": 5,
        "default": 0,
        "dst": {
            "type": "vif",
            "ip_range": [],
            "vif_ids": [
                "7c1dcebd-91a4-4e84-9e49-3d6d4e08a757"
            ]
        },
        "priority": 4,
        "action": "permit",
        "acl_type": "individual",
        "id": "f37d5ba4-e324-4ad5-9365-3f255a430784",
        "service_ids": [
            "8dbaf627-3a77-48e9-bb29-a964f5ccd2be",
            "2c75d860-ec2a-4109-b19f-8a840529ed33",
            "01f152f9-73fe-4106-8bb2-78c093d0cb8d",
            "21946329-b07e-419f-8cdb-51432ff34227"
        ]
    },
]
```


4.9.12.2.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.9.12.3. Query the ACL Details in VPC

4.9.12.3.1. Scenario

- Scenario: Query the ACL details in VPC
- Roles Supported: Super admin, system admin, tenant, and co-administrator with the Network Read permission
- Versions supported: SCP 6.8.0 custom version

4.9.12.3.2. URI

Table 4.9.222 API Path and Request Method

REST VERB	PATH	Description
GET	/janus/20180725/vpcs/{vpc_id}/acls/{acl_id}	Query the ACL Details in VPC

- Parameter Description:

Table 4.9.223 Path Parameters

Parameter	Location	Type	Description	Required
vpc_id	path	string	VPC ID	Yes
acl_id	path	string	ACL ID	Yes

4.9.12.3.3. Request

- Request Parameter

  None

- Request Example

  https://{SCP_IP}/janus/20180725/vpcs/c95cd26a-efaa-4fa8-aeb2-6b140689c16e/acls/cc6aaeda-0cf2-4acf-9237-dd1e0192f3fe

4.9.12.3.4. Response

- Response Parameter

Table 4.9.224 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	ACL ID	Yes
description	string	Description	Yes
action	string	View VPC ACL Action for ACL action details.	Yes
service_ids	list(string)	List of service IDs	Yes
src	dict	View Parameter Description - src for the source information details.	Yes
dst	dict	View Parameter Description - dst for the destination information details.	Yes
index	int	ACL index	Yes
default	int(0,1)	Whether the ACL is predefined	Yes
priority	int	ACL rule priority	Yes
enable	int(0,1)	Whether the tenant is enabled	Yes
acl_type	string	View ACL Type for VPC for ACL type details.	Yes

- Parameter Description - src:

Table 4.9.225 Parameter Description - src¶

Parameter	Type	Description	Value Returned
type	string	View Src or Dst Types of VPC for the source type details.	Yes
ip_range	list(string)	IP range	Yes
vif_ids	list(string)	List of VPC gateway IDs	Yes
network_name	string	Network name (returned for the specified network)	No

- Parameter Description - dst:

Table 4.9.226 Parameter Description - dst¶

Parameter	Type	Description	Value Returned
type	string	View Src or Dst Types of VPC for the destination type details.	Yes
ip_range	list(string)	IP range	Yes
vif_ids	list(string)	List of VPC gateway IDs	Yes
network_name	string	Network name (returned for the specified network)	No

Note

For common response parameters, view Response Format for details

- Response example:

```
{
    "src": {
        "network_name": "sunhao11",
        "type": "network",
        "ip_range": [
            "172.16.0.0/24"
        ],
        "vif_ids": []
    },
    "enable": 1,
    "description": "Test priority",
    "index": 4,
    "default": 0,
    "dst": {
        "network_name": "Subnet 1 ",
        "type": "network",
        "ip_range": [
            "192.168.0.0/24"
        ],
        "vif_ids": []
    },
    "priority": 3,
    "action": "drop",
    "acl_type": "individual",
    "id": "ad2d9cc2-cca7-46a7-88ec-67fd58444157",
    "service_ids": [
        "01f152f9-73fe-4106-8bb2-78c093d0cb8d",
        "2c75d860-ec2a-4109-b19f-8a840529ed33",
        "10000000-0000-0000-0000-b5bb7c29a3ab",
        "10000000-0000-0000-0000-60550aac10a2",
        "10000000-0000-0000-0000-38d48cbcff9c",
        "10000000-0000-0000-0000-c53ac35b3008",
        "10000000-0000-0000-0000-5115cb56b4ec"
    ]
}
```


4.9.12.3.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.9.12.4. Create ACL Rule for VPC

4.9.12.4.1. Scenario

- Create ACL Rule for VPC
- Roles supported: Super administrator, system administrator, tenant, and co-administrator with the Network Configuration permission
- Versions supported: SCP 6.8.0 custom version

4.9.12.4.2. URI

Table 4.9.227 API Path and Request Method¶

REST VERB	PATH	Description
POST	/janus/20180725/vpcs/{vpc_id}/acls	Create ACL Rule for VPC

- Parameter Description:

Table 4.9.228 Path Parameters¶

Parameter	Location	Type	Description	Required
vpc_id	path	string	VPC ID	Yes

4.9.12.4.3. Request

- Request Parameter

Table 4.9.229 Response Parameter

Parameter	Location	Type	Description	Required
action	body	string	View VPC ACL Action for ACL action details.	Yes
description	body	string	Description	Yes
service_ids	body	list(string)	List of service IDs	Yes
src	body	dict	View Parameter Description - src for source information details.	Yes
dst	body	dict	View Parameter Description - dst for destination information details.	Yes
acl_type	body	string	ACL type (The value can only be individual currently.)	Yes

- Parameter Description - src:

Table 4.9.230 Parameter Description - src

Parameter	Type	Description	Required
type	string	View Src or Dst Types of VPC for the source type details.	Yes
ip_range	list(string)	IP range	No

- Parameter Description - dst:

Table 4.9.231 Parameter Description - dst

Parameter	Type	Description	Required
type	string	View Src or Dst Types of VPC for the destination type details.	Yes
ip_range	list(string)	IP range	No

Note

Parameter description for src

If the type parameter is set to all or vif, there is no need to pass in a value for the ip_range parameter. If the type parameter is set to network, the value of ip_range must be the subnet CIDR of the VPC, such as 192.168.1.0/24. If the type parameter is set to ip, the value of the ip_range parameter must be a single IP address (such as 192.168.6.6) or an IP range (such as 192.168.6.7-192.168.6.66).

Parameter description for action

For HCI 6.7.0 and above, the value of the action parameter can be reject.

- Request Example

https://{SCP_IP}/janus/20180725/vpcs/365d8848-1243-4c3e-8c17-bad9d65c3b27/acls

```
{
    "action": "permit",
    "description": "test",
    "service_ids": [
        "10000000-0000-0000-0000-34929d1ccae9"
    ],
    "acl_type": "individual",
    "src": {
        "type": "ip",
        "ip_range": [
            "192.168.0.69."
        ]
    },
    "dst": {
        "type": "ip",
        "ip_range": [
            "192.168.0.70-192.168.0.88"
        ]
    }
}
```

4.9.12.4.4. Response

- Response Parameter

Table 4.9.232 Response Parameter

Parameter	Location	Type	Description	Value Returned
id	body	string	ACL ID	Yes
project_id	body	string	Tenant ID	Yes
enable	body	int(0,1)	Whether the tenant is enabled	No
description	body	string	Description	No
action	body	string	ACL action	No
service_ids	body	list(string)	List of service IDs	No
src	body	dict	View Parameter Description - src for the source information details.	Yes
dst	body	dict	View Parameter Description - dst for the destination information details.	Yes

- Parameter Description - src:

Table 4.9.233 Parameter Description - src

Parameter	Type	Description	Value Returned
type	string	View Src or Dst Types of VPC for the source type details.	Yes
ip_range	list(string)	IP range	Yes
vif_ids	list(string)	List of network egress IDs	Yes

- Parameter Description - dst:

Table 4.9.234 Parameter Description - dst¶

Parameter	Type	Description	Value Returned
type	string	View Src or Dst Types of VPC for the destination type details.	Yes
ip_range	list(string)	IP range	Yes
vif_ids	list(string)	List of network egress IDs	Yes

Note

For common response parameters, view Response Format for details

- Response example:

```
{
    "id": "ef3c048c-a7a6-4f0a-9deb-ce968eb5db0e",
    "project_id": "ef3c048c-a7a6-4f0a-9deb-ce968eb5db0e",
    "action": "permit",
    "enable": 1,
    "description": "test",
    "service_ids": [
        "10000000-0000-0000-0000-34929d1ccae9"
    ],
    "src": {
        "type": "ip",
        "ip_range": [
            "192.168.0.69."
        ]
    },
    "dst": {
        "type": "ip",
        "ip_range": [
            "192.168.0.70-192.168.0.88"
        ]
    }
}
```

4.9.12.4.5. Response Code

- Success: 201
- Errors: 400, 403, 500

4.9.12.5. Edit ACL Rule for VPC

4.9.12.5.1. Scenario

- Edit ACL Rule for VPC
- Roles supported: Super administrator, system administrator, and co-administrator with the Network Configuration permission
- Versions supported: SCP 6.8.0 custom version

4.9.12.5.2. URI

Table 4.9.235 API Path and Request Method¶

REST VERB	PATH	Description
PUT	/janus/20180725/vpcs/{vpc_id}/acls/{acl_id}	Edit ACL Rule for VPC

- Parameter Description:

Table 4.9.236 Path Parameters¶

Parameter	Location	Type	Description	Required
vpc_id	path	string	VPC ID	Yes
acl_id	path	string	ACL ID	Yes

4.9.12.5.3. Request

- Request Parameter

Table 4.9.237 Response Parameter¶

Parameter	Location	Type	Description	Required
action	body	string	View VPC ACL Action for ACL action details.	No
enable	body	int(0,1)	Whether the tenant is enabled	No
description	body	string	Description	No
service_ids	body	list(string)	List of service IDs	No
src	body	dict	View Parameter Description - src for the source information details.	No
dst	body	dict	View Parameter Description - dst for the destination information details.	No

- Parameter Description - src:

Table 4.9.238 Parameter Description - src¶

Parameter	Type	Description	Required
type	string	View Src or Dst Types of VPC for the source type details.	Yes
ip_range	list(string)	IP range	No

- Parameter Description - dst:

Table 4.9.239 Parameter Description - dst¶

Parameter	Type	Description	Required
type	string	View Src or Dst Types of VPC for the destination type details.	Yes
ip_range	list(string)	IP range	No

Note

Parameter description for src

If the type parameter is set to all or vif, there is no need to pass in a value for the ip_range parameter. If the type parameter is set to network, the value of ip_range must be the subnet CIDR of the VPC, such as 192.168.1.0/24. If the type parameter is set to ip, the value of the ip_range parameter must be a single IP address (such as 192.168.6.6) or an IP range (such as 192.168.6.7-192.168.6.66).

Parameter description for action

For HCI 6.7.0 and above, the value of the action parameter can be reject.

- Request Example

https://{SCP_IP}/janus/20180725/vpcs/365d8848-1243-4c3e-8c17-bad9d65c3b27/acls/ef3c048c-a7a6-4f0a-9deb-ce968eb5db0e

```
{
    "src": {
        "type": "network",
        "ip_range": [
            "192.168.0.6/24"
        ]
    },
    "dst": {
        "type": "vif"
    }
}
```

4.9.12.5.4. Response

- Response Parameter

Table 4.9.240 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Task ID	Yes

Note

For common response parameters, view Response Format for details

- Response example:

```
{
    "task_id": "6bf0a363-061d-4ebd-801f-2a21f218b2d2"
}
```

4.9.12.5.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.9.12.6. Delete ACL Rule for VPC

4.9.12.6.1. Scenario

- Delete ACL Rule for VPC

- Roles supported: Super administrator, system administrator, and co-administrator with the Network Configuration permission
- Versions supported: SCP 6.8.0 custom version

4.9.12.6.2. URI

Table 4.9.241 API Path and Request Method

REST VERB	PATH	Description
DELETE	/janus/20180725/vpcs/{vpc_id}/acls/{acl_id}	Delete ACL Rule for VPC

- Parameter Description:

Table 4.9.242 Path Parameters

Parameter	Location	Type	Description	Required
vpc_id	path	string	VPC ID	Yes
acl_id	path	string	ACL ID	Yes

4.9.12.6.3. Request

None

- Request Example

  https://{SCP_IP}/janus/20180725/vpcs/365d8848-1243-4c3e-8c17-bad9d65c3b27/acls/357fd33b-631f-4496-99b5-8883add63987

4.9.12.6.4. Response

- Response Parameter

Table 4.9.243 Response Parameter

Parameter	Type	Description	Value Returned
task_id	string	Task ID	Yes

Note

For common response parameters, view Response Format for details

- Response example:

```
{
    "task_id": "f0cd53e4-0b33-469e-93df-8d3e6fd18b71"
}
```

4.9.12.6.5. Response Code

- Success: 201
- Errors: 400, 403, 500

4.10. Datastores

4.10.1. Description of Common Datastore Management Parameters

4.10.1.1. Datastore Type

Table 4.10.1 Datastore Type¶

Value	Description
sharedisk	Shared disk

4.10.1.2. Datastore Status

Table 4.10.2 Datastore Status¶

Value	Description
creating	Creating
available	Available
deleting	Deleting
deleted	Moved to Recycle Bin
error	Abnormal
error_deleting	Error occurred while deleting datastore

4.10.2. Query Datastore List

4.10.2.1. Scenario

- Scenario: Query the datastore list
- Roles Supported: Super admin and system admin
- Versions supported: SCP6.3.70 and above

4.10.2.2. URI

Table 4.10.3 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/volumes	Query Datastore List

4.10.2.2.1. Description

Table 4.10.4 Path Parameters¶

Parameter	Location	Type	Description	Required
name	path	string	Datastore name	No
volume_type	path	string	Datastore type, view: Datastore Type for details.	No
status	path	string	Datastore status, view: Datastore Status for details.	No
user_id	path	string	Datastore creator ID	No
project_id	path	string	Tenant ID	No
az_id	path	string	Resource pool ID	No
storage_id	path	string	Storage ID	No
target_id	path	int	Target ID	No

Note

- When volume_type is sharedisk, az_id, storage_id, target_id can be used to identify the unique shared disk.
- For common path parameters, view Common Path Parameter Format for details

4.10.2.3. Request

4.10.2.3.1. Response Parameter

None

4.10.2.3.2. Request Example

- Query the list of tenant datastores: *https://{SCP_Endpoint}/janus/20180725/volumes?project_id=b9f644e5d4ed42358d287b19ea944e71*
- Query the list of available shared datastores: *https://{SCP_Endpoint}/janus/20180725/volumes?status=available&volume_type=sharedisk*

4.10.2.4. Response

4.10.2.4.1. Response Parameter

Table 4.10.5 Response Parameter¶

Parameter	Type	Description	Value Returned
id	string	Datastore ID	No
name	string	Datastore name	No
description	string	Description	No
size	int	Datastore size (GB)	No
volume_type	string	Datastore type, view: Datastore Type for details.	No
status	string	Datastore status, view: Datastore Status for details.	No
user_id	string	Datastore creator ID	No
project_id	string	Tenant ID	No
az_id	string	Resource pool ID	No
storage_id	string	Storage ID	No
target_id	string	Target ID	No
disk_id	string	Disk ID	No
created_at	string	Time created	No
updated_at	string	Time updated	No

Note

1. If you call Edit VM interface to mount a shared disk to the VM, the storage_id in disks can be passed in to the datastore ID or disk ID;
2. For common response parameters, view Response Format for details

4.10.2.4.2. Response Example

```
[
    {
        "status": "available",
        "user_id": "ac31898e787447c483b48d3e7832f697",
        "description": "Create shared disk-01 for testing",
        "created_at": "2021-04-25T04:17:30.000000",
        "target_id": "206",
        "updated_at": "2021-04-25T04:17:58.000000",
        "disk_id": "361f40f51b040ca0847f024f38cbebdae",
        "name": "Create_shared_disk-01_for_testing",
        "volume_type": "sharedisk",
        "storage_id": "71295054_vs_vol_rep2",
        "project_id": "b9f644e5d4ed42358d287b19ea944e71",
        "id": "9fda7ed9-0a2a-4fa3-8941-c6580e4a6f3e",
        "az_id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
        "size": 1
    },
    {
        "status": "available",
        "user_id": "ac31898e787447c483b48d3e7832f697",
        "description": "Create shared disk-02 for testing",
        "created_at": "2021-04-23T09:58:01.000000",
        "target_id": "205",
        "updated_at": "2021-04-23T09:58:38.000000",
        "disk_id": "360e903d6a04cad08f1101f12fd8228c4",
        "name": "Create_shared_disk-02_for_testing",
        "volume_type": "sharedisk",
        "storage_id": "71295054_vs_vol_rep2",
        "project_id": "b9f644e5d4ed42358d287b19ea944e71",
        "id": "c3ae3ac1-4da5-40a8-9cd3-1d28fdd6dd71",
        "az_id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
        "size": 1
    }
]
```


4.10.2.5. Response Code

- OK: 200
- Errors: 400, 403, 500

4.10.3. Query Datastore Details

4.10.3.1. Scenario

- Scenario: Query datastore details
- Roles Supported: Super admin and system admin
- Versions supported: SCP6.3.70 and above

4.10.3.2. URI

Table 4.10.6 API Path and Request Method¶

REST VERB	PATH	Description
GET	/janus/20180725/volumes/{volume_id}	Query Datastore Details

4.10.3.2.1. Description

Table 4.10.7 Path Parameters¶

Parameter	Location	Type	Description	Required
volume_id	path	string	Datastore ID	Yes

4.10.3.3. Request

4.10.3.3.1. Response Parameter

None

4.10.3.3.2. Request Example

- Query datastore details: *https://{SCP_Endpoint}/janus/20180725/volumes/c3ae3ac1-4da5-40a8-9cd3-1d28fdd6dd71*

4.10.3.4. Response

4.10.3.4.1. Response Parameter

Table 4.10.8 Response Parameter

Parameter	Type	Description	Value Returned
id	string	Datastore ID	No
name	string	Datastore name	No
description	string	Description	No
size	int	Datastore size (GB)	No
volume_type	string	Datastore type, view: Datastore Type for details.	No
status	string	Datastore status, view: Datastore Status for details.	No
user_id	string	Datastore creator ID	No
project_id	string	Tenant ID	No
az_id	string	Resource pool ID	No
storage_id	string	Storage ID	No
target_id	string	Target ID	No
disk_id	string	Disk ID	No
created_at	string	Time created	No
updated_at	string	Time updated	No

Note

1. If you call Edit VM interface to mount a shared disk to the VM, the storage_id in disks can be passed in to the datastore ID or disk ID;
2. For common response parameters, view Response Format for details

4.10.3.4.2. Response Example

```
{
    "status": "available",
    "user_id": "ac31898e787447c483b48d3e7832f697",
    "description": "create_share_disk",
    "created_at": "2021-04-23T09:58:01.000000",
    "target_id": "205",
    "updated_at": "2021-04-23T09:58:38.000000",
    "disk_id": "360e903d6a04cad08f1101f12fd8228c4",
    "name": "create_share_disk",
    "volume_type": "sharedisk",
    "storage_id": "71295054_vs_vol_rep2",
    "project_id": "b9f644e5d4ed42358d287b19ea944e71",
    "id": "c3ae3ac1-4da5-40a8-9cd3-1d28fdd6dd71",
    "az_id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b",
    "size": 1
}
```


4.10.3.5. Response Code

- OK: 200
- Errors: 400, 403, 404, 500

4.10.4. Create Datastore

4.10.4.1. Scenario

- Scenario: Create a datastore, only shared datastores can be created.
- Roles Supported: Super admin and system admin
- Versions supported: SCP6.3.70 and above

4.10.4.2. URI

Table 4.10.9 API Path and Request Method¶

REST VERB	PATH	Description
POST	/janus/20180725/volumes	Create Datastore

4.10.4.2.1. Description

None

4.10.4.3. Request

4.10.4.3.1. Response Parameter

Table 4.10.10 Response Parameter¶

Parameter	Location	Type	Description	Required
name	body	string	Datastore name	Yes
description	body	string	Description	No
az_id	body	string	Resource pool ID	Yes
size	body	int(1, 64512)	Datastore size (GB)	Yes
volume_type	body	string	Datastore type, view: Datastore Type for details.	Yes
project_id	body	string	Tenant ID	No
storage_id	body	string	Storage ID	No

Note

1. **When the datastore type is shared disk:**
   - If you do not specify a storage ID, the system will automatically schedule the VS storage ID to create a shared disk.
   - Only ID of VS storage can be specified to create datastores. The Query Storage List interface can be called to query the corresponding VS storage.
2. The selected resource pool must have a storage tag for VS storage, otherwise the creation will fail.

4.10.4.3.2. Request Example

- Create volume: *https://{SCP_Endpoint}/janus/20180725/volumes*

```
{
    "name": "create_share_disk",
    "description": "create_share_disk",
    "size": 1,
    "volume_type": "sharedisk",
    "az_id": "1c13ef3e-a049-4fc8-bc01-a2b5b279d46b"
}
```

4.10.4.4. Response

4.10.4.4.1. Response Parameter

Table 4.10.11 Response Parameter¶

Parameter	Type	Description	Value Returned
task_id	string	Task IDV	Yes

Note

1. After sending the request for creating datastores, you can call the Query Task Information interface to get task progress. The datastore ID is object_id of the corresponding response parameter.
2. For common response parameters, view Response Format for details

4.10.4.4.2. Response Example

```
{
    "task_id": "54b815be-d86d-41dc-902b-333c25a4b263"
}
```

4.10.4.5. Response Code

- Success: 201
- Errors: 400, 403, 404, 500

4.10.5. Delete Datastore

4.10.5.1. Scenario

- Scenario: Delete datastore
- Roles Supported: Super admin and system admin
- Versions supported: SCP6.3.70 and above

4.10.5.2. URI

Table 4.10.12 Request Path and Method

REST VERB	PATH	Description
DELETE	/janus/20180725/volumes/{volume_id}	Delete Datastore

4.10.5.2.1. Description

Table 4.10.13 Path Parameters

Parameter	Location	Type	Description	Required
volume_id	path	string	Datastore ID	Yes

4.10.5.3. Request

4.10.5.3.1. Response Parameter

Table 4.10.14 Response Parameter

Parameter	Location	Type	Description	Required
force	body	int(0, 1)	Whether to force deletion	Yes

> **Note**
>
> When force is 0, it indicates soft deletion. In this case, quota for datastores with specific storage tags will not be deallocated. Only when the datastore is completely deleted (force=1) will the corresponding quota be deallocated.

4.10.5.3.2. Request Example

```
{
    "force": 1
}
```

4.10.5.4. Response

4.10.5.4.1. Response Parameter

Table 4.10.15 Response Parameter

Parameter	Location	Type	Description	Required
task_id	path	string	Tenant ID	Yes

> **Note**
>
> For common response parameters, view Response Format for details

4.10.5.4.2. Response Example

```
{
    "task_id": "5957a45c-53a7-438c-9bce-4e7bf248c56c"
}
```

4.10.5.5. Response Code

- Normal: 204
- Errors: 400, 403, 404, 500

5. Appendix

5.1. Glossary

SCP
Sangfor Cloud Platform

VS
Virtual storage. To use virtual storage, the HCI cluster in the resource pool must contain at least two hosts.

VPC
Virtual Private Cloud

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